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Introduction

State-of-the-art: engineering in the era of big data

There is no real-world engineering without tolerances and managing uncertainty. Yet, the mathematical methods (such as differential geometry) used in engineering are mostly based in deterministic logic and variables, due to computability[1]. Whenever we deal with a non-separable space, there are issues with computability because some of the elements of a non-separable space cannot be approximated by a finite set of elements. And no one has found yet a systematic way to define a separable probability space with an arbitrary probability measure over real functions of real variables (and infinite-dimensional spaces in general). While there are separable spaces of real functions of real variables, whenever we add uncertainties/probabilities to such spaces we tend to create non-separable spaces[2]. As a consequence, there is a wide variety of ad hoc methods (which include Bayesian inference with priors which make ad hoc assumptions) to manage uncertainty in real-world engineering.

In the era of big data, the ad hoc methods which were inefficient before, cannot compete with specialized expertise developed by trial-and-error in a case-by-case basis (essentially capital intensive brute-force black boxes). This is unsustainable, since not only the crucial decisions are made exclusively by machines, but also these decisions cannot be reviewed and audited by humans (not even by the few rich human owners of these black boxes) because the mathematical language needed to communicate the reasoning behind the decisions made by the machines does not exist[3][4]. Paradoxically, in the era of big data, humans may have much less power and know much less about their world, "the devil is in the details" may be replaced by "the devil is in the machine".

The potential applications of mathematical methods for engineering which are based in probabilities and random variables (instead of deterministic logic and variables) cannot be overstated. For instance, neural networks are particularly good as a computer approximation of a vector in a separable Hilbert space[5]. Thus, neural networks and deep learning are not necessarily an alternative to Bayesian Inference and can instead be used as part of Bayesian Inference[6], specially when the posterior and prior probabilities can be parametrized by vectors in a Hilbert space.

A consequence of the lack of mathematical methods for engineering which are based in probabilities and random variables, is that Quantum Yang-Mills theory, Classical Statistical Field Theory (for dissipative systems, e.g. Navier-Sokes, and for Hamiltonians which are non-polynomial in the fields, e.g. General relativistic statistical mechanics) and Quantum Gravity all suffer from severe mathematical inconsistencies and produce unreliable predictions at best.

Why a simple solution exists

While the Lebesgue measure cannot be defined in an Euclidean-like infinite-dimensional space[7][8], it is well known since many decades that a uniform (Lebesgue-like) measure of an infinite-dimensional sphere can be defined using the Gaussian measure and the Fock-space (the Fock-space is a separable Hilbert space used in the second quantization of free quantum fields)[9]. Such a space can parametrize (we call it the free field parametrization) the probability distribution of another probability distribution with sample space given by the direct product of the base space and the field (say $\mathbb{R}^{n+1}$). The cost of the free field parametrization is that exact constraints (in the finite-dimensional space corresponding to each point of the base space) are needed to define a differentiable structure.

Thus, infinite-dimensions are not by themselves problematic. The problem is how to implement exact constraints in a separable probability space without attributing to the constrained space null probability measure. This is a much deeper problem that also affects statistical mechanics[10] and machine learning[11] (in spaces of finite dimensions).

A simple solution to this problem exists because the existence of a probability measure allows us to define a quantum Hamiltonian and quantum constraints, which are more general than the classical ones because they do not need to commute with the variables defining the sample space. As we will see, quantum constraints often allow us to implement exact constraints in a separable probability space without attributing to the constrained space null probability measure.

As a simple example of a motion which cannot be derived from a Lagrangian (and thus also not from a classical Hamiltonian without enlarging the system) we can cite[12] the oscillations of two classical coupled oscillators with different frequencies and different damping constants:

$$\begin{cases} \ddot{x}_1 + \lambda_1 \dot{x}_1 + \omega_1^2 x_1 - c_2 x_2 = 0 \\ \ddot{x}_2 + \lambda_2 \dot{x}_2 + \omega_2^2 x_2 - c_1 x_1 = 0 \end{cases}, \tag{1}$$

where c_1 and c_2 are constants. Nevertheless, if we consider a probability measure for this classical system then we can define a quantum Hamiltonian for it. Since this system is dissipative, it cannot have a conserved Energy but the pendulums do not disappear and thus the probability is conserved. Another example is when a classical Hamiltonian can be defined but it is much less useful than a

quantum Hamiltonian, due to non-linear dynamics[13].

The quantum formalism is the most general formalism whenever there is a conserved probability, thus it is expected that it can be applied even when classical Hamiltonians (due to energy dissipation[12], or non-linear dynamics[13]) or classical exact constraints (due to the fact that the constrained space has null probability measure) cannot.

Why Timepiece

Timepiece is about the mathematical inconsistencies of statistical field theory and its (unavoidable) consequences for the foundations of Quantum Mechanics: either the probability space is separable and then there are implications for foundations because quantum methods must be used in classical cases, or the probability space is not separable and then there are also implications for foundations because a non-separable probability space has compatibility issues with computability[1].

Moreover, if every quantum system is made of fundamental Particles, then Quantum Mechanics must at least be consistent with Quantum Field Theory. Moreover, in General Relativity the diffeomorphism transformations are time-dependent transformations and many of the predictions in Gravity are produced with the Hamiltonian formalism; however the symplectic form of conservative classical mechanics is not invariant under time-dependent transformations: the most obvious way to make non-relativistic mechanics consistent with General Relativity is to formulate it as a particular field theory whose phase-space is a fibred manifold over a "time" axis [14].

The name *Timepiece* refers to the above mentioned extra "time" axis (defined by the phase-space), appearing in a (quantum or classical) Hamiltonian formalism with time-dependent transformations, which is complementary to the standard time (defined by the time-evolution operator) of any Hamiltonian formalism.

Note that in any dynamical system there is (see Wikipedia):

1. a state which is a point in a state space (no time involved, the state is the present state). It may be simply a set, without the need of a smooth space-time structure defined on it. Usually we know what the state space is.

2. an evolution rule, which says how a new (future) state is generated from the present state. This rule may be non-deterministic and the notion of "time'" is determined by the evolution rule itself, it is not a pre-established concept. "Time'" can be measured by integers, by real or complex numbers or it can be a more general algebraic object, losing the memory of its physical origin.

Usually it is in the evolution rule that the discoveries happen. In the classical Lagrangian formalism,

time is a pre-established concept which may be in conflict with the evolution rule. That is why the definition of a dynamical system is very old and more general.

Why community science

First of all, the above mentioned problem (the lack of mathematical methods for engineering which are based in probabilities and random variables) is a hell of a problem. As discussed above, we believe that there is a simple solution to it, but nevertheless such simple solution could be considered the holy grail of High Energy Physics (specially to produce predictions for experiments), with important applications to Data Science in general (because Statistical Field Theory involves probability measures in infinite-dimensional spaces). This fact already justifies that we explore all available means to solve this problem, which includes community science.

But it turns out that community science is particularly well-suited to solve this problem. Because all the scientific articles in the main physics (and mathematics) journals are required to be in one of two categories (or in both):

1. Articles without implications to the interpretations of Quantum Mechanics;
2. Articles which do not study in a systematic way the mathematical inconsistencies of Field Theory.

As it was discussed above, the mathematical inconsistencies of statistical field theory have unavoidable consequences for the foundations of Quantum Mechanics. This implies that a scientific article addressing the subject of the Timepiece community in practice has no place in the main physics journals. Surely, one author can suppress or play with the words and try to fit one article in one of the two categories. But this is not a problem that can be completely solved by one author in just one article, this is a problem that must be addressed by a scientific community persistently for a few years.

These two categories make it impossible for such a scientific community to use the main physics journals.

The most obvious alternative is community science, specially if the solution to this very difficult problem is relatively simple, as we believe. Moreover, a technological ingredient which only appeared in 2011[15] is crucial to community science.

But now we address why these two categories exist in the first place. The ground reason is that the problem (the lack of mathematical methods for engineering which are based in probabilities and random variables) is very difficult and it was persistent since the end of World War II. Another reason is that if to solve the problem we would need quantum methods applied to statistical field

theory in a way that challenges the mainstream interpretations of quantum mechanics (as argued above), these two categories would work as a self-fulfilling prophecy: by excluding simple solutions to the problem, then only complex solutions could be found which then justifies the existence of the categories. Moreover, a crucial technological ingredient only appeared in 2011[15], without it most of the results presented here perhaps would not be possible. So there were five to six decades during which a strong scientific consensus built that in order for the scientific articles to be useful they cannot address head-on this very difficult problem. In some sense, other problems must be solved before such difficult problem can be successfully addressed. This was in some sense true until 2011[15], but in the meantime the technical foundation of far too many of the main scientific programs in theoretical physics (from string theory to quantum information) relies on the assumption that there will be no simple solution to this difficult problem soon.

Think about it: why would anyone study String Theory (which is full of conjectures) when there is a relatively simple mathematical definition of Field Theory for Hamiltonians which are non-polynomial in the fields? Why would anyone follow ad hoc ambiguous unification principles when designing hugely expensive High Energy Physics experiments, when there are relatively simple mathematical definitions for Quantum Yang-Mills theory and Quantum Gravity which are mutually consistent? Why would anyone differentiate between Quantum and Classical Information when this is inconsistent with the mathematical definitions of Quantum and Classical Statistical Field Theory? We cannot ask any human being to help us when his paycheck and of all of his colleagues is on the line, to happily consider the hypothesis that he allowed far too many years and far too many resources to be invested studying a subject that is worth next to nothing while a much simpler solution existed.

Consequences and applications

In Scientific Research as in Bayesian inference, there is always a prior probability distribution, and there is no prior which is better for all cases[16]: we always have to make assumptions. For instance, if we insist we can assume all objects in the night sky revolve around the Earth using epicycles and even consider that this is a very successful theory because it was a precursor of Fourier series and these have widespread applications today[17].

> *The Emperor realized that the people were right but could not admit to that. He though it better to continue the procession under the illusion that anyone who couldn't see his clothes was either stupid or incompetent. And he stood stiffly on his carriage, while behind him a page held his imaginary mantle.*
>
> —-Hans Christian Andersen (1837)

Thus, data, evidence and even mathematical proofs are not enough for us to abandon prior beliefs, there must be a moment when in order for us to achieve something we want we have to abandon our prior beliefs: in the case of assuming all objects in the night sky revolve around the Earth, it would make many relevant calculations in modern astronomy or in the global positioning system impossible.

In the case of Quantum Mechanics, it was always possible to see its mathematical formalism as a mere (but very useful) parametrization of probability and information theories. But, it was also possible to describe it (although in a forceful way) as a *new* and exotic alternative to (or generalization of) classical information theory, one that shook our sense of reality. It is not hard to believe that the mystery surrounding the exotic point of view can often convince the society to spend large amounts of money in research, at least in the short term.

The exotic point of view was sustained by a relatively constant flow of discoveries of new phenomena at higher and higher energies, which kept the mystery alive. But now, even optimists say that if the Large Hadron Collider finds nothing new, it will be harder to convince the governments of the world to build the next bigger, more expensive collider to keep research in High Energy Physics as we know it[18]. Thus, if the exotic point of view is not that mysterious nowadays and there exists a parametrization point of view, is there another way to convince the society to spend large amounts of money in research?

The answer is yes, there is another way. Because the parametrization point of view allows the mathematical formalism of Quantum Mechanics to be applied to probability and classical information theory (and then to artificial intelligence) not just to edge cases, but to the core of these theories. This would not be possible with the exotic point of view, which considered that Quantum Mechanics shook our sense of reality and thus must be an alternative to (or a generalization of) classical information theory and thus there would be no reason to expect applications to the core of classical information theory.

There are many potential applications beyond physics. For instance, since the supervised learning problems in machine learning can be thought of as learning a function from examples, these problems can be formulated as Bayesian modeling[19]. The free field parametrization may allow using any prior as an alternative to using the Gaussian process [19] or the conditional random field [20] as priors. It may also allow Random Fields in its full generality as an alternative to Markov Random Fields in Physics, Biology and Data Science [21], alternative to the semivariogram in spatial Data analysis [22], alternative to wavelets and renormalization in multi-resolution analysis [23] and alternative to Functional Data Analysis and Neural Networks in time series analysis [24]. Complex systems [25], sequential systems in machine learning [26] and Quantum Reinforcement Learning both in computer simulations and as a model for human decision-making [27], all share several features with statistical or quantum physics; (more) applications of quantum methods in these

subjects thus seems straightforward.

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Aim

Schrödinger described quantization as the consequence of solving an eigenvalue problem for the Hamiltonian[28]: in an infinite-dimensional linear space of functions, continuous and discrete (i.e. quantized) energy spectra may coexist. Thus, from the very beginning, there was a relation between the time evolution (defined by the Hamiltonian) and the notion of quantization.

There is no doubt that the best-known description of the experimental data collected so far is based on a quantum theory[29]. However, the notion of quantization is not much clearer than it was in 1926[30]. Up to now, there is no definition or an example of a Quantum Yang-Mills theory in four-dimensional space-time, with mathematical rigor and in the absence of approximations[31]. The status of Quantum Gravity is even worse since no reliable way to make predictions is known[32]. Classical Statistical Field Theory currently depends on the same renormalization formalism of Quantum Yang-Mills theory[33], without mathematical rigor for Hamiltonians which are non-polynomial in the fields (e.g. General relativistic statistical mechanics[34]) and a peculiar notion of "*continuum*"[1]. And Classical Field Theory does not suffice to deal with crucial dissipative systems such as those defined by the Navier-Stokes equation[35].

In this book we will change this status, proposing a simple and mathematically meaningful definition of quantization with applications to Quantum Yang-Mills theory, Quantum Gravity, Classical Statistical Field Theory and Classical Field Theory. We will also show that the mathematical inconsistencies of Field Theory do have consequences for the interpretation of the quantum mechanics of electrons and atoms and the interpretation of quantum mechanics in general. In summary, we argue that in fact there is a technical problem with the definition of Field Theories (with a solution), and there is no interpretation problem with Quantum Mechanics (beyond what is reasonable to expect for any theory of Physics).

Wave-function collapse versus Euler's formula

It is usually argued in the literature that the mathematical inconsistencies only affect Field Theory and not Quantum Mechanics; that there is no technical problem with Quantum Mechanics, only an interpretation problem. This is illustrated by Richard Feynman saying that:

> "*I think I can safely say that nobody understands Quantum Mechanics.*"

[1] See Section /pub/field/#a-translation-invariant-time-evolution-and-classical-statistical-field-theory.

But this argument is misleading because if every quantum system is made of fundamental Particles, then Quantum Mechanics must at least be consistent with Quantum Field Theory. Moreover, in the Hamiltonian formulation of time-dependent classical mechanics, the symplectic form is not invariant under arbitrary time-dependent coordinate and reference frame transformations; the most obvious way out is to formulate non-relativistic time-dependent mechanics as a particular field theory whose configuration space is a fibred manifold over a time axis [36][37].

Often quantum mechanics is interpreted as providing the probabilities of transition between different states of an individual system. This transition happens upon measurement, any measurement. The state of the system is defined by the wave-function which collapses to a different wave-function upon measurement.

This raises a number of interpretation problems as to what do we mean by state of a system. If the state before measurement is A, but after measurement the state is B, what is then the state during the measurement: A and/or B or something else? Strangely, despite that we don't know what happens during the measurement, we know very well the transition probabilities—because once we assume that the state of the system is defined by the wave-function, there are not (many) alternatives to the Born's rule defining the probabilities of transition as a function of the wave-function [38][39].

The ensemble interpretation where Quantum Mechanics is merely a formalism of statistical physics, describing a statistical ensemble of systems [40][41] avoids these interpretation problems: the state of the ensemble is unambiguously a standard measure space[2]—that is, a probability measure for the states of an individual system, as in classical statistical mechanics[3]. Then, there are several possible states of the ensemble, these different states are related by conditional probability measures which in turn are function of a group of symmetry transformations—such as a translation in space-time or a rotation in space. Note that in classical statistical mechanics, the conditional probability measures are deterministic (unlike in Quantum Mechanics), i.e. they transform deterministic ensembles into deterministic ensembles (by deterministic ensemble we mean that all systems in the statistical ensemble are exact copies of the same individual system).

Since we can always define a wave-function by taking the square-root of the probabilities[4], the

[2]Most probability spaces with real-world applications are *standard* measure spaces. See also Section /pub/paramet rization#the-unitary-operator-is-a-parametrization-of-any-conditional-probability-measure-in-a-standard-measure-space

[3]Gillies, D. (2000). *Philosophical Theories of Probability*. Routledge. Schwabl, F., & Brewer, W. D. (2006). *Statistical Mechanics* (2nd ed.). Springer Berlin Heidelberg.

See (Schwabl 2006). Note that the notion of probability distribution is not free from interpretation problems (Gillies 2000), but we believe these are intrinsic to all applications of probabilities.

[4]See Section /pub/parametrization/#the-unitary-operator-is-a-parametrization-of-any-conditional-probability-measure-in-a-standard-measure-space.

Koopman-von Neumann version of classical statistical mechanics [42] defines classical statistical mechanics as a particular case of quantum mechanics where the algebra of operators is necessarily commutative (because the conditional probability measures are deterministic).

Quantum mechanics in the ensemble interpretation, generalizes classical statistical mechanics by allowing conditional probability measures to be non-deterministic. For instance, the probability clock[5] involves a non-deterministic conditional probability measure. In classical statistical mechanics any non-deterministic conditional probability measure is an external foreign element to the theory, this is unnatural for a statistical theory. Thus Quantum Mechanics in the ensemble interpretation is a natural and unavoidable generalization of classical statistical mechanics.

However, the ensemble interpretation does not address the question why the wave-function plays a central role in the calculation of the probability distribution, unlike most other interpretations of quantum mechanics. By being compatible with most (if not all) interpretations of Quantum Mechanics, the ensemble interpretation is in practice a common denominator of most interpretations of Quantum Mechanics. It is useful, but it is not enough. For instance, the ensemble interpretation does not give any explanation as to why it looks like the electron's wave-function interferes with itself in the double-slit experiment[6]—that would imply that the wave-function describes (in some sense) an individual system. Also, the ensemble interpretation does not explain the role of Quantum Statistical Mechanics and the associated density matrix in the measurement process: due to the wave-function's collapse, the off-diagonal part of the density matrix is always set to zero, while the diagonal part of the density matrix containing the probabilities of transition is preserved—in a basis where the operators (corresponding to the measurable properties of the system) are diagonal[7]. Also, if a black-hole erases most information about an object that comes inside of it by turning this information to random, then it is not obvious how the symmetry of translation in time is conserved in the ensemble interpretation[8]. Moreover, the most prominent advocates of the ensemble interpretation were dissatisfied with the complementarity of position and momentum [43][44], convincing themselves and others that the complementarity of position and momentum could not be satisfactorily explained by the ensemble interpretation alone.

[5]Zhao, M.-J., & Jaeger, H. (2010). Norm-observable operator models. *Neural Computation*, *22*(7), 1927–1959. https://doi.org/10.1162/neco.2010.03-09-983

See also Section /pub/euler#eulers-formula-for-the-probability-clock

[6]Ananthaswamy, A. (2018). *Through two doors at once: The elegant experiment that captures the enigma of our quantum reality.* Penguin Publishing Group. Feynman, R. P., Leighton, R. B., & Sands, M. (2015). *The feynman lectures on physics, vol. III: The new millennium edition: Quantum mechanics.* Basic Books. https://www.feynmanlectures.caltech.edu/III_01.html

See also Section /pub/deterministic#the-youngs-double-slit-experiment

[7]See Section /pub/parametrization#quantum-mechanics-versus-a-non-commutative-generalization-of-probability-theory.

[8]See Section /pub/euler#black-hole-information-paradox-and-the-stern-gerlach-experiment.

In this book we show that the wave-function can be described as a multi-dimensional generalization of Euler's formula, and its collapse as a generalization of taking the real part of Euler's formula. The unitary operator in a separable Hilbert space is nothing else than one possible parametrization of any conditional probability measure in a standard measure space; the parametrization is a surjective map from the group of unitary operators to the set of all conditional probability measures in a standard measure space. The fact that the operators are unitary reflects the fact that the integral of the probability distribution is always 1. Two unitary operators are always related by another unitary operator, which is a linear transformation and it preserves the group. It is thus a good parametrization which leads us to represent a group of symmetry transformations using linear transformations. These conditional probability measures may be non-deterministic, thus quantum mechanics is a generalization of classical statistical mechanics (but not of probability theory).

This is ironic, since Feynman described Euler's formula as "our jewel" while the wave-function collapse certainly contributed for him to say "I think I can safely say that nobody understands Quantum Mechanics." Besides the irony, the fact that the wave-function is nothing else than one possible parametrization of any probability distribution means that the wave-function collapse is a feature of all random phenomena.

The above fact implies that alternatives to Quantum Mechanics motivated by a dissatisfaction with either the complementarity of position and momentum [44], or the wave-function collapse [45], may also feature complementarity of observables (possibly other than position and momentum) and wave-function collapse, once a parametrization with a wave-function is applied. The physical question is what are the conditional probability measures of the theory, in particular whether there are viable alternative theories to Quantum Mechanics where the time-evolution of the statistical ensemble is deterministic [46] or at least it is a stochastic process [47][48][49]; in such a case there would be other alternative parametrizations which also allows us to use methods of group theory and do not involve a wave-function and its collapse.

But the above fact also implies that the wave-function collapse is not provoked by the interaction with the environment[9]; the wave-function collapse does not emerge from some particular cases of classical statistics [50]; Quantum Mechanics is *not* a generalization of the concept of probability algebra from

[9]Haroche, S., & Raimond, J. M. (2006). *Exploring the Quantum: Atoms, Cavities, and Photons*. OUP Oxford. Mensky, M. (2000). *Quantum Measurements and Decoherence: Models and Phenomenology*. Springer Netherlands.

In a measurement there is always an interaction with the environment, therefore the environment necessarily affects the ensemble and it is possible that decoherence occurs. But such phenomena will be accounted for by the time-evolution of the ensemble. Note that a continuous (repeated) quantum measurement is a model of decoherence and thus decoherence does not avoid by itself the wave-function collapse (Mensky 2000, Haroche 2006). In case we opt for a model of decoherence which avoids collapse, then we are necessarily dealing with an alternative to Quantum Mechanics, such case was discussed above.

commutative to non-commutative algebra [51]; and thus quantum computation/information is not fundamentally different from classical computation/information[10]. The wave-function is a possible parametrization for any theory of Statistics, including Statistical Physics.

This is comforting, since it is consistent with the empirical facts that Quantum Mechanics applies to a very wide range of physical systems, from the Hydrogen atom, a neutron star or the Universe; and that the collapse occurs upon measurement, any measurement. This also opens the door into applying the wave-function parametrization not just to quantum mechanics and quantum statistical mechanics, but also to quantum field theory or even to other problems involving statistics other than quantum physics [25]. There are many potential applications[11]. For instance, since the supervised learning problems in machine learning can be thought of as learning a function from examples, these problems can be formulated as Bayesian modeling [19]. The wave-function parametrization may allow using any prior as an alternative to using the Gaussian process [19] or the conditional random field [20] as priors. It may also allow more general Random Fields as an alternative to Markov Random Fields in Physics, Biology and Data Science [21], alternative to the semivariogram in spatial Data analysis [22], alternative to wavelets and renormalization in multi-resolution analysis [23] and alternative to Functional Data Analysis and Neural Networks in time series analysis [24].

In Section 1, we show that the wave-function is one possible parametrization of any probability distribution[12]; in Section 2 we discuss the difference between the wave-function parametrization and the density matrix parametrization of Gleason's theorem; in Sections 3 and 4 we discuss symmetry transformations and deterministic transformations; in Section 5 we describe the relation between Euler's formula and the parametrization of a probability distribution by a real wave-function; in Sections 6 and 7 we describe the Stern-Gerlach experiment and Black Hole Information paradox using the Euler's formula and the ensemble interpretation; in Sections 8 and 9 we describe the parametrization of a probability distribution by a real wave-function (i.e. for a finite and generic number of states, respectively); in Section 10 we address complex and quaternionic wave-functions; in Section 11 we distinguish the time-evolution in quantum mechanics from a stochastic process; in Section 12 we show that the the time-evolution in quantum mechanics is a stochastic process if and only if it is deterministic; in Section 13 we discuss how the concept of (ir)reversible processes from thermodynamics applies to non-deterministic symmetry transformations; in Section 14 we show that quantum mechanics is a complete description of physical reality in the sense of Einstein-Podolsky-

[10]Wilde, M. M. (2013). *Quantum Information Theory*. Cambridge University Press. https://arxiv.org/abs/1106.1445

Different computers always have different properties, for instance different logic gates may enhance the performance of different algorithms (Wilde 2013). But neglecting performance, the quantum bits can be constructed using classical bits and quantum logic gates can also be constructed using classical logic gates, with the Hadamard transform as an example.

[11]See Section /pub/ml#wave-function-parametrization-in-machine-learning.

[12]

Rosen; in Section 15 we show that any deterministic theory compatible with relativistic Quantum Mechanics necessarily respects relativistic causality; in Section 16 we build an explicit example of a deterministic theory compatible with relativistic Quantum Mechanics; in Section 17 we describe the double-slit experiment using the ensemble interpretation; in Section 18 we argue that our results imply that the Bell inequalities merely establish a difference between Quantum Mechanics and a theory where time-evolution is a stochastic process, unlike what is often claimed in the literature; in Section 19 we will define constrained systems and the conditioned probability in particular; in Section 20 we will define an appropriate time-evolution to statistical field theory using constraints; we conclude in Section .

Note that in this book, the Hilbert space is always considered to be a separable Hilbert space [52] and unless otherwise stated it is a real Hilbert space.

Quantization due to time-evolution

For comparison purposes, we now address what quantization is not. Quantization is not replacing Poisson bracket's by canonical commutation relations. The method of replacing Poisson bracket's by canonical commutation relations can always be applied (for analytical functions), it is called prequantization [53][54]. However it doesn't lead by itself to useful results (hence the name prequantization). Of course, we can try to improve the method so that it leads to useful results (this is the geometrical quantization program [53]), however we end up with a definition of quantization which is so complex and arbitrary that it is not useful in practice, in particular in the presence of gauge symmetries.

Quantization is also not second quantization (based on a Fock-space), which relates a quantum description of a single-particle system to a quantum description of a many-particle system [54]. We can only apply second quantization to a quantum theory, hence the name "*second*".

Quantization is also not computing the Feynman's path integral, since we know that the Feynman's path integral does not have the property (sigma-additivity) which allows computation of the integral by approximating the integrand [55], and thus it is not an integral. Of course as in prequantization, we can try to improve the path integral [56], however we are very far from a consistent definition of path integral which is useful in practice.

Quantization is also not a perturbative expansion or a lattice regularization. These two different approximations are useful and have a clear definition, but since we know that they are complementary [55] then neither of them can be used to define quantization.

Note that there is enough experimental evidence to conclude that all the methods above mentioned—namely prequantization, second quantization, Feynman's path integral, perturbative expansion,

lattice reguralization—are related to the quantum phenomena and thus they are necessarily related with the definition of quantization. But we insist that it is also clear that none of them by itself can be used to define quantization.

There is a big conceptual problem with the notion of quantization: we are trying to relate a deterministic theory (classical mechanics) with a non-deterministic theory (quantum mechanics). From the point of view of (classical) information theory [57], the root of probabilities (i.e. non-determinism) is the absence of information. Statistical methods are required whenever we lack complete information about a system, as so often occurs when the system is complex [25]. Thus we can convert a deterministic theory to a statistical theory unambiguously (using trivial probability distributions); but we cannot convert a statistical theory into a deterministic theory unambiguously since we need new information[13].

On the other hand, the relation between quantum mechanics and a statistical theory (both are non-deterministic) is clear: the wave-function is nothing else than one possible parametrization of any probability distribution [58]; the parametrization is a surjective map from an hypersphere to the set of all possible probability distributions. Two wave-functions are always related by a rotation of the hypersphere, which is a linear transformation and it preserves the hypersphere. It is thus a good parametrization which allows us to represent a group of symmetry transformations (such as translations in space and time or rotations in space) using linear transformations of the hypersphere. These symmetry transformations may be non-deterministic, thus quantum mechanics is a generalization of classical statistical mechanics (but not of probability theory, thanks to the wave-function's collapse [58]).

The non-commutativity of operators is thus intrinsic to any statistical theory. This saves us from the need to "deform" commutative algebras into non-commutative ones upon quantization [59]. In our opinion, either the quantization of a classical theory or the classical limit of a quantum theory cannot go much beyond Koopman-von Neumann version of classical mechanics [42], i.e. a description of classical mechanics as a statistical theory (which is always possible, since a deterministic theory is a particular case of a statistical theory).

In Section [sec:parametrization] we will relate algebraic ideals with gauge symmetries, in such a way that it allows the definition of Hamiltonians which are non-polynomial in the fields using the Gaussian measure, such as in Classical Statistical Field Theory and Quantum Gravity; in Section 1 we will define gauge symmetry exclusively within the Hamiltonian formalism (that is, a definition independent from the Lagrangian formalism); in Section 2 we will show how to deal

[13]Hooft, G. 't. (2014). *The Cellular Automaton Interpretation of Quantum Mechanics. A View on the Quantum Nature of our Universe, Compulsory or Impossible?* https://arxiv.org/abs/1405.1548

E.g. the assumptions required by the deterministic models in (Hooft 2014) are new information.

with Lorentz covariance within the Hamiltonian formalism; in Section 3 we will define a Statistical Source Field Theory where the fields in phase-space also depend on a time coordinate; in Section 4 we show that the canonical commutation relations between momentum and position appear due to a non-deterministic time-evolution; in Section 5 we study the implications of using a gauge-variant gaussian measure for the phase diagram; in Section 6 we discuss spontaneous symmetry breaking within the framework defined in the previous sections; in Section 7 we discuss how to define operators which are effectively local, despite respecting the momentum constraint; in Section 8 we discuss the relation of our definition of Quantum Gauge Field Theory with the BRST formalism; in Section 9 we discuss Renormalization, the mass gap and the status of the problem from the Millennium prize; in Sections 10 and 11 we apply our formalism to the Yang-Mills and free electromagnetic field and the results are compatible with the existing literature; in Section 12 we show how diffeomorphisms can be incorporated within our formalism and we propose an Hamiltonian for quantum gravity; finally in Section we end with the Conclusions.

Resolution of the singularity of the ODE x'=x^2 when the initial x has uncertainties

Abstract

We show that the classical initial value problem x'=x^2 has a continuous solution for all time, when the initial x has any finite amount of uncertainties (which is unavoidable in engineering). This is in stark contrast with the singular solution when x has no uncertainties.

General picture

Consider a sample space defined by the real line. Since we can always define a wave-function by taking the square-root of the probability measure, the Koopman-von Neumann version of classical statistical mechanics[42][60][61][62] defines classical statistical mechanics as a particular case of quantum mechanics where the algebra of observable operators is necessarily commutative (because the time-evolution is deterministic).

However, it is well known that for two unbounded self-adjoint operators, the fact that they commute does not imply that their corresponding spectral measures commute[63], then we would say that they *strongly* commute. Thus in principle, it is possible to have a classical dynamical system defined by an *ordinary differential equation* where the unitary time-evolution is non-deterministic, that is, the algebra of observable operators is non-commutative. We will not study this possibility here, this is consistent with the fact that in a continuous standard probability space we can not neglect uncertainties, for that we would need to work in the larger space of finitely additive measures[2], which has very different properties than a standard probability space. Note that such solutions non-deterministic solutions do not appear if at the initial time, there is a joint probability distribution for all the random variables appearing in the ordinary differential equation, which is a common

assumption.

Since the real line is a simply connected group (for addition), then by Bargmann's theorem any strongly continuous projective representation of the real line is induced by some strongly continuous representation of the real line on the corresponding Hilbert space[64][65]. Then, Stone's theorem implies that such representation on the Hilbert space is the exponential of a time-independent Hamiltonian[64][65]. Thus, if there is a solution in a standard probability space of a differential equation without singularities for all times (including negative times), then there is a quantum Hamiltonian corresponding to such solution which is time-independent. Moreover, if such solution is deterministic then the fact that there are more Hamiltonians that generate the same differential equation does not change the unique solution, up to initial conditions: suppose that $x(t) = U(t)x(0)U^\dagger(t)$ is a deterministic solution to the differential equation, then $x(t) = f(x(0), t)$ is a measurable function of $x(0)$ and it can be rewritten as $c(t) = U_2^\dagger(t)x(t)U_2(t) = U_2^\dagger(t)f(x(0), t)U_2(t)$. It verifies $\dot{c} = U_2^\dagger(t)[H - H_2, f(x(0), t)]U_2(t) = 0$, because the difference between Hamiltonians must preserve the differential equation. Thus $x_2(t) = U_2(t)x(0)U_2^\dagger(t) = U_2(t)c(t)U_2^\dagger(t) = x(t)$. Then, $x(t) = U(t)x(0)U^\dagger(t)$ is the unique solution, up to initial conditions.

Moreover, any solution in a standard probability space is given by a time-dependent self-adjoint Hamiltonian in a larger probability space, then the singularities in a standard probability space are consequence of a time-dependent Hamiltonian which is non-integrable. But since any time-dependent Hamiltonian can be converted into a time-independent Hamiltonian in an even larger sample space, then we can always resolve the singularities in a larger sample space which better defines the dynamical system. Thus, solutions without singularities for all times (including negative times) defined by time-independent Hamiltonians are expressive enough, even to deal with singularities at the cost of choosing a larger sample space.

State-of-the-art

Consider the differential equation: $\dot{x} = x^2$. The general solution without uncertainties is:

$$x(t) = \frac{x(0)}{1 - t\ x(0)} \tag{2}$$

The general solution has a singularity at $t = \frac{1}{x(0)}$. There are no solutions without singularities, when the uncertainty is neglected[66].

When considering uncertainties, if we assume that the solution Equation 1 exists for some $t > 0$, then there is an isometry $T(t)$ such that $x(t) = T^\dagger(t)x(0)T(t)$ is a real number for all eigenvalues of $x(t)$, since the total probability must be conserved and the solution is deterministic. The eigenvalues

of the operator $x(0)$ such that $x(0) > \frac{1}{t}$ have no corresponding $x(t)$ which is a real number given by the isometry in agreement with the solution Equation 1, because the deterministic solution becomes ill-defined for $t \geq \frac{1}{x(0)}$. We can repeat the same reasoning for $t \geq 0$. Thus, we conclude that the solution Equation 1 is implemented by a non-unitary isometry when working with probability measures, for all $t \neq 0$. This is a time-dependent restriction on the allowed initial probability measures, for instance any initial Gaussian distribution is not allowed for any $t \neq 0$. The solution Equation 1 can also be implemented by a unitary operator at the cost of choosing a larger sample space, which we will do in the following.

We could think that a unitary time-evolution is somehow too demanding, that only with restrictions to the allowed initial probability measures we can find all solutions to the differential equation defined above. But as a matter of principle that should not be correct since we always work with such restrictions under the assumption that we will somehow be able to recover a corresponding result in the space of probability measures, otherwise such result cannot be tested experimentally or even in most numerical simulations (due to approximation uncertainties). Indeed, if we consider the space $L^2(\mathbb{R}^2)$ and the complex number $z(t) = x(t) + iy(t)$ satisfying $\dot{z}(t) = z^2(t)$ then the singularity condition $t \to \frac{1}{z(0)}$ cannot be satisfied except in a set of null measure where $y(0) = 0$, thus there is no singularity in $L^2(\mathbb{R}^2)$. But $y(0)$ can be as concentrated around zero as we want, so this unitary solution in a larger sample space recovers the corresponding non-unitary solution in the original sample space.

Changing the sample space, effectively changes the equation, since $y(0) = 0$ strictly is not possible in $L^2(\mathbb{R}^2)$. This has been the approach so far to resolve the singularities: modify the equation until no singularities exist up to some large finite time for some interval of interest for the initial value. For instance, the 3D Euler's equations appear to be appropriate for many problems, however since there is evidence that they have finite-time singularities[67] then we can instead use the more general 3D Navier-Stokes equations[68] in the hope that they don't have singularities for the parameters of interest. That is why it is so important to find out whether or not the 3D Navier-Stokes equations have singularities[67], because then we may need yet another set of equations. But this approach does not work in most real-world cases because it is very hard to predict whether or not a differential equation has singularities for the parameters of interest in a specific real-world problem[67]. So, in each specific real-world problem, engineers have to test and modify differential equations several times until they find one that fits their specific needs, which severely diminishes the predictive power of the calculations. In this article we propose a different approach without the need to change many differential equations, by using the uncertainties in x.

Resolution of the singularity using the uncertainties in x

The quantum Hamiltonian operator corresponding to $\dot{x} = x^2$ is $H = \frac{1}{2}(px^2 + x^2p) = (-ix + x^2p)$ (such that the commutator $[x, p] = i$ and the domain is the dense set of smooth functions with compact support), can be proved to be essentially self-adjoint[69][70] (using H^2 as a positive auxiliary operator in Corollary 1.1 of [69]), meaning that the unitary time-evolution $U(t)$ is uniquely defined. Even if $x(t) = U(t)x(0)U^\dagger(t)$ would commute with $x(0)$, these are unbounded self-adjoint operators so the relevant question is whether or not they *strongly* commute for all times. If they *strongly* commute for all times, then they can be simultaneously diagonalized and there is a deterministic solution without singularities. This would be in stark contrast with the fact that there are no deterministic solutions without singularities, when the uncertainty is neglected[66].

The unitary operator:

$$U(x,p) = \frac{e^{-i\frac{p}{x}}}{\sqrt{2\pi}x} \tag{3}$$

It diagonalizes the Hamiltonian, that is $H\ U(x,p) = p\ U(x,p)$. Then, the time evolution operator is given by:

$$\begin{aligned} e^{-iHt}\{\psi\}(x) &= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} dy\ U(x,p)e^{-ipt}U^*(y,p)\psi(y) \\ &= \int_{-\infty}^{+\infty} \frac{dy}{2\pi xy}\ 2\pi\delta\Big(t + \frac{1}{x} - \frac{1}{y}\Big)\psi(y) \\ &= \int_{-\infty}^{+\infty} \frac{dw}{xw}\ \delta\Big(t + \frac{1}{x} - w\Big)\psi\Big(\frac{1}{w}\Big) = \frac{1}{xt+1}\psi\Big(\frac{x}{xt+1}\Big) \end{aligned} \tag{4}$$

Where we did the change of variables $y \to \frac{1}{w}$. Given a self-adjoint operator ρ diagonal in the x basis (for instance, a probability measure):

$$\begin{aligned} &\rho\{\psi\}(x) = \rho(x)\psi(x) \\ &(e^{-iHt}\rho\ e^{iHt})\{\psi\}(x) = \\ &= \frac{1}{xt+1}\int_{-\infty}^{+\infty} \frac{dy}{1-yt}\ \delta\Big(\frac{x}{xt+1} - y\Big)\rho(y)\psi\Big(\frac{y}{1-yt}\Big) = \\ &= \rho\Big(\frac{x}{xt+1}\Big)\psi(x) \end{aligned} \tag{5}$$

A singularity at finite-time in the time-evolution of an initial value problem defined by an ordinary differential equation implies that the derivative in time of the corresponding wave-function diverges. In the case of x'=x^2, the coordinate goes instantly from $+\infty$ to $-\infty$ at some time, this is concerning but we need to consider a finite amount of uncertainties which allows us to define a wave-function.

Since the derivative in time corresponds to the Hamiltonian (whose spectral measure is conserved by the time-evolution) then if the initial condition only involves eigenfunctions corresponding to a spectrum up to some maximum absolute value, there is no way in which the time-evolution will produce a divergent derivative in time of the wave-function. Such initial condition also makes sense from a physics point of view, where we usually have limited energy available when we apply physical transformations to a system; it also makes sense from the point of view of numerical calculations, where the Krylov subspace methods for computations of matrix exponentials is guaranteed to converge for any wave-function defined in an Hilbert space[71][72] (even when the Hamiltonian is unbounded), that is, one where an initial condition with infinite energy has null measure (otherwise the well known phenomenon aliasing occurs).

Thus, the sample space of our initial system is usually not the coordinate space, since it verifies an energy limit. We can relate such initial sample space to the coordinate space through a physical transformation, which is only hypothetical, because it would require the availability of an arbitrarily large amount of energy. That is, we can only guess what the coordinate x would be (with an uncertainty) if we would do the necessary transformation, but never really do such transformation and thus never measure the coordinate.

Since the Hamiltonian differs from the translation in space by a change of variables $y \to \frac{1}{x}$, then using the Whittaker–Shannon interpolation (also called sinc interpolation) before the change of variables $y \to \frac{1}{x}$, we can completely define the wave-function in coordinate space through its values at discrete points in space $x_n = \frac{\Delta}{n}$ where n is an integer number (different from zero) and Δ is a constant proportional to the maximum energy. Note that between the points x_n and x_{n+1} the wave-function is fully determined (up to sets of null measure). Also, since the wave-function is square-integrable it must converge to zero when $x \to \pm\infty$, so we can approximate any wave-function for sufficiently large Δ, despite that the limit of the sequence of discrete points is not infinity (for infinite n).

Note that since the probability is the predual of the algebra of events, the meaning of the time-evolution of the probability is to define what is the probability in the past which would produce the system in the present. Thus, the time-evolution does not make an object cross the would-be singularity, since only probabilities resulting from an hypothetical physical transformation (which never occurs because we do not have enough energy available) are transferred from some points in space to others. We need to look for what happens in the sample space of the real system (and not

an hypothetical one), where the time-evolution is non-deterministic. The time-evolution would be deterministic in the coordinate space, but it is non-deterministic in the sample space which verifies an energy limit: that is the price to pay for resolving the singularity.

Note that the claim that limiting the energy of the initial condition suffices to solve the singularity is not by itself in conflict with the special relativity, because special relativity is expressed in Quantum Mechanics through an Hamiltonian that verifies extra conditions. By limiting the energy (Hamiltonian's spectrum) of the initial condition we do not break such extra conditions, because the spectrum of the Hamiltonian verifies the same extra conditions as the Hamiltonian itself.

Degenerate solutions

There are two different problems that singularities create. The first problem is its possible existence in problems where they cannot be detected: if the numerical method does not converge, we cannot know whether the cause is a singularity or an insufficient resolution of the numerical method. This problem is solved by the method discussed in this article.

The second problem is that the singularity allows for degenerate solutions, where the initial value problems before and after the singularity can have unrelated initial conditions. The method presented here chooses initial conditions compatible with analytic continuation from complex analysis. This may be useful for problems in complex analysis and one could argue that for problems related to physics this is the only possible solution, but certainly other resolutions of the degeneracy are more relevant in other contexts and the method presented here does not allow for such a choice.

In summary, the method presented here solves the first problem (distinguish singularity from insufficient resolution of the numerical methods), but it is not completely satisfactory for the second problem (access to all possible solutions, in particular degenerate solutions). Note that in Physics or complex analysis context, there is only one viable solution which is given by the method discussed in this article, so the second problem is important in general, but not important for the problems addressed by the present author.

Conjecture

We can easily see that for instance, all time-independent polynomial quantum Hamiltonians such that the domain is the dense set of smooth functions with compact support, can be proved to be essentially self-adjoint[69][70], using H^2 as a positive auxiliary operator in Corollary 1.1 of reference [69]. Thus, the conclusions apply to many more differential equations with would-be singularities and we can find non-deterministic solutions instead.

Wave-function parametrization of a probability measure

Abstract

We show that any conditional probability measure in a standard measure space is parametrized by a unitary operator on a separable Hilbert space. We propose unitary inference, a generalization of Bayesian inference. We study implications for classical statistical mechanics and machine learning.

1. Introduction

In Scientific Research as in Bayesian inference, there is always a prior probability distribution, and there is no prior which is better for all cases[16]: we always have to make assumptions. For instance, if we insist we can assume all objects in the night sky revolve around the Earth using epicycles and even consider that this is a very successful theory because it was a precursor of Fourier series and these have widespread applications today[17].

> *The Emperor realized that the people were right but could not admit to that. He though it better to continue the procession under the illusion that anyone who couldn't see his clothes was either stupid or incompetent. And he stood stiffly on his carriage, while behind him a page held his imaginary mantle.*
>
> —Hans Christian Andersen (1837)

Thus, data, evidence and even mathematical proofs are not enough for us to abandon prior beliefs, there must be a moment when in order for us to achieve something we want we have to abandon our prior beliefs: in the case of assuming all objects in the night sky revolve around the Earth, it would make many relevant calculations in modern astronomy or in the Global Positioning System impossible.

In the case of Quantum Mechanics, it was always possible to see its mathematical formalism as a

mere (but very useful) parametrization of probability and classical information theories[73]. But, it was also possible to describe it (although in a forceful way) as a *new* and exotic alternative to (or generalization of) classical information theory, one that shook our sense of reality. It is not hard to believe that the mystery surrounding the exotic point of view can often convince the society to spend large amounts of money in research, at least in the short term.

The exotic point of view was sustained by a relatively constant flow of discoveries of new phenomena at higher and higher energies, which kept the mystery alive. But now, even optimists say that if the Large Hadron Collider finds nothing new, it will be harder to convince the governments of the world to build the next bigger, more expensive collider to keep research in High Energy Physics as we know it[18]. Thus, if the exotic point of view is not that mysterious nowadays and there exists a parametrization point of view, is there another way to convince the society to spend large amounts of money in research?

The answer is yes, there is another way. Because the parametrization point of view allows the mathematical formalism of Quantum Mechanics to be applied to probability and classical information theory (and then to artificial intelligence) not just to edge cases, but to the core of these theories. This would not be possible with the exotic point of view, which considered that Quantum Mechanics shook our sense of reality and thus must be an alternative to (or a generalization of) classical information theory and thus there would be no reason to expect applications to the core of classical information theory.

2. Probability updates, machine learning and Quantum Mechanics

We call to the process of updating a state of knowledge with respect to new evidence as Probability updates[74]. There is little doubt that the Bayes rule should be applied to Probability updates, *whenever possible*[74]. But there is also little doubt that the Bayes rule is often not applicable[74] and thus Probability updates is a generalization of Bayesian inference (and of statistical inference). There is no consensus on what to do when the Bayes rule is not applicable.

Machine learning[75] (including numerical analysis in general[76]) and Quantum Mechanics[51] can be defined as a particular case of Probability updates. Thus, it is no surprise that there is a broad consensus that Machine learning and Quantum Mechanics are a generalization of Bayesian inference in some sense, but little consensus beyond that (such discussion is beyond the scope of this article, but it appears in reference[77]of the same author).

A Bayesian model[78] is defined by a regular conditional probability between standard measure spaces, also called a Markov kernel or likelihood. If we concatenate Bayesian models, such that the data (output) of one Bayesian model are the parameters (input) of another Bayesian model we

create a Markov process.

Markov processes cannot produce an arbitrary function of time, because there is an ordering (related with the concept of entropy) with respect to which all continuous-time Markov processes are monotonic[79]. Thus, Bayesian inference is irreversible.

In this article we propose reversible statistical models and a method of statistical inference defined by a quantum time-evolution which is reversible (we call it Unitary inference). The Bayes rule is applied to the final likelihood which results from the concatenation of several reversible models followed by the application of the Born rule.

Thus, such reversible inference is fully compatible with Bayesian inference, that is: given a Bayesian model, a prior probability and data that allows to produce a posterior probability through Bayesian inference, there is a reversible model that produces the same posterior probability as a function of the prior probability (see next section). Moreover, a quantum time-evolution can be used to define Probability updates, Machine Learning and Quantum Mechanics.

The crucial advantage of Unitary inference compared to Bayesian inference, besides the fact that it is more general, is that any prior wave-function can be transformed into any other prior wave-function through a change of basis, which also affects the unitary statistical models. In Bayesian inference, there are priors which are too wide, in the sense that the relevant sample space is too big to allow for a numerical integration. At first sight, unitary inference makes this problem worse because instead of one numerical integration we have to do many numerical integrations because we are dealing with matrices now. But it turns out that it also allows for basis changes and thus a prior wave-function that is too wide in one basis, can become extremely simple in another basis. Since, the basis change also affects the unitary statistical models, it is only the relation between the unitary statistical model and the prior probability distribution that is relevant. In any case, any approximation (Monte-Carlo, Neural networks, variational inference, etc.) can be mapped to the formalism of unitary inference (because it is the most general formalism), where the approximation can be seen as the choice of a favorable basis and prior probability distribution with respect to an approximation (through truncation, for instance) of the unitary statistical model.

Note that probabilities can always be seen as incomplete information, which in turn can be measured by entropy as defined in classical information theory. We claim that reversible models and unitary inference are not in themselves a source of entropy, but they still must deal with the entropy coming from other sources. This is guaranteed by the fact that unitary inference is fully compatible with Bayesian inference. Thus, we do not claim that unitary inference eliminates the entropy that comes from other sources.

3. Any conditional probability measure in a standard measure space is parametrized by a unitary operator

Measure spaces (in particular, algebraic measure theory as defined below) have many applications in classical mechanics and dynamical systems[80], when non-linear equations and/or complex sample spaces become an obstacle.

Any measure space can be defined as a probability space: a probability space is a measure space where a probability density was chosen, which is any measurable function normalized such that its measure is 1. In the historical (that is, Kolmogorov's) probability theory, a probability space has three parts: a sample space (which is the set of possible states of a system, also called phase space); a complete Boolean algebra of events (where each event is a subset of the set of possible states); and a probability measure which assigns a probability to each event. The probability is a map from complex random events to abstract random events, shifting all ambiguity with the notion of randomness to the abstract random events as described in the following: that the probability of an event is 0.32 means that our event has the same likelihood of finding a treasure that we know it was hidden in the sand of a 1 km wide beach, if we only search for it with a metal detector in a 320 m interval. While this treasure hunt is ambiguous (are there any clues for the location of the treasure? Etc.) the map from our complex events to this treasure hunt is unambiguous.

On the other hand, a *standard* measure space is isomorphic (up to sets with null measure) to the real Lebesgue measure in the unit interval or to a discrete (finite or countable) measure space or to a mixture of the two. Thus, topological notions such as dimension do not apply to *standard* measure spaces. Most probability spaces with real-world applications are *standard* measure spaces. Equivalently, a *standard* measure space can be defined such that the following correspondence holds: every commutative von Neumann algebra on a separable (complex or real) Hilbert space is isomorphic to $L^\infty(X,\mu)$ for some standard measure space (X,μ) and conversely, for every standard measure space (X,μ) the corresponding Lebesgue space $L^\infty(X,\mu)$ is a von Neumann algebra. As expected, the representation of an algebra of events in a (real or complex) Hilbert space uses projection-valued measures[81][82][83][84]. A projection-valued measure assigns a self-adjoint projection operator to each event, in such a way that the boolean algebra of events is represented by the commutative von Neumann algebra: intersection/union of events is represented by products/sums of projections, respectively. The state of the ensemble is a linear functional which assigns a probability to each projection. Thus, there is an algebraic measure theory[80][85][86][87] based on commutative von Neumann algebras on a separable Hilbert space which is essentially equivalent to measure theory (for standard measure spaces).

To be sure, the algebraic measure theory is based on *commutative* algebras, thus it is not a non-

commutative generalization of probability or information theory (see section 4). That is, there is no need for a conceptual or foundational revolution such as qubits replacing bits when switching from the historical to the algebraic probability theory [85]. Moreover, this is a common procedure in mathematics, as illustrated in the following quote [88] (note that a probability measure is related to integration):

> *The fundamental notions of calculus, namely differentiation and integration, are often viewed as being the quintessential concepts in mathematical analysis, as their standard definitions involve the concept of a limit. However, it is possible to capture most of the essence of these notions by purely algebraic means (almost completely avoiding the use of limits, Riemann sums, and similar devices), which turns out to be useful when trying to generalize these concepts*[...]
>
> T. Tao (2013)[88]

The relation of the algebraic measure theory with probability theory is the following: any joint probability density $p(x, y)$ between two *standard* measure spaces X, Y (possibly with continuous and discrete parts) can be written as $p(x, y) = |\mathcal{U}(y, x, 0)|^2$ where $\mathcal{U} : L^2(\mathbb{Z}) \to L^2(X \times Y)$ is a unitary operator between Hilbert spaces of complex square-integrable functions. Note that this result is not surprising and can be seen as a commutative version of Wigner's theorem[89]. The proof follows.

We have $p(y|x)p(x) = p(x, y) = |\mathcal{U}(y, x, 0)|^2 = p(x|y)p(y)$ up to sets with null measure, for any probability density $p(x)$ and some unitary $\mathcal{U}$ on the separable Hilbert space. Note that in a standard measure spaces it is always possible to define regular conditional probabilities[90] and to choose $p(x) = p_0(x) > 0$ for all $x \in X$, except in sets with null measure. Moreover, the choice of p_0 is independent of $p(y|x)$. Thus, we will parametrize $p(x, y) = p(y|x)p_0(x)$ the joint probability density for the tensor product $(x, y) \in X \times Y$ for a particular $p(x) = p_0(x) > 0$ for all $x \in X$, except in sets with null measure. Then, we can obtain a parametrization for any other $p(x)$ from the expression $p(y|x)p(x) = \frac{p(x,y)}{p_0(x)}p(x)$.

Since $p(x, y) \geq 0$ then there is always a normalized wave-function $\Psi \in L^2(X \times Y)$ such that $|\Psi(x, y)|^2 = p(x \otimes y)$. Let $\{e_j\}$ be an orthonormal basis of the Hilbert space $L^2(X \times Y)$. Then, we can build the unitary U through the Gram-Schmidt process, such that $\mathcal{U}(y, x, 0) = \Psi(x, y)$.

Note that the Cauchy-Schwarz inequality implies $|\int dx\Psi(x, y)\Phi(x)|^2 \leq \int dx|\Psi(x, y)|^2$ where the L^2 norm of $\Phi(x)$ is 1. Thus, $||\Psi\{\Phi\}|| = \int dy|\int dx\Psi(x, y)\Phi(x)|^2 \leq \int dxdy|\Psi(x, y)|^2 = 1$, this implies $\Psi(x, y)$ is bounded when defined as an operator $\Psi : L^2(X) \to L^2(Y)$.

Moreover,

Since Ψ is bounded, then it admits a Singular Value Expansion (through the polar decomposition),

that is, $\Psi = VDU^\dagger$, where D is a positive semi-definite diagonal operator, U is unitary and V is a partial isometry (with kernel where D is null and an isometry where D is non-null).

Then $Tp_0T^\dagger = \Psi\Psi^\dagger = VD^2V^\dagger$, but in general we have $TpT^\dagger = \Psi\frac{p}{p_0}\Psi^\dagger = VDU^\dagger\frac{p}{p_0}UDV^\dagger$.

Since the Hilbert space is separable, there is an orthonormal discrete basis, and we can then enlarge the discrete part of the sample space X to include the orthonormal elements e_k corresponding to the elements of the basis φ_k that are missing, setting $D_{kk} = 0$ for the new indices, replacing a would-be partial isometry V by the unitary W built through the Gram-Schmidt process such that $VV^\dagger W = V$. We get $\Psi = WDU^\dagger$.

The converse also holds. Given a bounded operator B, such that $tr(BB^\dagger) = 1$, then it defines a joint probability distribution of initial and final states $p(x, y) = |B(y, x)|^2$. From the joint probability, if $p(x) = \{B^\dagger B\}(x, x) > 0$ for all $x \in X$, then we can define a regular conditional probability density (since the measure space is standard).

The linearity of the commutative algebra; avoiding a fixed sample space *a priori*; and the fact that we can map complex random phenomena to an abstract random process unambiguously, are obvious advantages for algebraic measure theory when we want to compare probability theory with Quantum Mechanics, where the linearity of the canonical transformations is guaranteed by Wigner's theorem (it is essentially a consequence of the Born's rule applied to a non-commutative algebra of operators[89][91][92]); the Hilbert space of wave-functions replaces the sample space; and the canonical transformations are non-deterministic.

The algebraic measure theory is also different from defining the sample space as a reproducing kernel Hilbert space [93][94], since no sample space (whether it is a Hilbert space or not) is defined *a priori*. Note that defining the sample space as a Sobolev Hilbert space is common in classical field theory[95], but defining a general probability measure in such space is still an open problem.

4. Quantum Mechanics versus a non-commutative generalization of probability theory

The correspondence between geometric spaces and commutative algebras is important in algebraic geometry[14]. It is usually argued that the phase space in quantum mechanics corresponds to a non-commutative algebra, and thus it is a non-commutative geometric space in some sense[96]. It is a fact that Quantum Mechanics may inspire a non-commutative generalization of probability

[14]The correspondence between geometric spaces and commutative algebras is consequence of the Gelfand representation: there is an isomorphism between a commutative C^*-algebra A and the algebra of continuous functions of the spectrum of A.

theory, since the wave-function could also assign a probability to non-diagonal projections, these non-diagonal projections would generate a non-commutative algebra[51]. However, after the wave-function collapse, only a commutative algebra of operators remains. Thus, the phase space in quantum mechanics is a sample space of a standard measure space and the standard spectral theory (where the correspondence between geometric spaces and commutative algebras plays a main role [97]) suffices.

Consider for instance the projection P_X to a region of space X and a projection $UP_pU^\dagger$ to a region of momentum p, where P_X and P_p are diagonal in the same basis. The projections P_X and $UP_pU^\dagger$ are related by a Fourier transform U and thus are diagonal in different basis and do not commute (they are complementary observables). Since we can choose to measure position or momentum, it seems that Quantum Mechanics is a non-commutative generalization of probability theory[51].

But due to the wave-function collapse, Quantum Mechanics is not a non-commutative generalization of probability theory despite the appearances: the measurement of the momentum is only possible if a physical transformation of the statistical ensemble also occurs, as we show in the following.

Suppose that $E(P_X)$ is the probability that the system is in the region of space X, for the state of the ensemble E diagonal (i.e. verifying $E(O) = 0$ for operators O with null diagonal). Using the notation of [98], we have $E(A) = \mathrm{tr}(\rho A)$ where A is any operator and ρ is an (eventually unbounded) self-adjoint operator with $\mathrm{tr}(\rho) = 1$ and ρ is diagonal in the same basis where the projection operators P_X are diagonal. If an operator O has null diagonal in the same basis where P_X is diagonal, then $tr(\rho O) = 0$ for any ρ diagonal.

If we consider a unitary transformation U on the ensemble, then after the wave-function collapse[99] we have a new ensemble with state E_U given by:

$$E_U(A) = tr(\rho_U A) \tag{6}$$

Where ρ_U is diagonal (due to the wave-function collapse) in the same basis where the projection operators $UP_pU^\dagger$ are diagonal, defined such that:

$$\begin{aligned} E_U(D) &= \mathrm{tr}(\rho_U D) = \\ &= \mathrm{tr}(\rho U D U^\dagger) = E(UDU^\dagger) \end{aligned} \tag{7}$$

Where D is any diagonal operator and O is an operator with null diagonal in the same basis where P_X and P_p are diagonal. Thus, due to the wave-function collapse:

$$E_U(O) = 0 \tag{8}$$

Thus, $E_U(P_p) = E(UP_pU^\dagger)$ is the probability that the system is in the region of momentum p, for the state of the ensemble E_U. But the ensembles E and E_U are different, there is a physical transformation relating them.

Without collapse, we would have $E_U(O) = E(UOU^\dagger) \neq 0$ for operators O with null-diagonal, and we could talk about a common state of the ensemble E assigning probabilities to a non-commutative algebra. But the collapse keeps Quantum Mechanics as a Kolmogorov's probability theory, even when complementary observables are considered. We could argue that the collapse plays a key role in the consistency of the theory, as we will see below.

At first sight, our result that the wave-function is merely a parametrization of any probability measure, resembles Gleason's theorem [38][39]. However, there is a key difference: we are dealing with commuting projections and consequently with the wave-function, while Gleason's theorem says that any probability measure for all *non-commuting* projections defined in a Hilbert space (with dimension ≥ 3) can be parametrized by a density matrix. Note that a density matrix includes mixed states, and thus it is more general than a pure state which is represented by a wave-function.

We can check the difference in the 2-dimensional real case. Our result is that there is always a wave-function Ψ such that $\Psi^2(1) = \cos^2(\theta)$ and $\Psi^2(2) = \sin^2(\theta)$ for any θ.

However, if we consider non-commuting projections and a diagonal constant density matrix $\rho = \frac{1}{2}$, then we have:

$$\begin{cases} \mathrm{tr}(\rho \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]) = \frac{1}{2} \\ \mathrm{tr}(\rho \frac{1}{2} \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]) = \frac{1}{2} \end{cases} \tag{9}$$

Our result implies that there is a pure state, such that:

$$\mathrm{tr}(\rho \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]) = \frac{1}{2} \tag{10}$$

(e.g. $\rho = \frac{1}{2} \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$)

And there is another possibly different pure state, such that:

$$\mathrm{tr}(\rho \frac{1}{2} \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]) = \frac{1}{2} \tag{11}$$

(e.g. $\rho = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$)

But there is no ρ which is a pure state, such that:

$$\begin{cases} \mathrm{tr}(\rho \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]) = \frac{1}{2} \\ \mathrm{tr}(\rho \frac{1}{2} \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]) = \frac{1}{2} \end{cases} \tag{12}$$

On the other hand, Gleason's theorem implies that there is a ρ which is a mixed state, such that :

$$\begin{cases} \mathrm{tr}(\rho \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]) = \frac{1}{2} \\ \mathrm{tr}(\rho \frac{1}{2} \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]) = \frac{1}{2} \end{cases} \tag{13}$$

Gleason's theorem is relevant if we neglect the wave-function collapse, since it attaches a unique density matrix to non-commuting operators. However, the wave-function collapse affects differently the density matrix when different non-commuting operators are considered, so that after measurement the density matrix is no longer unique. In contrast, without the wave-function collapse, the wave-function parametrization of a probability measure would not be possible.

Another difference is that our result applies to standard probability theory, while Gleason's theorem applies to a non-commutative generalization of probability theory.

5. Free field parametrization in Bayesian inference and Statistical Mechanics

In Bayesian inference there is always a prior probability distribution, and there is no prior which is better for all cases[16]: we always have to make assumptions. For instance, if we choose a uniform prior in a continuous sample space then the maximum likelihood coincides with the maximum of the posterior (resulting from the prior once the data is taken into account). However, such maximum has null measure and thus no particular meaning. If we take a sample based in the posterior, we expect the sample to be somewhere near the maximum but never exactly at the maximum. Overfitting means that the inference process produced a sample which is inconsistent with our prior beliefs and thus could not be produced by Bayesian inference with an appropriate prior. Thus, overfitting means we need to choose another prior more consistent with our prior beliefs.

In Bayesian inference, the likelihood of the output data including correlations and variances fully determines the statistical model; then all statistical models can be seen as a particular case of one general statistical model for particular prior knowledge about the parameters of the general statistical model[100]. Thus, there is a probability distribution (the prior) of a probability distribution (the likelihood of the output data).

But functions (such as the likelihood of the output data) are in general infinite-dimensional spaces, so it makes sense to look for measures in infinite-dimensional spaces. While the Lebesgue measure cannot be defined in a Euclidean-like infinite-dimensional space[101][102], it is well known since many decades that a uniform (Lebesgue-like) measure of an infinite-dimensional sphere can be defined using the Gaussian measure and the Fock-space (the Fock-space is a separable Hilbert space used in the second quantization of free quantum fields)[9]. Such a space can parametrize (we call it the free field parametrization) the probability distribution of another probability distribution, which is exactly what we need: the infinite-dimensional sphere parametrizes the space of all likelihoods of the output data, while the wave-function whose domain is the sphere parametrizes a measure on the sphere.

In the free field parametrization, the uniform prior over the sphere defines a vector of the Hilbert space which when used as the prior for Bayesian inference with arbitrary data generates an orthogonal basis for the whole Fock-space. Such basis is related with a point process, with the number of points with a given feature corresponding to the number of modifications to the uniform prior (in the part of the sample space corresponding to such feature). Since Bayesian inference with any other prior can be seen as a combination of the results of different Bayesian inferences with the uniform prior for different data (eventually an infinite amount of data for the cases with null measure), then the uniform prior in the free field parametrization is in many cases (not in all cases[16]) appropriate for Bayesian inference in the absence of any other information.

Moreover, the frequentist view of probabilities is viable because many statistical problems can be solved by a Bayesian model where the parameters defining probabilities are inferred from Binomial processes with uniform prior (for instance, the Binomial converges to the Normal distribution under some conditions), which implies that it is often possible to incorporate prior knowledge efficiently by modifying the counting of events. This does not exclude of course just using an arbitrary prior (with methods such as preconditioned Crank–Nicolson MCMC for high-dimensional problems), but that might be computationally inefficient in some cases. In both cases (counting or MCMC), the free-field parametrization is crucial to the implementation of many constrained problems, this allows the uniform prior to be used in many cases where we are just interested in a subset of the sample space with null measure (as we discuss elsewhere).

Thus, ensemble forecasting[103]—with many applications and where an ensemble of different statistical models is built—can be seen as sampling from a Bayesian posterior corresponding to a particular Bayesian prior which selects which models constitute the ensemble.

This leads us to classical statistical mechanics: whatever system we study, we need a probability measure on the phase-space of such system corresponding to an ensemble which defines a Bayesian prior. When the system we are studying is itself an ensemble, and thus it is defined by another

probability distribution, then we can use the free field parametrization.

Quantum statistical mechanics is not different, since the Hilbert space on which the density matrix lives is merely a parametrization of a probability, due to the wave-function collapse. When the density matrix is not pure, then the probability defining the ensemble is a joint probability distribution of the initial and final states of the systems. We can always define the density matrix through a diagonal operator rotated by a unitary operator, with the diagonal operator defining the marginal probability of the initial state and the unitary operator defining the conditioned probability of the final state conditioned by the initial state.

We can also consider a statistical model which acts in the prior wave-function in a non-linear way, by redefining the Hilbert-space of the prior as a tensor product of two Fock-spaces and the corresponding statistical model as unitary (and thus linear), this is in principle possible for most non-linear actions we can think of, because a conditional probability measure includes all such actions. Note that every formalism has limits, so there will be always some exotic statistical model which cannot be redefined or at least not in a useful way.

This allows us to treat (classical or quantum) statistical processes as classical dynamical systems (where the system is itself an ensemble).

6. Free field parametrization for finite sample spaces and the spin-statistics theorem

The free field parametrization defined in the previous section depends on the fact that the sphere is infinite dimensional corresponding to a sample space with infinite degrees of freedom. But often we are interested in a tensor product of sample spaces, some of which have finite degrees of freedom.

Of course, we can always treat finite degrees of freedom as a subset of infinite degrees of freedom. But there might be an alternative as well.

We consider the sample space Z_2 with 2 degrees of freedom and fermionic creation and destruction operators for the pair of degrees of freedom. Then, the corresponding Fermionic Fock-space parametrizes the probability space.

If we consider now the sample space $Z_2 \times Z_2$, then naively there is a possibility of non-null products of creation operators corresponding to each pair of degrees of freedom. This is not a free field parametrization.

We need the space $\Gamma^s(L^2(\mathbb{Z}_2)) \otimes \Gamma^a(L^2(\mathbb{Z}_2))$ [104][105]. The bosonic free field corresponds to operators with bosonic number, while the fermionic free field corresponds to operators with fermionic number,

without the need for products of free fields. Instead of $Z_2 \times Z_2$ we could easily repeat the exercise for $Z_2^n \times \mathbb{R}^m$.

The advantage is that there are certain symmetry transformations (such as the rotation for different spins in the presence of a flavor symmetry[106]) which can only be represented by bosonic free fields while others can only be represented by fermionic free fields. This is what leads to the spin-statistics theorem[106] or to the fact that ghost fields verify an unusual spin-statistics correspondence. So, we better have an alternative to the bosonic free field parametrization discussed in the previous section.

7. Symmetries and unitary representations

Any conditional probability measure (in a standard measure space) which is function of a one-dimensional parameter (which we call time, without loss of generality) can be parametrized by a quantum process: a time-ordered product integral of unitary operators, defined by a time-dependent Hamiltonian operator, at the cost of choosing a larger sample space such that the corresponding isometry can be unitary. Such parametrization using a time-dependent Hamiltonian is particularly useful in systems where the words "time-dependent" have a clear meaning. For instance, a molecule in an ideal gas is constrained both theoretically (through its position operator and the effect that the Poincare group has on it) and experimentally to have conditional probability measures defined by a time-independent Hamiltonian operator; also, in a deterministic time-evolution, a time-independent classical Hamiltonian corresponds to a time-independent Hamiltonian operator. Nevertheless, such parametrization is always possible in a standard measure space, including for Markov processes; therefore quantum processes are not intrinsically exotic or weird, they are simply more general than Markov processes.

Moreover, as it was discussed in the previous two sections, a non-linear transformation of the wave-function can be redefined as a unitary operator (and thus linear) on a Fock-space. This is in principle possible for most non-linear transformations we can think of, because the Hilbert space is infinite-dimensional.

Note that Markov processes cannot produce an arbitrary function of time, because there is an ordering (related with the concept of entropy) with respect to which all continuous-time Markov processes are monotonic[79]. Moreover, a Markov process is a semigroup which is much harder to translate into an infinitesimal structure than the unitary operators which form a group, as discussed in reference[107]:

> "*The basic feature of Lie theory is that of using the group structure to translate global geometric and analytic problems into local and infinitesimal ones. These questions are solved by Lie algebra techniques which are essentially linear algebra and then translated*

back into an answer to the original problem. Surprisingly enough it is possible to follow this strategy to a large extend also for semigroups, but things become more intricate. Because of the missing inverses one has not only to deal with linear algebra but also with convex geometry at the infinitesimal level."

A quantum process can be classified by all unitary transformations of the wave-function that may occur at each time (called canonical transformations). When a subset of canonical transformations forms a group, such subset is always a linear unitary representation of a group which is then called symmetry group and the canonical transformations are then called symmetry transformations.

In conclusion, conditional probability measures defined by a linear unitary representation of a symmetry group are certainly natural and common.

8. Conservative transformations

A conservative transformation U is a canonical transformation that preserves the measure μ of the commutative von Neumann algebra $L^\infty(X,\mu)$. Note that the measure μ is an element of the dual space of $L^\infty(X,\mu)$ and thus an element of the spectrum of U, but not necessarily of the space $L^1(X,\mu)$ pre-dual of $L^\infty(X,\mu)$.

If there is a locally compact abelian group G with a transitive free action on the space X and a conservative transformation U conserves the equation $G=1$. Then there is a unitary time-evolution $U(t)=e^{iHt}$ with a self-adjoint operator H such that $[[H,P_A],P_B]=0$ where $A,B\subset X$ are any events and $P_A\in L^\infty(X,\mu)$ is a projection-valued measure, which is conservative. We just need to write $H=\sum_{j=1}^{n} p_j[H,x_j]+[H,x_j]p_j$, then:

$$U(t)gU^\dagger(t)|_{G=1}=1 \tag{14}$$

For all $g\in G$. Note that $|_{G=1}$ is a right ideal of the algebra of bounded operators on the Hilbert space and $[x_j,p_k]=i\delta_{jk}$, $[x_j,P_A]=0$ and p_j are the generators of the abelian group G. The operator H is uniquely defined by $[H,x_j]$ up to a constant.

In particular, a deterministic automorphism (defined in the next section) can be made conservative without modifying its effect on $L^\infty(X,\mu)$, if there is a locally compact abelian group G with a transitive free action on X. In that case, the deterministic conservative automorphism can be seen as just a change of variables. Note that any unitary automorphism can be defined as a time-evolution, for some Hamiltonian and time. However, a conservative time-evolution is not necessarily deterministic.

A particular case of a deterministic conservative automorphism is a measure-preserving transformation U, defined such that U is a change of variables with unit Jacobian.

9. Deterministic transformations

Crucially, the symmetry transformations include all the deterministic transformations, which will be defined in the following. Thus, the symmetry transformations are a generalization of the deterministic transformations.

A deterministic transformation $P_A \to U^\dagger P_A U$ where P_A is a projection operator and $A \subset X$ is any event, is such that $L^\infty \to U L^\infty(X,\mu) U^\dagger = L^\infty(Y,\nu)$, that is all (necessarily diagonal) elements of the abelian algebra are transformed in diagonal elements of another abelian algebra. Then, the corresponding conditional probability density is diagonal, and it is the Radon–Nikodym derivative of the measure μ with respect to ν[108].

In the particular case of a one-parameter continuous group, the deterministic transformations must be automorphisms. Thus, a deterministic automorphism $P_A \to U^\dagger P_A U$ is such that $U^\dagger P_A U$ commutes with $L^\infty = L^\infty(X,\mu)$, where P_A is a projection operator and $A \subset X$ is any event. Since $L^\infty(X,\mu)$ is maximal abelian, and it is a commutative unital C*-algebra[109] then $U^\dagger P_A U \in L^\infty$ can be written as a function of the spectrum of L^∞ (Gelfand representation). Such function is invertible because any element of the spectrum of L^∞ can be written as a function of the spectrum of $U^\dagger L^\infty U$ and the other way around. Thus, the points of the spectrum of L^∞ are mapped one-to one to other points of the spectrum of L^∞, irrespective of the probability distribution. Since we have effectively a deterministic case, then we call deterministic transformation to such transformation.

We conclude that an automorphism U is deterministic if and only if P_A and $U P_B U^\dagger$ commute for all events A, B. Thus, the complementarity of two observables (e.g. position and momentum) is due to the random nature of the symmetry transformation relating the two observables. This clarifies that probability theory has no trouble in dealing with non-commuting observables, as long as the collapse of the wave-function occurs. Note that Quantum Mechanics is not a generalization of probability theory, but it is definitely a generalization of classical statistical mechanics since it involves non-deterministic symmetry transformations. For instance, the time evolution may be non-deterministic unlike in classical statistical mechanics.

10. Ensemble forecasting allows the approximation of a non-linear infinite-dimensional model by a direct sum of linear models with few variables

> *The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve.* — Eugene Wigner (1960) [110]

There is no uniform countable measure. Thus, the rationals are not enough for Probability Theory. A standard probability space (which has countable and continuous measures) seems irreducible.

However, the rules to update the probability measure are not at all obvious, because we can always consider a probability space of probability spaces (ensemble forecasting). Assuming a quantum time-evolution (that is, linear) for the ensemble forecasting leads to no relevant restriction on the time-evolution of the probability spaces (it may be non-linear, although every formalism has a domain of application, so we still need to exclude a pathological time-evolution).

The Bayes rule applies only when we do a measurement, but this is a tautology. Since a measurement is not defined, then a measurement is applying the Bayes rule to update a probability: destroy the uncertainty (and thus the probability distribution) by converting probabilities into events, unpredictably and reproducing the probability distribution in the limit of infinite measurements. The limit of infinite measurements creates a joint probability distribution, with the results of the infinite measurements and the probability update after each measurement took place. It is not different from an unbiased sampling process taken over infinite time. Note that slightly biased sampling processes are common, but these can be converted to unbiased by enlarging the probability space (using ensemble forecasting, for instance).

Using the free field parametrization of ensemble forecasting[103] (see Sections 5, 6), we can diagonalize the quantum time-evolution, decomposing a non-linear infinite-dimensional model in a direct integral of linear models with only one boolean variable. This only solves part of the problem, since a direct integral can still be uncomputable. We have to show how to decompose an eventually continuous spectrum of the quantum time-evolution into a direct sum of sufficiently small intervals of energy, each of these intervals described by few variables. This is not obvious, since any continuous interval, no matter how small, is still uncountable. We will prove this in another article. But it is not unreasonable to assume it is possible to do it (for instance, adapting the Krylov subspace methods for computations of matrix exponentials[71][72]), as we do from now on.

Since the wave-function defines an ensemble of deterministic transformations (that is, functions), we can easily study and calculate each possible function when we are dealing with few variables.

This justifies why deterministic logic and deterministic mathematics find application in a world full of uncomputable functions.

This gives a mathematical definition on the renormalization process, which has been empirically observed to be possible to do in a wide range of problems (related or unrelated to Physics)[111][112]. That is, for reasonable initial conditions (because we do not have access to an infinite range of energies in the real-world, using man-made experimental apparatus) we can always approximately predict the time-evolution of a system using a model with few variables, using different unrelated models for different energy ranges.

The renormalization process is universal, and it applies to any statistical model (which defines a unitary transformation, playing the role of a quantum time-evolution, the "time" is abstract here). Thus, it allows efficient machine learning using Bayesian priors in the few most relevant variables, helping align models and incorporate prior knowledge.

11. Conditions for the classical limit of Quantum Mechanics

In the previous section, the few variables are related with the energy-momentum space, not with the coordinate space. Here we discuss the evidence about the necessary conditions to discard probabilities in the coordinate space, recovering Classical Mechanics. Note that these conditions are not rigorous, that is, often it works but not always, thus Classical Mechanics is never enough at any energy scale, Quantum Mechanics is always needed.

As far as we know the world is compatible with a continuum space-time, with rational functions of rational variable obtained through step functions and the partition of a continuum space-time. For instance, we can "see" that we have "five" fingers in our hand, but exactly why we can do this in a continuum space-time is not obvious. The mainstream mathematics depends on the fact that step, polynomials and smooth functions are relevant for applications. But it is important to look to the existing evidence on why these functions are relevant for applications.

Step, polynomial or smooth functions are dense in the L^2 measure. Any bounded function in a compact domain, can be approximated in the L^2 measure (that is, the average over a compact domain, which is common to do in classical physics when we partition a large domain in numerical approximations) by these calculable functions. That does not make the original function calculable, since often we do not want to average over a compact domain, specially when studying microscopic phenomena such as the double slit experiment (where it makes a macroscopic difference to open one or two microscopic slits for a particle to go through). This is consistent with the fact that calculable functions are useful in macroscopic phenomena, but not so much in microscopic phenomena, where only the probabilities admit calculable approximations, because they still can be approximated in

the L^2 measure.

Note that Chaos and singularities in Ordinary differential equations are phenomena of computable functions (computable at least for a small enough but finite time, thus they still admit the partition of a large domain in numerical approximations), despite that these are clues for the fact that many functions are uncomputable. Uncomputable functions are uncomputable for any finite time, unlike Chaos or singularities.

Gauge symmetry and dissipative dynamics in probability spaces

Abstract

We show that the wave-function parametrization allows us to implement gauge symmetries as exact constraints in a standard probability space without attributing to the constrained space null-measure. The wave-function parametrization of classical dissipative dynamics is also studied.

Introduction

Egison is a programming language based on patterns. The documentation[113] is a PhD thesis and it says:

> "Non-free data types are data types whose data have no canonical forms. For example, multisets are non-free data types because the multiset {a, b, b} has two other equivalent but literally different forms {b, a, b} and {b, b, a}. We developed the pattern-match facility for non-free data types."

Using the concept of non-free data type we can define manifolds or virtually any other mathematical object in the language Egison (thus also only conceptually or mathematically). This data type is called "non-free" because it is built from canonical forms (which are "free") by adding "patterns" which are actually gauge symmetries (which are a particular case of constraints, as we will see). For example, the constraint {a, b, b} = {b, a, b}. Since numbers always have canonical form, this is valid for defining manifolds not only symbolically but also numerically. The conclusion is that manifolds can be very useful in some cases, but if we are looking for a practical way to represent data on a computer then "patterns"/gauge symmetries are simpler and more general. Even if we only want manifolds, a good way to represent them on a computer (symbolically or numerically) is using "patterns"/gauge symmetries.

Now consider a continuous sample space (say, the real line). In this article we will show that quantum constraints allow us to implement exact gauge symmetries in a separable probability space without attributing to the constrained space null probability measure. Note that we can always define a wave-function by taking the square-root of the probability measure, the Koopman-von Neumann version of classical statistical mechanics[42][60][61][62] defines classical statistical mechanics as a particular case of quantum mechanics where the algebra of observable operators is necessarily commutative (because the time-evolution is deterministic).

In the next section we will study a simple example which shows another application of the wave-function parametrization of a probability measure, in the case of classical dissipative dynamics.

Wave-function parametrization of dissipative dynamics

The existence of a probability measure allows us to define a quantum Hamiltonian and quantum constraints, which are more general than the classical ones because they do not need to commute with the variables defining the sample space.

As a simple example of a motion which cannot be derived from a Lagrangian (and thus also not from a classical Hamiltonian without enlarging the system) we can cite[12] the oscillations of two classical coupled oscillators with different frequencies and different damping constants:

$$\begin{cases} \ddot{x}_1 + \lambda_1 \dot{x}_1 + \omega_1^2 x_1 - c_2 x_2 = 0 \\ \ddot{x}_2 + \lambda_2 \dot{x}_2 + \omega_2^2 x_2 - c_1 x_1 = 0 \end{cases} , \tag{15}$$

where c_1 and c_2 are constants. Nevertheless, if we consider a probability measure for this classical system then we can define a quantum Hamiltonian for it. Since this system is dissipative, it cannot have a conserved Energy but the pendulums do not disappear and thus the probability is conserved. Another example is when a classical Hamiltonian can be defined but it is much less useful than a quantum Hamiltonian, due to non-linear dynamics[13][2].

The quantum formalism is the most general formalism whenever there is a conserved probability, thus it is expected that it can be applied even when classical Hamiltonians (due to energy dissipation[12], or non-linear dynamics[13][2]) or classical exact constraints (due to the fact that the constrained space has null probability measure) cannot.

Gauge transformations, constrained systems and conditioned probability

Since in standard measure spaces it is always possible to define regular conditional probabilities[90], then in principle it is always possible to implement exact constraints in a separable probability space without attributing to the constrained space null probability measure.

But often the explicit solutions of the constraint equations cannot be found[114], as it is required to define conditional probabilities. In those cases, there is still the possibility of defining the probability measure of the constrained space as a pushforward measure from the unconstrained to the constrained space. Then, the constraint equations would not need to be solved explicitly, at the cost that the constraints are somehow special.

This includes the case of parametrizations, since these are surjective but often there are two or more point in the space of parameters which corresponds to the same point in the parametrized space[115], which is harmless when we evaluate functions at a point but it leads to problems when defining a measure for the parametrized space since the same point cannot be counted twice. As a matter of principle[116], for all parametrizations we can define a gauge group transforming points in the parameter space without modifying the corresponding point in the parametrized space. Thus all parametrizations are solutions to constraint equations requiring gauge invariance.

First-class constraints are the generators of a unitary gauge group (all operators of the group are constrained to be the identity operator). In case the gauge group is infinite-dimensional, there is some ambiguity in its definition [117][118]. In this article we will only address the finite-dimensional case, the infinite-dimensional case was studied in another article by the same author.

Note that it is a subalgebra of the commutative von Neumann algebra that is gauge-invariant and not the Hilbert space. In fact, in many cases it would be impossible for a cyclic state of the Hilbert space to be gauge-invariant, as it was noted long ago:

> "*So we have the situation that we cannot define accurately the vacuum state. We therefore have to work with a standard ket* $|S\rangle$ *which is ill-defined. One can, however, do many calculations without using the accurate conditions* [vacuum is gauge invariant] *and the successes of quantum electrodynamics are obtained in this way.*"
>
> Paul Dirac (1955)[119]

Indeed, there are some symmetries of the commutative von Neumann algebra of operators which necessarily the probability measure cannot have (see also [120]), since the probability measure is normalized to 1. For instance, consider an infinite-dimensional discrete basis $\{e_k\}$ of an Hilbert space (indexed by the integer numbers k) and the gauge symmetry group generated by the transformation

$e_k \to e_{k+1}$ (translation). There is no normalized wave-function (and thus no probability measure) which is translation-invariant. On the other hand, there is a translation-invariant subalgebra of the commutative von Neumann algebra (starting with the identity operator) where the expectation value of each operator is the same for an equivalence class of wave-functions defined up to a gauge symmetry.

The process of representing each equivalence class of operators (in a commutative von Neumann algebra) defined up to a gauge symmetry with at least one operator (in another commutative von Neumann algebra) is called gauge-fixing. We define gauge-fixing as complete whenever it crosses at most once each equivalence class, i.e. whenever there is no remnant gauge symmetry in the equivalence classes. We define gauge-fixing as unconstrained whenever the gauge generators are necessarily excluded from the commutative von Neumann algebra and thus do not impose constraints on the spectrum of the commutative algebra.

The Dirac brackets require the gauge-fixing to be both unconstrained and complete (as if the gauge symmetry could be eliminated), which is not possible in general due to the Gribov ambiguity [121]. Note that the Gribov ambiguity is not solved by the BRST cohomology [122]:

> "*Being gauge invariant, the BFV-PI necessarily reduces to an integral over modular space, irrespective of the gauge fixing choice. Nevertheless, which domain and integration measure over modular space are thereby induced are function of the choice of gauge fixing conditions. The BFV-PI is not totally independent of the choice of gauge fixing fermion* Ψ*.*"

Thus, the generators of the gauge group cannot be interpreted literally, that is, as mere constraints in a too large phase-space whose "non-physical" degrees of freedom need to be eliminated. Moreover, this picture makes little sense in standard measure spaces, where the dimension of the phase-space is not fixed *a priori*. Moreover, this picture makes little sense in infinite-dimensions: in gauge field theory, the gauge potentials can be fully reconstructed from the algebra of gauge-invariant functions, apart from the gauge potential and its derivatives at one specific arbitrary point in space-time [123]; thus the number of "non-physical" degrees of freedom would be finite at most, which clearly does not match with the uncountable infinite number of constraints.

If we consider instead a commutative von Neumann algebra and its spectrum[15], such that any non-trivial gauge transformation necessarily modifies any point of the spectrum while conserving the commutative von Neumann algebra (e.g. the commutative von Neumann algebra with spectrum given by the gauge field A_μ which is a function of space-time), then such commutative von Neumann algebra is one example of an incomplete unconstrained gauge-fixing. The gauge-fixing is incomplete

[15]The word *spectrum* is used in the sense of the Gelfand representation: there is an isomorphism between a commutative C*-algebra A and the algebra of continuous functions of the spectrum of A.

because the remnant gauge symmetry is a faithful (thus non-trivial) representation of the original gauge symmetry. Such commutative algebra has the crucial advantage that the gauge generators are necessarily excluded from the algebra, so that it can be used to define a separable Hilbert space compatible with the gauge group because the expectation value of any operator of the commutative algebra which commutes with the gauge generators is the same at each equivalence class, saving us the need to eliminate the "non-physical" degrees of freedom. Note that it is crucial that the von Neumann algebra used in the gauge-fixing is commutative.

Even when a solution exists, setting non-abelian gauge generators to zero in the wave-function would require to solve a non-linear partial differential equation with no obvious solution [124][125][126][127]. On the other hand, since bounded commuting normal operators can always be simultaneously diagonalized[63] there is always one basis where the gauge unitary transformations are a function of the spectrum and (if the gauge-fixing is unconstrained) there is another basis where the gauge unitary transformations are not a function of the spectrum and thus there are no constraints. The expectation values of the gauge-invariant operators in both basis are related by the unitary transformation relating both basis. Thus, for any wave-function for the unconstrained basis there is one corresponding wave-function for the constrained basis which gives the same expectation values for the gauge-invariant operators. This establishes a correspondence between gauge generators as a particular case of constraints, that is, whenever there is a basis where the gauge generators do not depend only on the spectrum then such basis effectively solves the constraint equations when at least one solution exists. We now just need to check that the constrained spectrum is not empty: for a locally compact gauge group (a Lie group, for instance), a constant measure (Haar measure) always exists which allows to create a functional which is gauge invariant.

Finally, since the gauge-invariant operators commute with all diagonal operators in the unconstrained basis, then the diagonal operators in the unconstrained basis may modify the value of the constraint operator in the constrained basis but they conserve the values of the gauge-invariant operators at every point of the spectrum. Thus, we can define a measurable function which projects the spectrum in the constrained basis to another spectrum where the constraints are null (built using the Haar measure, which is a limit of the average of the action of the constraint operators on the diagonal operators in the unconstrained basis). Then, the pushforward measure using such measurable function implements the exact constraints in a separable probability space without attributing to the constrained space null probability measure (which is our goal).

Note that it suffices to constrain to zero the Casimir operators of the (eventually non-commutative) Lie algebra of constraints, this imposes the constraints without the need for the constraints to be part of the commutative von Neumann algebra, only the Casimir operators are included in the commutative algebra.

The gauge symmetry is different from anomalies. An anomaly is a failure of a symmetry of the wave-function to be restored in the limit in which a symmetry-breaking parameter (usually introduced due to the mathematical consistency of the theory) goes to zero. On the other hand, in the case of a gauge symmetry, there is no way to introduce a symmetry-breaking parameter because we only consider expectation values of gauge-invariant operators, thus we can never observe an anomaly.

Quantization of a classical Gauge Mechanics system

An example of the quantization of a classical system is the following. The Hilbert space is $L^2(\mathbb{R}^2 \times \mathbb{Z}_2))$ [104][105]. The $\mathbb{Z}_2$ corresponds to the ghost degree of freedom (k) which can be $k = 0$ or $k = 1$. The $\mathbb{R}^2$ degrees of freedom correspond to one complex operator (ϕ).

The Hamiltonian for the gauge mechanics quantum theory has the same form as the classical Hamiltonian Action:

$$\begin{aligned} & H = -\frac{1}{2}\left[\frac{1}{2}|\pi|^2 + V(|\phi|^2) + \text{ h.c.}\right] \\ & \Omega = (\pi\phi + \pi^*\phi^*)\psi^\dagger \\ & \psi^\dagger\{\Psi\}(\phi, \phi^*, k) = \Psi(\phi, \phi^*, 1)\delta_{k0} \\ & \psi\{\Psi\}(\phi, \phi^*, k) = \Psi(\phi, \phi^*, 0)\delta_{k1} \\ & [\phi, \pi] = \phi\pi - \pi\phi = i \\ & [\phi, \pi^*] = \phi\pi^* - \pi^*\phi = 0 \\ & \{\psi, \psi^\dagger\} = \psi\psi^\dagger + \psi^\dagger\psi = 1 \end{aligned} \tag{16}$$

where (ϕ, π) is a complex field/conjugate momentum field (correspondingly). $V(|\phi|^2)$ is a functional (a polynomial or other) on $|\phi|^2$. The BRST charge is Ω, where ψ is the ghost field.

The gauge generator in the gauge mechanics system is the charge operator:

$$Q = \pi\phi + \pi^*\phi^* \tag{17}$$

The wave-function needs not be gauge-invariant, just the operators corresponding to observables need to commute with the gauge generator. We need to separate the gauge generator from the gauge-invariant algebra, that is, not only the gauge-invariant algebra must commute with the gauge generator, but also the gauge generator cannot be included in the gauge-invariant algebra. This is guaranteed by unconstrained gauge-fixing: the gauge-invariant algebra is a sub-algebra of the commutative von Neumann algebra with spectrum given by the operators ϕ, ϕ^*, k, such that

the generator necessarily modifies any point of the spectrum up to a set with null measure. The conjugate fields π, π^* or the gauge generator are not part of the gauge-invariant algebra, since they do not commute with the corresponding operators ϕ, ϕ^* which are included in the commutative algebra.

Balancing discretization and locality

There is no doubt that any numerical calculation must involve some form of discretization of continuum sets. And there are many efficient ways to do it, including ways (wavelets) that account for multiple relevant scales/resolutions. But all of them destroy locality, which in many applications (in particular involving Physics) is also important[115].

For instance, in the definition of a trajectory of an object under uncertainty in the context of autonomous driving, it is important not only the position at each time but also the velocity at each time. Such velocity at each time is locally constrained by the position at each time and thus it is not a fully independent variable. The discretization turns these local constraints into global constraints which are much more complex[115] and then it is hard to add uncertainty after the discretization in a coherent way[61].

Of course, sometimes complex approximations are unavoidable but these should be kept somehow under control and the best way to do it is to have a profound knowledge of the model without approximations as a reference, so that we can distinguish what are inherent features of a model and what is noise introduced by an approximation. Thus, it is advantageous to build a continuous model whenever locality is relevant. But a continuous model with a differentiable structure and under uncertainty was never built before, even for something as simple as the trajectory of an object which includes position and velocity over time.

The main difficulty lies in defining a probability measure when the velocity is constrained by the position at each time, since the subset defined by the constraints has null measure when considering the Gaussian measure for the position and (unconstrained) velocity.

Reconstructing the classical trajectory of any isolated quantum system*

[128][129][130]

Introduction: Comparing the time evolution with a stochastic process

Quantum Mechanics is not a generalization of probability theory, but it is definitely a generalization of classical mechanics since it involves non-deterministic transformations to the state of the system. For instance, the time evolution may be non-deterministic unlike in classical mechanics.

There are three major metaphysical views of time [131]: presentism, eternalism and possibilism. The possibilism consists in considering the presentism for the future and the eternalism for the past, so it is inconsistent with a time translation symmetry. The presentism view coincides with the Hamiltonian formalism of physics, that the state of the system is defined by a point in the phase space. When the time evolution of the system is deterministic it traces a phase space trajectory for the system, however the definition of the state of the system does not involve time, i.e. only the present exists[16]. The eternalism view coincides with the Lagrangian formalism of physics, that the state of the system is defined by a function of time. When the time evolution of the system is deterministic, this function of time coincides with the phase-space trajectory of the classical Hamiltonian formalism and so which metaphysical view of time we use is irrelevant from an experimental point of view (in the deterministic case).

But when the time-evolution of the system is non-deterministic, we may have a hard time studying the time-evolution from the Lagrangian formalism and/or eternalism metaphysical view. The key fact about Quantum Mechanics which makes it incompatible with the eternalism/Lagrangian

[16]The presentism view is compatible with (non-quantum) special or general relativity\cite{diracdynamics,adm}. There are difficulties with the quantum versions of relativity (more with general relativity than with special relativity~\cite{wigner}), but these difficulties have little to do with the presentism as we will discuss in forthcoming articles.

point of view is that the time-evolution is not necessarily a stochastic process, i.e. there is not necessarily a collection of random events indexed by time[17]. We only apply one non-deterministic transformation of the state of the system, however there are many different transformations we can choose from and the set of choices is indexed by a parameter we call time, which is fine from the presentism/Hamiltonian point of view since only the present exists.

Note that a random experiment always involves a preparation followed by a measurement. For instance, we shake a dice in our hand and throw it over a table until it stops (preparation), then we check the position where it stopped (measurement).

If we just throw the dice without shaking our hand, the probability distribution for the measurement outcome is different than if we shake our hand. There is nothing mysterious about this: two different preparations lead to two different probability distributions. Whether or not we actually do the measurement does not change anything, what changes the probability distribution is the preparation.

Then we can think about a preparation which is function of an element of a symmetry group, for instance translation in time. From the point of view of probability theory or experimental physics, this is a valid option. However, it is important to note that this preparation function of time is not a stochastic process in time. A stochastic process in time is a set of random experiments indexed by time, while in the preparation which is function of time we have a single random experiment dependent on the parameter time. As an example, consider a) throwing the dice 10 times, one time per minute during 10 minutes and b) shake the dice in our hand for a number of minutes T between 0 and 10 and then throw the dice once. The preparation in b) is dependent on the time parameter T, while in a) the time selects the one of the many identically prepared experiments which was done at the selected time.

Note that the experiments a) and b) above are different but can be combined: we could do many random experiments, each of them would be dependent on a parameter. This fact is important in Section 12.

In the remaining of this section, we comment on conditioned probability and the random walk. It is well-known that quantum mechanics can be described as the Wick-rotation of a Wiener stochastic process [55]. In other words, the time evolution in Quantum Mechanics is a Wiener process for imaginary time. This is the origin of the Feynman's path integral approach to Quantum Mechanics and Quantum Field Theory.

Since the Wiener process is one of the best known Lévi processes—a Lévi process is the continuous-

[17]It is possible to insist that the time-evolution is always a stochastic process, but then we must consider a different theory than Quantum Mechanics, e.g. described by the Lindblad equation instead of the Schrödinger equation~\cite{WeinbergLindblad}.

time analog of a random walk—this fact often leads to an identification of Quantum Mechanics with a random walk. In particular, it often leads to an identification of the probabilities calculated in Quantum Mechanics with conditioned probabilities—the next state in a random walk is conditioned by the previous state.

Certainly, the usefulness of group theory is common to both a random walk and to Quantum Mechanics and this unavoidably leads to similarities between a random walk and Quantum Mechanics. However, imaginary time is very different from real time and thus the probabilities calculated in Quantum Mechanics are not necessarily conditioned probabilities in a random walk.

In order to relate a random walk (or any other stochastic process) with Quantum Mechanics correctly, we need the probability distribution for the complete paths of the random walk. Then, we can use a wave-function parametrization of the probability distribution for the complete paths of the random walk. Finally, we can apply quantum methods to this wave-function. The result is a Quantum Stochastic Process [132], which is not a generalization of a stochastic process due to the wave-function collapse, but merely the parametrization of a stochastic process with a wave-function.

Time translation is a stochastic process if and only if it is deterministic

Now we are able to prove one of the main results of this paper, namely that there is a group action of a Wigner's symmetry group on the probability distribution for the state of a system, if and only if the Wigner's symmetry group transforms deterministic (probability) distributions into deterministic (probability) distributions. A corollary is that time translation in Quantum Mechanics is a stochastic process if and only if it is deterministic. This mathematical fact is overlooked by the assumptions of both the Bell's theorem and the Einstein-Podolsky-Rosen (EPR) paradox.

As it was discussed in Section 3, Wigner's theorem [133][91][92] implies that the action of a symmetry group on the wave-function is necessarily linear and unitary. In Section 4, we showed that the action of a symmetry group on the wave-function is deterministic if and only if P_A and $U_g P_B U_g^\dagger$ commute for all events A, B and for all the elements g of the group, where P_A is a projection-valued-measure.

This means that U is a deterministic transformation if and only if $U_{la} U^*_{ma} = 0$ for all a, l, m such that $l \neq m$.

Now we check the necessary and sufficient conditions for the action of a symmetry group on the wave-function to correspond to an action on the corresponding probability distribution.

That is, if we start with some probability distribution $\mathrm{diag}(\rho_1)$, then the action of each element g of the group on the wave-function will produce (after the collapse) a different probability distribution $\mathrm{diag}(\rho_g)$. The composition of the actions of two group elements g, h on the probability distribution is given by the succession of the two random experiments corresponding to g and h: $P(A) =$

$\mathrm{tr}(\mathrm{diag}(\rho_g)U_h P_A U_h^\dagger)$.

However, Wigner's theorem [133][91][92] implies that the action of a symmetry group on the wave-function is necessarily linear and unitary, thus $P(A) = \mathrm{tr}(\rho_g U_h P_A U_h^\dagger)$.

Thus there is a group action of the symmetry group on the probability distribution if and only if $\mathrm{tr}(\mathrm{diag}(\rho_g)U_h P_A U_h^\dagger) = \mathrm{tr}(\rho_g U_h P_A U_h^\dagger)$ for any pure density matrix ρ_g and any event A and group element h.

The equality above is equivalent to $\sum_{k,b:k\neq b} U^*_{ka}\Psi_k\Psi^*_b U_{ba} = 0$, where U_{ba} are the elements of the matrix U_h. We can see that if U is a deterministic transformation, then the equality is satisfied, since $U^*_{ja}U_{jl} = 0$ for all a, l, j such that $a \neq l$. On the other hand, if U is a non-deterministic transformation then for some a, l, m such that $l \neq m$, we have $U^*_{ma}U_{la} \neq 0$. Then for $\Psi_k = \frac{1}{\sqrt{2}}(\delta_{km} + \delta_{kl})$, we get $\sum_{k,b:k\neq b} U^*_{ka}\Psi_k\Psi^*_b U_{ba} = U^*_{ma}U_{la} \neq 0$, i.e. there is no group action of the symmetry group on the probability distribution.

Symmetries as irreversible processes

The concept of (ir)reversible process from thermodynamics also needs a careful discussion in quantum mechanics. A non-deterministic symmetry transformation, when acting on a deterministic ensemble increases the entropy of the ensemble after the wave-function collapse and therefore must be an irreversible transformation. Yet, a symmetry transformation always has an inverse symmetry transformation, because it is included in a symmetry group, so it must be considered reversible in some sense.

The way out of this apparent contradiction is the role of time in the quantum formalism, which was discussed in Sections 11 and 12. In the ensemble interpretation, the individual system is entirely defined by a standard phase-space, which implies that the time plays no fundamental role in quantum mechanics nor in classical Hamiltonian mechanics. Then, time-evolution in quantum mechanics is not a stochastic process unless it is deterministic. Therefore, there is not a probability distribution for each time (or for other parameter corresponding to the symmetry group).

If we consider a stochastic process with only two probability distributions corresponding to the initial and final times, then the complete symmetry transformation is irreversible (if it is non-deterministic and it acts in a deterministic ensemble). However, this does not imply that it is a "bad" symmetry, because no stochastic process can be defined in between the initial and final times. On the other hand, if the symmetry group contains only deterministic transformations then a stochastic process can be defined in between the initial and final times and such process is reversible, as expected.

Quantum Mechanics is EPR-complete

The Einstein-Podolsky-Rosen (EPR) main claim [43] (namely, that Quantum Mechanics is an incomplete description of physical reality), is defended by reducing to absurd the negation of the main claim, i.e. by reducing to absurd that position (Q) and momentum (P) are not simultaneous elements of reality. In the EPR article it is stated: "*one would not arrive at our conclusion if one insisted that two or more physical quantities can be regarded as simultaneous elements of reality only when they can be simultaneously measured or predicted.*[...] *This makes the reality of P and Q depend upon the process of measurement carried out on the first system, which does not disturb the second system in any way. No reasonable definition of reality could be expected to permit this.*"

The reduction to absurd of the negation of the claim, could only be a satisfactory argument if the claim itself (namely, the quantities position and momentum of the same particle are simultaneous elements of reality, despite they cannot be simultaneously measured or predicted) would not be absurd as well. But the claim itself raises eyebrows to say the least, once we remember that (in Quantum Mechanics, by definition) measuring the position with infinite precision completely erases any knowledge about the momentum of the same particle.

In Quantum Mechanics as in classical Hamiltonian mechanics, the state of an individual system is a point in a phase space, and the phase space is both the domain and image of the deterministic physical transformations. As in any statistical theory, we may know only the probability distribution for the state of the individual system, instead of knowing the state of the individual system. The relation between quantum mechanics and a statistical theory is clear: the wave-function is a parametrization for any probability distribution [58].

There are two kinds of incompleteness in a non-Markov stochastic process. The two kinds of incompleteness are in correspondence with the two concepts: stochastic and non-Markov, respectively.

1) Stochastic: From the point of view of (classical) information theory [57], the root of probabilities (i.e. non-determinism) is by definition the absence of information. Statistical methods are required whenever we lack complete information about a system, as so often occurs when the system is complex [25]. Thus we can convert a deterministic process to a stochastic process unambiguously (using trivial probability distributions); but we cannot convert a stochastic process into a deterministic process unambiguously since we need new information [18].

2) non-Markov: any non-Markov stochastic process can be described as a Markov stochastic process where some variables defining the state of the system are hidden (i.e. unknown) [134][135]. Conversely, by definition any irreducible [19] Markov process where some variables defining the

[18] E.g. the assumptions required by the deterministic models in reference~\cite{automaton} are new information.

[19] The word irreducible avoids the case where the state of the system is the direct product of two states corresponding

state of the system are hidden will give rise to a non-Markov process. For instance, the physical phenomena which generates examples of Brownian motion is deterministic and thus Markov, but real-world Brownian motion is often non-Markov (because we cannot measure the state of the system completely [136][137]) despite the fact that the Brownian motion is one of the most famous examples of a Markov process.

In reference [138] (authored by A. Einstein and contemporary of the EPR paradox) the two kinds of incompleteness are clearly distinguished:

> "[...] *I believe that the* [quantum] *theory is apt to beguile us into error in our search for a uniform basis for physics, because, in my belief, it is an incomplete representation of real things, although it is the only one which can be built out of the fundamental concepts of force and material points (quantum corrections to classical mechanics). The incompleteness of the representation is the outcome of the statistical nature (incompleteness) of the laws. I will now justify this opinion."*

The incompleteness of the representation corresponds to the non-Markov kind, while the incompleteness of the laws corresponds to the stochastic kind. By definition, in Quantum Mechanics any sequence of measurements is a Markov stochastic process (thus it has the stochastic kind of incompleteness) [20]. Note that any non-Markov stochastic process can be described as a Markov stochastic process where some variables defining the state of the system are hidden (i.e. unknown) [134][135].

Since Quantum Mechanics does not have the non-Markov kind of incompleteness, position and momentum can only be simultaneous elements of reality in another theory very different from Quantum Mechanics. That both the claim and its negation are absurd, is strong evidence that some of the assumptions leading to the Einstein-Podolsky-Rosen (EPR) paradox [43] do not hold.

So, why did the author tried to justify (using the EPR paradox [43], among other arguments) that in Quantum Mechanics the stochastic kind of incompleteness necessarily leads to a non-Markov kind of incompleteness?

The following paragraph from the same reference [138] suggests that the author was trying to favor the cause that any future theoretical basis should be deterministic, not just Markov (since statistical mechanics is often Markov).

> "*There is no doubt that quantum mechanics has seized hold of a beautiful element of*

to 2 irreducible Markov processes, in such case the fact that some variables are hidden does not imply that the resulting stochastic process is non-Markov.

[20] The term (non-)Markov stochastic process is used in this paper in the classical sense as in reference~\cite{non_markov}. We do not mean a quantum (non-)Markov stochastic process as in reference~\cite{non_markov}, despite the fact that Quantum Mechanics is used to calculate the transition probabilities between measurements.

truth, and that it will be a test stone for any future theoretical basis, in that it must be deducible as a limiting case from that basis, just as electrostatics is deducible from the Maxwell equations of the electromagnetic field or as thermodynamics is deducible from classical mechanics. However, I do not believe that quantum mechanics will be the starting point in the search for this basis, just as, vice versa, one could not go from thermodynamics (resp. statistical mechanics) to the foundations of mechanics."

However and as discussed in Section 4, there is no mathematical argument that suggests that in general a deterministic model is more fundamental than a stochastic one, quite the opposite. Since the wave-function is merely a possible parametrization of any probability distribution [58], we also cannot claim that a deterministic model is more fundamental than Quantum Mechanics. Thus, the stochastic kind of incompleteness is harmless.

So, the EPR paradox appears as an attempt to justify a mathematical statement (that a deterministic model is more fundamental than Quantum Mechanics) with arguments from physics (trying to link to the non-Markov kind of incompleteness), for which no mathematical arguments could be found. Note that a statement referring to any future theoretical basis is essentially a mathematical statement because the physical model is any (since the theoretical basis is any).

However, it is a failed attempt because it missed the fact discussed in Section 12, that the time evolution is a stochastic process if and only if it is deterministic.

In the EPR paradox, there is no probability distribution for the state of system after the spatial separation of the entangled particles and before the transformation involved in the measurement takes place, because the time evolution (being in this case non-deterministic) is not a stochastic process. We can only consider the probability distribution for the state of system after the spatial separation of the entangled particles and after the transformation involved in the measurement takes place. This is overall a non-local physical transformation since it involves the spatial separation of the entangled particles. But it does not violate relativistic causality, since both the spatial separation of the entangled particles and the transformation involved in the measurement do not by themselves violate relativistic causality, so their composition does not violate causality either.

Unlike many popular no-go arguments [139], we are not arguing against the requirement that a physical theory should be complete, in fact we claim that Quantum Mechanics is a complete statistical theory (as defined by EPR).

Note that Bohr already declared Quantum Mechanics as a "complete" theory, however he did it at the cost of a radical revision of the classical notions of causality and physical reality [140]. He wrote: "Indeed the finite interaction between object and measuring agencies conditioned by the very existence of the quantum of action entails —because of the impossibility of controlling the reaction

of the object on the measuring instruments if these are to serve their purpose—the necessity of a final renunciation of the classical ideal of causality and a radical revision of our attitude towards the problem of physical reality." [140] Such notion of a "complete" theory mostly favors the EPR claim: the only way that Quantum Mechanics could be complete is if it is incompatible with the classical notions of causality and physical reality. Thus from a logic point of view, there is no disagreement between Einstein and Bohr, their disagreement is about what basic features an acceptable theory should have, whether or not it should be compatible with the classical notions of causality and physical reality.

In contrast, the fact—that the time evolution is a stochastic process if and only if it is deterministic—which was overlooked is perfectly compatible with the classical notions of physical reality (because Quantum Mechanics has a standard phase-space) and causality (as we will show in Section 15). We claim that Quantum Mechanics—being non-deterministic and thus a generalization of classical mechanics—does not entail a radical departure from the basic features that an acceptable theory should have, according to EPR [43]. In fact in Quantum Mechanics and in classical Hamiltonian mechanics, the state of an individual system is a point in a phase space, and the phase space is both the domain and image of the deterministic physical transformations.

Any deterministic theory compatible with relativistic Quantum Mechanics necessarily respects relativistic causality

The only known theory consistent with the experimental results in high energy physics [29] is a quantum gauge field theory which is mathematically ill-defined [31]. Due to the mathematically illness, the relation of such a theory with Quantum Mechanics is still object of debate and it will be addressed soon in another article by the present author.

In the mean time we will have to consider a free system, which suffices to address the EPR paradox. For a free system, we know well what is relativistic Quantum Mechanics [81]. The time evolution of the wave-function is described by the Dirac equation for a free particle, which is a real (i.e. non-complex) equation.

Relativistic causality is satisfied in relativistic Quantum Mechanics, meaning that there is a propagator which vanishes for a space-like propagation [81]. In other words, the probability that the system moves faster than light is null.

A deterministic theory compatible with relativistic Quantum Mechanics is one which when applied to an ensemble of free systems, will reproduce the statistical predictions of Quantum Mechanics.

Since in relativistic Quantum Mechanics the probability that the system moves faster than light is null, then no system (described by the deterministic theory) in the ensemble moves faster than

light. Thus any deterministic theory compatible with relativistic Quantum Mechanics necessarily respects relativistic causality. The question we left open here and address in the next section, is whether one such deterministic theory exists.

A deterministic theory compatible with relativistic Quantum Mechanics

Does a deterministic theory—consistent with the non-deterministic time evolution of Quantum Mechanics—exists?

The answer is yes, and we will build one example of such deterministic theory in this section.

In an experimental setting, we always have a discrete set of possible outcomes and thus Quantum Mechanics always predicts a cumulative distribution function. This allows us to apply the inverse-transform sampling method [141] for generating pseudo-random numbers consistently with the probability distribution predicted by Quantum Mechanics.

An experiment in Quantum mechanics always involves the repetition of an experimental procedure many times. In the deterministic theory however, each time we execute the experimental procedure we are not executing exactly the same experimental procedure. We consider a number (any number will do) which will be the seed of the pseudo-random number generator and then we generate pseudo-random numbers consistently with the probability distribution predicted by Quantum Mechanics. The experimental procedure is: 1) generate one pseudo-random number and 2) modify the state of the system accordingly with the pseudo-random number.

In the case of relativistic Quantum Mechanics, the probability of violating relativistic causality is null. Thus, the experimental procedure never violates relativistic causality. The modifications of the state of the system are however necessarily not infinitesimal since the phase space of the experimental setting is discrete. This does not violate relativistic causality, since the finite modifications to the state of the system occur in finite intervals of time.

We can however consider intervals of time as small as we like and thus modifications to the state of the system as small as we like. The only requirement for this is that the computational resources involved in the pseudo-random number generation are as large as needed (which is valid from a logical point of view). Note that since time evolution in quantum mechanics is not necessarily a stochastic process, we will often have that a sequence of experimental procedures executed at regular and small intervals of time produces different statistical data than than just one experimental procedure executed at once after the same total time has passed (e.g. in the double-slit experiment). But this cannot be considered a radical departure of the classical notion of physical reality, since in the (very old) presentism view of classical Hamiltonian mechanics, the phase space (i.e. the physical reality) does not involve the notion of time [131]. Moreover when the time evolution is

deterministic then it is a stochastic process, therefore if we study only deterministic transformations then we can recover the eternalism view of classical Lagrangian mechanics without any conflict with relativistic causality. For instance, this implies that in the double-slit experiment we can in principle reconstruct the trajectory of each particle and conclude about which slit the particle has went through (see the next section).

From a logical point of view, this deterministic theory is valid and by definition it always agrees with the experimental predictions of Quantum Mechanics, thus it is experimentally indistinguishable from Quantum Mechanics.

From the metaphysics point of view, this deterministic theory is unacceptable, since it involves pseudo-random number generation. For instance, in the double-slit experiment we (or some super-natural entity) would need to somehow "program" each particle to follow a different path determined by a different number, which is absurd. However, the present author has no interest in building a nice deterministic theory compatible with Quantum Mechanics, for the reasons exposed in Section 4.

Note that this deterministic theory is not super-deterministic, i.e. the experimental physicists are free to choose which measurements and which transformations of the state of the system to do [142][143]. However, an experimental procedure involves a symmetry transformation of the state of the system. Since the symmetry transformation in this deterministic theory is reproduced by the pseudo-random number generation, then when we apply the inverse-transform sampling method we need to know already what is the symmetry transformation. Thus there is a kind of conspiracy between the symmetry transformation and the pseudo-random generator, but such conspiracy is part of the definition of the deterministic symmetry transformation itself. There are assumptions about freedom of choice in the literature which exclude our deterministic (but not super-deterministic) theory, because the authors erroneously consider that an experimental procedure which involves a transformation of the state of the system is instead an observation without consequences to the system [142][143].

Reconstruction of the trajectory

In presentism, then there is no such thing as "doing nothing". When we let the "time pass" we choose a particular transformation to act on the system that depends on a parameter (time) of our choice. Thus time becomes part of the random experiment, not of the reality we are studying. We can then reconstruct a differentiable continuous trajectory, however such trajectory depends on the choices we made for the random experiments. This provides a mechanism through which the measurement (defined without time) necessarily affects what it is being measured (defined with time). Such mechanism only makes sense when considering a reality which includes time.

Note that to define a differentiable trajectory with uncertainties, we need a rigorous definition of

a classical statistical field theory, this will be done in the next chapters. In the following we will focus on a subset of the trajectory in a discrete set of time instants (which does not imply that the trajectory is differentiable or continuous).

Considering an arbitrary trajectory for a wave-function that starts in the state $n = 0$ (not arbitrary), evolves to an arbitrary initial wave-function, and from there it follows an arbitrary trajectory. Such arbitrary trajectory can be parametrized by a quantum time-evolution. Despite we can only measure the final time of the trajectory, post-selection[144][145] using probabilities conditional on the final state and the same quantum time-evolution, allows us to repeat the experiment in the same conditions and predict the results of a measurement at another time between the initial and final times. This process can be repeated many times until we reconstruct the arbitrary quantum time-evolution between time intervals of our choice.

The divided time must be compared with the conditional probability, not with the probability for the total time. Using conditional probability may mimic super-determinism, so we may evade some assumptions of relevant no-go theorems.

The Young's double slit experiment

The ensemble interpretation does not give any explanation as to why it looks like the electron's wave-function interferes with itself in the Young's double-slit experiment [146][147][148]—that would imply that the wave-function describes (in some sense) an individual system. We will fill that gap in this section.





The key to understand the results of the double-slit experiment is the role of time in the quantum formalism, which was discussed in detail in Section 12. In the ensemble interpretation the individual system is entirely defined by a standard phase-space, which implies that the time plays no fundamental role in quantum mechanics nor in classical Hamiltonian mechanics. Moreover, the time-evolution in quantum mechanics is not a stochastic process unless it is deterministic. Therefore, there is not a probability distribution for each time (or for other parameter corresponding to the symmetry

group).

In the double-slit experiment, the time-evolution of the electron after being fired (S1) is a product of two non-deterministic symmetry transformations: first, going through one or another slit with a 50/50 probability (S2); and second, a non-deterministic propagation from (S2) until (F). If at least one of these two symmetry transformations would be deterministic, then we could define a stochastic process including the 3 instants in time (S1), (S2) and (F). But since both transformations are nondeterministic, the only stochastic process that can be defined only includes the 2 instants in time (S1) and (F), and the corresponding transformations from (S1) to (S2) and from (S2) to (F) have never occurred.

The only "mystery" that needs to be clarified is the fact that the non-deterministic propagation of the electron from (S2) until (F) is such that it appears that the electron interferes with itself, just like a classical wave would do. To simplify the discussion we will only consider the electrons that reach the detector along 2 different angles $\sin(\theta_1) = \frac{2\pi}{pd}$ and $\sin(\theta_2) = \frac{\pi}{pd}$ where p is the electron's linear momentum. So, a selected electron can only go through one of these 2 angles, the electrons that go through other angles are discarded.

The wave-function at (S1) is $\Psi = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$. The time-evolution from (S1) until (S2) may be the identity matrix $U = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ or $U = \frac{1}{\sqrt{2}} \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, depending on whether the second slit is closed or open, respectively. If the second slit is open, then $U\Psi = \frac{1}{\sqrt{2}} \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ meaning that the electron may go through both slits with equal probability.

The time-evolution from (S2) until (F) is given by the unitary transformation $U' = \frac{1}{\sqrt{2}} \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$, that is, it sums the wave-functions from both slits for the first angle and it subtracts the wave-functions from both slits for the second angle.

Thus, if the second slit is closed, we have at (F) the wave-function $\Psi = \frac{1}{\sqrt{2}} \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$ meaning that the electron may come along angles 1 or 2 with equal probability[21]. But if the second slit is open, we have at (F) the wave-function $\Psi = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$ meaning that the electron will only come along angle 1; since the electron would have come through both slits with equal probability if we would see what happened at (S2), it appears that from (S2) until (F) it interferes with itself constructively(destructively) along the angle 1(2) respectively.

The "mystery" is therefore similar to the probability clock 5: How is it possible that a 50/50 probability becomes 100/0? It is possible because precisely because the time plays no fundamental role in quantum mechanics nor in classical Hamiltonian mechanics. There is not a probability distri-

[21]Strictly speaking, the larger the angle the less probability, but this fact can be compensated by collecting the electrons coming along the larger angle within a larger angular width, such that the total probabilities for angles 1 and 2 are normalized to be equal.

bution for each time (or for other parameter corresponding to the symmetry group). The symmetry transformation $U'U$ is different from a stochastic process where the symmetry transformations U' and then U are applied, and there is no reason why it should not be different.

Do the Bell inequalities hold?

The Bell inequalities [149] do not hold—since Quantum Mechanics cannot be distinguished from a complete statistical theory—because the assumptions of the Bell inequalities overlooked the fact that time-evolution is a stochastic process if and only if it is deterministic. As long as the time-evolution of the phase-space is a symmetry and it respects relativistic causality, there is no reasonable argument why a complete statistical theory should be a stochastic process. The whole point of the Bell inequalities is to distinguish Quantum Mechanics from a "standard" statistical theory, but a "standard" statistical theory means that the theory is completely defined by a probability distribution in a phase-space (which is the case of Quantum Mechanics and classical statistical mechanics).

One could argue instead that the inequalities do hold, but there is an implicit assumption that the theory which is being compared to Quantum Mechanics has a time-evolution which is a stochastic process. Even in that case (see Section 12) we have that for any set of experimental results supporting relativistic Quantum Mechanics, there is a deterministic theory (and so the time-evolution is a stochastic process) which is also compatible with the same experimental results. So, to save the Bell inequalities we would need now to find fundamental arguments against such deterministic theory. But, which arguments? Such deterministic theory is compatible with any experimental test about relativistic causality and it is not super-deterministic. These arguments would need to be somehow against the existence of pseudo-random number generators in Nature, but such generators *do* exist in Nature because we humans built some of them and we are part of Nature.

To be sure, the present author does not expect that a reasonable deterministic theory will in the future replace Quantum Mechanics. But once it is established that Quantum Mechanics is a complete statistical theory, the idea that we can rule out a reasonable deterministic theory, is also an absurd: it would imply affirming the Bayesian point of view and ruling out the Frequentist point of view. Two logical constructions can always be mutually incompatible, despite being both consistent when considered independently of each other (e.g. the Bayesian and Frequentist points of view). In the Bayesian point of view, the probability expresses a degree of belief, and so the probability *is* an entity which exists by itself. In the Frequentist point of view the root of probabilities is the absence of deterministic information that does exist somehow and is revealed through events. But if such information exists, then we cannot rule out that there is a reasonable deterministic theory which describes such information.

In summary, either we can say that the Bell inequalities do not hold or instead, we can also say that the Bell inequalities (despite being mathematically valid inequalities) involve unrealistic assumptions which render them innocuous.

Wave-function collapse versus Euler's formula*

[150]

Euler's formula for the probability clock

The previous sections established that the ensemble interpretation is self-consistent. However, the ensemble interpretation does not address the question why the wave-function plays a central role in the calculation of the probability distribution, unlike most other interpretations of quantum mechanics. By being compatible with most (if not all) interpretations of Quantum Mechanics, the ensemble interpretation is in practice a common denominator of most interpretations of Quantum Mechanics. It is useful, but it is not enough.

In this and in the following sections we will show that the wave-function is nothing else than one possible parametrization of any probability distribution. The wave-function can be described as a multi-dimensional generalization of Euler's formula, and its collapse as a generalization of taking the real part of Euler's formula. The wave-function plays a central role because it is a good parametrization that allows us to represent a group of transformations using linear transformations of the hypersphere. It is precisely the fact that the hypersphere is not the phase-space of the theory that implies the collapse of the wave-function. Without collapse, the wave-function parametrization would be inconsistent.

Suppose that we have an oscillatory motion of a ball, with position $x = \cos(t)$ and we want to make a translation in time[22], $\cos(t) \to \cos(t + a)$. This is a non-linear transformation. However, if we consider not only the position but also the velocity of the ball, we have the "wave-function" given by the Euler's formula $q(t) = e^{it}$ and x is the real part of q. Then, a translation is represented by a

[22]Here the time t is merely a parameter and not necessarily the physical time. We call this arbitrary parameter "time" t for pedagogical reasons, as is common practice in the pedagogical literature about periodic functions (e.g. about Fourier analysis).

rotation $q(t+a) = e^{ia}q(t)$. To know x after the translation, we need to take the real part of the wave-function $e^{ia}q(t)$, *after* applying the translation operator.

Of course, $\cos(t)$ is not positive and so it has nothing to do with probabilities. However, we can easily apply Euler's formula to a probability clock. A probability clock [151] is a time-varying probability distribution for a phase-space with 2 states, such that the probabilities are $\cos^2(t)$ and $\sin^2(t)$, for the first and second states respectively.

A 2-dimensional real wave-function allows us to apply the Euler's formula to the probability clock:

$$\Psi(t) = \exp\left(\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right] t\right) \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} \cos(t) \\ \sin(t) \end{smallmatrix}\right] \tag{18}$$

The Euler's formula for the density matrix is:

$$\Psi\Psi^\dagger = \left[\begin{smallmatrix} \cos^2(t) & \cos(t)\sin(t) \\ \cos(t)\sin(t) & \sin^2(t) \end{smallmatrix}\right] = \frac{1}{2} + \left[\begin{smallmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{smallmatrix}\right] (\cos(2t) + J\sin(2t)) \tag{19}$$

Where $J = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ plays the role of the imaginary unit in the Euler's formula for the probability clock. A measurement using a diagonal projection triggers the collapse of the wave-function, such that a new density matrix is obtained by setting the off-diagonal part (i.e. the part proportional to J) of the original density matrix to zero. The probability distribution is given by the diagonal part of the density matrix, i.e. by taking the "real part" of the "complex number" $\cos(2t) + J\sin(2t)$:

$$\mathrm{diag}(\Psi\Psi^\dagger) = \left[\begin{smallmatrix} \cos^2(t) & 0 \\ 0 & \sin^2(t) \end{smallmatrix}\right] = \frac{1}{2} + \left[\begin{smallmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{smallmatrix}\right] \cos(2t) \tag{20}$$

Since $\cos^2(t) + \sin^2(t) = 1$ and $0 < \cos^2(t) < 1$, we can confirm that the wave-function parametrizes all probability distribution functions for a phase-space with 2 states, i.e. for any probability p there is an angle t such that the cosine $\cos(t)$ of that angle verifies $\cos^2(t) = p$. Moreover, two wave-functions are always related by a rotation $\Psi(t+a) = \exp(Ja)\,\Psi(t)$, for some a.

Note that the rotation is an invertible linear transformation that preserves the space of wave-functions. This does not happen with probability distributions: the most general linear transformation of a probability distribution that preserves the space of probability distributions is:

$$M(a,b) = \left[\begin{smallmatrix} \cos^2(a) & \cos^2(b) \\ \sin^2(a) & \sin^2(b) \end{smallmatrix}\right] \qquad \text{(where a, b are real numbers)} \tag{21}$$

because if we apply M to a deterministic distribution $\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$ or $\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$ we must obtain probability distributions which leads to the constraints $\cos^2(a) + \sin^2(a) = \cos^2(b) + \sin^2(b) = 1$ and $\cos^2(a), \sin^2(a), \cos^2(b), \sin^2(b) \geq 0$; the matrix M such that:

$$M\frac{1}{2}\left[\begin{smallmatrix}1\\1\end{smallmatrix}\right] = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right] \tag{22}$$

is necessarily singular and so it is not suitable to represent a symmetry group.

The wave-function is thus a good parametrization which allows us to represent a group of transformations using linear transformations of the points of a circle. The collapse of the wave-function is nothing more than taking the real part of a complex number as in most applications of Euler's formula in Engineering, reflecting the fact that the circle is not the phase-space of the theory. Thus the wave-function is nothing more than a parametrization of the probability distribution.

The Stern-Gerlach experiment

We follow reference [152] for the description of the Stern-Gerlach experiment, first carried out in Frankfurt by O. Stern and W. Gerlach in 1922. This experiment makes a strong case in favor of generalizing the symmetry transformations to become non-deterministic, moreover the theoretical predictions only require a phase-space with two states like the one already discussed in the previous section. Note that we only make measurements along the z and x-axis, but if we also had made measurements along the y-axis then the phase space would require four states or a parametrization with a complex wave-function, see Section 10. Some articles such as reference [153] argue that a "full quantum" analysis of the Stern-Gerlach experiment must involve the position degrees of freedom and thus a phase-space with more than two states. But as in every theoretical model for any real experiment we should consider only a phase-space which is as large as it is strictly necessary to compute all predictions for all practical purposes and do not waste time with redundant calculations which only add complexity and increase the likelihood of committing mistakes. Of course, the real Stern-Gerlach experiment involves much more than two states, for instance if the electrical power feeding the experiment is shutdown due to an earthquake or if the man managing the experiment has a heart attack it will affect the experimental results, but all predictions for all practical purposes can be computed using a phase-space with only two degrees of freedom.

In the Stern-Gerlach experiment, a beam of silver atoms is sent through a magnetic field with a gradient along the z or x-axis and their deflection is observed. The results show that the silver atoms possess an intrinsic angular momentum (spin) that takes only one of two possible values (here represented by the symbols + and -). Moreover in sequential Stern-Gerlach experiments (see figure 1), the measurement of the spin along the z-axis destroys the information about a atom's spin along the x axis.



We consider in the phase-space, not only the spin of one atom of the beam, but also the angle of orientation of a macroscopic object which serves as a reference, for pedagogical purposes. The corresponding complete wave-function is thus a reducible representation of the rotation group. When we apply a rotation to the phase-space, the rotation is a non-deterministic transformation of the spin of the atom and a deterministic transformation of the macroscopic object. Thus, to keep track of the part of the wave-function corresponding to the angle of orientation of the reference macroscopic object we only need the central value of the probability distribution for such angle, which we will call simply "the angle" for brevity. And then we only consider the part of the wave-function corresponding to the spin of the atom.

In , $cos^2(t)$ is the probability for the spin to be in the state $+$, while $sin^2(t)$ is the probability for the spin to be in the state $-$. The non-deterministic symmetry transformation given by a rotation of the spin along the $x - z$ plane is parametrized by the parameter t and its linear representation on the wave-function is described in .

In the first measurement, the angle of the reference macroscopic object is 0 with respect to the z-axis; and we know for sure that the spin is in the state $+$ ($t = 0$) because we are measuring the spin along the z-axis of atoms that were previously filtered to be in the state $+$ when measuring the spin along the z-axis (see the first graph in figure 1).

A second sequential measurement along the x-axis means that we rotate the reference macroscopic object 90 degrees along the x-z plane so the new angle is 90 degrees; for the atom we first make a 45-degrees rotation along the x-z plane ($t = \pi/4$)[23] and then we determine whether the spin is in the $+$ or $-$ state (i.e. the wave-function collapses, see the second graph in figure 1). The probability for the spin to be in the states $+/-$ is now 50%/50%, because the rotation is a non-deterministic symmetry transformation.

A third sequential measurement along the z-axis means that we rotate the reference macroscopic object -90 degrees along the x-z plane so the new angle is again 0 degrees; for the atom we first apply a -45-degrees rotation along the x-z plane ($t = -\pi/4$) to the atoms with spin $+$ and then we determine whether the spin is in the $+$ or $-$ state (i.e. the wave-function collapses one more time, see the third graph in figure 1). Despite that in the first measurement the spin was in the state $+$,

[23]Note that the angle is 45-degrees instead of 90-degrees because the spin group is a double cover of the orthogonal group, for which the angle would be 90-degrees.

the probability for the spin to be in the states $+/-$ is 50%/50% in the third measurement, because the rotation is a non-deterministic symmetry transformation and we applied it in the second and third measurements to switch from the z to the x-axis and then to switch again from the x to the z-axis.

As we have seen in the previous sections, generalizing the symmetry transformations to be non-deterministic suffices to account for all experimental results described by Quantum Mechanics, with the Stern-Gerlach experiment being one example. The question remaining is whether the Euler's formula applies for phase-spaces with more than 2 states, which would imply that the collapse of the wave-function is merely a mathematical artifact of the wave-function parametrization.

Black hole information paradox and the Stern-Gerlach experiment

What is exactly a black hole from the point of view of a quantum theory? That's a tough question. Because of that, the black hole information paradox[154][155] is not necessarily related with *real* black holes.

Nevertheless, we can always think of the Stern-Gerlach experiment, described in the previous section. The argument here is that there is always a unitary transformation such that the corresponding probability distribution is necessarily the constant distribution, for all initial states in the same orthogonal basis. Thus, if a black-hole erases most information about an object that comes inside of it by turning this information to random, that is not incompatible with a unitary time-evolution. We have seen an analogous case in the previous section for a 2-state phase space.

Certainly, the collapse of the wave-function is not unitary and thus the transformation on the ensemble is also not unitary. If we measure the properties of the black-hole immediately after the object comes inside, the information is erased. However since the time-evolution is unitary, if the transformation is not only about the object coming inside but about more events then the information is not necessarily lost. If such events do not affect the degrees of freedom that were erased (which is expected since a black-hole is defined by few parameters), then the information will remain erased. Only with a quantum theory for black holes we can know for sure which events can happen after an object comes inside a black-hole.

In any case, a transformation which erases information is compatible with a unitary time-evolution.

Euler's formula for a phase-space with 4 states

We address now a system with 4 possible states. A real normalized wave-function φ_1 can be parametrized in terms of Euler angles (i.e. standard hyper-spherical coordinates and following

reference [156]) as:

$$\begin{aligned}\varphi_1 =& c_1\ l_1 + s_1\ \varphi_2\\ \varphi_2 =& c_2\ l_2 + s_2\ \varphi_3\\ \varphi_3 =& c_3\ l_3 + s_3\ \varphi_4\\ \varphi_4 =& l_4\end{aligned} \tag{23}$$

Where $c_n = \cos(\theta_n)$ and $s_n = \sin(\theta_n)$ stand for the cosine and sine of an arbitrary angle θ_n (i.e. θ_n is an arbitrary real number), respectively; and n is an integer number verifying $1 \leq n < 4$. The set l_1, l_2, l_3, l_4 are normalized vectors forming an orthonormal basis of a -dimensional real vector space.

The Euler's formula for the corresponding density matrices is:

$$\begin{aligned}\varphi_1\varphi_1^\dagger =& \frac{1}{2} + \frac{1}{2}(l_1 l_1^\dagger - \varphi_2\varphi_2^\dagger)(\cos(2\theta_1) + J_1\sin(2\theta_1))\\ \varphi_2\varphi_2^\dagger =& \frac{1}{2} + \frac{1}{2}(l_2 l_2^\dagger - \varphi_3\varphi_3^\dagger)(\cos(2\theta_2) + J_2\sin(2\theta_2))\\ \varphi_3\varphi_3^\dagger =& \frac{1}{2} + \frac{1}{2}(l_3 l_3^\dagger - \varphi_4\varphi_4^\dagger)(\cos(2\theta_3) + J_3\sin(2\theta_3))\\ \varphi_4\varphi_4^\dagger =& l_4 l_4^\dagger\end{aligned} \tag{24}$$

Where $J_n = (l_n\varphi_{n+1}^\dagger - \varphi_{n+1}l_n^\dagger)$ plays the role of the imaginary unit in the Euler's formula, in the subspace generated by the vectors $\{l_n, \varphi_{n+1}\}$. Thus, the collapse of the wave-function for a phase-space with states is a recursion of collapses of 2-dimensional real wave-functions. The conditional probabilities are given by the diagonal part of the density matrix, i.e. by taking the "real part" of the "complex numbers" $\cos(2\theta_n) + J_n\sin(2\theta_n)$:

$$\begin{aligned}P(1|1\text{ or above})) &= \frac{1}{2} + \frac{1}{2}\cos(2\theta_1) \qquad P((2\text{ or above})|(1\text{ or above})) = \frac{1}{2} - \frac{1}{2}\cos(2\theta_1)\\ P(2|2\text{ or above})) &= \frac{1}{2} + \frac{1}{2}\cos(2\theta_2) \qquad P((3\text{ or above})|(2\text{ or above})) = \frac{1}{2} - \frac{1}{2}\cos(2\theta_2)\\ P(3|3\text{ or above})) &= \frac{1}{2} + \frac{1}{2}\cos(2\theta_3) \qquad P((4\text{ or above})|(3\text{ or above})) = \frac{1}{2} - \frac{1}{2}\cos(2\theta_3)\\ P(4|(4\text{ or above})) &= 1\end{aligned} \tag{25}$$

where $P(2|(2\text{ or above}))$ stands for probability for the state to be $n = 2$ knowing that the state is either $n = 2$, or $n = 3$, ... or $n = 4$. Note that these conditional probabilities are arbitrary, i.e. for any probability p there is an angle θ_n such that the cosine $c_n = \cos(\theta_n)$ of that angle verifies $c_n^2 = p$.

The fact that the previous conditional probabilities are arbitrary, implies that the probability distribution is arbitrary, since for any probability distribution we have:

$$
\begin{aligned}
P(1) =& P(1|(1 \text{ or above})) \\
P(2) =& P((2 \text{ or above})|(1 \text{ or above}))* \\
& P(2|(2 \text{ or above})) \\
P(3) =& P((2 \text{ or above})|(1 \text{ or above}))* \\
& P((3 \text{ or above})|(2 \text{ or above}))* \\
& P(3|(3 \text{ or above})) \\
P(4) =& P((2 \text{ or above})|(1 \text{ or above}))* \\
& P((3 \text{ or above})|(2 \text{ or above}))* \\
& P((4 \text{ or above})|(3 \text{ or above}))* \\
& P(4|(4 \text{ or above}))
\end{aligned}
\tag{26}
$$

Moreover, two wave-functions are always related by a rotation. Thus we can confirm that any probability distribution for states, can be reproduced by the Born rule for some wave-function:

$$
\begin{aligned}
P(n) =& |\varphi^\dagger l_n|^2 \\
P(1) =& (c_1)^2 \\
P(2) =& (s_1 c_2)^2 \\
P(3) =& (s_1 s_2 c_3)^2 \\
P(4) =& (s_1 s_2 s_3)^2
\end{aligned}
\tag{27}
$$

Euler's formula for a generic phase-space

A probability distribution can be discrete or continuous. A continuous probability distribution is a probability distribution that has a cumulative distribution function that is continuous. Thus, any partition of the phase-space (where each part of the phase-space has a non-null Lebesgue measure) is countable.

Consider now a countable (possibly infinite) partition of the phase-space. The corresponding countable orthonormal basis for the separable Hilbert space is $\{l_n\}$, where each index $n > 0$ corresponds to an element of the partition of the phase-space. We can parametrize a normalized vector in the Hilbert space [156], as $v_n = c_n l_n + s_n v_{n+1}$, where $c_n = \cos(\theta_n)$ and $s_n = \sin(\theta_n)$ stand for the cosine and sine of an arbitrary angle θ_n (i.e. θ_n is an arbitrary real number), respectively; and $n > 0$ is an integer number. The first vector v_1 is the wave-function of the full phase-space. Note that the parametrization is valid for infinite dimensions, because in the recursive equation all we need to assume about the vector v_{n+1} is that it is normalized and orthogonal to $\{l_1, l_2, ...l_n\}$,

which is a valid assumption in infinite dimensions. Then we define v_{n+1} in terms of v_{n+2} in the same way, and so on. The recursion does not need to stop.

Then, the projection to the linear space generated by v_n is:

$$v_n v_n^\dagger = \frac{1}{2} + \frac{1}{2}(l_n l_n^\dagger - \varphi_{n+1}\varphi_{n+1}^\dagger)(\cos(2\theta_n) + J_n \sin(2\theta_n)) \tag{28}$$

Where $J_n = (l_n \varphi_{n+1}^\dagger - \varphi_{n+1} l_n^\dagger)$ plays the role of the imaginary unit in the Euler's formula, in the subspace generated by the vectors $\{l_n, \varphi_{n+1}\}$. Thus, the collapse of the wave-function for a generic phase-space is a recursion of collapses of 2-dimensional real wave-functions. The conditional probabilities are given by the diagonal part of the density matrix, i.e. by taking the "real part" of the "complex numbers" $\cos(2\theta_n) + J_n \sin(2\theta_n)$: The operator $v_n v_n^\dagger$ is a projection thanks to the off-diagonal[24] terms $c_n s_n (l_n v_{n+1}^\dagger + v_n l_{n+1}^\dagger)$.

Defining (n or above) $= \{k : k \geq n\}$ as the event which contains all parts of the phase-space with index starting at n, we can write the probability distribution as:

$$\begin{aligned} P(n) &= P((n \text{ or above}))P(n|(n \text{ or above})) \\ &= \left(\prod_{k=1}^{n-1} P((k+1 \text{ or above})|(k \text{ or above}))\right) P(n|(n \text{ or above})) \end{aligned} \tag{29}$$

That is, as a product of the probabilities

$$P(n|(n \text{ or above})) \text{ and } P((n+1 \text{ or above})|(n \text{ or above})) \tag{30}$$

which verify:

$$P(n|(n \text{ or above})) \ + \ \ P((n+1 \text{ or above})|(n \text{ or above})) = 1. \tag{31}$$

If the off-diagonal terms are suppressed (collapsed), we obtain a diagonal operator which represents the probability distribution $P(n)$ in the Hilbert space:

$$\text{diag}(v_n v_n^\dagger) = c_n^2 l_n l_n^\dagger + s_n^2 v_{n+1} v_{n+1}^\dagger \tag{32}$$

That is, $P(n) = \text{tr}(\text{diag}(v_1 v_1^\dagger) l_n l_n^\dagger)$ and $P(O) = 0$ for operators O with null-diagonal. Note that $c_n^2 = P(n|(n \text{ or above}))$ and $s_n^2 = P((n+1 \text{ or above})|(n \text{ or above}))$ and these probabilities are

[24] In a basis where all $l_n l_n{}^\dagger$ are diagonal.

arbitrary, i.e. for any probability p there is an angle θ_n such that the co $c_n = \cos(\theta_n)$ of that angle verifies $c_n^2 = p$.

The fact that these conditional probabilities are arbitrary, implies that the probability distribution is arbitrary, since the probability distribution can be written in terms of these conditional probabilities as shown in Equation [eq:cond].

Complex and Quaternionic Hilbert spaces

While the parametrization with a real wave-function is always possible, it may not be the best one. As we have seen, the wave-function parametrization allows us to apply group theory to the states of the ensemble, since unitary transformations (i.e. a multi-dimensional rotation) preserve the properties of the parametrization (in particular the conservation of total probability).

The union of a set of projection operators and the unitary representation of a group, is a set of normal operators. Suppose that there is no non-trivial closed subspace of the Hilbert space left invariant by this set of normal operators. The (real version of the) Schur's lemma [81][82][83] implies that the set of operators commuting with the normal operators forms a real associative division algebra—such division algebra is isomorphic to either: the real numbers, the complex numbers or the quaternions.

If we do a parametrization by a real wave-function and consider only expectation values of operators that commute with a set of operators isomorphic to the complex or the quaternionic numbers, then we can equivalently define wave-functions in complex and quaternionic Hilbert spaces [157][81][82].

Let us consider the quaternionic case (it will be then easy to see how is the complex case). We have a discrete state space defined by two real numbers n, m, with $1 \leq m \leq 4$ and we only consider the probabilities for n independently on m, $P(n) = \sum_{m=1}^{4} P(n, m)$.

Then a more meaningful parametrization—reflecting by construction the restriction on the operators we are considering—uses a quaternionic wave function v_1. Let $\{l_n\}$ be an orthonormal basis of quaternionic wave-functions and we have:

$$v_n v_n^\dagger = c_n^2 l_n l_n^\dagger + s_n^2 v_{n+1} v_{n+1}^\dagger + c_n s_n (l_n v_{n+1}^\dagger + v_{n+1} l_n^\dagger) \tag{33}$$

Note that there is a basis where $l_n l_n^\dagger$ is real diagonal and thus upon collapse $v_n v_n^\dagger$ becomes real diagonal as well.

The complex case is just the above case with complex numbers replacing quaternions and a state space which is the union of 2 identical spaces. The continuous case is analogous, since there is a partition of the phase-space which is countable.

Free field parametrization in Classical Statistical Field Theory and Navier-Stokes equations

Free field parametrization in Statistical Field Theory

As it was seen in /pub/parametrization, the wave-function of two variables can be written as the product of a wave-function in one variable and the square root of a conditional probability. To parametrize a field we want a conditional probability, because that's the most general non-deterministic function. And so we can parametrize a conditional probability with the wave-function of two variables.

Moreover, since the Lebesgue measure cannot be defined in infinite-dimensions[7][8], then we cannot simply define the probability distribution of a classical field (a deterministic function of a space, say $\mathbb{R}^n$). We consider instead the probability distribution of another probability distribution with sample space given by the direct product of the space and the field (say $\mathbb{R}^{n+1}$). There is no loss of generality because the boolean algebra of the field corresponds to particular probability distributions concentrated around the field values at each point in space. Applying a wave-function parametrization to the probability distribution in $\mathbb{R}^{n+1}$, we get an infinite-dimensional sphere where a uniform (Lebesgue-like) measure can be defined using the Gaussian measure and the Fock-space (the Fock-space is used in the second quantization of free quantum fields)[9]. The probability densities which are not strictly positive have null measure.

Since the second-quantization procedure can be applied recursively, we can assume without loss of generality that the Hamiltonian is quadratic in the creation/annihilation operators and thus no further regularization is needed (the wave-function parametrization can be considered a regularization by itself). For instance, in case the Hamiltonian acting in the Fock-space is not quadratic in the creation/annihilation operators, then we can consider instead a new Fock-space where the base Hilbert

space is the original Fock-space. The new Hamiltonian is quadratic in the creation/annihilation operators.

We can also assume the Hamiltonian to be local, at the cost of enlarging the Fock-space. This is consistent with the fact that any non-Markov stochastic process can be described as a Markov stochastic process where some variables defining the state of the system are hidden (i.e. unknown)[134][135]. That we can assume the Hamiltonian to be local and quadratic is consistent with the fact that a complete physical system is also a free system. The free field associated to a free system can be made an orthogonal(fermion)/symplectic(boson) complex representation of the Poincare group depending on whether its spin is semi-integer/integer respectively, regardless of the interactions occurring within the free system[158][159]. Thus in field theory the wave-function parametrization includes a free field parametrization. The remaining question now is how to define the notion of a field within the Fock-space (a separable Hilbert space).

As it was discussed before, every commutative von Neumann algebra on a separable complex Hilbert space is isomorphic to $L^\infty(X,\mu)$ for some standard measure space (X,μ) and conversely, for every standard measure space (X,μ) the corresponding Lebesgue space $L^\infty(X,\mu)$ is a commutative von Neumann algebra. Thus, it may be tempting to base a theory in the Lebesgue space $L^\infty(X,\mu)$ instead of the standard measure space (X,μ). The problem is that $L^\infty(X,\mu)$ is dual to the standard measure space (X,μ), but not the other way around. The dual of $L^\infty(X,\mu)$ is a space of finitely additive measures[2], which is mathematically much more complex than a standard measure space. Since the Lebesgue measure does not exist in infinite-dimensions, most of the approaches to field theory are based in deterministic functions (including functional calculus and classical field theory [160]) which is equivalent to $L^\infty(X,\mu)$ where the standard measure space is the Fock-space. But the Fock-space does not contain deterministic functionals in general (that is, when there is a continuous set of possibilities for the value of a function at a point), thus any space which is an extension of deterministic functionals is a space of finitely additive measures much more complex than a standard measure space.

While there is a mathematically rigorous definition of classical field theory [160], so far the definition of a (classical) statistical field theory [33] is tied to the definition of a quantum field theory [161], which involves a lattice spacing necessary to regularize and renormalize the ultraviolet divergences of the field theory. The notion of continuum limit in a discrete lattice is that for a large enough energy scale the predictions of the theory are independent of the type of discrete regularization used [161], thus the lattice in the regularized theory is always discrete. The regularization and renormalization are related to the decomposition of a field defined in the continuum through discrete wavelets and it is roughly the translation of the products of fields into products of wavelet components[23] (a related approach involves a semivariogram [22]). Such translation of products of fields only allows

polynomial Hamiltonians; in particular, when using the Fock space without regularization (which is a possible way to implement a continuous tensor product[162]), only quadratic Hamiltonians are allowed (i.e. for free fields).

This excludes a rigorous definition of the classical statistical version of many classical field theories (such as General Relativity) since so far there is no reason why the Hamiltonian of a classical field theory should be polynomial in the fields, not to mention the problems with Quantum Gravity [163]. This is unacceptable: for most classical field theories, the definition of the corresponding classical statistical field theories should be straightforward, because the real-world measurements are never fully accurate.

The above is an indication that an alternative definition of Statistical Field Theory which allows the definition of non-polynomial Hamiltonians should not be too hard to find. Indeed, the essential obstruction to an infinite-dimensional Lebesgue measure is its σ-finite property (to be the countable direct sum of finite measures)[7][8]. Once we drop the σ-finite property, several relatively simple candidates exist [7][8]. In our case, we are not looking for an infinite-dimensional Lebesgue measure (no one expects the probability measure itself to be translation-invariant), but only for a translation-invariant time-evolution of the probability measure (i.e. the time-evolution is an operator, not a real number) and thus there is no reason to expect such operator to be σ-finite until it is evaluated against a probability measure when it becomes another probability measure—which cannot be translation-invariant.

The way out is that the basis of the theory is the Fock space (a separable Hilbert space, thus associated to a standard measure space) which crucially, it does not include deterministic functionals. A probability distribution with sample space given by the direct product of the base space and the field can model a non-deterministic function since we can always choose a measure (the Gaussian measure for instance) which covers the whole base space, with the non-deterministic function evaluated at each point of the base-space given by the conditional probability distribution conditioned on each point of the base-space. Note that due to the relation between Fock-spaces and tensor products, we can consider regions of the sample space as small as we want and there is null measure for a null marginal probability in each one of these regions, and for each component of the Fourier series corresponding to such region; thus the conditional probability distribution conditioned on each point of the base-space is well defined everywhere except in sets with null measure.

Thus, the free field parametrization is appropriate for non-deterministic functions but not for deterministic functions, since the prior (uniform measure of an infinite-dimensional sphere, we call it the uniform prior from now on) attributes null measure to deterministic functions. If we would be interested in deterministic functions, we would be in trouble[7][8]. This does not come for free either, since we need to sacrifice deterministic functionals in general and thus classical field theory (which

may be welcome from a quantum foundations point of view). That includes sacrificing the evaluation of a function in a point and thus replacing a classical field by a quantum field (in the Fock-space sense). Without the evaluation of a function in a point, continuity and differentiability notions must be redefined for quantum fields, including the generalized Stokes theorem (or fundamental theorem of calculus). Fortunately, as we will see in the following, this is relatively straightforward because the smooth wave-functions are dense in the Hilbert space.

In the quantum version, the integration domain in the Stokes theorem is replaced by a uniform measure or any other probability measure, while a set of constraints define a non-deterministic geometry which is not significantly more complex than differential geometry (which is great for High Energy Physics and Gravity).

On the other hand, differential geometry and most of the machine learning methods are in the dual space of a commutative von Neumann algebra but not in the pre-dual space (which corresponds to the Fock-space). Such dual space has some good properties but not all those necessary to define an information theory (that is, some consistent way of defining how much we know and not know about anything; on the other hand, the Fock-space has all required properties). For instance, differential geometry would be of little use whenever there is uncertainty about the value of a parameter of a system (which happens most of the time), if there would not exist another similar version of differential geometry which can be applied to non-deterministic systems.

The (bosonic or fermionic) Fock-space can be used as the wave-function parametrization of an arbitrary probability distribution of a classical field over space (for instance the 3D space), in the following way. For a Hamiltonian involving at most first-order derivatives of the fields, the propositions about a (non-free) field at an arbitrary discrete finite number n of points in space are parametrized by the Fock-space over $S = \mathbb{R}^{d(1+m)+m}$, where d is the number of space dimensions and m is the number of bosonic fields (to include fermionic fields we replace $\mathbb{R}$ by the discrete set $\{0, 1\}$). The operator corresponding to a proposition is:

$$\pi(\phi(x_1) \cap ... \cap \phi(x_n)) = a^\dagger(\phi_1, x_1)a(\phi_1, x_1)...a^\dagger(\phi_n, x_n)a(\phi_n, x_n) \tag{34}$$

Above, we assume no coinciding points such as $x_n = x_1$. Note that this operator still needs to be normalized to be considered a proposition (such that its integral is the unit), since it is composed of number operators and not projections. Moreover, the commutative algebra generated by number operators is consistent with a point process and its wave-function parametrization can be a bosonic or a fermionic Fock-space. The normalization is done using the property of tensor-product of the Fock-space, until each Fock-space appearing in the tensor-product either has at most one point or it has a discrete base-space, in both-cases it is straightforward to remove the base-space from each

Fock-space.

In (quantum or classical) statistical field theory, the problem we want to solve is about a probability distribution so it is about an eigenvalue problem and diagonalizing a time-evolution operator, because the eigenfunction needs not even exist and we use ideals instead. This allows us to know approximately the probability distribution of the initial and final state through numerical methods for partial differential equations and regression (Gaussian process regression [19] or statistical finite element method [164], for instance). On the other-hand, in classical field theory (including in numerical calculations such as the finite element method [165][166]) it is about the fields themselves and so the solution must be part of an Hilbert space (because completeness of the space is crucial for the existence proofs) and we need an alternative to the L^2 measure since the differential operator is unbounded with respect to the L^2 measure: such alternative is the Sobolev Hilbert space [167] which is harder to work with because it involves some correlation between different points in space due to differentiability requirements[168][169].

The cost of the free field parametrization is that we need to implement derivatives and coordinates in continuum space as an extra structure at the local level using constraints, which allows well-defined products of fields and its derivatives at the same point in the continuum space. For an Hamiltonian which depends on the field derivatives up to first order, the constraints are $iD_x = 0$ where:

$$\begin{aligned}
&[p_{(j)}, \phi^{(k)}(\chi)] = i\delta_j^k \\
&[p_{(j)}, \phi^{(1)}(x)] = i\delta_j^1 \\
&\phi(x) = \phi^{(0)}(\chi) + \int_\chi^x dy\ \phi^{(1)}(y) \\
&D_x = ip_{(0)}\phi^{(1)}(x) + \partial_x
\end{aligned} \tag{35}$$

Crucially, due to the Bianchi identity and that the Hamiltonian commutes with the constraints, we have:

$$\begin{aligned}
&\int d\phi dx a^\dagger(\phi, x)[[iD_x, \phi^{(0)}], H]a(\phi, x) = \\
&= -\int d\phi dx a^\dagger(\phi, x)[iD_x, [H, \phi^{(0)}]]a(\phi, x) \\
&\int d\phi dx a^\dagger(\phi, x)[[iD_x, p_{(1)}], H]a(\phi, x) = \\
&= -\int d\phi dx a^\dagger(\phi, x)[iD_x, [H, p_{(1)}]]a(\phi, x)
\end{aligned} \tag{36}$$

The Hamiltonians that depend on the second order (or more, even infinite order) derivative, can be redefined as Hamiltonians that depend on the first order derivative of the fields, by introducing

more fields and constraints which will relate some fields with the derivative of other fields. Also, a space-dependent Hamiltonian can be redefined as space-independent one, by introducing extra fields with space-derivatives which are constrained to be the unit in one of the space coordinates.

Note that in classical Lagrangian (and Hamiltonian) Field Theory there are also derivatives of fields (through jets of modules [170]), which are consistent with the commutation relations [171] for operator fields and constraints defined above.

We can easily check that these constraints are verified for a wave-function with one creation operator and the field derivatives are given by $p\psi = \phi^{(1)}\partial_{\phi^{(0)}}\psi$ (up to sets with null measure) where ψ is the (Fourier transformed) wave-function defining the creation operator and p is the eigenvalue of the derivative operator resulting from the Fourier transform, thus the constrained spectrum is not empty and so the constraints are well defined.

Note that in classical Lagrangian (and Hamiltonian) Field Theory there are also derivatives of fields (through jets of modules [170][171]), which are consistent with the commutation relations [171] for operator fields defined in the following.

Thus we can define the Hamiltonian dependent on only the first-oder derivative of the field $\phi(x)$ and from it derive the time-evolution operator, relating derivatives of the field $\phi(x)$ to derivatives of the wave-function. Note that the smooth functions are dense in the Lebesgue square-integrable space, thus the fields (and the Hamiltonian H) can be treated as if they would be be smooth. The constraints lead to commutation relations of the fields and its derivatives, similar to the commutation relations appearing in the Lagrangian formalism [172].

These constraints still allow for spontaneous symmetry breaking, which is about the correlation of the fields at two points in space separated by an infinite distance[173].

The resulting time-evolution operator is translation-invariant. There is still no infinite-dimensional, translation-invariant and σ-finite measure (as expected [7][8]).

We may now define Hamiltonians which are non-polynomial functions of the field $\phi(x)$ with respect to the spectral measure $d\phi(x)$, just like it happens in Quantum Mechanics (e.g. the Hamiltonian for the Hydrogen atom). Our formalism is much more powerful and general than the peculiar notion of "continuum" through renormalization [161] also because it involves the continuum independently from renormalization and it can also be renormalized (to reproduce the standard perturbative calculations, for instance).

Local operators and the momentum constraint

The momentum constraint also generates a gauge symmetry, once we consider a spectral measure where the fields are functions of space. Then the momentum constraint always modifies the spectral measure and so we have a complete unconstrained gauge-fixing.

Since all operators must be invariant under a translation in space, how can we define local operators? In rigor we can't, but we can define operators which are linear combinations of local operators, which behave effectively as local operators. Consider the operator $\int d\vec{x}\ l(\vec{x})$ which is translation invariant, where $l(\vec{x})$ is a local operator. The translation-invariant operator behaves effectively as a local operator when the wave-function is concentrated around one point in the 3-dimensional space. Note that then the wave-function is not translation invariant which is fine, as long as the operators are translation invariant.

Quadratic Ordering

The (symmetric) Weyl ordering of operators [174][175] conserves the exponential of operators, unlike normal-ordering [176]. This is an important property of the ordering, because we use often the Trotter product formula [177][178] (e.g. in the time-evolution operator).

However, the Weyl ordering is not appropriate for the second-quantization of an infinite-dimensional Hilbert space, since it leads to infinite vacuum expectation values of quadratic operators. Thus, we introduce the Quadratic Ordering which is a kind of "incomplete" normal ordering, which conserves the exponential of operators. That is, we only order operators which conserve the number operator such that these operators are a product of quadratic operators (where each quadratic operator conserves the number operator and has null vacuum expectation value). All other operators can be generated from these products of quadratic operators and operators with exclusively creation (or exclusively annihilation) operators which are automatically ordered.

The quadratic operators are normalized by normal operators acting in the Hilbert space H which generates the Fock-space, while the operators with exclusively creation (or exclusively annihilation) operators are normalized by wave-functions (elements of the tensor-products of H). In this way, the commutation rules in the Fock-space are related to the commutation rules of the Hilbert space H.

Mass gap

In the following, we solve the hardest part of the Millenium Prize problem "Yang-Mills existence and mass gap", by showing that if the Hamiltonian of the Yang-Mills theory is positive-definite, then Yang-Mills theory theory can be reformulated with or without a mass gap without any observable

consequences. To solve the remaining part of the Millenium Prize problem "Yang-Mills existence and mass gap", we need to define with mathematical rigor the Yang-Mills theory within our formalism, prove that it exists and it is non-trivial and show that the Hamiltonian of the Yang-Mills theory is positive-definite, all this we will do in another chapter.

The mass gap is the eigenvalue of the Hamiltonian which is closer to the eigenvalue null corresponding to the vacuum state. We now show that the mass gap (in a field theory defined through the wave-function parametrization as above) cannot be physically measured and it can be modified at will. That is, a bounded from below Hamiltonian with a null mass gap (the free Electromagnetic field, for instance) can be modified to a bounded from below Hamiltonian with an arbitrary mass gap without any observable consequence.

This is so because the commutative algebra of observables is generated by hermitian operators $a^\dagger(\phi, x)a(\phi, x)$ (due to the infinite-dimensional sphere with a uniform measure) and the time-evolution does not modify this fact, since the Hamiltonian commutes with the number operator. Thus, the number operator commutes with the algebra of observables which implies that the number operator can be added to the Hamiltonian, modifying the mass gap without observable consequences. This fact is not in conflict with the Casimir effect[179].

Note that in our formalism as in Quantum Mechanics, the Hamiltonian can be bounded from below because it is defined a priori with a correct (Quadratic) ordering. On the other hand, already in the second quantization of free fields (and in perturbative Quantum Field Theory in general) the Hamiltonian cannot be bounded from below in an infinite space-time because it is defined using space coordinates where it cannot have a correct ordering of operators[180] (only in momentum coordinates can normal ordering be applied) and some kind of renormalization is needed.

Holomorphic fields

The Hilbert space is the symmetric Fock space $\Gamma^s(L^2(\mathbb{R}^3))$, where we have the real and imaginary parts of the holomorphic functions and the imaginary coordinate, for a fixed real coordinate which plays the role of time in the time evolution. The Cauchy-Riemann equations define the Hamiltonian. Constraints are required because the Hamiltonian depends on the partial derivative of the holomorphic function on the imaginary coordinate.

Note that if a complex function of complex variable is locally integrable in an open domain, and satisfies the Cauchy–Riemann equations weakly, then the function agrees almost everywhere with an analytic function in such open domain.

The Hamiltonian is:

$$H = \pi_v \partial_x u - \pi_u \partial_x v \tag{37}$$

The constraints are:

$$[D_x, u + iv] = 0 \tag{38}$$

There is no need to define a gauge symmetry for these constraints, because there is an explicit solution

Free field parametrization in Navier-Stokes equations

[68]

The Hilbert space is the tensor product of the symmetric and antisymmetric Fock spaces $\Gamma^s(L^2(\mathbb{R}^{15} \times \mathbb{Z}_2^3)) \otimes \Gamma^a(L^2(\mathbb{R}^{15} \times \mathbb{Z}_2^3))$ [104](*Fock Space*, n.d.). This gives us a graded Lie superalgebra of creation and annihilation operators, of both bosonic and fermionic types. The total fermionic number is defined as the sum of all number operators corresponding to fermionic degrees of freedom. The fermionic creation and annihilation operators correspond to a field configuration where the total fermionic number is odd, while the bosonic creation and annihilation operators correspond to a field configuration where the total fermionic number is even.

The $\mathbb{R}^{33}$ degrees of freedom correspond to 3 space coordinates (x), 3 fields (u_k) and its corresponding space-derivatives ($\partial_j u_k = u_{k,j}, \partial_i \partial_j u_k = u_{k,ij}$). The $\mathbb{Z}_2^3$ degrees of freedom correspond to 1 ghost corresponding to the divergence constraint (ψ) and its corresponding space-derivatives ($\partial_j \psi = \psi_j$), minus one because we are taking the tensor product of two Fock-spaces which introduces an extra $\mathbb{Z}_2$ degree of freedom.

We define:

$$
\begin{aligned}
&a(\vec{x},...) = a(\vec{x},(\vec{u},\mu,\oplus_k(\vec{u}_k,\mu_k)))\\
&\int d^3\vec{x}... = \int d^3\vec{x}(d^3\vec{u}\sum_{\mu=0}^{1}\prod_k(d^3\vec{u}_k\sum_{\mu_k=0}^{1}))\\
&\{[a(\vec{x},...,\mu),a^\dagger(\vec{y},...,\nu)]\} = a(\vec{x},...,\mu)a^\dagger(\vec{y},...,\nu) + (-1)^{(\mu \mathrm{mod} 2)(\nu \mathrm{mod} 2)}a^\dagger(\vec{y},...,\nu)a(\vec{x},...,\mu) =\\
&= \delta((\vec{x},...)-(\vec{y},...))\\
&[u_j,\pi^k] = u_j\pi^k - \pi^k u_j = i\delta_j^k\\
&[u_{j,k},\pi^{mn}] = u_{j,k}\pi^{mn} - \pi^{mn}u_{j,k} = i\delta_j^n\delta_k^m\\
&[u_{i,jk},\pi^{lmn}] = u_{i,jk}\pi^{lmn} - \pi^{lmn}u_{o,jk} = i\delta_i^l\delta_j^m\delta_k^n\\
&\{\psi,\psi^\dagger\} = \psi\psi^\dagger + \psi^\dagger\psi = 1\\
&\{\psi_j,\psi_k^\dagger\} = \psi_j\psi_k^\dagger + \psi_k^\dagger\psi_j = \delta_{jk}
\end{aligned}
\tag{39}
$$

where

$$
\begin{aligned}
&\psi^\dagger\{a\}(t,...,j) = a(t,...,1)\delta_{j0}\\
&\psi\{a\}(t,...,j) = a(t,...,0)\delta_{j1}
\end{aligned}
\tag{40}
$$

The Hamiltonian for the Navier-Stokes equations is defined as:

$$
\begin{aligned}
&H = \int d^3\vec{x}...a^\dagger(\vec{x},...)H(\vec{x},...)a(\vec{x},...)\\
&H(\vec{x},...) = \pi^i(u_ju_{i,j} - \nu u_{i,jj}) + (h.c.)\\
&\Omega = \int d^3\vec{x}...a^\dagger(\vec{x},...)\Omega(\vec{x},...)a(\vec{x},...)\\
&\Omega(\vec{x},..) = \left[u_{j,j}\psi^\dagger\right]
\end{aligned}
\tag{41}
$$

where $\nu > 0$ and $(h.c.)$ means hermitian conjugate. The BRST charge is Ω, where ψ is the ghost field. Note that the divergence constraint can be easily solved (e.g. using the replacement $u_{3,3} = u_{1,1}+u_{2,2}$) and there is a wave-function which verifies it, so the BRST formalism is used only to allow a more elegant formulation without distinguishing one of the space indices and not to define the theory itself.

We can check that since $H(\vec{x})$ is a time-independent polynomial quantum Hamiltonians such that the domain is the dense set of smooth functions with compact support, it can be proved to be essentially self-adjoint[69][70], using $H^2(\vec{x})$ as a positive auxiliary operator in Corollary 1.1 of [69]. Following the same reasoning, we can prove that H is also essentially self-adjoint. The Hamiltonian $H(\vec{x})$ is uniquely defined up to a constant if we want the time-evolution to be conservative (that

is, to conserve the constraints $\pi^k = 0$), such constant does not modify the time-evolution of the abelian von Neumann algebra.

In conclusion, we proved that the solution to the Navier-Stokes differential equation in a standard probability space exists and it is unique. Since a smooth deterministic field can be approximated with an uncertainty as small as we want by the initial conditions in a standard probability space, then we solved the problem of existence and uniqueness of the Navier-Stokes equations (at least as far as it can be solved).

Real representations, CPT theorem and the relativistic position operator*

[181]

Introduction

When we talk about Lorentz covariance in Quantum Physics, we refer to two different formalisms that must be mutually compatible:

1. a theory which is diffeomorphism-invariant (which includes Poincare invariance) with particular tetrad fields corresponding to a Minkowski space-time;

2. a theory which only needs to be Poincare invariant but without tetrad fields not even a space-time (both could otherwise be helpful implementing the Poincare invariance).

Assuming that we can define[25] the diffeomorphism-invariant theory (1), there are still many predictions which involve the Minkowski space-time only or not even any fields (such as in the non-relativistic limit of Quantum Mechanics) and which would be impractical to calculate in the theory (1). For all those predictions, the theory (2) is a useful stepping stone.

There are then two different foundations for the theory (2):

- just define the theory[26] and show that it is Poincare invariant, postponing foundational questions to when the diffeomorphism-invariant theory (1) is defined[27];

- establish the foundations of the Poincare-invariant theory (2) by itself, without references to a space-time, as we will do in this chapter.

[25] As we will do in Chapter /pub/timepiece.
[26] The theory (2) will be defined in Chapter /pub/quantization.
[27] As we will do in Chapter /pub/timepiece.

Clearly, the foundations of theory (2) by itself are not indispensable and so this chapter can be skipped in a first reading. Nevertheless, since the foundations of theories (1) and (2) must be mutually consistent, establishing both of these foundations is an important consistency check, especially because not much is known about Quantum Gravity theoretically and experimentally.

Concerning the relation between the Hamiltonian formalism and special relativity, there are two kind of questions we can ask: 1) is the Hamiltonian formalism compatible with Lorentz symmetry? 2) based on the space-time "philosophy", why should the time-evolution play a distinguished role in the Hamiltonian formalism? The first question is technical, while the second question is conceptual. We do not have an answer to the second question, which is expected given the difficulties with Lorentz-symmetry of other approaches to quantization [182]. But in this section we will answer explicitly and positively to the first question. In short, the fact that time evolution plays a special role allows us to use only Poincare representations with positive squared mass. Considering Poincare representations with positive squared mass is self-consistent and it is in no way in conflict with Lorentz symmetry.

A complete physical system is a free system. If we neglect gravity, the wave-function associated to the free system is a unitary representation of the Poincare group, regardless of the interactions occurring within the free system [159].

When the Hilbert space is the direct sum of irreducible representations of a symmetry group, then these representations will be defined by numbers (e.g. mass and spin) which are invariant under the symmetry group. Thus there will be a set of operators whose diagonal form corresponds to those invariants, we will call them Casimir operators. When the symmetry group is abelian and continuous (e.g. translation in time), then the generator of the group (e.g. the Hamiltonian) is a Casimir operator, and the invariant numbers defining the representations are called the constants of motion. Certainly, when we move from non-relativistic quantum mechanics and consider instead the Poincare group, then the Hamiltonian is no longer a Casimir operator and the notion of constants of motion needs to be reviewed. Nevertheless, the Casimir operators can be chosen arbitrarily (just like the Hamiltonian in non-relativistic quantum mechanics).

For a positive squared mass, the spin and the sign of the Energy are also Poincare invariants. The sign of the Energy times the modulus of the mass is the center-of-mass Energy, while the spin is the center-of-mass angular momentum. Thus, the Casimir operator whose eigenvalues are the center-of-mass Energy may have negative eigenvalues and it will be the analogous operator to the Hamiltonian of the non-relativistic formalism. As will be seen in Section 4, such operator has the formal form of the Hamiltonian action (i.e. it is the difference between the generator of translations in time and the Hamiltonian operator). The Casimir operators necessarily commute with the momentum operator and thus they do not change the 3-momentum eigenstate. Thus we

can solve the problem in a basis where the 3-momentum operator is diagonal.

In such a basis, the translations in space-time can be written as $T(x)\Psi(\gamma\vec{v}) = e^{iM\tau(\gamma\vec{v},x)}\Psi(\gamma\vec{v})$, where $\tau(\gamma\vec{v},x) = \gamma x_0 - \gamma\vec{v}\cdot\vec{x}$.

Note that $\gamma = \sqrt{1+\gamma^2\vec{v}^2}$ is a function of $\gamma\vec{v}$. Thus in a basis where the 3-momentum is diagonal, the translations in space-time have the same structure as the time-evolution in non-relativistic space-time, with M playing the role of the non-relativistic Hamiltonian and the numerical factor $\tau(\gamma\vec{v},x)$ playing the role of the time (it is indeed the proper-time).

For each 3-momentum eigenstate, there is a corresponding inertia referential where the 3-momentum is null, i.e. the referential of the center-of-mass. In such referential, the modulus of the energy is the invariant mass, the signal of the energy is also a Lorentz invariant and the angular momentum is the spin. Thus, the eigenvalues of the Hamiltonian and angular momentum operator in the center-of-mass define three Lorentz invariants which define the Poincare representation completely.

Despite we do not know a priori the diagonal form of the Hamiltonian, we know that it is either continuous or discrete in the neighborhood of the eigenvalue 0 (in the referential of the center-of-mass). If it is continuous then the zero energy has null measure. If it is discrete, we can modify the Hamiltonian adding an appropriate constant such that the zero energy is not one of the eigenvalues. Note that this is only possible in a complex Hilbert space and this is equivalent to adding to the system a free massive particle with null 3-momentum relative to the system. In any case, we can assume without loss of generality that our system is a quantum superposition of massive free systems with null 3-momentum. Then, the Lorentz transformations become known and are given by the Wigner irreducible massive representations of the Poincare group[159]. If the Hamiltonian is bounded from below then the vacuum state is not Lorentz invariant, as it was already suggested[182].

In the center-of-mass, the relevant group is not the Poincare group, but the little group of spatial rotations and the translation in time[159]. Thus the spatial and time coordinates of space-time, become separated. The fields are no longer representations of the Lorentz group, but only of the rotation group and the canonical commutation relations are not in conflict with the little group of spatial rotations.

Note that we use Wigner's convention for the definition of the 3-momentum of the free complete system: it is the eigenvalue of the generator of the translations in space for the complete system (i.e. all fields defining the phase-space are translated in space). Thus in the center-of-mass, the algebra of operators has a constraint imposing that the operators are translation invariant.

The cyclic vector defining the Hilbert space needs not be translation-invariant (in the center-of-mass), just the operators need to be translation-invariant in the center-of-mass[28]. This gives us a big

[28]As it was discussed in Chapter /pub/parametrization.

freedom to choose the cyclic vector defining the Hilbert space (which is related with the initial state of the system).

We assume that the translations in space of the complete system conserve the Hamiltonian and the constraints equations, such that setting the total 3-momentum to zero in no way conflicts with the constrained Hamiltonian system. Nevertheless, the restriction that there is a referential where the total 3-momentum is null, excludes the free complete system from traveling at the speed of light (e.g. a photon with non-null energy). Then the dynamics determined by the Hamiltonian becomes linked with the time coordinate (for a photon this would not be the case[183]).

Therefore and unlike what it is often claimed in the literature, it is false that (canonical) quantization is incompatible with Lorentz covariance. Note that the phenomenologically successful (but ill defined) path integral formalism based on the Lagrangian is in fact equivalent to a path integral based on the Hamiltonian[184]. In our formalism, the only restriction is that we need to consider representations with positive squared mass, then the dynamics determined by the Hamiltonian becomes linked with the time coordinate[183]. The question why only positive squared masses are relevant is a reformulation of question 2) which will be left open in this paper. Similar assumptions concerning the energy-momentum of the full system are also done in the Källén-Lehmann representation of a non-perturbative two-point correlation function, where it is assumed that the eigenvalues of the 3-momentum squared are not larger than those of the squared energy[185].

The special role of the little group of rotations and the time evolution in our definition of a (special relativistic) Quantum Field Theory may seem to be a step back in the road towards a general relativistic quantum theory. In Chapter /pub/timepiece we show that this is not the case, in fact our rigorous definition of a Quantum Yang-Mills theory marks *"a turning point in the mathematical understanding of quantum field theory, with a chance of opening new horizons for its applications"* in quantum gravity, as requested in reference[31].

We also make a comment on relativistic causality: the fact that we are considering only Poincare representations with non-negative squared mass leaves us in a good position to guarantee relativistic causality. However, we are working only in the 3-momentum space of the (free) complete quantum system. In order to study relativistic causality we need to make a unitary transformation to the 3-coordinate space, thus we need to define a position operator for a free quantum system.

Motivation

The Poincare group was first introduced as the set of transformations that leave invariant the Maxwell equations for the classical electromagnetic field. The complex representations of the Poincare group were systematically studied[159][186][187][188] and used in the definition of quantum

fields[189][190].

The formulation of quantum mechanics with a complex Hilbert space is equivalent to a formulation with a real Hilbert space and particular density matrix and observables[157]. Moreover, for time-independent Hamiltonian, quantum mechanics can be defined as the eigenvalue problem $(H(\vec{x})-E)\Psi(\vec{x})=0$[191], in the relativistic version, the energy may be replaced by the mass squared in the equation $((\eta^{\mu\nu}\partial_\mu\partial_\nu)-m^2)\Psi(x)=0$. Quantum Theory on real Hilbert spaces was investigated before[192][193][193][194][195], the main conclusion was that the formulation of non-relativistic Quantum Mechanics with a real Hilbert space is necessarily equivalent to the complex formulation. We could not find in the literature a systematic study on the real representations of the Poincare group, as it seems to be common assumptions:
1) since non-relativistic Quantum Mechanics is necessarily complex then the relativistic version must also be—it is hard to accept this as relativistic causality requires the existence of anti-particles[196];
2) the energy positivity implies complex Poincare representations—it is not necessarily true as only in a many-particles description the energy positivity is well defined, remember the Feynman–Stueckelberg or the Dirac sea interpretations of anti-particles[197][198];
3) Quantum Field Theory based on the Wightman axioms (which assume complex Poincare representations) is the most general framework incorporating the physics principles of Quantum Mechanics and Poincare covariance— the quantization of gauge fields does not respect Wightman axioms[55], attempts to define non-perturbatively a Quantum Field Theory with gauge interactions involve string theory or space-times with dimensions lower than 4, Euclidean metric or toroidal topology, in this context studying the real Poincare representations cannot be considered a departure from physics principles.

The reasons motivating the study of the real representations of the Poincare group are:

1) The real representations of the Poincare group play an important role in the classical electromagnetism and general relativity[199][200] and in Quantum Theory— e.g. the Higgs boson, Majorana fermion or quantum electromagnetic fields transform as real representations under the action of the Poincare group.

2) The parity—included in the full Poincare group—and charge-parity transformations are not symmetries of the Electroweak interactions[201]. It is not clear why the charge-parity is an apparent symmetry of the Strong interactions[202] or how to explain the matter-antimatter asymmetry[203] through the charge-parity violation. Since the self-conjugate finite-dimensional representations of the identity component of the Lorentz group are also representations of the parity, this work may be useful in future studies of the parity and charge-parity violations.

3) The localization of complex irreducible unitary representations of the Poincare group is incom-

patible with causality, Poincare covariance and energy positivity[204][205][206][207][208][209], while the complex representation corresponding to the photon is not localizable[210][211]. In contrast to the classical theory, in Quantum Field Theory with gauge interactions it impossible to define the electric charge localization of a large family of charged states in a meaningful way[55]. Since the free Dirac equation is self-conjugate in the Majorana basis, this study may be useful to the definition of a Poincare covariant position operator as a projection-valued measure (which Wightman considered to express the physical idea of localizability[212]).

Systems on real and complex Hilbert spaces

The position operator in Quantum Mechanics is mathematically expressed using a system of imprimitivity: a set of projection operators— associated with the coordinate space—acting on a Hilbert space; a group of symmetries acting both on the Hilbert space and on the coordinate space in a consistent way[211][213].

Many representations of a group—such as the finite-dimensional representations of semisimple Lie groups[214] or the unitary representations of separable locally compact groups[215]—are direct sums (or integrals) of irreducible representations, hence the study of these representations reduces to the study of the irreducible representations.

If the set of normal operators commuting with an irreducible real unitary representation of the Poincare group is isomorphic to the quaternions or to the complex numbers, then the most general position operator that the representation space admits is not complex linear, but real linear. Therefore, in this case, the real irreducible representations generalize the complex ones and these in turn generalize the quaternionic ones.

The study of irreducible representations on complex Hilbert spaces is in general easier than on real Hilbert spaces, because the field of complex numbers is the algebraic closure —where any polynomial equation has a root— of the field of real numbers. There is a well studied map, one-to-one or two-to-one and surjective up to equivalence, from the complex to the real linear finite-dimensional irreducible representations of a real Lie algebra[216][217].

Section 2 reviews and extends that map from the complex to the real irreducible representations—finite-dimensional or unitary—of a Lie group on a Hilbert space. Using Mackey's imprimitivity theorem, we extend the map further to systems of imprimitivity. This section follows closely the reference[216], with the addition that we will also use the Schur's lemma for unitary representations on a complex Hilbert space[218].

Related studies can be found in the references[219][220].

Finite-dimensional representations of the Lorentz group

The Poincare group, also called inhomogeneous Lorentz group, is the semi-direct product of the translations and Lorentz Lie groups[214]. Whether or not the Lorentz and Poincare groups include the parity and time reversal transformations depends on the context and authors. To be clear, we use the prefixes full/restricted when including/excluding parity and time reversal transformations. The Pin(3,1)/SL(2,C) groups are double covers of the full/restricted Lorentz group[221]. The semi-direct product of the translations with the Pin(3,1)/SL(2,C) groups is called IPin(3,1)/ISL(2,C) Lie group — the letter (I) stands for inhomogeneous.

A projective representation of the Poincare group on a complex/real Hilbert space is an homomorphism, defined up to a complex phase/sign, from the group to the automorphisms of the Hilbert space. Since the IPin(3,1) group is a double cover of the full Poincare group, their projective representations are the same[221]. All finite-dimensional projective representations of a simply connected group, such as SL(2,C), are usual representations[196]. Both SL(2,C) and Pin(3,1) are semi-simple Lie groups, and so all its finite-dimensional representations are direct sums of irreducible representations[214]. Therefore, the study of the finite-dimensional projective representations of the restricted Lorentz group reduces to the study of the finite-dimensional irreducible representations of SL(2,C).

The Dirac spinor is an element of a 4 dimensional complex vector space, while the Majorana spinor is an element of a 4 dimensional real vector space[222][223][224][225]. The complex finite-dimensional irreducible representations of SL(2,C) can be written as linear combinations of tensor products of Dirac spinors.

In Section 2.3 we will review the Pin(3,1) and SL(2,C) semi-simple Lie groups and its relation with the Majorana, Dirac and Pauli matrices. We will obtain all the real finite-dimensional irreducible representations of SL(2,C) as linear combinations of tensor products of Majorana spinors, using the map from Section 2. Then we will check that all these real representations are also projective representations of the full Lorentz group, in contrast with the complex representations which are not all projective representations of the full Lorentz group. We could not find these results explicitly in the literature but they are straightforward to derive and so probably known by some people, the results are derived here for completeness and explicitness.

Unitary representations of the Poincare group

According to Wigner's theorem, the most general transformations, leaving invariant the modulus of the internal product of a Hilbert space, are: unitary or anti-unitary operators, defined up to a complex phase, for a complex Hilbert space; unitary, defined up to a signal, for a real Hilbert space[92][211]. This motivates the study of the (anti-)unitary projective representations of the full

Poincare group.

All (anti-)unitary projective representations of ISL(2,C) are, up to isomorphisms, well defined unitary representations, because ISL(2,C) is simply connected[196]. Both ISL(2,C) and IPin(3,1) are separable locally compact groups and so all its (anti-)unitary projective representations are direct integrals of irreducible representations[215]. Therefore, the study of the (anti-)unitary projective representations of the restricted Poincare group reduces to the study of the unitary irreducible representations of ISL(2,C).

The spinor fields, space-time dependent spinors, are solutions of the free Dirac equation[226]. The real/complex Bargmann-Wigner fields[227][228], space-time dependent linear combinations of tensor products of Majorana/Dirac spinors, are solutions of the free Dirac equation in each tensor index. The complex unitary irreducible projective representations of the Poincare group with discrete spin or helicity can be written as complex Bargmann-Wigner fields.

In Section 2.4, we will obtain all the real unitary irreducible projective representations of the Poincare group, with discrete spin or helicity, as real Bargmann-Wigner fields, using the map from Section 2. For each pair of complex representations (of ISL(2,C)) with positive/negative energy, there is one real representation. We will define the Majorana-Fourier and Majorana-Hankel unitary transforms of the real Bargmann-Wigner fields, relating the coordinate space with the linear and angular momenta spaces. We show that any localizable unitary representation of the Poincare group (ISL(2,C)), compatible with Poincare covariance, verifies: 1) it is a direct sum of irreducible representations which are massive or massless with discrete helicity. 2) it respects causality; 3) if it is complex it contains necessarily both positive and negative energy subrepresentations 4) it is an irreducible representation of the Poincare group (including parity) if and only if it is: a)real and b)massive with spin 1/2 or massless with helicity 1/2. If a) and b) are verified the position operator matches the coordinates of the Dirac equation.

The free Dirac equation is diagonal in the Newton-Wigner representation[210], related to the Dirac representation through a Foldy-Wouthuysen transformation[229][230] of Dirac spinor fields. The Majorana-Fourier transform, when applied on Dirac spinor fields, is related with the Newton-Wigner representation and the Foldy-Wouthuysen transformation. In the context of Clifford Algebras, there are studies on the geometric square roots of -1 [231] and on the generalizations of the Fourier transform[232], with applications to image processing[233]. It was showed before that point localized local quantum fields—operator valued distributions satisfying the Wightman axioms— cannot be a massless infinite spin representation [234].

The current literature related with the position operator of representations of the Poincare group include: modular and string-like localization in the context of local quantum field

theory [235][236][237][238]; non-commutative coordinates [239][240]; coordinates based on equations [241][242]; unsharp (fuzzy) localization using positive operator valued measures [243][244][245][246][247]; localization of the energy density [248][249][250][251][252]; two dimensional[253] or axial symmetric[254] or space-time[255] localization of photons; pseudo-hermitian representations of Quantum Mechanics[256]. All the above mentioned approaches departure, in one way or another, from using the system of imprimitivity to implement the position operator for a unitary representation of the Poincare group. Finally taking the Newton-Wigner position seriously has the problem that for spin one-half the position does not coincide with the coordinates appearing in the Dirac equation, with all the phenomenological consequences that it implies[257][258]. The results presented in Section 2.4 are a motivation to not departure from the systems of imprimitivity to describe the position of relativistic systems.

Energy Positivity

While it seems that the localization of particles in either relativistic quantum mechanics[235] or relativistic quantum field theory[259] is not possible (at least with the properties we would expect), it is not clear whether this is a limitation of the relativistic Quantum framework itself, or due to some properties of our definition of particles which are incompatible with a proper definition of localization. For instance, it is expected that by specifying the energy-momentum properties of the vacuum we then may have troubles to define the localization of all the states related with the vacuum—a consequence of the Reeh-Schlieder theorem[260]—, since momentum and position do not commute; that does not imply that the localization cannot be defined at all within the relativistic Quantum framework once we relax the energy-momentum properties of the vacuum.

In non-relativistic Quantum Mechanics the time is invariant under the Galilean transformations —excluding the time reversal transformation—and so the generator of translations in time is also invariant. Therefore, the positivity of the Energy and the localization in space of a state can be defined simultaneously. In relativistic Quantum Mechanics, the time is not invariant under Lorentz transformations, as a consequence the positivity of the Energy and the localization in space of a state cannot be defined simultaneously— the corresponding projection operators do not commute.

In the framework of Algebraic QFT(related with the Wightman axioms), there is a definition for Quantum Field Theories with interactions in terms of formal power series in the coupling constants[235], which is based on the canonical quantization where the positivity of Energy is well defined by construction and the localization problem is handled by introducing anti-particles—causality implies the existence of anti-particles[196], a related approach led Dirac to predict the positron[197]. Yet, it is also possible to build a description of a many particles system where the localization in space of a state is well defined by construction and the Energy positivity problem can

be handled with the Feynman–Stueckelberg interpretation for anti-particles, as Energy positivity and localization are complementary. Dirac himself was the first to consider an approach which do not assume the positivity of Energy by construction[261] and quantization in de Sitter space-time may be achieved in a related approach[262].

If we need to revisit known results at a deep level to non-perturbatively define a Quantum Field Theory with interactions[31], defining the position operator with a projection-valued measure(which Wightman considered to represent the physical idea of localizability) in a many-particles system seems to be useful. In the known formulations of Algebraic QFT there are no pure states (potentially preferring an ensemble interpretation of Quantum Mechanics)[235] and so the position operator is not defined with a projection-valued measure.

The description of a many-particles system based on the definition of the position operator with a projection-valued measure will be discussed in the section5.

Systems on real and complex Hilbert spaces

Definition 1 (System). A system (M, V) is defined by:
1) the (real or complex) Hilbert space V;
2) a set M of bounded endomorphisms on V.

The representation of a symmetry is an example of a system: a representation space plus a set of operators representing the action of the symmetry group in the representation space[263].

Definition 2 (Complexification). Consider a system (M, W) on a real Hilbert space. The system (M, W^c) is the complexification of the system (M, W), defined as $W^c \equiv \mathbb{C} \otimes W$, with the multiplication by scalars such that $a(bw) \equiv (ab)w$ for $a, b \in \mathbb{C}$ and $w \in W$. The internal product of W^c is defined—for $u_r, u_i, v_r, v_i \in W$ and $< v_r, u_r >$ the internal product of W—as:

$$< v_r + iv_i, u_r + iu_i >_c \equiv < v_r, u_r > + < v_i, u_i > + i < v_r, u_i > - i < v_i, u_r > \tag{42}$$

Definition 3 (Realification). Consider a system (M, V) on a complex Hilbert space. The system(M, V^r) is the realification of the system (M, V), defined as $V^r \equiv V$ is a real Hilbert space with the multiplication by scalars restricted to reals such that $a(v) \equiv (a + i0)v$ for $a \in \mathbb{R}$ and $v \in V$. The internal product of V^r is defined—for $u, v \in V$ and $< v, u >$ is the internal product of V—as:

$$< v, u >_r \equiv \frac{< v, u > + < u, v >}{2} \tag{43}$$

Note 4. *Let H_n, with $n \in \{1,2\}$, be two Hilbert spaces with internal products $<,>: H_n \times H_n \to \mathbb{F}, (\mathbb{F} = \mathbb{R}, \mathbb{C})$. A (anti-)linear operator $U : H_1 \to H_2$ is (anti-)unitary iff:*
1) it is surjective;
2) for all $x \in H_1$, $< U(x), U(x) >=< x, x >$.

Proposition 5. *Let H_n, with $n \in \{1,2\}$, be two complex Hilbert spaces and H_n^r its complexification. The following two statements are equivalent:*

1) The operator $U : H_1 \to H_2$ is (anti-)unitary;

2) The operator $U^r : H_1^r \to H_2^r$ is (anti-)unitary, where $U^r(h) \equiv U(h)$, for $h \in H_1$.

Proof. Since $< h, h >=< h, h >_r$ and $U^r(h) = U(h)$, for $h \in H_1$, we get the result. □

Definition 6 (Equivalence). Consider the systems (M, V) and (N, W):
1) A normal endomorphism of (M, V) is a bounded endomorphism $S : V \to V$ commuting with $S^\dagger$ and m, for all $m \in M$; an anti-endomorphism in a complex Hilbert space is an anti-linear endomorphism;
2) An isometry of (M, V) is a unitary operator $S : V \to V$ commuting with m, for all $m \in M$;
3) The systems (M, V) and (N, W) are unitary equivalent iff there is a isometry $\alpha : V \to W$ such that $N = \{\alpha m \alpha^\dagger : m \in M\}$.

We use the trivial extension of the definition of irreducibility from representations to systems.

Definition 7 (Irreducibility). Consider the system (M, V) and let W be a linear subspace of V:
1) (M, W) is a (topological) subsystem of (M, V) iff W is closed and invariant under the system action, that is, for all $w \in W$:$(mw) \in W$, for all $m \in M$;
2) A system (M, V) is (topologically) irreducible iff their only sub-systems are the non-proper (M, V) or trivial $(M, \{0\})$ sub-systems, where $\{0\}$ is the null space.

Definition 8 (Structures). 1) Consider a system (M, V) on a complex Hilbert space. A C-conjugation operator of (M, V) is an anti-unitary involution of V commuting with m, for all $m \in M$;
2) Consider a system (M, W) on a real Hilbert space. A R-imaginary operator of (M, W), J, is an isometry of (M, W) verifying $J^2 = -1$.

The map from the complex to the real systems

Definition 9. Consider an irreducible system (M, V) on a complex Hilbert space:
1) The system is C-real iff there is a C-conjugation operator;
2) The system is C-pseudoreal iff there is no C-conjugation operator but there is an anti-unitary

operator of (M, V);
3) The system is C-complex iff there is no anti-unitary operator of (M, V).

Definition 10. Consider the system (M, W) on a real Hilbert space and let (M, W^c) be its complexification: 1) (M, W) is R-real iff (M, W^c) is C-real irreducible;
2) (M, W) is R-pseudoreal iff (M, V) is C-pseudoreal irreducible, with $W^c = V \oplus \bar{V}$; 3) (M, W) is R-complex iff (M, V) is C-complex irreducible, with $W^c = V \oplus \bar{V}$.

Proposition 11. *Any irreducible real system is R-real or R-pseudoreal or R-complex.*

Proof. Consider an irreducible system (M, W) on a real Hilbert space. There is a C-conjugation operator of (M, W^c), θ, defined by $\theta(u + iv) \equiv (u - iv)$ for $u, v \in W$, verifying $(W^c)_\theta = W$.

Let (M, X^c) be a proper non-trivial subsystem of (M, W^c). Then θ is a C-conjugation operator of the subsystems (M, Y^c) and (M, Z^c), where $Y^c \equiv \{u + \theta v : u, v \in X^c\}$ and $Z^c \equiv \{u : u, \theta u \in X^c\}$. Therefore, $Y^c = \{u + iv : u, v \in Y\}$ and $Z^c = \{u + iv : u, v \in Z\}$, where $Y \equiv \{\frac{1+\theta}{2} u : u \in Y^c\}$ and $Z \equiv \{\frac{1+\theta}{2} u : u \in Z^c\}$, are invariant closed subspaces of W. If $Y = \{0\}$ then $Z = \{0\}$ and $Y^c = X^c = \{0\}$, in contradiction with X^c being non-trivial. If $Z = W$ then $Y = W$ and $Z^c = X^c = W^c$, in contradiction with X^c being proper. Therefore $Z = \{0\}$ and $Y = W$, which implies $Z^c = \{0\}$ and $Y^c = W^c$.

So, (M, W) is equivalent to $(M, (X^c)^r)$, due to the existence of the bijective linear map $\alpha : (X^c)^r \to W$, $\alpha(u) = u + \theta u$, $\alpha^{-1}(u + \theta u) = u$, for $u \in (X^c)^r$. Suppose that there is a C-conjugation operator of (M, X^c), θ'. Then $(M, W_\pm)$ is a proper non-trivial subsystem of (M, W), where $W_\pm \equiv \{\frac{1 \pm \theta'}{2} w : w \in W\}$, in contradiction with (M, W) being irreducible. □

Proposition 12. *Any real system which is R-real or R-pseudoreal or R-complex is irreducible.*

Proof. Consider an irreducible system on a complex Hilbert space (M, V). There is a R-imaginary operator J of the system (M, V^r), defined by $J(u) \equiv iu$, for $u \in V^r$.

Let (M, X^r) be a proper non-trivial subsystem of (M, V^r). Then J is an R-imaginary operator of (M, Y^r) and (M^r, Z^r), where $Y^r \equiv \{u + Jv : u, v \in X^r\}$ and $Z^r \equiv \{u : u, Ju \in X^r\}$. Then (M, Y) and (M, Z) are subsystems of (M, V), where the complex Hilbert spaces $Y \equiv Y^r$ and $Z \equiv Z^r$ have the scalar multiplication such that $(a + ib)(y) = ay + bJy$, for $a, b \in \mathbb{R}$ and $y \in Y$ or $y \in Z$. If $Y = \{0\}$, then $Z = X^r = \{0\}$ which is in contradiction with X^r being non-trivial. If $Z = V$, then $Y = V$ and $X^r = V^r$ which is in contradiction with X^r being non-trivial. So $Z = \{0\}$ and $Y = V$, which implies that $V = (X^r)^c$.

Then there is a C-conjugation operator of (M, V), θ, defined by $\theta(u + iv) \equiv u - iv$, for $u, v \in X^r$. We have $X^r = V_\theta$. Suppose there is a R-imaginary operator of (M, V_θ), J'. Then $(M, V_\pm)$, where $V_\pm \equiv \{\frac{1 \pm iJ'}{2} v : v \in V\}$, are proper non-trivial subsystems of (M, V), in contradiction with (M, V)

being irreducible.

Therefore, if (M, V) is C-real, then (M, V_θ) is R-real irreducible. If (M, V) is C-pseudoreal or C-complex, then (M, V_θ^r) is R-pseudoreal or R-complex, irreducible. □

Schur Systems

Definition 13 (Schur System). A system (M, V), on a complex Hilbert space V, is a Schur system if the set of normal operators of (M, V) is isomorphic to $\mathbb{C}$.
Consider an irreducible system (M, W), on a real Hilbert space W and let (M, W^c) be its complexification: 1) (M, W) is Schur R-real iff (M, W^c) is Schur C-real;
2) (M, W) is Schur R-pseudoreal iff (M, V) is Schur C-pseudoreal, with $W^c = V \oplus \bar{V}$;
3) (M, W) is Schur R-complex iff (M, V) is Schur C-complex, with $W^c = V \oplus \bar{V}$.

Lemma 14. *Consider a Schur system (M, V) on a complex Hilbert space. An anti-isometry of (M, V), if it exists, is unique up to a complex phase.*

Proof. Let θ_1,θ_2 be two anti-isometries of (M, V). The product $(\theta_2\theta_1)$ is an isometry of (M, V); since (M, V) is irreducible, $(\theta_2\theta_1) = e^{i\phi}$; with $\phi \in \mathbb{R}$.

Therefore $\theta_2 = \alpha\theta_1\alpha^{-1}$; where $\alpha \equiv e^{i\frac{\phi}{2}}$ is a complex phase. □

Proposition 15. *Two R-real Schur systems are isometric iff their complexifications are isometric.*

Proof. Let (M, V) and (N, W) be C-real Schur systems, with θ_M and θ_N the respective C-conjugation operators. If there is an isometry $\alpha : V \to W$ such that $\alpha M = N\alpha$, then $\vartheta \equiv \alpha\theta_M\alpha^{-1}$ is an anti-isometry of (N, W). Since it is unique up to a phase, then $\theta_N = e^{i\phi}\vartheta$. Therefore $e^{i\frac{\phi}{2}}\alpha$ is an isometry between (M, V_θ) and (N, W_θ), where $V_{\theta_M} \equiv \{(1 + \theta_M)v : v \in V\}$. □

Proposition 16. *Two C-complex or C-pseudoreal Schur systems are isometric or anti-isometric iff their realifications are isometric.*

Proof. Let (M, V) and (N, W) be R-complex or R-pseudoreal Schur systems, with J_M and J_N the respective R-imaginary operators. If there is an isometry $\alpha : V \to W$ such that $\alpha M = N\alpha$, then $K \equiv \alpha J_M\alpha^{-1}$ is a R-imaginary operator of (N, W). When considering (N, W_{J_N}) and (M, V_{J_M}), where $W_{J_N} \equiv \{(1 - iJ_N)w : w \in W\}$, we get that $(1 - J_NK)(1 - KJ_N) = r$ as an operator of W_{J_N}, where r is a non-negative null real scalar. If $c = 0$ then $K = -J_N$ and α defines an anti-isometry between (M, V_{J_M}) and (N, W_{J_N}). If $c \neq 0$ then $(1 - J_NK)\alpha c^{-\frac{1}{2}}$ is an isometry between (M, V_{J_M}) and (N, W_{J_N}). □

Proposition 17. *The space of normal operators of a R-real Schur system is isomorphic to $\mathbb{R}$.*

Proof. Let (M, V) be a C-real Schur system, with θ the C-conjugation operator. If there is an

endomorphism $\alpha : V \to V$ such that $\alpha M = M\alpha$, we know that $\alpha = re^{i\varphi}$. Then the endomorphism of V_θ is a real number. □

Proposition 18. *The space of normal operators of a R-complex Schur system is isomorphic to* $\mathbb{C}$.

Proof. Let (M, V) be a R-complex Schur system, with J the R-imaginary operator. If there is a normal operator α of (M, V), then $KK^\dagger$ is a normal operator of the C-complex Schur system (M, V_J), where $K \equiv (\alpha + J\alpha J)$ and $V_J \equiv \{(1 - iJ)v : v \in V\}$. If $KK^\dagger = r > 0$, then $\frac{K}{\sqrt{r}}$ is unitary and V_J is equivalent to $\overline{V}_J$ which would imply that (M, V) is C-pseudoreal. Therefore $K = 0$ and hence α is a normal operator of (M, V_J), so $\alpha = re^{J\theta}$. □

Proposition 19. *The space of normal operators of a R-pseudoreal Schur system is isomorphic to* $\mathbb{H}$ *(quaternions).*

Proof. Let (M, V) be a R-pseudoreal Schur system, with J the R-imaginary operator. If there is an endomorphism α of (M, V), then $SS^\dagger$ and $TT^\dagger$ are a self-adjoint endomorphisms of the C-complex Schur system (M, V_J), where $S \equiv (\alpha - J\alpha J)/2$, $T \equiv (\alpha + J\alpha J)/2$ and $V_J \equiv \{(1 - iJ)v : v \in V\}$. Let K be an unitary operator of (M, V) and anti-commuting with J, then $K^2 = e^{J\theta}$ and $Ke^{J\theta} = K(K^2) = (K^2)K = e^{J\theta}K$, therefore $K^2 = -1$. If $TT^\dagger = t > 0$, then $\frac{T}{\sqrt{t}}$ is unitary and anti-commutes with J, TK is a normal endomorphism of (M, V_J) and therefore $T = Kc + KJd$; if $TT^\dagger = 0$ then $c = d = 0$. If $SS^\dagger = s > 0$, then $\frac{S}{\sqrt{s}}$ is unitary and commutes with J, S is a normal endomorphism of (M, V_J) and therefore $S = a + Jb$; if $SS^\dagger = 0$ then $a = b = 0$.

Therefore $\alpha = S + T = a + Jb + Kc + KJd$, which is isomorphic to the quaternions. □

Finite-dimensional representations

Lemma 20 (Schur's lemma for finite-dimensional representations[218]). *Consider an irreducible finite-dimensional representation* (M_G, V) *of a Lie group* G *on a complex Hilbert space* V*. If the representation* (M_G, V) *is irreducible then any endomorphism* S *of* (M_G, V) *is a complex scalar.*

Lemma 21. *Consider an irreducible complex finite-dimensional representation* (M, V) *on a complex Hilbert space. Then there is internal product such that: 1) The system is C-real iff there is an anti-linear involution of* (M, V)*;*
2) The system is C-pseudoreal iff there is not an anti-linear bounded involution of (M, V)*, but there is an anti-isomorphism of* (M, V)*;*
3) The system is C-complex iff there is no anti-isomorphism of (M, V)*.*

Proof. Let S be an anti-isomorphism of an irreducible representation (M, V). Then $S^2 = re^{i\varphi}$. But S^2 commutes with S which is anti-linear, so $S^2 = \pm r$. So, there is an internal product such that S is anti-unitary. □

Definition 22. A finite-dimensional system is completely reducible iff it can be expressed as a direct sum of irreducible systems.

Note 23 (Weyl theorem). *All finite-dimensional representations of a semi-simple Lie group (such as SL(2,C)) are completely reducible.*

Unitary representations and Systems of Imprimitivity

Definition 24 (Normal System). A System (M, V) is normal iff M is a set M of normal operators on V closed under Hermitian conjugation—for all $m \in M$ there is $n \in M$ such that $n = m^\dagger$.

A unitary representation or a System of Imprimitivity are examples of a normal System.

Note 25. *$W^\perp$ is the orthogonal complement of the subspace W of the Hilbert space V if:*
1) $V = W \oplus W^\perp$, that is, all $v \in V$ can be expressed as $v = w + x$, where $w \in W$ and $x \in W^\perp$;
2) if $w \in W$ and $x \in W^\perp$, then $x^\dagger w = 0$.

Lemma 26. *Consider a normal system (M, V). Then, for all subsystem (M, W) of (M_G, V), $(M_G, W^\perp)$ is also a subsystem of (M, V), where $W^\perp$ is the orthogonal complement of the subspace W.*

Proof. Let (M, W) be a subsystem of (M, V). $W^\perp$ is the orthogonal complement of W.

For all $x \in W^\perp$, $w \in W$ and $m \in M$, $< mx, w >=< x, m^\dagger w >$.

Since W is invariant and there is $n \in M$, such that $n = m^\dagger$, then $w' \equiv (m^\dagger w) \in W$.

Since $x \in W^\perp$ and $w' \in W$, then $< x, w' >= 0$.

This implies that if $x \in W^\perp$), also $(mx) \in W^\perp$, for all $m \in M$. □

Lemma 27. *Any Schur normal system on a complex Hilbert space is irreducible.*

Proof. Let (M, W) and $(M, W^\perp)$ be sub-systems of the complex Schur system (M, V), where $W^\perp$ is the orthogonal complement of W.

There is a bounded endomorphism $P : V \to V$, such that, for $w, w' \in W$, $x, x' \in W^\perp$, $P(w+x) = w$. $P^2 = P$ and P is hermitian:

$$< w' + x', P(w + x) >=< w', w >=< P(w' + x'), w + x > \tag{44}$$

Let $w' \equiv mw \in W$ and $x' \equiv mx \in W^\perp$:

$$
\begin{aligned}
mP(w+x) &= mw = w' \\
Pm(w+x) &= P(w'+x') = w'
\end{aligned}
\tag{45}
$$

Which implies that P commutes with all $m \in M$, so $P \in \{0,1\}$. If $P = 1$, then $W = V$, if $P = 0$, then W is the null space. □

So a complex Schur normal system is irreducible, and hence, from Defns.Definition 10,Definition 13 and Prop.Proposition 12, a real Schur normal system is also irreducible.

Lemma 28 (Schur's lemma for unitary representations[218]). *Consider an irreducible unitary representation (M, V) of a Lie group G on a complex Hilbert space V. If the representation (M, V) is irreducible then any normal operator N of (M, V) is a scalar.*

Definition 29. A unitary system is completely reducible iff it can be expressed as a direct integral of irreducible systems.

Note 30. *All unitary representations of a separable locally compact group (such as the Poincare group) are completely reducible.*

Systems of Imprimitivity

Definition 31. Consider a measurable space (X, M), where M is a σ-algebra of subsets of X. A projection-valued-measure, π, is a map from M to the set of self-adjoint projections on a Hilbert space H such that $\pi(X)$ is the identity operator on H and the function $< \psi, \pi(A)\psi >$, with $A \in M$ is a measure on M, for all $\psi \in H$.

Definition 32. Suppose now that X is a representation of G. Then, a system of imprimitivity is a pair (U, π), where π is a projection valued measure and U an unitary representation of G on the Hilbert space H, such that $U(g)\pi(A)U^{-1}(g) = \pi(gA)$.

Note 33 (Imprimitivity Theorem (thrm 6.12 [264][265][266][267][211])). *Let G be a Lie group, H its closed subgroup. Let a pair (V, E) be a system of imprimitivity for G based on G/H on a separable complex Hilbert space. Then there exists a representation L of H such that (V, E) is equivalent to the canonical system of imprimitivity (V_L, E_L). For any two representations L, L' of the subgroup H the corresponding canonical systems of imprimitivity are equivalent if and only if L, L' are equivalent. The sets of normal operators commuting with (V_L, E_L) and L are isomorphic.*

Lemma 34 (Schur's lemma for systems of imprimitivity[218]). *Let a pair (V, E) be a system of imprimitivity for G based on G/H on a separable complex Hilbert space. If (V, E) is irreducible then then any normal operator N commuting with (V, E) is a scalar.*

Proof. Consider a representation L of H such that (V, E) is equivalent to the canonical system of imprimitivity (V_L, E_L). If L would be reducible then there would be a non-trivial normal projection operator commuting with L, but then the imprimitivity theorem implies that there would also be a non-trivial normal projection operator commuting with (V, E) which is in contradiction with the irreducibility of (V, E), therefore L is irreducible. The Schur's lemma for unitary representations then implies that any normal operator commuting with L is a scalar, the imprimitivity theorem then implies the result. □

So we can define a map from the real to the complex systems of imprimitivity—analogous to the one for unitary representations. So we extended an existing map from the complex to the real linear finite-dimensional irreducible representations of a real Lie algebra[216][217] to the infinite-dimensional (unitary) case.

Finite-dimensional representations of the Lorentz group

We could not find the following results explicitly in the literature but they are straightforward to derive and so probably known by some people, the results are derived here for completeness and explicitness.

Majorana, Dirac and Pauli Matrices and Spinors

Definition 35. $\mathbb{F}^{m\times n}$ is the vector space of $m \times n$ matrices whose entries are elements of the field $\mathbb{F}$.

In the next remark we state the Pauli's fundamental theorem of gamma matrices. The proof can be found in the reference[268].

Note 36 (Pauli's fundamental theorem). *Let A^μ, B^μ, $\mu \in \{0, 1, 2, 3\}$, be two sets of 4×4 complex matrices verifying:*

$$\begin{aligned} A^\mu A^\nu + A^\nu A^\mu &= -2\eta^{\mu\nu} \\ B^\mu B^\nu + B^\nu B^\mu &= -2\eta^{\mu\nu} \end{aligned} \tag{46}$$

Where $\eta^{\mu\nu} \equiv diag(+1, -1, -1 - 1)$ is the Minkowski metric.

1) There is an invertible complex matrix S such that $B^\mu = SA^\mu S^{-1}$, for all $\mu \in \{0, 1, 2, 3\}$. S is unique up to a non-null scalar.

2) If A^μ and B^μ are all unitary, then S is unitary.

Proposition 37. *Let α^μ, β^μ, $\mu \in \{0, 1, 2, 3\}$, be two sets of 4×4 real matrices verifying:*

$$\begin{aligned}\alpha^\mu\alpha^\nu+\alpha^\nu\alpha^\mu&=-2\eta^{\mu\nu}\\ \beta^\mu\beta^\nu+\beta^\nu\beta^\mu&=-2\eta^{\mu\nu}\end{aligned}\tag{47}$$

Then there is a real matrix S, with $|detS|=1$, such that $\beta^\mu=S\alpha^\mu S^{-1}$, for all $\mu\in\{0,1,2,3\}$. S is unique up to a signal.

Proof. From remark Note 36, we know that there is an invertible matrix T', unique up to a non-null scalar, such that $\beta^\mu=T'\alpha^\mu T'^{-1}$. Then $T\equiv T'/|det(T')|$ has $|detT|=1$ and it is unique up to a complex phase.

Conjugating the previous equation, we get $\beta^\mu=T^*\alpha^\mu T^{*-1}$. Then $T^*=e^{i2\theta}T$ for some real number θ. Therefore $S\equiv e^{i\theta}T$ is a real matrix, with $|detS|=1$, unique up to a signal. □

Definition 38. The Majorana matrices, $i\gamma^\mu$, $\mu\in\{0,1,2,3\}$, are 4×4 complex unitary matrices verifying:

$$(i\gamma^\mu)(i\gamma^\nu)+(i\gamma^\nu)(i\gamma^\mu)=-2\eta^{\mu\nu}\tag{48}$$

The Dirac matrices are $\gamma^\mu\equiv-i(i\gamma^\mu)$.

In the Majorana bases, the Majorana matrices are 4×4 real orthogonal matrices. An example of the Majorana matrices in a particular Majorana basis is:

$$\begin{aligned}i\gamma^1&=\begin{bmatrix}+1&0&0&0\\0&-1&0&0\\0&0&-1&0\\0&0&0&+1\end{bmatrix}\quad i\gamma^2=\begin{bmatrix}0&0&+1&0\\0&0&0&+1\\+1&0&0&0\\0&+1&0&0\end{bmatrix}\quad i\gamma^3=\begin{bmatrix}0&+1&0&0\\+1&0&0&0\\0&0&0&-1\\0&0&-1&0\end{bmatrix}\\ i\gamma^0&=\begin{bmatrix}0&0&+1&0\\0&0&0&+1\\-1&0&0&0\\0&-1&0&0\end{bmatrix}\quad i\gamma^5=\begin{bmatrix}0&-1&0&0\\+1&0&0&0\\0&0&0&+1\\0&0&-1&0\end{bmatrix}\quad=-\gamma^0\gamma^1\gamma^2\gamma^3\end{aligned}\tag{49}$$

In reference [269] it is proved that the set of five anti-commuting 4×4 real matrices is unique up to isomorphisms. So, for instance, with 4×4 real matrices it is not possible to obtain the euclidean signature for the metric.

Definition 39. The Dirac spinor is a 4×1 complex column matrix, $\mathbb{C}^{4\times1}$.

The space of Dirac spinors is a 4 dimensional complex vector space.

Lemma 40. *The charge conjugation operator Θ, is an anti-linear involution commuting with the Majorana matrices $i\gamma^\mu$. It is unique up to a complex phase.*

Proof. In the Majorana bases, the complex conjugation is a charge conjugation operator. Let Θ and Θ' be two charge conjugation operators operators. Then, $\Theta\Theta'$ is a complex invertible matrix

commuting with $i\gamma^\mu$, therefore, from Pauli's fundamental theorem, $\Theta\Theta' = c$, where c is a non-null complex scalar. Therefore $\Theta' = c^*\Theta$ and from $\Theta'\Theta' = 1$, we get that $c^*c = 1$. $\square$

Definition 41. Let Θ be a charge conjugation operator.

The set of Majorana spinors, denoted here by $Pinor$, is the set of Dirac spinors verifying the Majorana condition (defined up to a complex phase):

$$Pinor \equiv \{u \in \mathbb{C}^{4\times 1} : \Theta u = u\} \tag{50}$$

The set of Majorana spinors is a 4 dimensional real vector space. Note that the linear combinations of Majorana spinors with complex scalars do not verify the Majorana condition.

There are 16 linear independent products of Majorana matrices. These form a basis of the real vector space of endomorphisms of Majorana spinors, $End(Pinor)$. In the Majorana bases, $End(Pinor)$ is the vector space of 4×4 real matrices.

Definition 42. The Pauli matrices $\sigma^k,\ k \in \{1, 2, 3\}$ are 2×2 hermitian, unitary, anti-commuting, complex matrices. The Pauli spinor is a 2×1 complex column matrix. The space of Pauli spinors is denoted by $Pauli$.

The space of Pauli spinors, denoted here by $Pauli$, is a 2 dimensional complex vector space and a 4 dimensional real vector space. The realification of the space of Pauli spinors is isomorphic to the space of Majorana spinors.

On the Lorentz, SL(2,C) and Pin(3,1) groups

Note 43. *The Lorentz group, $O(1,3) \equiv \{\lambda \in \mathbb{R}^{4\times 4} : \lambda^T\eta\lambda = \eta\}$, is the set of real matrices that leave the metric, $\eta = diag(1, -1, -1, -1)$, invariant.*

The proper orthochronous Lorentz subgroup is defined by $SO^+(1,3) \equiv \{\lambda \in O(1,3) : det(\lambda) = 1, \lambda^0{}_0 > 0\}$. It is a normal subgroup. The discrete Lorentz subgroup of parity and time-reversal is $\Delta \equiv \{1, \eta, -\eta, -1\}$.

The Lorentz group is the semi-direct product of the previous subgroups, $O(1,3) = \Delta \ltimes SO^+(1,3)$.

Definition 44. The set Maj is the 4 dimensional real space of the linear combinations of the Majorana matrices, $i\gamma^\mu$:

$$Maj \equiv \{a_\mu i\gamma^\mu : a_\mu \in \mathbb{R},\ \mu \in \{0, 1, 2, 3\}\} \tag{51}$$

Definition 45. $Pin(3,1)$ [221] is the group of endomorphisms of Majorana spinors that leave the space Maj invariant, that is:

$$Pin(3,1) \equiv \left\{ S \in End(Pinor) : \ |detS| = 1, \ S^{-1}(i\gamma^{\mu})S \in Maj, \ \mu \in \{0,1,2,3\} \right\} \tag{52}$$

Proposition 46. *The map* $\Lambda : Pin(3,1) \to O(1,3)$ *defined by:*

$$(\Lambda(S))^{\mu}{}_{\nu} i\gamma^{\nu} \equiv S^{-1}(i\gamma^{\mu})S \tag{53}$$

is two-to-one and surjective. It defines a group homomorphism.

Proof. 1) Let $S \in Pin(3,1)$. Since the Majorana matrices are a basis of the real vector space Maj, there is an unique real matrix $\Lambda(S)$ such that:

$$(\Lambda(S))^{\mu}{}_{\nu} i\gamma^{\nu} = S^{-1}(i\gamma^{\mu})S \tag{54}$$

Therefore, Λ is a map with domain $Pin(3,1)$. Now we can check that $\Lambda(S) \in O(1,3)$:

$$\begin{aligned} &(\Lambda(S))^{\mu}{}_{\alpha} \eta^{\alpha\beta} (\Lambda(S))^{\nu}{}_{\beta} = -\frac{1}{2}(\Lambda(S))^{\mu}{}_{\alpha}\{i\gamma^{\alpha}, i\gamma^{\beta}\}(\Lambda(S))^{\nu}{}_{\beta} = \\ &= -\frac{1}{2} S\{i\gamma^{\mu}, i\gamma^{\nu}\}S^{-1} = S\eta^{\mu\nu}S^{-1} = \eta^{\mu\nu} \end{aligned} \tag{55}$$

We have proved that Λ is a map from $Pin(3,1)$ to $O(1,3)$.

2) Since any $\lambda \in O(1,3)$ conserve the metric η, the matrices $\alpha^{\mu} \equiv \lambda^{\mu}{}_{\nu} i\gamma^{\nu}$ verify:

$$\{\alpha^{\mu}, \alpha^{\nu}\} = -2\lambda^{\mu}{}_{\alpha}\eta^{\alpha\beta}\lambda^{\nu}{}_{\beta} = -2\eta^{\mu\nu} \tag{56}$$

In a basis where the Majorana matrices are real, from Proposition Proposition 37 there is a real invertible matrix S_{λ}, with $|detS_{\Lambda}| = 1$, such that $\lambda^{\mu}{}_{\nu} i\gamma^{\nu} = S_{\lambda}^{-1}(i\gamma^{\mu})S_{\lambda}$. The matrix S_{Λ} is unique up to a sign. So, $\pm S_{\lambda} \in Pin(3,1)$ and we proved that the map $\Lambda : Pin(3,1) \to O(1,3)$ is two-to-one and surjective.

3) The map defines a group homomorphism because:

$$\begin{aligned} &\Lambda^{\mu}{}_{\nu}(S_1)\Lambda^{\nu}{}_{\rho}(S_2) i\gamma^{\rho} = \Lambda^{\mu}{}_{\nu} S_2^{-1} i\gamma^{\nu} S_2 \\ &= S_2^{-1} S_1^{-1} i\gamma^{\mu} S_1 S_2 = \Lambda^{\mu}{}_{\rho}(S_1 S_2) i\gamma^{\rho} \end{aligned} \tag{57}$$

□

Note 47. *The group* $SL(2,\mathbb{C}) = \{e^{\theta^j i\sigma^j + b^j\sigma^j} : \theta^j, b^j \in \mathbb{R},\ j \in \{1,2,3\}\}$ *is simply connected. Its projective representations are equivalent to its ordinary representations*[196].

There is a two-to-one, surjective map $\Upsilon : SL(2,\mathbb{C}) \to SO^+(1,3)$*, defined by:*

$$\Upsilon^\mu_{\ \nu}(T)\sigma^\nu \equiv T^\dagger \sigma^\mu T \tag{58}$$

Where $T \in SL(2,\mathbb{C})$*,* $\sigma^0 = 1$ *and* σ^j*,* $j \in \{1,2,3\}$ *are the Pauli matrices.*

Lemma 48. *Consider that* $\{M_+, M_-, i\gamma^5 M_+, i\gamma^5 M_-\}$ *and* $\{P_+, P_-, iP_+, iP_-\}$ *are orthonormal basis of the 4 dimensional real vector spaces Pinor and Pauli, respectively, verifying:*

$$\gamma^0\gamma^3 M_\pm = \pm M_\pm,\ \sigma^3 P_\pm = \pm P_\pm \tag{59}$$

The isomorphism $\Sigma : Pauli \to Pinor$ *is defined by:*

$$\begin{aligned} \Sigma(P_+) = M_+,\ \Sigma(iP_+) = i\gamma^5 M_+ \\ \Sigma(P_-) = M_-,\ \Sigma(iP_-) = i\gamma^5 M_- \end{aligned} \tag{60}$$

The group $Spin^+(3,1) \equiv \{\Sigma \circ A \circ \Sigma^{-1} : A \in SL(2,\mathbb{C})\}$ *is a subgroup of* $Pin(1,3)$*. For all* $S \in Spin^+(1,3)$*,* $\Lambda(S) = \Upsilon(\Sigma^{-1} \circ S \circ \Sigma)$*.*

Proof. From remark Note 47, $Spin^+(3,1) = \{e^{\theta^j i\gamma^5\gamma^0\gamma^j + b^j\gamma^0\gamma^j} : \theta^j, b^j \in \mathbb{R},\ j \in \{1,2,3\}\}$. Then, for all $T \in SL(2,C)$:

$$-i\gamma^0\Sigma \circ T^\dagger \circ \Sigma^{-1} i\gamma^0 = \Sigma \circ T^{-1} \circ \Sigma^{-1} \tag{61}$$

Now, the map $\Upsilon : SL(2,\mathbb{C}) \to SO^+(1,3)$ is given by:

$$\Upsilon^\mu_{\ \nu}(T) i\gamma^\nu = (\Sigma \circ T^{-1} \circ \Sigma^{-1}) i\gamma^\mu (\Sigma \circ T \circ \Sigma^{-1}) \tag{62}$$

Then, all $S \in Spin^+(3,1)$ leaves the space Maj invariant:

$$S^{-1} i\gamma^\mu S = \Upsilon^\mu_{\ \nu}(\Sigma^{-1} \circ S \circ \Sigma) i\gamma^\nu \in Maj \tag{63}$$

Since all the products of Majorana matrices, except the identity, are traceless, then $det(S) = 1$. So, $Spin^+(3,1)$ is a subgroup of $Pin(1,3)$ and $\Lambda(S) = \Upsilon(\Sigma^{-1} \circ S \circ \Sigma)$. □

Definition 49. The discrete Pin subgroup $\Omega \subset Pin(3,1)$ is:

$$\Omega \equiv \{\pm 1, \pm i\gamma^0, \pm\gamma^0\gamma^5, \pm i\gamma^5\} \tag{64}$$

The previous lemma and the fact that Λ is continuous, implies that $Spin^+(1,3)$ is a double cover of $SO^+(3,1)$. We can check that for all $\omega \in \Omega$, $\Lambda(\pm\omega) \in \Delta$. That is, the discrete Pin subgroup is the double cover of the discrete Lorentz subgroup. Therefore, $Pin(3,1) = \Omega \ltimes Spin^+(1,3)$

Since there is a two-to-one continuous surjective group homomorphism, $Pin(3,1)$ is a double cover of $O(1,3)$, $Spin^+(3,1)$ is a double cover of $SO^+(1,3)$ and $Spin^+(1,3) \cap SU(4)$ is a double cover of $SO(3)$. We can check that $Spin^+(1,3) \cap SU(4)$ is equivalent to $SU(2)$.

Finite-dimensional representations of SL(2,C)

Note 50. *Since SL(2,C) is a semisimple Lie group, all its finite-dimensional (real or complex) representations are direct sums of irreducible representations.*

Note 51. *The finite-dimensional complex irreducible representations of SL(2,C) are labeled by (m,n), where $2m, 2n$ are natural numbers. Up to equivalence, the representation space $V_{(m,n)}$ is the tensor product of the complex vector spaces V_m^+ and V_n^-, where $V_m^\pm$ is a symmetric tensor with $2m$ Dirac spinor indexes, such that $\gamma^5{}_k v = \pm v$, where $v \in V_m^\pm$ and $\gamma^5{}_k$ is the Dirac matrix γ^5 acting on the k-th index of v.*

The group homomorphism consists in applying the same matrix of $Spin^+(1,3)$, correspondent to the $SL(2,C)$ group element we are representing, to each index of v. $V_{(0,0)}$ is equivalent to $\mathbb{C}$ and the image of the group homomorphism is the identity.

These are also projective representations of the time reversal transformation, but, for $m \neq n$, not of the parity transformation, that is, under the parity transformation, $(V_m^+ \otimes V_n^-) \to (V_m^- \otimes V_n^+)$ and under the time reversal transformation $(V_m^+ \otimes V_n^-) \to (V_m^+ \otimes V_n^-)$.

Lemma 52. *The finite-dimensional real irreducible representations of SL(2,C) are labeled by (m,n), where $2m, 2n$ are natural numbers and $m \geq n$. Up to equivalence, the representation space $W_{(m,n)}$ is defined for $m \neq n$ as:*

$$\begin{aligned} W_{(m,n)} &\equiv \{\frac{1 + (i\gamma^5)_1 \otimes (i\gamma^5)_1}{2} w : w \in W_m \otimes W_n\} \\ W_{(m,m)} &\equiv \{\frac{1 + (i\gamma^5)_1 \otimes (i\gamma^5)_1}{2} w : w \in (W_m)^2\} \end{aligned} \tag{65}$$

where W_m is a symmetric tensor with m Majorana spinor indexes, such that $(i\gamma^5)_1 (i\gamma^5)_k w = -w$,

where $w \in W_m$; $(i\gamma^5)_k$ is the Majorana matrix $i\gamma^5$ acting on the k-th index of w; $(W_m)^2$ is the space of the linear combinations of the symmetrized tensor products $(u \otimes v + v \otimes u)$, for $u, v \in W_m$.

The group homomorphism consists in applying the same matrix of $Spin^+(1,3)$, correspondent to the $SL(2,C)$ group element we are representing, to each index of the tensor. In the $(0,0)$ case, $W_{(0,0)}$ is equivalent to $\mathbb{R}$ and the image of the group homomorphism is the identity.

These are also projective representations of the full Lorentz group, that is, under the parity or time reversal transformations, $(W_{m,n} \to W_{m,n})$.

Proof. For $m \neq n$ the complex irreducible representations of SL(2,C) are C-complex. The complexification of $W_{(m,n)}$ verifies $W^c_{(m,n)} = (V^+_m \otimes V^-_n) \oplus (V^-_m \otimes V^+_n)$.

For $m = n$ the complex irreducible representations of SL(2,C) are C-real. In a Majorana basis, the C-conjugation operator of $V_{(m,m)}$, θ, is defined as $\theta(u \otimes v) \equiv v^* \otimes u^*$, where $u \in V^+_m$ and $v \in V^-_m$. We can check that there is a bijection $\alpha : W_{(m,m)} \to (V_{(m,m)})_\theta$, defined by $\alpha(w) \equiv \frac{1-i(i\gamma^5)_1 \otimes 1}{2} w$; $\alpha^{-1}(v) \equiv v + v^*$, for $w \in W_{(m,m)}$, $v \in (V_{(m,m)})_\theta$.

Using the map from Section 2, we can check that the representations $W_{(m,n)}$, with $m \geq n$, are the unique finite-dimensional real irreducible representations of SL(2,C), up to isomorphisms.

We can check that $W^c_{(m,n)}$ is equivalent to $W^c_{(n,m)}$, therefore, invariant under the parity or time reversal transformations. □

As examples of real irreducible representations of $SL(2,C)$ we have for $(1/2,0)$ the Majorana spinor, for $(1/2,1/2)$ the linear combinations of the matrices $\{1, \gamma^0\vec{\gamma}\}$, for $(1,0)$ the linear combinations of the matrices $\{i\vec{\gamma}, \vec{\gamma}\gamma^5\}$. The group homomorphism is defined as $M(S)(u) \equiv Su$ and $M(S)(A) \equiv SAS^\dagger$, for $S \in Spin^+(1,3)$, $u \in Pinor$, $A \in \{1, \vec{\gamma}\gamma^0\}$ or $A \in \{i\vec{\gamma}, \vec{\gamma}\gamma^5\}$.

We can check that the domain of M can be extended to $Pin(1,3)$, leaving the considered vector spaces invariant. For $m = n$, we can define the "pseudo-representation" $W'_{(m,m)} \equiv \{((i\gamma^5)_1 \otimes 1)w : w \in W_{(m,m)}\}$ which is equivalent to $W_{(m,m)}$ as an $SL(2,C)$ representation, but under parity transforms with the opposite sign. As an example, the "pseudo-representation" $(1/2,1/2)$ is defined as the linear combinations of the matrices $\{i\gamma^5, i\gamma^5\vec{\gamma}\gamma^0\}$.

Unitary representations of the Poincare group

Bargmann-Wigner fields

Definition 53. Consider that $\{M_+, M_-, i\gamma^0 M_+, i\gamma^0 M_-\}$ and $\{P_+, P_-, iP_+, iP_-\}$ are orthonormal basis of the 4 dimensional real vector spaces $Pinor$ and $Pauli$, respectively, verifying:

$$\gamma^3\gamma^5 M_\pm = \pm M_\pm,\ \sigma^3 P_\pm = \pm P_\pm \tag{66}$$

Let H be a real Hilbert space. For all $h \in H$, the bijective linear map $\Theta_H : Pauli \otimes_{\mathbb{R}} H \to Pinor \otimes_{\mathbb{R}} H$ is defined by:

$$\begin{aligned} &\Theta_H(h \otimes_{\mathbb{R}} P_+) = h \otimes_{\mathbb{R}} M_+,\ \Theta_H(h \otimes_{\mathbb{R}} iP_+) = h \otimes_{\mathbb{R}} i\gamma^0 M_+ \\ &\Theta_H(h \otimes_{\mathbb{R}} P_-) = h \otimes_{\mathbb{R}} M_-,\ \Theta_H(h \otimes_{\mathbb{R}} iP_-) = h \otimes_{\mathbb{R}} i\gamma^0 M_- \end{aligned} \tag{67}$$

Definition 54. Let H_n, with $n \in \{1, 2\}$, be two real Hilbert spaces and $U : Pauli \otimes_{\mathbb{R}} H_1 \to Pauli \otimes_{\mathbb{R}} H_2$ be an operator. The operator $U^\Theta : Pinor \otimes_{\mathbb{R}} H_1 \to Pinor \otimes_{\mathbb{R}} H_2$ is defined as $U^\Theta \equiv \Theta_{H_2} \circ U \circ \Theta_{H_1}^{-1}$.

The space of Majorana spinors is isomorphic to the realification of the space of Pauli spinors.

Definition 55. The real Hilbert space $Pinor(\mathbb{X}) \equiv Pinor \otimes L^2(\mathbb{X})$ is the space of square integrable functions with domain $\mathbb{X}$ and image in $Pinor$.

Definition 56. The complex Hilbert space $Pauli(\mathbb{X}) \equiv Pauli \otimes L^2(\mathbb{X})$ is the space of square integrable functions with domain $\mathbb{X}$ and image in $Pauli$.

Definition 57. The real vector space $Pinor_j$, with $2j$ a positive integer, is the space of linear combinations of the tensor products of $2j$ Majorana spinors, symmetric on the spinor indexes. The real vector space $Pinor_0$ is the space of linear combinations of the tensor products of 2 Majorana spinors, anti-symmetric on the spinor indexes.

Definition 58. The real Hilbert space $Pinor_j(\mathbb{X}) \equiv Pinor_j \otimes L^2(\mathbb{X})$ is the space of square integrable functions with domain $\mathbb{X}$ and image in $Pinor_j$.

Definition 59. The space of (real) Bargmann-Wigner fields $BW_j(\mathbb{R}^3)$ is defined as:

$$BW_j \equiv \{\Psi \in Pinor_j(\mathbb{R}^3) : \left(e^{iH(\vec{x})t}\right)_k \Psi = \left(e^{iH(\vec{x})t}\right)_1 \Psi; 1 \leq k \leq 2j; t \in \mathbb{R}\} \tag{68}$$

Note that if the equality $e^{-iH_1 t}\Psi = e^{-iH_2 t}\Psi$ holds for all differentiable $\Psi \in H$ then for the continuous linear extension the equality holds for all $\Psi \in H$, by the bounded linear transform theorem.

Definition 60. The complex Hilbert space $Dirac_j(\mathbb{X}) \equiv Pinor_j(\mathbb{X}) \otimes \mathbb{C}$ is the complexification of $Pinor_j(\mathbb{X})$. The space of complex Bargmann-Wigner fields is the complexification of the space of real Bargmann-Wigner fields.

Proposition 61. *Consider a unitary operator $U : Pinor_j(\mathbb{R}^3) \to Pinor_j(\mathbb{X})$ such that $U \circ H^2 = E^2 \circ U$, where*

$$iH\{\Psi\}(\vec{x}) \equiv \left(\gamma^0\vec{\partial} + i\gamma^0 m\right)_k \Psi(\vec{x}) \tag{69}$$

the Majorana matrices act on some Majorana index k; $E^2\{\Phi\}(X) \equiv E^2(X)\Phi(X)$ with $E(X) \geq m \geq 0$ a real number.

Then the operator $U' : Pinor(\mathbb{R}^3) \to Pinor(\mathbb{X})$ is unitary, where U' is defined by:

$$U' \equiv \frac{E + UH\gamma^0 U^\dagger}{\sqrt{E+m}\sqrt{2E}} \tag{70}$$

Proof. Note that since $E^2 = U^\dagger H^2 U$, $E = \sqrt{E^2}$ commutes with $UH\gamma^0 U^\dagger$. We have that

$$(U')^\dagger(U') = \frac{E + U\gamma^0 HU^\dagger}{\sqrt{E+m}\sqrt{2E}} \frac{E + UH\gamma^0 U^\dagger}{\sqrt{E+m}\sqrt{2E}} = 1 \tag{71}$$

We also have that $(U')(U')^\dagger = 1$. Therefore, U' is unitary. □

Fourier-Majorana Transform

Note 62. *The Fourier Transform $\mathcal{F}_P : Pauli(\mathbb{R}^3) \to Pauli(\mathbb{R}^3)$ is an unitary operator defined by:*

$$\mathcal{F}_P\{\psi\}(\vec{p}) \equiv \int d^n\vec{x} \frac{e^{-i\vec{p}\cdot\vec{x}}}{\sqrt{(2\pi)^n}} \psi(\vec{x}),\ \psi \in Pauli(\mathbb{R}^3) \tag{72}$$

Where the domain of the integral is $\mathbb{R}^3$.

Note 63. *The inverse Fourier transform verifies:*

$$\begin{aligned} -\vec{\partial}^2\ \mathcal{F}_P^{-1}\{\psi\}(\vec{x}) &= (\mathcal{F}_P^{-1} \circ R)\{\psi\}(\vec{x}) \\ i\vec{\partial}_k\ \mathcal{F}_P^{-1}\{\psi\}(\vec{x}) &= (\mathcal{F}_P^{-1} \circ R'_k)\{\psi\}(\vec{x}) \end{aligned} \tag{73}$$

Where $\psi \in Pauli(\mathbb{R}^3)$ and $R, R'_k : Pauli(\mathbb{R}^3) \to Pauli(\mathbb{R}^3)$, with $k \in \{1,2,3\}$, are linear maps defined by:

$$\begin{aligned} R\{\psi\}(\vec{p}) &\equiv (\vec{p})^2\psi(\vec{p}) \\ R'_k\{\psi\}(\vec{p}) &\equiv \vec{p}_k\ \psi(\vec{p}) \end{aligned} \tag{74}$$

Definition 64. The Fourier-Majorana transform $\mathcal{F}_M : Pinor_j(\mathbb{R}^3) \to Pinor_j(\mathbb{R}^3)$ is an unitary operator defined by:

$$\mathcal{F}_M\{\Psi\}(\vec{p}) \equiv \int d^3\vec{x}\Big(\frac{e^{-i\gamma^0\vec{p}\cdot\vec{x}}}{\sqrt{(2\pi)^3}}\Big)_1 \prod_{k=1}^{2j}\Big(\frac{E_p + H(\vec{x})\gamma^0}{\sqrt{E_p+m}\sqrt{2E_p}}\Big)_k \Psi(\vec{x}),\ \Psi \in Pinor_j(\mathbb{R}^3) \tag{75}$$

The matrices with the index k apply on the corresponding spinor index of Ψ.

The inverse Fourier-Majorana transform verifies:

$$\begin{aligned}(iH(\vec{x}))_k\ \mathcal{F}_M^{-1}\{\psi\}(\vec{x}) &= (\mathcal{F}_M^{-1}\circ R)\{\psi\}(\vec{x})\\ \vec{\partial}_l\ \mathcal{F}_M^{-1}\{\psi\}(\vec{x}) &= (\mathcal{F}_M^{-1}\circ R')\{\psi\}(\vec{x})\end{aligned} \tag{76}$$

Where $\psi \in Pinor_j(\mathbb{R}^3)$ and $R, R' : Pinor_j(\mathbb{R}^3) \to Pinor_j(\mathbb{R}^3)$ are linear maps defined by:

$$\begin{aligned}R\{\psi\}(\vec{p}) &\equiv (i\gamma^0)_k E_p\psi(\vec{p})\\ R'\{\psi\}(\vec{p}) &\equiv (i\gamma^0)_1\vec{p}_l\ \psi(\vec{p})\end{aligned} \tag{77}$$

Hankel-Majorana Transform

Definition 65. Let $\vec{x} \in \mathbb{R}^3$. The spherical coordinates parametrization is:

$$\vec{x} = r(\sin(\theta)\sin(\varphi)\vec{e_1} + \sin(\theta)\sin(\varphi)\vec{e_2} + \cos(\theta)\vec{e_3}) \tag{78}$$

where $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ is a fixed orthonormal basis of $\mathbb{R}^3$ and $r \in [0, +\infty[$, $\theta \in [0, \pi]$, $\varphi \in [-\pi, \pi]$.

Definition 66. Let

$$\mathbb{S}^3 \equiv \{(p, l, \mu) : p \in \mathbb{R}_{\geq 0}; l, \mu \in \mathbb{Z}; l \geq 0; -l \leq \mu \leq l\} \tag{79}$$

The Hilbert space $L^2(\mathbb{S}^3)$ is the real Hilbert space of real Lebesgue square integrable functions of $\mathbb{S}^3$. The internal product is:

$$< f, g >= \sum_{l=0}^{+\infty}\sum_{\mu=-l}^{l-1}\int_0^{+\infty} dp f(p, l, \mu)g(p, l, \mu),\ f, g \in L^2(\mathbb{S}^3) \tag{80}$$

Definition 67. The Spherical transform $\mathcal{H}_P : Pauli(\mathbb{R}^3) \to Pauli(\mathbb{S}^3)$ is an operator defined by:

$$\mathcal{H}_P\{\psi\}(p, l, \mu) \equiv \int r^2 dr d(\cos\theta) d\varphi \frac{2p}{\sqrt{2\pi}} j_l(pr) Y_{l\mu}(\theta, \varphi)\psi(r, \theta, \varphi),\ \psi \in Pauli(\mathbb{R}^3) \tag{81}$$

The domain of the integral is $\mathbb{R}^3$. The spherical Bessel function of the first kind j_l [270], the spherical harmonics $Y_{l\mu}$[271] and the associated Legendre functions of the first kind $P_{l\mu}$ are:

$$\begin{aligned} j_l(r) \equiv & r^l\Big(-\frac{1}{r}\frac{d}{dr}\Big)^l\frac{\sin r}{r} \\ Y_{l\mu}(\theta,\varphi) \equiv & \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l^\mu(\cos\theta)e^{i\mu\varphi} \\ P_l^\mu(\xi) \equiv & \frac{(-1)^\mu}{2^l l!}(1-\xi^2)^{\mu/2}\frac{\mathrm{d}^{l+\mu}}{\mathrm{d}\xi^{l+\mu}}(\xi^2-1)^l \end{aligned} \tag{82}$$

Note 68. *Due to the properties of spherical harmonics and Bessel functions, the Spherical transform is an unitary operator. The inverse Spherical transform verifies:*

$$\begin{aligned} -\vec{\partial}^2\ \mathcal{H}_P^{-1}\{\psi\}(\vec{x}) &= (\mathcal{H}_P^{-1}\circ R)\{\psi\}(\vec{x}) \\ (-x^1 i\partial_2 + x^2 i\partial_1)\ \mathcal{H}_P^{-1}\{\psi\}(\vec{x}) &= (\mathcal{H}_P^{-1}\circ R')\{\psi\}(\vec{x}) \end{aligned} \tag{83}$$

Where $\psi \in Pauli(\mathbb{S}^3)$ and $R, R' : Pauli(\mathbb{S}^3) \to Pauli(\mathbb{S}^3)$ are linear maps defined by:

$$\begin{aligned} R\{\psi\}(p,l,\mu) &\equiv p^2\psi(p,l,\mu) \\ R'\{\psi\}(p,l,\mu) &\equiv \mu\ \psi(p,l,\mu) \end{aligned} \tag{84}$$

Definition 69. The Hilbert space $Pinor_{j,n}$, with $(j-\nu)$ an integer and $-j \leq n \leq j$ is defined as:

$$Pinor_{j,n} \equiv \{\Psi \in Pinor_j : \sum_{k=1}^{k=2j}(\gamma^0)_1\Big(\gamma^0\gamma^3\gamma^5\Big)_k\Psi = 2n\Psi\} \tag{85}$$

Where $\Big(\gamma^3\gamma^5\Big)_k$ is the matrix $\gamma^3\gamma^5$ acting on the Majorana index k.

Definition 70. The Spherical transform $\mathcal{H}'_P : Pinor_j(\mathbb{R}^3) \to Pinor_j(\mathbb{S}^3)$ is an operator defined by:

$$\mathcal{H}'_P\{\psi\}(p,l,J,\nu) \equiv \sum_{\mu=-l}^{l}\sum_{n=-j}^{j} < l\mu jn|J\nu > \Big(\mathcal{H}_P^\Theta\Big)_1\{\psi\}(p,l,\mu,n),\ \psi \in Pinor_j(\mathbb{R}^3) \tag{86}$$

$< l\mu jn|J\nu >$ are the Clebsh-Gordon coefficients and $\psi(p,l,\mu,n) \in Pinor_{j,n}$ such that $\psi(p,l,\mu) = \sum_{n=-j}^{j}\psi(p,l,\mu,n)$. $(j-n)$, $(J-\nu)$ and $(J-j)$ are integers, with $-J \leq \nu \leq J$ and $|j-l| \leq J \leq j+l$. $\Big(\mathcal{H}_P^\Theta\Big)_1$ is the realification of the transform $\mathcal{H}_P$, with the imaginary number replaced by the matrix $i\gamma^0$ acting on the first Majorana index of ψ.

Definition 71. The Hankel-Majorana transform $\mathcal{H}_M : Pinor_j(\mathbb{R}^3) \to Pinor_j(\mathbb{S}^3)$ is a unitary operator defined by:

$$\begin{aligned}\mathcal{H}_M\{\Psi\}(p,l,J,\nu) \equiv \sum_{\mu=-l}^{l}\sum_{n=-j}^{j} < l\mu jn|J\nu > \int d^3\vec{x} \\ \Big(\frac{2p}{\sqrt{2\pi}} j_l(pr) Y_{l\mu}(\theta,\varphi)\Big)_1 \prod_{k=1}^{2j} \Big(\frac{E_p + H(\vec{x})\gamma^0}{\sqrt{E_p+m}\sqrt{2E_p}}\Big)_k \Psi(\vec{x},n)\end{aligned} \tag{87}$$

The matrices with the index k apply on the corresponding spinor index of $\Psi \in Pinor_j(\mathbb{R}^3)$. $< l\mu jn|J\nu >$ are the Clebsh-Gordon coefficients and $\Psi(\vec{x},n) \in Pinor_{j,n}$ such that $\Psi(\vec{x}) = \sum_{n=-j}^{j} \Psi(\vec{x},n)$.

The inverse Hankel-Majorana transform verifies:

$$\begin{aligned}(iH(\vec{x}))_k \ \mathcal{H}_M^{-1}\{\psi\}(\vec{x}) = (\mathcal{H}_M^{-1} \circ R)\{\psi\}(\vec{x}) \\ (-x^1\partial_2 + x^2\partial_1 + \sum_{k=1}^{2j}(i\gamma^0\gamma^3\gamma^5)_k) \ \mathcal{H}_M^{-1}\{\psi\}(\vec{x}) = (\mathcal{H}_M^{-1} \circ R')\{\psi\}(\vec{x})\end{aligned} \tag{88}$$

Where $\psi \in Pinor_j(\mathbb{S}^3)$ and $R, R' : Pinor_j(\mathbb{S}^3) \to Pinor_j(\mathbb{S}^3)$ are linear maps defined by:

$$\begin{aligned}R\{\psi\}(p,l,J,\nu) \equiv (i\gamma^0)_k E_p \psi(p,l,J,\nu) \\ R'\{\psi\}(p,l,J,\nu) \equiv (i\gamma^0)_1 \nu \ \psi(p,l,J,\nu)\end{aligned} \tag{89}$$

Application to the momentum of Majorana spinor fields

Definition 72. The Majorana-Fourier Transform $\mathcal{F}_M : Pinor(\mathbb{R}^3) \to Pinor(\mathbb{R}^3)$ is an operator defined by:

$$\mathcal{F}_M\{\Psi\}(\vec{p}) \equiv \int d^3\vec{x} \ \frac{e^{-i\gamma^0\vec{p}\cdot\vec{x}}}{\sqrt{(2\pi)^3}} \frac{\not{p}\gamma^0 + m}{\sqrt{E_p+m}\sqrt{2E_p}} \Psi(\vec{x}), \ \Psi \in Pinor(\mathbb{R}^3) \tag{90}$$

Where the domain of the integral is $\mathbb{R}^3$, $m \geq 0$, $E_p \equiv \sqrt{\vec{p}^2 + m^2}$ and $\not{p} = E_p\gamma^0 - \vec{p}\cdot\vec{\gamma}$.

Proposition 73. *The Majorana-Fourier Transform is a unitary operator.*

Proof. The proof is immediate using Prop. Proposition 61, but we will do it in an independent more explicit way. The Majorana-Fourier Transform can be written as:

$$\begin{aligned}\mathcal{F}_M\{\Psi\}(\vec{p}) \equiv & \sqrt{\frac{E_p+m}{2E_p}}\Big(\int d^3\vec{x}\ \frac{e^{-i\gamma^0\vec{p}\cdot\vec{x}}}{\sqrt{(2\pi)^3}}\Psi(\vec{x})\Big) \\ & -\sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|}\Big(\int d^3\vec{x}\ \frac{e^{+i\gamma^0\vec{p}\cdot\vec{x}}}{\sqrt{(2\pi)^3}}\Psi(\vec{x})\Big)\end{aligned} \tag{91}$$

So, one gets:

$$\mathcal{F}_M\{\Psi\} = S \circ \mathcal{F}_P^{\Theta}\{\Psi\} \tag{92}$$

Where $S : Pinor(\mathbb{R}^3) \to Pinor(\mathbb{R}^3)$ is a bijective linear map defined by:

$$\begin{bmatrix} S\{\Psi\}(+\vec{p}) \\ S\{\Psi\}(-\vec{p}) \end{bmatrix} \equiv \begin{bmatrix} \sqrt{\frac{E_p+m}{2E_p}} & -\sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|} \\ \sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|} & \sqrt{\frac{E_p+m}{2E_p}} \end{bmatrix} \begin{bmatrix} \Psi(+\vec{p}) \\ \Psi(-\vec{p}) \end{bmatrix} \tag{93}$$

We can check that the 2×2 matrix appearing in the equation above is orthogonal. Therefore S is an unitary operator. Since $\mathcal{F}_P^{\Theta}$ is also unitary, $\mathcal{F}_M$ is unitary. □

Proposition 74. *The inverse Majorana-Fourier Transform verifies:*

$$\begin{aligned}(\gamma^0\vec{\gamma}\cdot\vec{\partial} + i\gamma^0 m)\mathcal{F}_M^{-1}\{\Psi\}(\vec{x}) &= (\mathcal{F}_M^{-1} \circ R)\{\Psi\}(\vec{x}) \\ \vec{\partial}_j\mathcal{F}_M^{-1}\{\Psi\}(\vec{x}) &= (\mathcal{F}_M^{-1} \circ R_j)\{\Psi\}(\vec{x})\end{aligned} \tag{94}$$

Where $\Psi \in Pinor(\mathbb{R}^3)$ and $R, R_j : Pinor(\mathbb{R}^3) \to Pinor(\mathbb{R}^3)$ are linear maps defined by $R\{\Psi\}(\vec{p}) = i\gamma^0 E_p\Psi(\vec{p})$ and $R_j\{\Psi\}(\vec{p}) = i\gamma^0\vec{p}_j\Psi(\vec{p})$.

Proof. We have $\mathcal{F}_M^{-1} = (\mathcal{F}_P^{\Theta})^{-1} \circ S^{-1}$. Then:

$$(\gamma^0\vec{\gamma}\cdot\vec{\partial} + i\gamma^0 m)(\mathcal{F}_P^{\Theta})^{-1}\{\Psi\}(\vec{x}) = ((\mathcal{F}_P^{\Theta})^{-1} \circ Q)\{\Psi\}(\vec{x}) \tag{95}$$

Where $Q : Pinor(\mathbb{R}^3) \to Pinor(\mathbb{R}^3)$ is a linear map defined by:

$$\begin{bmatrix} Q\{\Psi\}(+\vec{p}) \\ Q\{\Psi\}(-\vec{p}) \end{bmatrix} \equiv \begin{bmatrix} i\gamma^0 m & i\vec{p}\cdot\vec{\gamma} \\ -i\vec{p}\cdot\vec{\gamma} & i\gamma^0 m \end{bmatrix} \begin{bmatrix} \Psi(+\vec{p}) \\ \Psi(-\vec{p}) \end{bmatrix} \tag{96}$$

Now we show that $Q \circ S^{-1} = S^{-1} \circ R$:

$$\begin{array}{l}\left[\begin{array}{cc} i\gamma^0 m & i\vec{p}\cdot\vec{\gamma} \\ -i\vec{p}\cdot\vec{\gamma} & i\gamma^0 m \end{array}\right]\left[\begin{array}{cc} \sqrt{\frac{E_p+m}{2E_p}} & \sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|} \\ -\sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|} & \sqrt{\frac{E_p+m}{2E_p}} \end{array}\right] = \\ = \left[\begin{array}{cc} \sqrt{\frac{E_p+m}{2E_p}} & \sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|} \\ -\sqrt{\frac{E_p-m}{2E_p}}\frac{\vec{p}\cdot\vec{\gamma}\gamma^0}{|\vec{p}|} & \sqrt{\frac{E_p+m}{2E_p}} \end{array}\right]\left[\begin{array}{cc} i\gamma^0 E_p & 0 \\ 0 & i\gamma^0 E_p \end{array}\right]\end{array} \tag{97}$$

We also have that:

$$\vec{\partial}_j(\mathcal{F}_P^{\Theta})^{-1}\{\Psi\}(\vec{x}) = ((\mathcal{F}_P^{\Theta})^{-1}\circ R_j)\{\Psi\}(\vec{x}) \tag{98}$$

Where $R_j : Pinor(\mathbb{R}^3) \to Pinor(\mathbb{R}^3)$ is the linear map defined by:

$$\left[\begin{array}{c} R_j\{\Psi\}(+\vec{p}) \\ R_j\{\Psi\}(-\vec{p}) \end{array}\right] \equiv \left[\begin{array}{cc} i\gamma^0\vec{p}_j & 0 \\ 0 & -i\gamma^0\vec{p}_j \end{array}\right]\left[\begin{array}{c} \Psi(+\vec{p}) \\ \Psi(-\vec{p}) \end{array}\right] \tag{99}$$

It verifies $R_j \circ S^{-1} = S^{-1} \circ R_j$. □

Definition 75. The Energy Transform $\mathcal{E} : Pinor(\mathbb{R}) \to Pinor(\mathbb{R})$ is an operator defined by:

$$\mathcal{E}\{\Psi\}(p^0) \equiv \int dx^0 \ \frac{e^{i\gamma^0 p^0 x^0}}{\sqrt{2\pi}}\Psi(x^0),\ \Psi \in Pinor(\mathbb{R}) \tag{100}$$

Where the domain of the integral is $\mathbb{R}$, $m \geq 0$.

Proposition 76. *The Energy transform is an unitary operator.*

Proof. The Energy transform can be written as:

$$\mathcal{E}\{\Psi\}(p^0) = \Theta_{L^2} \circ \mathcal{F}_P(-p^0) \circ \Theta_{L^2}^{-1}\{\Psi\} \tag{101}$$

Where $\mathcal{F}_P(-p^0)$ is a Pauli-Fourier transform over $\mathbb{R}$ and Θ was defined in Definition Definition 53. Since the Pauli-Fourier transform is unitary, so is the Energy transform. □

The energy transform can be applied in the time coordinate of a Majorana spinor field, x^0, after a (linear or spherical) momentum transform on the space coordinates, $\vec{x}$, to define an unitary energy-momentum transform:
- for the linear case $\mathcal{E} \circ \mathcal{F}_M : Pinor(\mathbb{R}^4) \to Pinor(\mathbb{R}^4)$;
- for the spherical case $\mathcal{E} \circ \mathcal{H}_M : Pinor(\mathbb{R}^4) \to Pinor(\mathbb{R} \times \mathbb{S}^3)$.

Real unitary representations of the Poincare group

Definition 77. The $IPin(3,1)$ group is defined as the semi-direct product $Pin(3,1) \ltimes \mathbb{R}^4$, with the group's product defined as $(A,a)(B,b) = (AB, a + \Lambda(A)b)$, for $A, B \in Pin(3,1)$ and $a, b \in \mathbb{R}^4$ and $\Lambda(A)$ is the Lorentz transformation corresponding to A.

The $ISL(2,C)$ group is isomorphic to the subgroup of $IPin(3,1)$, obtained when $Pin(3,1)$ is restricted to $Spin^+(1,3)$. The full/restricted Poincare group is the representation of the $IPin(3,1)/ISL(2,C)$ group on Lorentz vectors, defined as $\{(\Lambda(A), a) : A \in Pin(3,1), a \in \mathbb{R}^4\}$.

Definition 78. Given a Lorentz vector l, the little group G_l is the subgroup of $SL(2,C)$ such that for all $g \in G_l$, $g\not{l} = \not{l}g$.

Proposition 79. *Given a Lorentz vector l, consider a set of matrices $\alpha_k \in SL(2,C)$ verifying $\alpha_k \not{l} = \not{k}\alpha_k$. Let $H_k \equiv \{\alpha^{-1}_{\Lambda_S(k)} S \alpha_k : S \in SL(2,C)\}$. Then $H_k = G_l$.*

Proof. We can check that $H_k \subset G_l$. For any $s \in G_l$, there is $S = \alpha_{\Lambda_S(k)} s \alpha_k^{-1}$ such that $s \in H_k$. □

For $i\not{l} = i\gamma^0$, we can set $\alpha_p = \frac{\not{p}\gamma^0 + m}{\sqrt{E_p + m}\sqrt{2m}}$ and $G_l = SU(2)$. For $i\not{l} = (i\gamma^0 + i\gamma^3)$, we can set $\alpha_p = B_v R_p$, where the boost velocity is $v = \frac{E_p^2 - 1}{E_p^2 + 1}$ along $\vec{p}$ and $R_p = e^{-\gamma^2\gamma^1\theta/2} e^{-\gamma^1\gamma^3\phi/2}$ is a rotation from the z axis to the axis $\frac{\vec{p}}{E_p} = (\sin\phi\cos\theta\gamma_1 + \sin\phi\sin\theta\gamma_2 + \cos\phi\gamma_3)$; $G_l = SE(2)$

$$SE(2) = \{(1 + i\gamma^5(\gamma^1 a + \gamma^2 b)(\gamma^0 + \gamma^3)) e^{i\gamma^0\gamma^3\gamma^5\theta} : a, b, \theta \in \mathbb{R}\}. \tag{102}$$

Note 80. *The complex irreducible projective representations of the Poincare group with finite mass split into positive and negative energy representations, which are complex conjugate of each other. They are labeled by one number j, with $2j$ being a natural number. The positive energy representation spaces V_j are, up to isomorphisms, written as a symmetric tensor product of Dirac spinor fields defined on the 3-momentum space, verifying $(\gamma^0)_k \Psi_j(\vec{p}) = \Psi_j(\vec{p})$. The matrices with the index k apply in the corresponding spinor index of Ψ_j.*

The representation space V_0 is, up to isomorphisms, written in a Majorana basis as a complex scalar defined on the 3-momentum space.

The representation map is given by:

$$\begin{aligned} L_S\{\Psi\}(\vec{p}) &= \sqrt{\frac{(\Lambda^{-1})^0(p)}{E_p}} \prod_{k=1}^{2j} (\alpha^{-1}_{\Lambda(p)} S \alpha_p)_k \Psi(\vec{\Lambda}^{-1}(p)) \\ T_a\{\Psi\}(\vec{p}) &= e^{-ip\cdot a}\Psi(\vec{p}) \end{aligned} \tag{103}$$

Where $\alpha_p = \frac{\not{p}\gamma^0 + m}{\sqrt{E_p + m}\sqrt{2m}}$.

Proposition 81. *The real irreducible projective representations of the Poincare group with finite mass are labeled by one number* j*, with* $2j$ *being a natural number. The representation spaces* W_j *are, up to isomorphisms, written as a symmetric tensor product of Majorana spinor fields defined on the 3-momentum space, verifying* $(i\gamma^0)_k \Psi_j(\vec{p}) = (i\gamma^0)_1 \Psi_j(\vec{p})$*. The matrices with the index* k *apply in the corresponding spinor index of* Ψ_j*.*

The representation space V_0 *is, up to isomorphisms, written in a Majorana basis as a real scalar defined on the 3-momentum space, times the identity matrix of a Majorana spinor space.*

The representation map is given by:

$$\begin{aligned} L_S\{\Psi\}(\vec{p}) &= \sqrt{\frac{(\Lambda^{-1})^0(p)}{E_p}} \prod_{k=1}^{2j} (\alpha^{-1}_{\Lambda(p)} S \alpha_p)_k \Psi(\vec{\Lambda}^{-1}(p)) \\ T_a\{\Psi\}(\vec{p}) &= e^{-i\gamma^0 p \cdot a} \Psi(\vec{p}) \end{aligned} \tag{104}$$

Note 82. *The complex irreducible projective representations of the Poincare group with null mass and discrete helicity split into positive and negative energy representations, which are complex conjugate of each other. They are labeled by one number* j*, with* $2j$ *being an integer number. The positive energy representation spaces* V_j *are, up to isomorphisms, written as a symmetric tensor product of Dirac spinor fields defined on the 3-momentum space, verifying* $(\gamma^0)_k \Psi_j(\vec{p}) = \Psi_j(\vec{p})$ *and* $(\gamma^3\gamma^5)_k \Psi_j(\vec{p}) = \pm \Psi_j(\vec{p})$*, with the plus sign if* j *is positive and the minus sign if* j *is negative.*

The representation space V_0 *is, up to isomorphisms, written in a Majorana basis as a scalar defined on the 3-momentum space.*

The representation map is given by:

$$\begin{aligned} L_S\{\Psi\}(\vec{p}) &= \sqrt{\frac{(\Lambda^{-1})^0(p)}{E_p}} \prod_{k=1}^{2j} (e^{i\gamma^0\gamma^3\gamma^5\theta})_k \Psi(\vec{\Lambda}^{-1}(p)) \\ T_a\{\Psi\}(\vec{p}) &= e^{-ip \cdot a} \Psi(\vec{p}) \end{aligned} \tag{105}$$

Where θ *is the angle of the rotation of the little group* $SE(2)$*.*

Note 83. *The real irreducible projective representations of the Poincare group with null mass and discrete helicity are labeled by one number* j*, with* $2j$ *being an integer number. The positive energy representation spaces* V_j *are, up to isomorphisms, written as a symmetric tensor product of Majorana spinor fields defined on the 3-momentum space, verifying* $(i\gamma^0)_k \Psi_j(\vec{p}) = (i\gamma^0)_1 \Psi_j(\vec{p})$ *and* $(\gamma^3\gamma^5)_k \Psi_j(\vec{p}) = \pm \Psi_j(\vec{p})$*, with the plus sign if* j *is positive and the minus sign if* j *is negative.*

The representation space V_0 is, up to isomorphisms, written in a Majorana basis as the realification of the complex functions defined on the 3-momentum space, with the operator correspondent to the imaginary unit given by the matrix $i\gamma^0$ of a Majorana spinor space.

The representation map is given by:

$$\begin{aligned} L_S\{\Psi\}(\vec{p}) &= \sqrt{\frac{(\Lambda^{-1})^0(p)}{E_p}} \prod_{k=1}^{2j} (e^{i\gamma^0\gamma^3\gamma^5\theta})_k \Psi(\vec{\Lambda}^{-1}(p)) \\ T_a\{\Psi\}(\vec{p}) &= e^{-i\gamma^0 p\cdot a}\Psi(\vec{p}) \end{aligned} \tag{106}$$

Where θ is the angle of the rotation of the little group $SE(2)$.

Localization

Note 84 (Theorem 6.12 of [211]). *There is a one-to-one correspondence between the complex system of imprimitivity (U,P), based on $\mathbb{R}^3$, and the representations of $SU(2)$. The system (U,P) is equivalent to the system induced by the representation of $SU(2)$.*

Definition 85. A covariant system of imprimitivity is a system of imprimitivity (U,P), where U is a representation of the Poincare group and P is a projection-valued measure based on $\mathbb{R}^3$, such that for the Euclidean group $U(g)\pi(A)U^{-1}(g) = \pi(gA)$ and for the Lorentz group, for a state at time null at point $\vec{x} = 0$, $L\{\Psi\}(0) = S\Psi(0)$.

Definition 86. A localizable real unitary representation of the Poincare group, compatible with Poincare covariance, consists of a system of imprimitivity on R^3 for which at time null and $\vec{x} = 0$, the Lorentz transformations do not act on the space coordinates.

So, the localization of a state in $x = 0$ is a property invariant under relativistic transformations.

Proposition 87. *Any localizable unitary representation of the Poincare group, compatible with Poincare covariance, is a direct sum of irreducible representations which are massive or massless with discrete helicity.*

Proof. Since the system is a unitary Poincare representation, it is a direct sum of irreducible unitary Poincare representations and so there must be an unitary transformation U, such that:

$$\begin{aligned} \Psi(x+a) &= (Ue^{-JP\cdot a}U^{-1})\{\Psi\}(x) \\ S\Psi(\Lambda(x)) &= (ULU^{-1})\{\Psi\}(x) \end{aligned} \tag{107}$$

Where J is the operator corresponding to the imaginary unit after the realification of the Poincare representation and L is the representation of the Lorentz group, so L commutes with J. P is the

energy-momentum operator and S acts only on the index of Ψ.

The system of imprimitivity is a representation of $SU(2)$, hence the operator $i\gamma^0$ is well defined. If we make a Fourier transformation, then we get that:

$$(Ue^{J\vec{P}\cdot\vec{a}}U^{-1})\{\Psi\}(\vec{p}) = e^{i\gamma^0\vec{p}\cdot\vec{a}}\Psi(\vec{p}) \tag{108}$$

Note that this equation is valid for all $\vec{p}$. The system is a direct sum of irreducible unitary Poincare representations. Then, for $m^2 < 0$ only the subspace $\vec{p}^2 \geq |m^2|$ is valid. For $p = 0$ only the subspace $\vec{p} = 0$ is valid. Since the other types of irreducible representations verify $p \neq 0$ and $m^2 \geq 0$, the complementary subspaces $\vec{p}^2 < |m^2|$ or $\vec{p} \neq 0$ cannot be representation spaces and hence the representations with $m^2 < 0$ and $p = 0$ cannot be subspaces of a localizable representation.

So we are left with $p \neq 0$ and $m^2 \geq 0$. Now we can define a subspace for each m^2, such that the square of the generator of translations in time is given by $\vec{\partial}^2 + m^2$. In each subspace there is a localizable representation.

Given a subspace with $p \neq 0$ and $m^2 \geq 0$, M, we consider the subspace N of the representation $M \oplus M_0$ verifying $e^{iHt}\Psi = e^{iH_0t}\Psi$, where M_0 is a spin-0 representation and e^{iH_0t} is the translation in time acting on M_0. Then, $e^{iH_0(\vec{\partial})t}U = Ue^{iH_0(J\vec{P})}$. Multiplying U by $\alpha_p\sqrt{m/E_p}$ we can check that $J\Psi = i\gamma^0\Psi$ and so N is equivalent to M.

Now we define the unitary transformation $\Lambda\{\Psi\}(p) = \sqrt{\frac{E_p}{\Lambda^0(p)}}\Psi(\Lambda^{-1}(p))$. Then, we can check that $S \equiv L\Lambda^{-1}$ and it does not depend on $\vec{p}$. If we redefine $U\{\Psi\}(\vec{p}) = \alpha_p\sqrt{\frac{1}{\Lambda^0(p)}}U'\{\Psi\}(\vec{p})$, then we get that $\Lambda S\alpha_p U'\{\Psi\}(\vec{p}) = \alpha_p\Lambda Q_p U'\{\Psi\}(\vec{p})$ and so U' commutes with the Poincare representation.

If we look for subspaces where $m^2 = 0$ and the representation of Q_p has infinite spin, then the boost in the z direction for a momenta in the z direction multiplies the modulus of the translations of $SE(2)$ by E_p, which is in contradiction with the fact that $S \equiv L\Lambda^{-1}$ does not depend on $\vec{p}$.

So, we are left with a direct sum of massive representations and massless with discrete helicity. □

Proposition 88. *For any complex localizable unitary representation of the Poincare group, compatible with Poincare covariance, it if contains as a subspace a positive energy representation then it also contains the corresponding negative energy representation.*

Proof. The subspaces defined by the projectors involving the $i\gamma^0$s in the Q_p representation are not conserved by the system of imprimitivity because γ^0 does not commute with the matrices $\vec{\gamma}\gamma^0$ present in the transformation from momenta to coordinate space. When we go back to coordinate space, the projector on the $i\gamma^0$s can be written as an equality of the time translations which is not

part of the commuting ring of the SU(2) representation and hence it does not commute with the system of imprimitivity on R^3. □

Corollary 1. *A localizable Poincare representation is an irreducible representation of the Poincare group (including parity) if and only if it is: a)real and b)massive with spin 1/2 or massless with helicity 1/2.*

Proof. Since the subspaces defined by the projectors involving the $i\gamma^0$s in the Q_p representation are not conserved by the system of imprimitivity, then the condition for irreducibility cannot involve such projectors, which only happens for real representations with one spinor index. □

Notice that the condition of irreducibility of the representations admits localized solutions— the derivative of a bump function is a bump function, so we can find bump functions in the representation space—but it does not admit a position operator—the subspace of bump functions is not closed. Hence, we can say that a particular spin 1 state is in an arbitrarily small region of space, but the measurement of the position of an arbitrary spin 1 state might make it no longer a spin 1 state.

However, we can always define a frame field composed of a frame of 4 orthogonal Majorana spinors, then any spin state can be defined with respect to such frame field because we can generate any spin from algebraic combinations of 4 orthogonal Majorana spinors. Thus, we can always localize the frame fields carrying arbitrary spin states and a position operator can be defined in this way. Note that frame fields are also used in the tetrad formalism in general relativity to allow for diffeomorphisms while in this case the frame fields allow for arbitrary spins, so this is not a revolutionary result despite being relevant.

Going to complex systems, we can check that in the massive case, the condition of irreducibility does not admit localized solutions—given a localized solution Ψ in a region of space, then the result of the application of the projection operator to Ψ is not localized in a region of space. As for the massless representation, the condition of positive energy does not admit localized solutions either—for the same region as above—, but the condition for a chiral irreducible representation does admit localized solutions. The parity operator for such a chiral irreducible representation is anti-linear.

The localizable Poincare representation is Poincare covariant because for time $x^0 = 0$ at point $\vec{x} = 0$, we have for the Lorentz group $L\{\Psi\}(0) = S\Psi(0)$. The localizable Poincare representation is compatible with causality because the propagator $\Delta(x) = 0$ for $x^2 < 0$ (space-like x), where the propagator is defined for spin or helicity 1/2 as:

$$\Delta(x) \equiv \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} \frac{\not{p}\gamma^0 + m}{\sqrt{E_p + m}} e^{-i\gamma^0 p\cdot x} \frac{\not{p}\gamma^0 + m}{\sqrt{E_p + m}} \tag{109}$$

And verifies:

$$\Psi(x) = \int d^3\vec{y}\Delta(x-y)\Psi(y) \tag{110}$$

To show it we just need to do a Lorentz transformation such that $x^0 = 0$ and then show that $\Delta((0,\vec{x})) = 0$ for $\vec{x} \neq 0$.

Spinor frame and CPT theorem

As we have seen above, irreducibility is compatible with localizability only for real representations with spin 1/2. But that is all we need to localize arbitrary spins because we can build any spin out of tensor products of spin 1/2.

To do that we need to define a spinor frame, which is an orthonormal basis of 4 spinors. Thus our irreducible sample space will be $\mathbb{R}^3 \times Pin(3,1)$, it is a principal homogeneous space associated to the group $Pin(3,1)$ defining the spinor frame. Then we can enlarge the sample space with quantum numbers which are Poincare invariant, for instance a different mass or dressed spin. By dressed spin we mean a spin which is defined with respect to the spinor frame and thus it is Poincare invariant.

In the space coordinates, we can check that the most general mass in the Hamiltonian iH which is covariant under the connected component of the Lorentz group is $iH = \vec{\partial} \cdot \vec{\gamma}\gamma^0 + i\gamma^0 m_1 + \gamma^0\gamma^5 m_2$, thus it is also invariant under a parity-time reversal transformation (PT). This is essentially the CPT theorem, since CPT is a particular parity-time reversal transformation. Then we just need to define all the fields and the corresponding reparametrization invariance, to check if the parity breaking term can be absorbed by a reparametrization in which case we do not have parity violation (which is often called CP violation, since CP is a particular parity reversal transformation).

Quantization due to time-evolution: Yang-Mills and Classical Statistical Field Theory

Quantum Yang-Mills theory, Classical Statistical Field Theory (for Hamiltonians which are non-polynomial in the fields, e.g. General relativistic statistical mechanics) and Quantum Gravity all suffer from severe mathematical inconsistencies and produce unreliable predictions at best.

We define a class of statistical field theories where the probability distribution of the infinite-dimensional phase-space is mathematically defined by a wave-function in a Fock-space, allowing Hamiltonians which are non-polynomial in the fields. In this formalism, the wave-function is crucial already for classical statistical field theory due to mathematical reasons.

By defining gauge symmetries through algebraic ideals, we propose definitions for Quantum Yang-Mills and Classical Statistical Field Theory. Finally, we test the consistency of our formalism with the quantization of the free Electromagnetic field.

Lorentz covariance

Concerning the relation between the Hamiltonian formalism and special relativity, in this section we will just define the theory and show that it is Poincare invariant, postponing foundational questions to when a diffeomorphism-invariant (which includes Poincare invariance) theory is defined[29]. In short, the fact that time evolution plays a special role allows us to use only Poincare representations with positive squared mass. Considering Poincare representations with positive squared mass is self-consistent and it is in no way in conflict with Lorentz symmetry.

A complete physical system is a free system. If we neglect gravity, the wave-function associated to the free system is a unitary representation of the Poincare group, regardless of the interactions

[29]As we will do in Chapter /pub/timepiece. See also Chapter /pub/position for a different, Poincare-only approach.

occurring within the free system [159].

When the Hilbert space is the direct sum of irreducible representations of a symmetry group, then these representations will be defined by numbers (e.g. mass and spin) which are invariant under the symmetry group. Thus there will be a set of operators whose diagonal form corresponds to those invariants, we will call them Casimir operators. When the symmetry group is abelian and continuous (e.g. translation in time), then the generator of the group (e.g. the Hamiltonian) is a Casimir operator, and the invariant numbers defining the representations are called the constants of motion. Certainly, when we move from non-relativistic quantum mechanics and consider instead the Poincare group, then the Hamiltonian is no longer a Casimir operator and the notion of constants of motion needs to be reviewed. Nevertheless, the Casimir operators can be chosen arbitrarily (just like the Hamiltonian in non-relativistic quantum mechanics).

For a positive squared mass, the spin and the sign of the Energy are also Poincare invariants. The sign of the Energy times the modulus of the mass is the center-of-mass Energy, while the spin is the center-of-mass angular momentum. Thus, the Casimir operator whose eigenvalues are the center-of-mass Energy may have negative eigenvalues and it will be the analogous operator to the Hamiltonian of the non-relativistic formalism[30]. The Casimir operators necessarily commute with the momentum operator and thus they do not change the 3-momentum eigenstate. Thus we can solve the problem in a basis where the 3-momentum operator is diagonal.

In such a basis, the translations in space-time can be written as:

$$T(x)\Psi(\gamma\vec{v}) = e^{iM\tau(\gamma\vec{v},x)}\Psi(\gamma\vec{v}) \tag{111}$$

Where $\tau(\gamma\vec{v},x) = \gamma x_0 - \gamma\vec{v}\cdot\vec{x}$.

Note that $\gamma = \sqrt{1+\gamma^2\vec{v}^2}$ is a function of $\gamma\vec{v}$. Thus in a basis where the 3-momentum is diagonal, the translations in space-time have the same structure as the time-evolution in non-relativistic space-time, with M playing the role of the non-relativistic Hamiltonian and the numerical factor $\tau(\gamma\vec{v},x)$ playing the role of the time (it is indeed the proper-time).

For each 3-momentum eigenstate, there is a corresponding inertial referential where the 3-momentum is null, i.e. the referential of the center-of-mass. In such referential, the modulus of the energy is the invariant mass, the signal of the energy is also a Lorentz invariant and the angular momentum is the spin. Thus, the eigenvalues of the Hamiltonian and angular momentum operator in the center-of-mass define three Lorentz invariants which define the Poincare representation completely.

[30] As will be seen in Chapter /pub/timepiece, such operator has the formal form of the Hamiltonian action (i.e. it is the difference between the generator of translations in time and the Hamiltonian operator).

Despite we do not know a priori the diagonal form of the Hamiltonian, we know that it is either continuous or discrete in the neighborhood of the eigenvalue 0 (in the referential of the center-of-mass). If it is continuous then the zero energy has null measure. If it is discrete, we can modify the Hamiltonian adding an appropriate constant such that the zero energy is not one of the eigenvalues. Note that this is only possible in a complex Hilbert space and this is equivalent to adding to the system a free massive particle with null 3-momentum relative to the system. In any case, we can assume without loss of generality that our system is a quantum superposition of massive free systems with null 3-momentum. Then, the Lorentz transformations become known and are given by the Wigner irreducible massive representations of the Poincare group[159]. If the Hamiltonian is bounded from below then the vacuum state is not Lorentz invariant, as it was already suggested[182].

In the center-of-mass, the relevant group is not the Poincare group, but the little group of spatial rotations and the translation in time[159]. Thus the spatial and time coordinates of space-time, become separated. The fields are no longer representations of the Lorentz group, but only of the rotation group and the canonical commutation relations are not in conflict with the little group of spatial rotations.

Note that we use Wigner's convention for the definition of the 3-momentum of the free complete system: it is the eigenvalue of the generator of the translations in space for the complete system (i.e. all fields defining the phase-space are translated in space). Thus in the center-of-mass, the algebra of operators has a constraint imposing that the operators are translation invariant. This constraint is the same needed to define the free field parametrization, the constraint is $iD_x = 0$ as defined in chapter /pub/field, so no generality is lost and the free field parametrization is consistent with Poincare covariance.

The cyclic vector defining the Hilbert space needs not be translation-invariant (in the center-of-mass), just the operators need to be translation-invariant in the center-of-mass[31]. This gives us a big freedom to choose the cyclic vector defining the Hilbert space (which is related with the initial state of the system).

We assume that the translations in space of the complete system conserve the Hamiltonian and the constraints equations, such that setting the total 3-momentum to zero in no way conflicts with the constrained Hamiltonian system. Nevertheless, the restriction that there is a referential where the total 3-momentum is null, excludes the free complete system from traveling at the speed of light (e.g. a photon with non-null energy). Then the dynamics determined by the Hamiltonian becomes linked with the time coordinate (for a photon this would not be the case[183]).

Therefore and unlike what it is often claimed in the literature, it is false that (canonical) quantization

[31]As it was discussed in Chapter /pub/parametrization.

is incompatible with Lorentz covariance. Note that the phenomenologically successful (but ill defined) path integral formalism based on the Lagrangian is in fact equivalent to a path integral based on the Hamiltonian[184]. In our formalism, the only restriction is that we need to consider representations with positive squared mass, then the dynamics determined by the Hamiltonian becomes linked with the time coordinate[183]. The question why only positive squared masses are relevant is a reformulation of question 2) which will be left open in this paper. Similar assumptions concerning the energy-momentum of the full system are also done in the Källén-Lehmann representation of a non-perturbative two-point correlation function, where it is assumed that the eigenvalues of the 3-momentum squared are not larger than those of the squared energy[185].

Gauge-variant gaussian measure and the phase diagram

Unlike in non-relativistic Quantum Mechanics, there is no rigorous definition of what is a Quantum Yang-Mills theory to compare our results with, then our approach in field theory is different than in the non-relativistic case. Our goal is to build a self-consistent rigorous theory which after some approximations (e.g. perturbative expansion, or a ultra-violet cutoff) reproduces the successful predictions of the Standard Model of Particle Physics.

The Feynman's path-integral produces the notion of a gauge-invariant vacuum state, when the canonical formalism is "derived" from the path-integral formalism. However, the Feynman's path integral assumes the existence of a translation-invariant σ-finite (i.e. Lebesgue like) measure which is therefore gauge-invariant. Yet, it is proved that in rigor such infinite-dimensional Lebesgue measure cannot exist. As a consequence, the notion of a gauge-invariant vacuum state is inconsistent. On the other hand, a gauge-variant probability measure for an infinite-dimensional phase-space is mathematically consistent (e.g. a gaussian measure).

Some symmetries of the algebra of operators cannot be symmetries of the cyclic state defining the Hilbert space[32]. In the case of gauge symmetry, the gauge potentials can be fully reconstructed from the algebra of gauge-invariant operators [123]. Moreover, the Fock space (defined on a 3-dimensional space) produces well-defined expectation functionals for the gauge-invariant operators [272]. The expectation-values of the gauge-invariant operators fully define the statistical gauge field theory (since the gauge potentials can be fully reconstructed [123]), thus the gauge-variant operators can be neglected. Of course, gauge-variant operators can act on the Fock-space, but the link between these operators and the underlying manifold of gauge potentials is destroyed since the expectation-value is not gauge-invariant.

Since only (fully) gauge-invariant operators are allowed and the wave-functions necessarily break the gauge-symmetry, in scattering theory we always need to work in the in-in formalism. Of course, we

[32]As it was discussed in Chapter /pub/parametrization.

can use the more common in-out formalism in intermediate steps. Explicitly, any bounded normal operator can be expressed in diagonal form using projection-valued measures. These projections are built using a basis of the Hilbert space, which can be expressed using gauge-invariant operators acting on the cyclic state (initial state). Thus, the complex amplitudes of the in-out formalism can be expressed as expectation values of the in-in formalism. The amplitudes are complex to allow that a constant can be added to the Hamiltonian without observable consequences, which is crucial to ensure the Lorentz covariance of the theory as it was discussed in the previous section.

Thus the gauge-variant initial (cyclic) states are perfectly fine, even at the non-perturbative level. In the canonical formalism the same gauge-variant initial states are allowed due to the adiabatic approximation and the Gell-Mann and Low's theorem[273][274], although this issue is more subtle than in our formalism[275].

This does not imply that the mean-field approximation (that is, the usual choice of exact wave-function around which perturbative corrections are applied) always works, but the non-perturbative problems of the mean-field approximation are not exclusive to the gauge-symmetry (e.g. the mean-field approximation also breaks global symmetries in the two-Higgs-doublet model). So it can happen that within our (non-perturbative) mathematical definition of Quantum Gauge Field Theory, some exact predictions differ substantially from the corresponding perturbative approximation, being the predictions involving non-null non-abelian global charges obvious candidates for this difference to show up [124][125](Chru\ifmmode \acute{\vphantom}s\vphantom{}\else ś\fi{\vphantom}\vphantom{}ciel & Kondracki, 1987)[127]. However, the non-perturbative validity of our definition also shows that there is no fundamental reason why all non-abelian global charges must be null as claimed recently [276].

The global charge operator is different from a linear combination of gauge generators which are constrained to be zero (due to surface terms at infinity). In the in-in formalism, we can characterize an initial state with a non-null global charge using only gauge-invariant operators (e.g. the Casimir operators of the algebra generated by the global charge of an abelian or non-abelian gauge-theory, are gauge-invariant operators).

Spontaneous symmetry breaking

There are several definitions of spontaneous symmetry breaking in the context of statistical mechanics, based on: a long-range order parameter which is the expectation value of a function $f(A)$ invariant under a group G (e.g. the modulus $f(A) = |A|$) of an operator A which is translation invariant and breaks G; or a conditional expectation value of some operator A given some condition $C = 0$ that breaks the symmetry; or a two-point correlation function with the points at an infinite distance from each other (related with boundary conditions)[173].

In the translation-invariant time-evolution discussed in Chapter /pub/field, the probability distribution is asymptotically constant in the continuum space so that spontaneous symmetry breaking is possible.

As it was discussed in the previous section, the wave-function necessarily breaks the full gauge symmetry. But it may or not break the global gauge symmetry (remnant of the full gauge symmetry). However, there is no spontaneous symmetry breaking of the (full or global) gauge symmetry, since the expectation-values of the gauge-variant operators are null due to the constraints[33]. Note that the gauge-invariant operators are not sensitive to the long-range correlations which could signal spontaneous symmetry breaking[277].

Still the phase diagram of the theory may be sensitive to whether or not the field configuration at infinity conserves the global gauge symmetry. Since only gauge-invariant operators are allowed[34], we must distinguish between the concrete manifold appearing in the phase-space and the family of manifolds (obtained from the concrete manifold through different choices of transition maps between local charts) to which the expectation values correspond. Thus, the concrete manifold appearing in the phase-space may feature spontaneous symmetry breaking of the global gauge symmetry and this may cause effects in the phase-diagram (such as the existence of a phase transition), despite that there is no spontaneous symmetry breaking of the (full or global) gauge symmetry; the Higgs mechanism is an example of such a case.

Local operators and the momentum constraint

The momentum constraint also generates a gauge symmetry, once we consider a spectral measure where the fields are functions of space. Then the momentum constraint always modifies the spectral measure and so we have a complete unconstrained gauge-fixing.

Since all operators must be invariant under a translation in space, how can we define local operators? In rigor we can't, but we can define operators which are linear combinations of local operators, which behave effectively as local operators. Consider the operator $\int d\vec{x}\ l(\vec{x})$ which is translation invariant, where $l(\vec{x})$ is a local operator. The translation-invariant operator behaves effectively as a local operator when the wave-function is concentrated around one point in the 3-dimensional space. Note that then the wave-function is not translation invariant which is fine, as long as the operators are translation invariant.

[33] As it was discussed in Chapter /pub/parametrization.
[34] As it was discussed in Chapter /pub/parametrization.

Quantization due to time evolution

We introduce now the procedure of quantization due to unitary time evolution. The idea is that every operator corresponding to an observable is the time evolution of an operator belonging to a common commutative algebra of observables. The time-evolution transforms a sequence of time-ordered operators [278] (which commute algebraically but the time-ordering is non-commutative) into a sequence of (algebraically) non-commuting operators acting on a single slice of time of the wave-function. For a class of time evolutions (of the type of non-relativistic Quantum Mechanics), the canonical commutation relation of position and momentum are reproduced (strictly speaking, it is the Weyl relation that is reproduced, i.e. the exponenciated version of the canonical commutation relation).

We use again the Trotter exponential product approximation [178], verifying for small ϵ:

$$e^{i\epsilon A}e^{i\epsilon B} = e^{i\epsilon(A+B)-\frac{\epsilon^2}{2}[A,B]+i\mathcal{O}(\epsilon^3)} \tag{112}$$

This is a good approximation since it works for unbounded self-adjoint operators A, B.

Let now $\epsilon = \frac{1}{n}$ with n arbitrarily large. Then,

$$e^{i\epsilon A}e^{iB}e^{-i\epsilon A} = (e^{i\epsilon A}e^{i\epsilon B}e^{-i\epsilon A})^n = e^{iB-\epsilon[A,B]+i\mathcal{O}(\epsilon^2)} \tag{113}$$

Therefore, for small enough ϵ

$$e^{i\epsilon p^2}e^{ix}e^{-i\epsilon p^2} = e^{i(x+\epsilon p)} \tag{114}$$

Now we need a definition of covariant derivative in time of the position operator x, consistent with the fact that only $F(a) = U(a)e^{ix}U^\dagger(a)$ (where $U(a)$ is the time-evolution) is bounded while x is unbounded. If we would be dealing with a commutative algebra, then the natural definition would be:

$$\lim_{\epsilon\to 0} F^{\frac{1}{\epsilon}}(0)(F^{\frac{1}{\epsilon}}(\epsilon))^\dagger \tag{115}$$

For a trivial parallel transport, we would get as required:

$$\lim_{\epsilon\to 0} F^{\frac{1}{\epsilon}}(0)(F^{\frac{1}{\epsilon}}(\epsilon))^\dagger = \lim_{\epsilon\to 0} e^{i\frac{x(0)-x(-\epsilon)}{\epsilon}} \tag{116}$$

But since we are dealing with a non-commutative algebra, we need to use the Trotter exponential product approximation formula, to define the exponential version of the time derivative:

$$\lim_{\epsilon\to 0}\lim_{n\to\infty}(F^{\frac{1}{n\epsilon}}(0)(F^{\frac{1}{n\epsilon}}(\epsilon))^\dagger)^n \tag{117}$$

And so for the parallel transport $U(\epsilon)=e^{i\epsilon(p^2+V(x))}$ where $V(x)$ is a potential only dependent on the position operator, the exponential version of the time derivative of the position operator x is:

$$\begin{aligned}&\lim_{\epsilon\to 0}\lim_{n\to\infty}(F^{\frac{1}{n\epsilon}}(0)(F^{\frac{1}{n\epsilon}}(\epsilon))^\dagger)^n=\\&\lim_{\epsilon\to 0}\lim_{n\to\infty}(e^{i\frac{1}{n\epsilon}x}U(\epsilon)e^{-i\frac{1}{n\epsilon}x}U^\dagger(\epsilon))^n=e^{ip}\end{aligned} \tag{118}$$

The result is the exponential of the momentum operator, which verifies the Weyl relations with respect to the exponential of the position operator. With some abuse of language, we can say that for this type of time-evolution (and thus for this type of Hamiltonian $p^2+V(x)$, which is most common in non-relativistic Quantum Mechanics), the time derivative of the position operator is the momentum operator. Thus the quantization (i.e. the Weyl relations) may appear in a statistical field theory due to a particular non-deterministic time-evolution.

One major advantage of the quantization due to time-evolution is that it applies not only to the variables of position and momentum which have an obvious correspondence in classical statistical mechanics, but it also applies to variables without an obvious correspondence in classical statistical mechanics such as fermions and ghosts (we just apply the time-evolution to the variables) and also in the presence of gauge symmetries as we will see in the remaining of this work.

Relation with the BRST formalism

The results of reference [279], are consistent with our formalism:
1) there is no fundamental reason why BRST-like gauge-fixing should be Lorentz invariant;
2) but it is crucial that the BRST-like gauge-fixing term involves the time derivative of the fields which have an arbitrary time-evolution, otherwise the theory becomes non-local and the perturbative expansion becomes inconsistent (as it happens in the Coulomb gauge, see also [280][281]);
3) for technical reasons that (apparently) are not related with Lorentz covariance, the R_ξ gauges are better suited for perturbation theory than any other gauge (Lorentz covariant or non-covariant).

The result 1) is consistent with our formalism where the fields are representations of the little group of rotations and not of the Lorentz group. In our formalism, we do not need the BRST-like gauge-fixing to define the theory, however to go from a phase-space defined in space-time to a phase-space defined in space only we need to fix the time-evolution of all fields. As we will see

in this Section, this is done using a BRST-like gauge fixing in agreement with result 2). Finally, there does not seem to exist an obvious reason for result 3), but in any case our formalism is not incompatible with the result 3).

We are working from the start with a self-consistent statistical field theory.

The BRST charge is useful for a non-commutative algebra, because when we multiply the right and left ideal the result would not be an ideal if the BRST charge would not square to 0. The alternatives would be to use (standard) gauge-fixing, which suffers from nonlocality in general and suffers from the Gribov problem in non-abelian gauge theories; or to work only with a commutative sub-algebra of the algebra of operators satisfying the constraints, which is challenging for a non-deterministic time-evolution.

The algebra of operators is then enlarged from gauge-invariant operators to BRST-invariant (not necessarily gauge-invariant) operators. These BRST-invariant operators are divided into equivalence classes. The BRST cohomology maps (in a bijective way) each equivalence class with a corresponding gauge-invariant operator satisfying the constraints. When performing algebraic manipulations in a gauge-invariant operator, we can convert the gauge-invariant operator into BRST-invariant, then do the manipulations in the space of BRST-invariant operators and then convert the resulting BRST-invariant operator again into a gauge-invariant operator.

Note that there are two possible inner-products: one degenerate where the ghost fields are self-adjoint and another non-degenerate where the ghost fields are not self-adjoint and behave like standard fermioninc creation and annihilation operators [282]. We use the non-degenerate inner-product, so that we are always working within an Hilbert space formalism. We can do this, because the vacuum state in our formalism is not Lorentz invariant if it exists and the existence of such vacuum state is not mandatory for our formalism. This would not possible in the covariant operator formalism [283] where the vaccuum is Lorentz invariant. Since our vacuum breaks Lorentz invariance, then the spin-statistics theorem does not hold and the ghosts are as consistent as a Schrödinger field. Moreover, since the ghosts only appear in intermediate stepts and the ghosts never appear in the definition of expectation values of observables (neither in the operators nor in the wave-function), then the complete system when treated as a free particle respects the spin-statistics theorem.

Crucially, the BRST cohomology merely simplifies the expression defining a gauge-invariant operator into another equivalent expression, it does *not* affect the gauge-variant wave-function and therefore the Gribov problem does not arise.

There is a well-known subtlety with the BRST cohomology that we will address in chapter /pub/timepiece: the BRST cohomology is itself gauge-invariant and mathematically well-defined, but it is merely a dispensable auxiliary step in a calculation performed in the context of a quantum

formalism. If the quantum formalism is mathematically inconsistent, if the formulation of the calculation crucially depends on the BRST-invariant algebra (not our case in chapter /pub/timepiece, but it is the case of the path integral), surprises are possible. In particular, if all the details of the calculation are only known for the BRST-invariant algebra and not before for the gauge-invariant operators (not our case in chapter /pub/timepiece, but it is the case of the quantum BRST formalism), then the gauge-invariance of the BRST cohomology does not imply that the calculation would be the same in all gauges or that the quantum formalism is logically consistent (and in fact it is not due to the Gribov problem). This is discussed in reference [284]:

"Being gauge invariant, the BFV-PI necessarily reduces to an integral over modular space, irrespective of the gauge fixing choice. Nevertheless, which domain and integration measure over modular space are thereby induced are function of the choice of gauge fixing conditions. The BFV-PI is not totally independent of the choice of gauge fixing fermion Ψ."

In the (our) case of the quantization due to time-evolution (see chapter /pub/timepiece), the quantum formalism is mathematically well-defined and all details of the calculations are known, regardless of whether we apply the BRST cohomology or not. Since we use the BRST cohomology to merely simplify the expression defining a gauge-invariant operator into another equivalent expression, the Gribov problem does not affect us.

Pure SU(3) Yang-Mills theory

In the following, we follow reference [282] where it is possible. The structure constants f_{abc} are related with the $SU(N)$ generators T_a by:

$$\begin{aligned} &= if_{abc}T_c \\ \mathrm{tr}(T_aT_b) &= \frac{1}{2}\delta_{ab} \end{aligned} \tag{119}$$

The $SU(N)$ index $\{a, b, c\}$ allows us to cover also the electromagnetism by eliminating the $SU(N)$ index and thus setting the structure constants f_{abc} to zero. The covariant derivative is given by:

$$\begin{aligned} D_j &= \partial_j - igT_aA_{ja} \\ [D_j, D_k] &= -igT_aF_{jka} \end{aligned} \tag{120}$$

The magnetic components of the gauge field strength tensor are given by:

$$\begin{aligned} B_{ia} &= \frac{1}{2}\epsilon_{ijk}(\partial_jA_{ka} - \partial_kA_{ja} - gf_{abc}A_{jb}A_{kc}) \\ B_i &= i\frac{1}{2g}\epsilon_{ijk}[D_j, D_k] = B_{ia}T_a \end{aligned} \tag{121}$$

Where the indices i,j,k correspond to the spatial dimensions only, i.e. from 1 to 3. It verifies the Jacobi relation

$$= i\frac{1}{2g}\epsilon_{ijk}[D_i,[D_j,D_k]] = 0 \tag{122}$$

The Hilbert space is the tensor product of the symmetric and antisymmetric Fock spaces $\Gamma^s(L^2(\mathbb{R}^{99}\times\mathbb{Z}_2^{31}))\otimes\Gamma^a(L^2(\mathbb{R}^{99}\times\mathbb{Z}_2^{31}))$ [104](*Fock Space*, n.d.). This gives us a graded Lie superalgebra of creation and annihilation operators, of both bosonic and fermionic types. The total fermionic number is defined as the sum of all number operators corresponding to fermionic degrees of freedom. The fermionic creation and annihilation operators correspond to a field configuration where the total fermionic number is odd, while the bosonic creation and annihilation operators correspond to a field configuration where the total fermionic number is even.

The $\mathbb{R}^{99}$ degrees of freedom correspond to 3 space coordinates (x), 3x8=24 fields (A_{ka}) and its corresponding space-derivatives ($\partial_j A_{ka}$). The $\mathbb{Z}_2^{31}$ degrees of freedom correspond to 8 ghosts corresponding to each SU(3) generator (ψ_a) and its corresponding space-derivatives ($\partial_j\psi_a$), minus one because we are taking the tensor product of two Fock-spaces which introduces an extra $\mathbb{Z}_2$ degree of freedom.

We define:

$$\begin{aligned}
&a(\vec{x},...) = a(\vec{x},\oplus_a(\vec{A}_a,\mu_a,\oplus_k(\vec{A}_{a,k},\mu_{ak})))\\
&\int d^3\vec{x}... = \int d^3\vec{x}\prod_a(d^3\vec{A}_a\sum_{\mu_a=0}^{1}\prod_k(d^3\vec{A}_{a,k}\sum_{\mu_{ak}=0}^{1}))\\
&\{[a(\vec{x},...,\mu),a^\dagger(\vec{y},...,\nu)]\} = a(\vec{x},...,\mu)a^\dagger(\vec{y},...,\nu) + (-1)^{(\mu\text{mod}2)(\nu\text{mod}2)}a^\dagger(\vec{y},...,\nu)a(\vec{x},...,\mu) =\\
&= \delta((\vec{x},...)-(\vec{y},...))\\
&[A_{ja},\pi_b^k] = A_{ja}\pi_b^k - \pi_b^k A_{ja} = i\delta_j^k\delta_{ab}\\
&[A_{ja,k},\pi_b^{mn}] = A_{ja,k}\pi_b^{mn} - \pi_b^{mn}A_{ja,k} = i\delta_j^n\delta_k^m\delta_{ab}\\
&\{\psi_a,\psi_b^\dagger\} = \psi_a\psi_b^\dagger + \psi_b^\dagger\psi_a = \delta_{ab}\\
&\{\psi_{aj},\psi_{bk}^\dagger\} = \psi_{aj}\psi_{bk}^\dagger + \psi_{bk}^\dagger\psi_{aj} = \delta_{ab}\delta_{jk}
\end{aligned} \tag{123}$$

where

$$\begin{aligned}
\psi^\dagger\{a\}(t,...,j) &= a(t,...,1)\delta_{j0}\\
\psi\{a\}(t,...,j) &= a(t,...,0)\delta_{j1}
\end{aligned} \tag{124}$$

The Hamiltonian for the Yang-Mills theory in the Weyl gauge is defined as:

$$\begin{aligned} H &= \int d^3\vec{x}...a^\dagger(\vec{x},...)H(\vec{x},...)a(\vec{x},...) \\ H(\vec{x},...) &= -\frac{1}{2}\pi_a^i\pi_a^i - \frac{1}{2}B_{ia}B_{ia} \\ \Omega &= \int d^3\vec{x}...a^\dagger(\vec{x},...)\Omega(\vec{x},...)a(\vec{x},...) \\ \Omega(\vec{x},..) &= \left[\pi_a^k\partial_k\psi_a^\dagger - \pi_a^k f_{abc}A_{kb}\psi_c^\dagger - i\frac{1}{2}f_{abc}\psi_a^\dagger\psi_b^\dagger\psi_c\right] \end{aligned} \tag{125}$$

where (ϕ, π) is a complex field/conjugate momentum field over timepiece t (correspondingly). $V(|\phi|^2)$ is a functional (a polynomial or other) on $|\phi|^2$. The BRST charge is Ω, where ψ is the ghost field, λ is the (real) gauge field and p_λ its conjugate momentum.

We need to separate the ideal (gauge generator) from the gauge-invariant algebra. That is, not only the gauge-invariant algebra must commute with the ideal, but also the ideal cannot be included in the gauge-invariant algebra. This is guaranteed by (unconstrained) gauge-fixing as it was discussed in Chapter /pub/parametrization. For instance, in the case of the free Electromagnetic field, while the gauge generator commutes with the ideal (because the ideal is the gauge generator itself) it cannot be included in the gauge-invariant algebra; thus, the Electromagnetic field as a whole is not part of the gauge-invariant algebra. And that is a good thing, since then we always work with local field operators, unlike those of the Coulomb gauge.

Since the Weyl gauge is an incomplete gauge-fixing, this Hamiltonian is invariant under a remnant gauge symmetry generated by $\theta_a(D_i\pi^i)_a$, where the function θ is constant in time (the remnant gauge transformations are constant in time). We can now apply the BRST cohomology for the remnant gauge symmetry (as discussed in reference [121]) and choose the Lorentz gauge.

The first step is to introduce a new field Φ and a new constraint $\Pi = 0$ where Π is the conjugate field of Φ. Such a field is completely spurious because it allows a complete and unconstrained gauge-fixing, thus we can remove it from the gauge-invariant operators (as expected, otherwise we would have a different theory from the one we started before introducing the new field).

Timepiece and the Gribov ambiguity

We then define a subclass of statistical field theories in Minkowski space-time where the (classical) canonical coordinates when modified by a non-deterministic time evolution, verify the canonical commutation relations. By defining gauge symmetries through algebraic ideals, we propose definitions for Quantum Yang-Mills and Quantum Gravity. Finally, we test the consistency of our formalism with the quantization of the free Electromagnetic field.

Quantization due to time-evolution

Gauge symmetry: definition within Quantum Field Theory

The concept of gauge symmetry is clear in classical field theory: it is a particular symmetry of a singular Lagrangian. A singular Lagrangian is a Lagrangian which does not determine the time-evolution of all fields. Consequently, there are fields whose time-evolution is arbitrary and so these fields cannot be physical fields. The gauge symmetries are the symmetries of the Lagrangian which transform the fields with one time-evolution to fields with another time-evolution. The singular Lagrangian only determines the time-evolution of the gauge-invariant algebraic combinations of fields, thus only these algebraic combinations are observable.

However, the gauge symmetry clashes at a fundamental level with the canonical commutation relations of Quantum Field Theory.

When the Lagrangian is singular, in the classical Hamiltonian formalism there are conjugate momenta constrained to be null while the corresponding field has a definite value. This is incompatible with the canonical commutation relations of the corresponding quantum fields: the fact that the conjugate momentum is null implies that the field value is undefined. Note that the BRST formalism does not solve this problem for non-abelian gauge symmetries due to the Gribov ambiguity[121].

Moreover, in General Relativity the diffeomorphism transformations are time-dependent transformations and many of the predictions in Gravity are produced with the Hamiltonian formalism; however the symplectic form of conservative classical mechanics is not invariant under time-dependent

transformations: the most obvious way to make non-relativistic mechanics consistent with General Relativity is to formulate it as a particular field theory whose phase-space is a fibred manifold over a "time" axis [14].

Since the wave-function is a parametrization of any probability distribution, the canonical commutation relations of Quantum Field Theory should be used as a step towards an Hamiltonian formulation of classical gauge theory (and also of quantum gauge theory, of course) and not the other way around. At the cost of using non-commutative operators (and thus of using a non-deterministic formulation), we get the crucial advantage that the phase-space *coincides* with the coordinate space, i.e. every coordinate comes equipped with a conjugate momentum operator *without* enlarging the phase-space. Note that the conjugate momentum operator is not part of the phase-space in this case, since it does not commute with the corresponding coordinate.

Once we choose to use the canonical commutation relations, the remaining question is merely what is the correct phase-space for our problem. In Source Field Theory (defined in the next Section), the source fields do have a time coordinate and are also part of the phase-space, so that the Lagrangian becomes dispensable. We also discuss the definition of gauge symmetry within Quantum Field Theory in Section /pub/quantization#relation-with-the-brst-formalism.

Note that a formulation of classical gauge theories within a Quantum formalism (besides classical statistical and quantum gauge theories), may also be advantageous. Theories such as classical electrodynamics or more generally classical non-abelian gauge theories [285] involve a system of non-linear partial differential equations. It is a very hard problem to study in general the space of classical solutions of such systems[35]. Even when a few solutions can be found, they may not be the ones that describe the physical system correctly. A consistent theory covering many cases only exists (at the moment) for systems of linear partial differential equations[286]. Thus to solve many non-linear deterministic theories we may not have better alternative (at the moment) than to consider them as a particular case of a statistical theory and apply linear quantum methods on its wave-function parametrization [42][287]—then the building blocks of the overall deterministic theory are non-deterministic.

[35]Fefferman, Charles L. (2007). *Existence and smoothness of the Navier-Stokes equation*. http://www.claymath.org/sites/default/files/navierstokes.pdf. Retrieved from http://www.claymath.org/sites/default/files/navierstokes.pdf Tao, Terence. (2007). *Why global regularity for Navier-Stokes is hard*. https://terrytao.wordpress.com/2007/03/18/why-global-regularity-for-navier-stokes-is-hard/. Retrieved from https://terrytao.wordpress.com/2007/03/18/why-global-regularity-for-navier-stokes-is-hard/

A well known example is the Navier-Stokes equation.

Statistical Source Field Theory

The method of quantization described in the next section is inspired by the Source formalism of Schwinger [288] which is itself both an alternative to and inspired by the Feynman's path integral, where time-ordering [278][289] plays a key role.

In this and in the next sections we will consider fields defined in a one dimensional time, neglecting the space dimensions. The extension of the results of this section to fields defined in four dimensional space-time is straightforward, since time and space are separated in the Hamiltonian formalism (see Section).

We here use the term field meaning a function of a parameter t which we also call timepiece. The timepiece has a close relation with the time, the main difference is that our fields are part of the phase-space of the theory: the state of the system is given by the functions of the timepiece t. Thus the phase-space has similarities to the space of trajectories in time of the Lagrangian formalism. Then, the time-evolution will modify the state of the system as a function of the time parameter τ.

Therefore, our fields are best described as source fields and we are dealing with a statistical source field theory. Using a wave-function, we can parametrize the probability distribution for a source field in timepiece. The linear space generated by all wave-functions is a Fock space [287]. The Fock space has the properties of a continuous tensor product of Fock-spaces corresponding to fields defined in infinitesimal timepiece-intervals, i.e. $\varphi(t)dt$. The time-evolution will not only advance the timepiece-intervals forward, but it will modify the wave functions corresponding to each timepiece interval accordingly to an Hamiltonian which plays here the role of the connection in a covariant derivative. With abuse of language, we can describe the situation as a continuous tensor product of initial-value (i.e. Cauchy) problems, instead of just one initial value problem as in standard Quantum Mechanics.

For consistency with the free field parametrization (see Section ...), we also impose a constraint for the algebra of operators to be translation invariant in the timepiece t. This constraint is the same needed to define the free field parametrization, the constraint is $iD_t = 0$ as defined in chapter /pub/field, so no generality is lost and the free field parametrization is consistent with the Hamiltonian formalism, as we will see next. As it was discussed in chapter /pub/field, the cyclic vector defining the Hilbert space needs not be translation-invariant, just the operators need to be translation-invariant. This gives us a big freedom to choose the cyclic vector defining the Hilbert space (which is related with the initial state of the system).

When the Lagrangian is non-singular, the constraint $iD_t = 0$ allows us to define an Hamiltonian which is consistent not only when the timepiece is considered, but also when we want to reduce the phase-space by getting rid of the timepiece (through a pushforward measure) and obtain the same

Hamiltonian we would get from the Lagrangian formalism directly.

The main advantages of this more general formalism will be discussed in the next sections.

Pure Yang-Mills theory

The Hilbert space (in the particular case of SU(3) Yang-Mills theory) is the tensor product of the symmetric and antisymmetric Fock spaces $\Gamma^s(L^2(\mathbb{R}^{164} \times \mathbb{Z}_2^{39})) \otimes \Gamma^a(L^2(\mathbb{R}^{164} \times \mathbb{Z}_2^{39}))$ [104](*Fock Space*, n.d.). This gives us a graded Lie superalgebra of creation and annihilation operators, of both bosonic and fermionic types. The total fermionic number is defined as the sum of all number operators corresponding to fermionic degrees of freedom. The fermionic creation and annihilation operators correspond to a field configuration where the total fermionic number is odd, while the bosonic creation and annihilation operators correspond to a field configuration where the total fermionic number is even.

The $\mathbb{R}^{164}$ degrees of freedom correspond to 4 space coordinates (x), 4x8=32 fields ($A_{\mu a}$) and its corresponding space-derivatives ($\partial_\mu A_{\nu a}$). The $\mathbb{Z}_2^{39}$ degrees of freedom correspond to 8 ghosts corresponding to each SU(3) generator (ψ_a) and its corresponding spacetime-derivatives ($\partial_\mu \psi_a$), minus one because we are taking the tensor product of two Fock-spaces (bosonic and fermionic) which introduces an extra $\mathbb{Z}_2$ degree of freedom.

We define:

$$
\begin{aligned}
&a(x,...) = a(x, \oplus_a(\alpha_a, (\oplus_\mu A_{\mu,a}), \oplus_\nu(\alpha_{a\nu}, (\oplus_\mu A_{\mu a,\nu}))))\\
&\int d^4x... = \int d^4x \prod_a ((\prod_\mu dA_{\mu a}) \sum_{\alpha_a=0}^{1} \prod_\nu ((\prod_\mu dA_{\mu a,\nu}) \sum_{\alpha_{a\nu}=0}^{1}))\\
&\{[a(x,...,\mu), a^\dagger(y,...,\nu)]\} = a(x,...,\alpha)a^\dagger(y,...,\beta) + (-1)^{(\alpha \mathrm{mod} 2)(\beta \mathrm{mod} 2)} a^\dagger(y,...,\alpha)a(x,...,\beta) =\\
&= \delta((x,...) - (y,...))\\
&[A_{\mu a}, \pi_b^\nu] = A_{\mu a}\pi_b^\nu - \pi_b^\nu A_{\mu a} = i\delta_\mu^\nu \delta_{ab}\\
&[A_{\nu a,\mu}, \pi_b^{\alpha\beta}] = A_{\nu a,\mu}\pi_b^{\alpha\beta} - \pi_b^{\alpha\beta} A_{\nu a,\mu} = i\delta_\mu^\alpha \delta_\nu^\beta \delta_{ab}\\
&\{\psi_a, \psi_b^\dagger\} = \psi_a\psi_b^\dagger + \psi_b^\dagger\psi_a = \delta_{ab}\\
&\{\psi_{a\mu}, \psi_{b\nu}^\dagger\} = \psi_{a\mu}\psi_{b\nu}^\dagger + \psi_{b\nu}^\dagger\psi_{a\mu} = \delta_{ab}\delta_{\mu\nu}
\end{aligned}
\tag{126}
$$

where

$$
\begin{aligned}
&\psi^\dagger\{a\}(x,...,j) = a(x,...,1)\delta_{j0}\\
&\psi\{a\}(x,...,j) = a(x,...,0)\delta_{j1}
\end{aligned}
\tag{127}
$$

The Hamiltonian for the Yang-Mills theory in our formalism is related to the classical Hamiltonian Action[290]:

$$\begin{aligned}
&H = \int d^4x ... a^\dagger(x,...) H(x,...) a(x,...) \\
&H(x,...) = i\partial_0 - \pi_a^i A_{ia,0} + \pi_a^i (D_i A_{0a}) - \frac{1}{2}\pi_a^i \pi_a^i - \frac{1}{2} B_{ia} B_{ia} \\
&\Omega = \int d^4x ... a^\dagger(x,...) \Omega(x,...) a(x,...) \\
&\Omega(x,..) = \left[\pi_a^\mu \partial_\mu \psi_a^\dagger - \pi_a^\mu f_{abc} A_{\mu b} \psi_c^\dagger - i\frac{1}{2} f_{abc} \psi_a^\dagger \psi_b^\dagger \psi_c \right]
\end{aligned} \tag{128}$$

Due to the BRST cohomology, we can add to the Hamiltonian the term:

$$\{\Omega, i\psi_a A_{0a}\} = -\pi_a^\mu A_{0a,\mu} - i\psi_a^\dagger \partial_0 \psi_a \tag{129}$$

Using the constraint imposing timepiece translation invariance, we obtain the BRST invariant 3D Hamiltonian in the Weyl gauge:

$$H_W(x,...) = H(x,...) + \{\Omega, \psi_a A_{0a}\} \approx -\frac{1}{2}\pi_a^i \pi_a^i - \frac{1}{2} B_{ia} B_{ia} \tag{130}$$

Considering only operators respecting time-ordering, we can move all operators to a single slice of time and then the predictions coincide with the Dirac quantization of the Yang-Mills theory (and also for the Electromagnetism). A pushforward measure allows us to reduce the phase-space, integrating 3D space only.

Since the Weyl gauge is an incomplete gauge-fixing, this new Hamiltonian H_W is invariant under a remnant gauge symmetry generated by $\theta_a (D_i \pi^i)_a$, where the function θ is constant in time. We can now reapply the BRST cohomology for the remnant gauge symmetry (as discussed in reference [291]). The remnant gauge transformations are constant in time.

The time-evolution and the remnant gauge transformations of A_{0a} decouple from the time-evolution and the remnant gauge transformations of A_{ia} (respectively); A_{0a} commutes with the remnant gauge transformations. Thus, the gauge fields A_{0a} can be studied first, without taking into account the A_{0i} fields. We do this by integrating only the A_{0a} fields, while keeping the A_{ia} fields as external variables to the theory and within the theory it is as if we would only consider operators which do not depend on A_{ia}. Then, we are left with the A_{0a} fields only, which commute with H_W and also with the remnant gauge transformations, then the corresponding gauge theory is that of a gauge-invariant field which is constant in time, which has an easy solution.

After integrating the A_{0a} fields, we obtain operators which depend on A_{ia} only and do not depend on A_{0a}. Note that in the literature (and in reference [121] in particular) it is usually claimed that the operators do not depend on A_{0a} because they were chosen not to depend on A_{0a} (due to an argument related to gauge-invariance as we will see below) and not because the fields A_{0a} were integrated first. This is not incompatible with our results because the remnant-gauge-invariant operators involving A_{ia} only, are also fully gauge-invariant. Thus our framework is more general but it is compatible with reference [121] because we can also simply choose remnant-gauge-invariant operators involving A_{ia} only, which will be automatically fully gauge-invariant even if we just ignore the A_{0a} fields.

Considering the remnant gauge-symmetry and only a single slice of time, we can then just follow reference [121] on how to choose the Lorentz gauge. The first step is to introduce a new field Φ and a new constraint $\Pi = 0$ where Π is the conjugate field of Φ. Such a field is completely spurious because it allows a complete and comprehensive gauge-fixing, thus we can remove it from the gauge-invariant operators (as expected, otherwise we would have a different theory from the one we started before introducing the new field).

Free electromagnetic field: an exact example

While the free electromagnetic field is an abelian gauge theory and thus it is somewhat simpler than a Yang-Mills theory (for instance, there is no Gribov ambiguity), it has the crucial advantage that the time-evolution has an exact solution, because the Hamiltonian is quadratic in the fields.

The extrapolation of the results of the previous section to the electromagnetism is straightforward since the Yang-Mills theory is an obvious generalization of the electromagnetism. The contraction of the index is just an integration in a 4-dimensional space-time and we set the structure constants f_{abc} to zero.

Due to the complete non-comprehensive gauge-fixing, we always start from the magnetic components of the gauge field strength tensor B_i (for local self-adjoint operators). Using a (remnant from the Weyl gauge) gauge-invariant unitary transformation (such as time-evolution) we can generate other operators, since the unitary transformation can be transferred from the operators to the wave-function without modifying the constraints. Thus, through time-evolution (in the Weyl gauge), we also have the local self-adjoint operator $\vec{\partial} \times \vec{\pi}$, i.e. the curl of the Electric Field. It is not possible to use just the Electric Field or its divergence as an operator acting on the Hilbert space, which is a sign of consistency because the divergence of the Electric Field is a constraint set to zero before the definition of the Hilbert space. This allows us to always work with local operators, which avoids the localization problems which arise when working with non-local transverse fields [281].

Renormalization, the mass gap and the Millennium prize

We couldn't propose a mathematical definition of a Quantum gauge theory without discussing renormalization and the Millennium prize (Clay Mathematics Institute) [31], which are related.

We can easily conclude that the Hamiltonian in the Weyl gauge is positive definite and thus, as discussed in Section /pub/field#mass-gap, Yang-Mills theory can be reformulated with or without a mass gap without any observable consequences. As for the existence of the Yang-Mills theory, we just need to prove that there is a point in the phase-space where all constraints are satisfied, which we did already.

The Millennium problem *"Yang-Mills and Mass Gap"* defined by the Clay Mathematics Institute [31], consists essentially in defining a Quantum Yang-Mills theory in a mathematically rigorous and useful way. By useful, we mean that the definition must mark *"a turning point in the mathematical understanding of quantum field theory, with a chance of opening new horizons for its applications"* (as stated in reference [31]).

Note that there is a monetary prize associated to the problem. If the requirement would only be that the definition would be mathematically rigorous, then (as so often happens in mathematics) the definition could have no implications to theoretical physics. No one doubts that many mathematical entities resembling a quantum Yang-Mills theory exist, the relevance of finding "the" definition lies in the hope that it will lead to progress not only in mathematics but also in physics.

On the other hand, "useful" is not an objective criterium. Since the authors of reference [31] believed that any useful definition of a Quantum Yang-Mills theory would necessarily establish the existence of a mass gap, then they added the requirement that the definition of Quantum Yang-Mills must be such that it establishes to the existence of a mass gap, that is, that the spectrum of the Hamiltonian is discrete close to the ground state.

The reason that the authors of reference [31] believed so, is that the existence of a mass gap implies clustering which is a locality property required (from a technical point-of-view) to extrapolate many results obtained in simplified theories to a realistic Quantum Field Theory.

The propagation of an excitation (a photon, a gluon) in a medium is best described in an effective field theory framework[292]. Renormalization is an essential part of an effective field theory and it describes the quasi-(free) particle properties of the excitation (such as the effective mass), whether the associated perturbation theory needs to be regularized or not due to ultra-violet divergencies [196].

But without perturbative expansions, the initial state needs not to be an eigenfunction of the Hamiltonian (and usually it is not), so that we do not have to eliminate divergences due to tadpoles [293]; any initial state is a good asymptotic state so we have no infrared divergences [294]; and we do not

have loops introduced by the perturbative expansion so no ultra-violet divergences [294].

That is, our theory needs no regularization. Note that a regularized theory is necessarily an effective field theory, where the renormalization plays a key role.

Our definition of Quantum Yang-Mills theory is thus not an effective field theory. Thus the renormalization group in our theory cannot be distinguished from a regular background symmetry group (that is, a symmetry group that acts not just on the fields but also on the parameters of the theory, leaving the observables invariant). The renormalization group plays no fundamental role in our definition of Quantum Yang-Mills theory. This is the ultimate reason why establishing the existence of a mass gap has nothing to do with the problem of defining the theory (at least in the way we do it).

On the physical parity transformation and antiparticles*

On the physical parity transformation and antiparticles

Introduction

[295][296]

It was argued a long time ago [297], that there is a basis where CP becomes a parity transformation (where CP and P are the transformations usually identified as charge-parity and parity transformations). Majorana spinors and the Majorana basis for the Dirac matrices were used. It was explicitly said: "*Any non-Hermitian field can always be decomposed into a sum of two Hermitian fields. The use of Hermitian fields simplifies some of the general discussions, and there is no loss of generality*". In such a basis there are no fields which are complex-conjugate representations (usually associated with the charge conjugation), all particles are their own antiparticles (from a canonical quantization point of view) and the difference between CP and P dissolves.The authors of reference [297] were well aware that Weyl spinors can be used to parametrize a Majorana spinor as much as the Majorana spinor can be used to parametrize a Weyl spinor, stating "*the use of a Hermitian field to describe a two-component spin–$\frac{1}{2}$ particle was made by E. Majorana*". The advantage of the Majorana spinor is that it is a representation of parity, while the Weyl spinor is not, thus Majorana spinors are relevant when studying parity violation.

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Note that the Poincare group is the quotient group of the group of symmetries by the normal subgroup of internal symmetries. The charge-conjugation is not a distinguished transformation in Quantum Field Theory, and depending on the Lagrangian it may not even make sense to define it (this is the case of the Standard Model, as we will see). Thus, CP is a candidate for a physical

parity transformation (included in the Poincare group) [297]. If charge conjugation is defined, it is part of the group of internal symmetries and there will be parity transformations which are charge conjugating and parity transformations which are not charge conjugating. If the group of internal symmetries is linearly reductive (as is usually the case), we can without loss of generality [298] use the composite operators which are invariant under the internal symmetries to determine whether or not parity is conserved.

However, reference [297] is ambiguous about the role of fields which are complex representations (where particles are not their anti-particles) at the quantum level and it does not address the Standard Model (which was still being developed at the time). In this paper, we address these questions very explicitly. In particular, it is clear nowadays that fields which are complex representations play no role at the quantum level (except as an intermediate step in defining the real representations).

The notion of "generalized" parity has been used since then, but only in part as "generalized" CP or "generalized" P without ever relating CP and P [299]. The reason is that in order to relate CP and P we need to define the fields as real representations, e.g. use Majorana spinors instead of Weyl spinors [225]. Note that classical fields can indeed be defined as complex representations, but these classical fields cannot be quantum fields. At the quantum level, all fields are real representations of the group of symmetries, because CP is a linear transformation. In the classical Action (used in the path integral quantization), if we would consider the fields as complex representations then CP would be necessarily anti-linear [300] instead of a linear transformation as it should be [301].

This happens even if the vacuum is treated as a complex representation in canonical quantization, since the fields are self-adjoint operators and not quantum states. More generally, a quantum system is defined by the real numbers which are the expectation values of self-adjoint operators representing observables; thus a physical transformation of the system transforms the self-adjoint operators into other self-adjoint operators and the linear space of self-adjoint operators is a real representation space (otherwise the condition of self-adjointness is not preserved). Of course, we can do linear combinations of the self-adjoint fields using complex coefficients, but the group action always preserves the condition of self-adjointness (so they are real representations, since a real structure is preserved [216]). The situation is analogous to the fact that in a complex vector space, the adjoint representation of a compact Lie group preserves a real structure and thus it is equivalent to a real representation.

This implies that a representation theory which is consistent at the quantum level for the Standard Model is more complicated than what appears in most introductory books, for instance: complex irreducible representations of the group $G \times H$ are a direct product of complex irreducible representations of G and of H respectively, while this does not happen for real irreducible representations. Thus, addressing the Standard Model with real representations is not trivial and (to the knowledge

of the author) it was not done before. References such as [297][298][302][303], where it is understood that real representations are the most suitable for general discussions are rare in the physics literature, while literature making sound claims [304] which are explicitly false (and do not even make sense at the quantum level) is abundant. Even books dedicated to group theory in physics [305] only discuss classical fields which are complex representations, because they are easier to discuss [36].

On the other hand, with complex representations we cannot define all linear operators available in Quantum Field Theory, some of these operators have important phenomenological consequences, for instance: the approximate custodial symmetry of the Higgs potential [29][306][302], pseudo-goldstone bosons [302], Majorana neutrinos [307], or the parity transformation [297]. Fortunately, there is a well known map from the complex to the real irreducible representations of any Lie group [37], so that using real representations requires just one correction with respect to the complex representations.

The fact that all particles are their own antiparticles (from a canonical quantization point of view), does not exclude the existence of conserved quantum numbers. Namely, global $U(1)$ symmetries corresponding to the lepton or baryon quantum numbers may or may not be conserved, depending on the Lagrangian [308] and the path-integral measure [309]. Note that in the Standard Model, the baryon number is anomalous (it is not conserved by the path-integral measure) which implies that the baryon asymmetry of the Universe is a non-perturbative problem and thus necessarily complicated [309] and beyond the scope of this paper.

In section 2, we review why the quantum fields are self-adjoint operators and thus real representations, despite that the vacuum may be an element of a complex Hilbert space. In section 3, we check explicitly what this implies for the Standard Model. In section https://timepiece.pubpub.org/pub/m0qt90d9/draft#majorana-spinors-in-canonical-quantization-and-antiparticles

[309]

4, we state some consequences for extensions to the Standard Model.

[36] Olive, K. A., & others. (2014). Review of Particle Physics. *Chin. Phys. C*, *38*, 090001. https://doi.org/10.1088/1674-1137/38/9/090001 Schrödinger, E. (1926). An Undulatory Theory of the Mechanics of Atoms and Molecules. *Phys. Rev.*, *28*, 1049–1070. https://doi.org/10.1103/PhysRev.28.1049

In the particular case of the book (Olive & others), the reason invoked to avoid real representations is that "Schur's lemma does not hold for a real representation." which is misleading, what it happens is that Schur's lemma is simpler for complex representations than for real representations. This is one of the many examples where Wikipedia is more trustworthy than the physics literature.

[37]

Majorana spinors in canonical quantization and antiparticles

Given a real Hilbert space V with inner product $<,>$ we can always construct an associated complex Clifford C* algebra [220]. Let $C(V)$ be the associated complex Clifford algebra, i.e. $C(V)$ is a unital associative complex algebra such that there is an injective linear map $a : V \to C(V)$, verifying $a^2(v) =< v, v > 1$ and $a^*(v) = a(v)$, $C(V)$ admits a unique involution $*$ and it is generated by the operators $a(v)$, for all $v \in V$. The algebra has a natural norm given by $||a(v_1)...a(v_n)|| \equiv \sqrt{||a(v_n)...a(v_1)a(v_1)...a(v_n)||} = ||v_1||...||v_n|| \cdot ||1||$. The C* algebra $C[V]$ is the completion $C(V)$ with respect to its natural norm.

Consider now a skew-symmetric operator J acting on the Hilbert space V, with $J^2 = -1$. The vacuum functional verifies $< O\ a(v + iJv) >= 0$ for any operator O and $v \in V$. The condition defining the vacuum involves complex numbers, despite that the Hilbert space V is real. The operator $a(v + iJv)$ is an annihilation operator, while $a(v - iJv)$ is a creation operator. The operators $a(v) = a(v + iJv) + a(v - iJv)$ are self-adjoint and represent a particle which is its own antiparticle, for any $v \in V$. Thus we see that it is always possible to do the canonical fermionic quantization of a real Hilbert space, with the corresponding operators $a(v)$ being self-adjoint.

For bosons, we have a similar situation, except that a commutation relation holds $[a(v), a(w)] =< v, Jw > i$ instead of $\{a(v), a(w)\} =< v, w > 1$; a sympletic product [310][311][312] $< v, Jw > i$ replaces the inner product $< v, w > 1$; and the space V is a real symplectic space instead of a real Hilbert space. The operators $a(v) = a(v + iJv) + a(v - iJv)$ are again self-adjoint and represent a particle which is its own antiparticle.

We stress once again that we can do linear combinations of the self-adjoint fields using complex coefficients, but the action of the group of symmetries always preserves the condition of self-adjointness (so they are real representations, since a real structure is preserved [216]), that is $a(v) \to a(Tv)$ for some real operator T acting on the real space V.

We can then relate the canonical quantization with the path integral quantization. In the path integral, the real fermionic fields (Majorana spinors) are Grassmann (anti-commuting) fields while the bosons are classical (commuting) fields.

Majorana spinors in the Standard Model

The group of gauge symmetries is the semi-direct product $SU(2)_L \times U(1)_Y \times SU(3)_C$. Consider[38] for the moment just the weak-Higgs sector: there are one weak $SU(2)_L$ Higgs doublet ϕ and the gauge field W_μ^j with $j, k, l = 1, 2, 3$. The Lagrangian is:

[38]

$$
\begin{aligned}
\mathcal{L}_{\rm W} &\equiv ((D^\mu\phi)^\dagger(D_\mu\phi) - V(\phi) - \frac{1}{4}W^j_{\mu\nu}W^{j\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu},\\
D_\mu\phi &\equiv \left(\partial_\mu + igW^j_\mu\frac{\tau_j}{2} + ig'\frac{\sigma_3}{2}B^\mu\right)\phi\\
W^j_{\mu\nu} &\equiv -\frac{i}{g}{\rm tr}([D_\mu, D_\nu]\tau^j) = \partial_\mu W^j_\nu - \partial_\nu W^j_\mu - g\epsilon^{jkl}W^k_\mu W^l_\nu,\\
B_{\mu\nu} &\equiv -\frac{i}{g'}{\rm tr}([D_\mu, D_\nu]\sigma_3) = \partial_\mu B_\nu - \partial_\nu B_\mu
\end{aligned}
\tag{131}
$$

where $V(\phi)$ is the Higgs Potential, D_μ is the covariant derivative dependent on the gauge fields W^j_μ and B_μ. The gauge field strength tensors are $W^j_{\mu\nu}$ and B_μ, while g and g' are coupling constants, ϵ^{jkl} is the Levi-Civita tensor. The Pauli matrices $i\tau_j$ are the generators of the $SU(2)_L$ gauge group and $i\sigma_j$ are the generators of the $SU(2)$ custodial group. The Higgs field verifies a Majorana condition $i\sigma_2\phi = i\tau_2\phi^*$ and thus it is a real representation of the symmetries.

Consider now a Majorana spinor field which is a fermionic (Grassmannian) field Q_L it has $3{\times}3{\times}2{\times}4 = 72$ real components, containing 3 flavours, it is a triplet of $SU(3)_C$ (with the imaginary unit replaced by $i\gamma_5$), it is an $SU(2)_L$ doublet and it verifies $i\gamma_5 Q_L = i\sigma_3 Q_L$, and the Majorana condition $i\sigma_2 Q_L = i\tau_2 Q_L^*$. Introduce also Majorana fermions d_R, u_R which have 36 real components each, containing 3 flavours, they are anti-triplets of $SU(3)_C$ (with the imaginary unit replaced by $i\gamma_5$) and invariant under $SU(2)_L$. We set the hyper-charges of the gauge symmetry $U(1)_Y$ as $Q_L(1/6_Y)$, $i\gamma_0 d_R(1/3_Y)$, $i\gamma^0 u_R(-2/3_Y)$, i.e. for $\phi \to e^{i\frac{\sigma_3}{2}\vartheta}\phi$ then $Q_L \to e^{i\gamma_5\frac{\vartheta}{6}}Q_L$ and $u_R(x) \to e^{-i\gamma_5\frac{2\vartheta}{3}}u_R(x)$. Hence, these are quarks.

The most general $SU(2)_L$ gauge invariant products of ϕ and Q_L are linear combinations of $\overline{Q_L}\phi$, $\overline{Q_L}i\sigma_2\phi$[39]. The most general gauge-invariant form for the Yukawa couplings with the quarks is then

$$
\begin{aligned}
-\frac{1}{2}\mathcal{L}_{\rm Y} &= Q_L^\dagger\gamma^0\ M_d\phi\ d_R + Q_L^\dagger\gamma^0\ i\sigma_2 M_u\phi\ u_R\\
M_w &\equiv M_{w\,r} - M_{w\,i}i\gamma_5
\end{aligned}
\tag{132}
$$

with M_{wr} and M_{wi} real 3×3 matrices, where $w = u, d$.

The matrices $M_d \equiv U_L{\rm diag}(m_d, m_s, m_b)U_R^{d\dagger}$ and $M_u \equiv U_L V^\dagger{\rm diag}(m_u, m_c, m_t)U_R^{u\dagger}$ are the quark mass matrices. The conventional CKM matrix (except with $-i\gamma_5$ replacing the imaginary unit) is given by V.

The Lagrangian for the quarks is then:

[39] the basis of matrices commuting with the generators of $SU(2)_L$ is $\{1, i\sigma_j\}$, with $j, = 1, 2, 3$, for a total of 4 matrices. Due to the projector in Q_L, we must divide the total by 2 which leaves us with 2 linearly independent products.

$$\mathcal{L}_{\mathrm{Q}} = iQ_L^\dagger \gamma^0 \gamma^\mu D_\mu \; Q_L + iu_R^\dagger \gamma^0 \gamma^\mu D_\mu \; u_R + id_R^\dagger \gamma^0 \gamma^\mu D_\mu \; d_R - \frac{1}{4} G^j_{\mu\nu} G^{j\mu\nu} \tag{133}$$

Where D_μ is the gauge-covariant derivative with gauge couplings corresponding to each fermion, and $G_{\mu\nu}$ is the $SU(3)_C$ gauge field strength tensor corresponding to the gauge field G^a_μ (gluons, with $a = 1, ..., 8$).

The lepton sector with three right handed neutrinos is analogous in the absence of Majorana masses. There is a global symmetry $U(1)_{nb} \times U(1)_{nl}$ related to the baryon and lepton (no Majorana masses) numbers.

The outer automorphism group of $SU(3)$ or $U(1)_Y$ is Z_2, while the outer automorphism group of $SU(2)_L$ is the trivial group. The parity symmetry is violated by the CKM matrix. Promoting the CKM matrix to a background field, the Lagrangian is invariant under the background symmetry given by the semi-direct product $SU(2)_L \times (SU(3)_C \times U(1)_Y) \rtimes Z_4$ and the Z_4 group is generated by the (generalized) parity reversal transformation $\phi(t, \vec{x}) \to i\sigma_2 \phi(t, -\vec{x})$, $Q_L(t, \vec{x}) \to -\sigma_2 \gamma_0 Q_L(t, -\vec{x})$, $u_R(t, \vec{x}) \to i\gamma_0 u_R(t, -\vec{x})$, $d_R(t, \vec{x}) \to i\gamma_0 d_R(t, -\vec{x})$. The gauge fields B_μ and G^a_μ transform under parity reversal according to $B_\mu(t, \vec{x}) \to -B_\mu(t, -\vec{x})$ and $G^a_\mu(t, \vec{x}) \to s^a G^a_\mu(t, -\vec{x})$ (with $s^1, s^3, s^4, s^6, s^8 = -1$ and $s^2, s^5, s^7 = 1$ in the usual basis for the Gell-Mann matrices).

Note that this is a natural parity transformation, since it acts on the Majorana spinors with $i\gamma_0$ and $\vec{x} \to -\vec{x}$. What it is uncommon is the fact that the generator of the gauge group $U(1)_Y$ is $i\gamma_5$, but this is not a transformation of the Lorentz group so it shouldn't affect our definition of parity. As for the Higgs scalar field, we have a generalized parity transformation of order-4 and the generator of the gauge group $U(1)_Y$ is also not a transformation of the Lorentz group.

Since the parity transformation is order-4 for u_R, then it is order-4 after Electroweak "symmetry breaking", since u_R is $SU(2)_L$-gauge-invariant. See reference [303] for more details on the Electroweak "symmetry breaking"[40].

Consequences for extensions to the Standard Model

https://timepiece.pubpub.org/pub/parity/draft#consequences-for-extensions-to-the-standard-model

Since CP and P can be both identified as parity transformations, then parity is broken only by the CKM matrix in the Standard Model. Thus, the popular Left-Right symmetric models [299][313] accomplish the same as any other model where CP is broken spontaneously and not explicitly, at least with respect to the parity violation. The Left-Right symmetric models may have many virtues,

[40]

perhaps they can be motivated using the definition of charge conjugation (which is not defined in the Standard Model), but clearly the theoretical motivation usually stated for these models is misleading with respect to parity violation. Such theoretical motivation should also take into account that spontaneous parity violation can be described as a particular case of explicit parity violation [303].

Also, the parity transformation is order-4 both on the Higgs field and in the fermion fields. When considering the $SU(2)_L$-gauge-invariant operators (or after electroweak "symmetry breaking"), the parity transformation will still be order-4 on the fermions, despite that it acts trivially on the Higgs boson. Thus the claim that models of the Higgs sector featuring an order-4 CP symmetry, broken to order-2 CP symmetry (in the Higgs sector) after electroweak "symmetry breaking" are unique with respect to its CP properties [314], since the Standard Model already has these features. That does not imply that these models do not have (other) many virtues.

This also contradicts a previous claim that depending on how the neutrinos acquire a mass, the double-cover of the Lorentz group could be $Pin(1,3)$ or $Pin(3,1)$ [221], but in fact it must be $Pin(3,1)$ since the parity is order-4 in the right-handed quarks (and thus in the right-handed neutrinos if they are introduced).

Since the vacuum is not CP or even CPT invariant due to the spinor frame, there is no reason to expect the universe to be composed in equal parts by matter and anti-matter.

Diffeomorphisms and gravity

Quantum gravity

https://math.stackexchange.com/questions/3197969/manifolds-that-admit-lorentzian-metrics

[315]

The first step towards Quantum Gravity, is to choose which version of (classical) General Relativity we want to quantize and the corresponding definition of space-time. For that purpose, it is useful to review the case of the Yang-Mills theory exposed in the previous chapters: in a non-abelian gauge-theory, the Gribov ambiguity forces us to consider a phase-space formed by fields defined on not only space but also time. This is related to the fact that in a fibre bundle (the mathematical formulation of a classical gauge theory) the time cannot be factored out from the total space because the topology of the total space is not a product of the base-space (time) and the fibre-space, despite that the total space is *locally* a product space. Thus, the Hamiltonian constraints are in fact a tool to define a probability measure for a manifold with a non-trivial topology (a principal fibre bundle for the gauge group) such that the only measurable functions are the gauge-invariant functions[272], because a phase-space of gauge fields defined *globally* on a 4-dimensional space-time (i.e. a fibre bundle with a trivial topology, when the base space is the Minkowski space-time) produces well-defined expectation functionals for the gauge-invariant operators acting on a fibre bundle with a non-trivial topology[272].

Since only gauge-invariant operators are allowed, we must distinguish between the concrete manifold appearing in the phase-space and the family of manifolds (obtained from the concrete manifold through different choices of transition maps between local charts) to which the expectation values correspond. Regarding General Relativity, the tetrad fields can be reconstructed from diffeomorphism- and local Lorentz-invariant operators[316], as in the reconstruction theorem of Yang-Mills theory[123]. The concrete space-time has a spinor structure, otherwise it would be necessary to define spinor fields which transform non-linearly under the diffeomorphism group [317]. Also, a noncompact (4 dimensional) space-time admits a spinor structure if and only if it has a global field of orthonormal

tetrads [318]; meaning that the concrete space-time is parallelizable[318] and so the Levi-Civita and the Weizenbock connections coexist[319] and General Relativity may be equivalently defined as Teleparallel Gravity[320]. Many well-known solutions of the Einstein equations do have a spinor structure[318]. However, if only diffeomorphism- and local Lorentz-invariant operators are allowed, then the expectation values would correspond to any space-time with a Lorentz metric as if we would have defined the tetrads only *locally*.

In fact, the space-time in the Hamiltonian formalism is time-orientable since it requires a global time-like vector field giving the direction of the time-evolution [321]; the space-time also requires a volume form (to define the Hamiltonian) which is an orientation [322]; the space-time is also space-orientable, because it is orientable and time-orientable [321]. Both the volume form and the global time-like vector field are invariant under local Lorentz transformations. Crucially, the diffeomorphism invariance of the Hamiltonian formalism remains intact.

Moreover, the Einstein-Hilbert action with the metric as a dynamical variable has second-order derivatives of the metric, which requires adding extra terms to the action in the presence of some boundary conditions [323].

We stress that a parallelizable space-time doesn't need to have a Cauchy surface because the derivatives corresponding to different tetrads do not need to commute, thus the parallelizable space-time doesn't need to be globally hyperbolic (i.e. topologically a product space of time and a 3-dimensional manifold)[324][325][318][326].

If we assume that there is also a global scalar field corresponding to time (which implies that the space-time has stable causality[321]), then we can define a family of 3-dimensional achronal (but not necessarily space-like) hypersurfaces corresponding to constant times[321]. However, the step from local existence of solutions to the Einstein equations to the existence of a maximal development is non-trivial, due to the diffeomorphism invariance of the equations [168][327]. It turns out that given initial data (that is, one 3-dimensional hypersurface corresponding to a specific time), there is a unique maximal globally hyperbolic development [168]. Despite that the maximal development is only a subset of the total space-time, it is nevertheless a space-time in itself which can be defined by an Hamiltonian formalism. This justifies the success of the predictions of numerical relativity based on the ADM formalism [328][329]) which assumes an orientable globally hyperbolic spacetime (that is, all of the 3-dimensional achronal hypersurfaces are space-like).

Thus, there is no guarantee "a priori" that in Quantum Gravity the space-time is globally hyperbolic as it is assumed in the quantum ADM formalism. Moreover, the classical ADM formalism is a weakly hyperbolic system of partial differential equations and thus it is divergent and not well-posed[327]. Thus, there is no classical Hamiltonian formalism which is well-posed. Note that the "Hamiltonian

formalism" proposed in reference[327] is admittedly only defined locally and thus it is not a solution to the Cauchy problem in General Relativity as a complete Hamiltonian formalism should be.

The second step towards Quantum Gravity is to choose the mathematical tools needed to define the theory. There is a widespread belief that the sequence of generalizations on the descriptions of space and time 1) Galilean invariance, 2) special relativity, 3) general relativity; which happened for deterministic theories should also happen for quantum theories [330] (the present author also shared this belief in the past [331]). Following this belief, the special role of the little group of rotations and the time evolution in our definition of a (special relativistic) Quantum Field Theory seems to be a step back in the road towards a general relativistic quantum theory. Thus, our definition would not be of much value since a rigorous definition of a Quantum Yang-Mills theory should mark *"a turning point in the mathematical understanding of quantum field theory, with a chance of opening new horizons for its applications"*, as stated in reference [31].

But in the Hamiltonian formalism (and so in any quantum theory), the diffeomorphisms are generated by constraints [290] while the Poincare transformations act in a non-trivial way in the algebra of operators. Hence, the diffeomorphisms cannot be a generalization of a non-trivial Poincare symmetry and so, *a priori* diffeomorphisms are not incompatible with the special role of the little group of rotations and the time evolution[290] . *A priori*, the formalism of gauge-invariant Quantum Field Theory may be enough to define quantum gravity.

The crucial fact that allows us to distinguish the Poincare symmetry from diffeomorphisms and gauge symmetry is the fact that the time-evolution generated by the Hamiltonian is *not* a diffeomorphism or a gauge symmetry. There are indeed diffeomorphisms which advance the time coordinate of the phase-space (as we would expect from the time-evolution), however they differ from the time-evolution in how they affect the wave-function as a whole. Thus, the Poincare symmetry is not a diffeomorphism or a gauge symmetry and it is not generated by constraints. The algebra of operators is a non-trivial representation of the Poincare group, while it is diffeomorphism invariant and gauge invariant. The generators of the (global) Poincare group are also diffeomorphism invariant and gauge invariant.

We define mathematically a general relativistic Quantum Field Theory, verifying all the properties we expect from Quantum Gravity, without adding complexity with respect to the mathematical definition of gauge-invariant Quantum Field Theory presented in the previous sections.

Unlike in Yang-Mills theory, the Hamiltonian of the Einstein-Cartan theory involves canonical commutation relations of analytic functions of operators, which is no challenge by itself when there is a non-polynomial function of only one of the operators involved [171]. With the Weyl ordering of products of operators and the measure defined in the previous section, there is no challenge or

ambiguity if the kinetic part of the Hamiltonian is polynomial both in the fields and in its canonical conjugates while the potential is rational in the fields only (with no dependence on the canonical conjugates), just like in the Hydrogen atom in non-relativistic Quantum Mechanics. Note that since the Hamiltonian is defined in a 4-dimensional space-time, the diffeomorphisms (and all other constraints) are polynomial in the fields and its canonical conjugates, unlike in Loop Quantum Gravity [332][333].

Moreover, no vaccuum state needs to be defined (the Hamiltonian needs not to be bounded from below), thus there is no need for reguralization and so renormalization plays no essential role (at the non-perturbative level).

Moreover, the operator ordering ambiguity in Loop Quantum Gravity may limit it to the same peculiar notion of "continuum" through renormalization as in (Wilson's) Quantum Field Theory, as suggested in reference [333]:

> *"[...] the ultimate goal is to use Hamiltonian renormalisation to find a continuum theory for canonical Quantum Gravity. Here we can use the LQG [Loop Quantum Gravity] candidate as a starting point because it is rather far developed, but of course the flow scheme developed can be applied to any other canonical programme".*

We propose instead that Quantum Gravity at all stages of the quantization programme, should be seen as a particular case of a (special relativistic) Quantum Field Theory defined in the continuum, which is a major simplification with respect to Loop Quantum Gravity.

Therefore, in the quantization due to time-evolution, there is no obvious advantage of trying to rewrite the Hamiltonian as a polynomial in the canonical operators, in contrast with Loop Quantum Gravity. We can still do it after the definition of the quantum theory[163], for the purpose of numerical computations in perturbation theory for instance.

We will start by defining the Hamiltonian for Classical General Relativity within the formalism of Quantization due to the time-evolution. Just for the purpose of solving initial-value problems within the classical theory, it is already advantageous since it allows us to define an Hamiltonian which is diffeomorphism invariant (unlike in the ADM formalism[328][329]), avoiding the technical complexity of the definition of chains of constraints[327]. Moreover, the low-energy limit of Quantum Gravity must anyway be studied and it should match the classical theory.

Classical Hamiltonian

The classical Action for the Einstein–Cartan theory of gravitation is:

$$
\begin{aligned}
S =& \int d^4x \mathcal{L} \\
\mathcal{L} =& eR \\
e =& \det e^a_\mu \\
R =& g^{\mu\nu} R_{\mu\nu} \\
R_{\mu\nu} =& R^\alpha_{\mu\alpha\nu} \\
R^\alpha_{\beta\mu\nu} =& e^\alpha_a e^b_\beta \partial_\mu \omega^a_{\nu b} - e^\alpha_a e^b_\beta \partial_\nu \omega^a_{\mu b} + e^\alpha_a e^b_\beta \omega^a_{\mu c} \omega^c_{\nu b} - e^\alpha_a e^b_\beta \omega^a_{\nu c} \omega^c_{\mu b} \\
\omega^a_{\mu b} =& 1/2(\eta^{ac} e^\nu_c \eta_{bk} \delta^\alpha_\mu \delta^\beta_\nu - e^\nu_b \delta^\alpha_\mu \delta^\beta_\nu \delta^a_k - \eta^{ac} e^\alpha_c e^j_\mu \eta_{jk} e^\beta_b)(\partial_\alpha e^k_\beta - \partial_\beta e^k_\alpha) \\
g^{\mu\nu} =& e^\mu_a e^\nu_b \eta^{ab}
\end{aligned}
\tag{134}
$$

Where $a, b, c, j, k \in \{0, 1, 2, 3\}$.

Then we integrate by parts in the variable $\partial_\nu e^a_\alpha$ and replace:

$$
\begin{aligned}
\partial_\alpha e =& - e e^a_\mu \partial_\alpha e^\mu_a \\
\partial_\alpha e^a_\mu =& - e^a_\nu \partial_\alpha e^\nu_b e^b_\mu
\end{aligned}
\tag{135}
$$

We also define:

$$
\begin{aligned}
\chi_a{}^b =& \delta_a{}^b + v_a v^b \\
e_{abc} =& \partial_\alpha e^\beta{}_b e^\alpha{}_a e_\beta{}^k \eta_{kc} \\
E_{ab} =& \chi_a{}^{a_1} v^c \chi_b{}^{a_2} e_{a_1 cb} \\
E_a =& \chi_a{}^{a_1} v^c v^b e_{a_1 cb} \\
T_{abc} =& e_{abc} - e_{bac} \\
T_{ab} =& v^c \chi_a{}^{a_1} T_{ca_1 b} \\
A_{ab} =& \chi_a{}^{a_1} \chi_b{}^{a_2} (T_{a_1 a_2} - T_{a_2 a_1}) \\
T =& \chi_a{}^{a_1} \chi_b{}^{a_2} T_{a_1 a_2} \eta^{ab} \\
S_{ab} =& \chi_a{}^{a_1} \chi_b{}^{a_2} (T_{a_1 a_2} + T_{a_2 a_1} - \frac{2}{3} \eta_{a_1 a_2} T) \\
T_a =& v^c \chi_a{}^{a_2} v^b T_{ca_2 b} \\
\mathcal{T}_{abc} =& \chi_a{}^{a_1} \chi_b{}^{a_2} \chi_c{}^{a_3} T_{a_1 a_2 a_3} \\
\mathcal{T}_{ab} =& \chi_a{}^{a_1} \chi_b{}^{a_2} v^c T_{a_1 a_2 c}
\end{aligned}
\tag{136}
$$

Where $v^a = v^\mu e^a_\mu$ are the tetrad components of the globally defined time-like vector verifying $v^\mu v_\mu = -1$, $a, b, c, a_1, a_2, a_3 \in \{0, 1, 2, 3\}$ and $T_{cba} = -T_{cab}$.

Up to a divergence term, the Lagrangian density is given by:

$$\begin{aligned}
&\mathcal{L} \approx e\left(T_{ab}{}^{b}T^{ac}{}_{c} - \frac{1}{2}T_{abc}T^{acb} - \frac{1}{4}T_{abc}T^{abc}\right) = \\
&= \frac{1}{4}S_{ab}S^{ab}e - \frac{2}{3}(T)^2 e - \frac{1}{2}A_{ab}\mathcal{T}^{ab}e - 2\mathcal{T}^{ab}{}_{a}T_{b}e + \\
&- \frac{1}{2}\mathcal{T}_{abc}\mathcal{T}^{acb}e - \frac{1}{4}\mathcal{T}_{abc}T^{abc}e + \mathcal{T}_{ba}{}^{a}\mathcal{T}^{bc}{}_{c}e + \frac{1}{4}\mathcal{T}_{ab}\mathcal{T}^{ab}e
\end{aligned} \tag{137}$$

This is in agreement with reference [320], up to a global sign of the Lagrangian density (which does not affect the classical equations of motion). Then we vary the resulting Lagrangian density in the variables $\partial_\alpha e^\beta_a$, to obtain the polymomentum [334]:

$$\begin{aligned}
&p^{ab} = \frac{\delta\mathcal{L}}{\delta\partial_\alpha e^\beta_a}v_\alpha e^\beta_c \eta^{cb} = \\
&= (\frac{\delta\mathcal{L}}{\delta S_{a_3a_4}}\frac{\delta S_{a_3a_4}}{\delta T_{a_1a_2}} + \frac{\delta\mathcal{L}}{\delta T}\frac{\delta T}{\delta T_{a_1a_2}} + \frac{\delta\mathcal{L}}{\delta A_{a_3a_4}}\frac{\delta A_{a_3a_4}}{\delta T_{a_1a_2}} + \frac{\delta\mathcal{L}}{\delta T_{a_3}}\frac{\delta T_{a_3}}{\delta T_{a_1a_2}})\frac{\delta T_{a_1a_2}}{\delta\partial_\alpha e^\beta_a}v_\alpha e^\beta_c\eta^{cb} \\
&= e\Big(\frac{1}{2}S_{a_3a_4}(\eta^{a_3a_1}\eta^{a_4a_2} + \eta^{a_4a_1}\eta^{a_3a_2} - \frac{2}{3}\eta^{a_3a_4}\eta^{a_1a_2}) - \frac{4}{3}T\eta^{a_1a_2} + \\
&- \frac{1}{2}\mathcal{T}_{a_3a_4}(\eta^{a_3a_1}\eta^{a_4a_2} - \eta^{a_4a_1}\eta^{a_3a_2}) + 2\mathcal{T}_{a_3a_4}{}^{a_4}\eta^{a_3a_1}v^{a_2}\Big)\delta_{a_1}{}^{a}\delta_{a_2}{}^{b} \\
&= e\Big(S^{ab} - \frac{4}{3}T\eta^{ab} - \mathcal{T}^{ab} + 2\mathcal{T}^{ac}{}_{c}v^b\Big) \\
&\mathcal{A}^{ab} = \chi^{a}{}_{a_1}\chi^{b}{}_{a_2}(p^{a_1a_2} - p^{a_2a_1}) = -2e\mathcal{T}^{ab} \\
&\mathcal{P} = \eta_{ab}\chi^{a}{}_{a_1}\chi^{b}{}_{a_2}p^{a_1a_2} = -4eT \\
&\mathcal{S}^{ab} = \chi^{a}{}_{a_1}\chi^{b}{}_{a_2}(p^{a_1a_2} + p^{a_2a_1} - \frac{2}{3}\eta^{a_1a_2}\mathcal{P}) = 2eS^{ab} \\
&p^a = v_b p^{ab} = 2e\mathcal{T}^{ac}{}_{c}
\end{aligned} \tag{138}$$

The Hamiltonian density in 3-dimensional space is:

$$\begin{aligned}
&\mathcal{H} = v_\alpha\frac{\delta\mathcal{L}}{\delta\partial_\alpha e^\beta_a}v^\mu\partial_\mu e^\beta_a - \mathcal{L} = p^{ab}T_{ab} + p^{ab}E_{ab} - \mathcal{L} = -\mathcal{L} + \frac{1}{4}\mathcal{A}^{ab}A_{ab} + \frac{1}{4}\mathcal{S}^{ab}S_{ab} + \\
&+ \frac{1}{3}\mathcal{P}T + p^aT_a + \frac{1}{2}\mathcal{A}^{ab}E_{ab} + \frac{1}{2}\mathcal{S}^{ab}E_{ab} + \frac{1}{3}\mathcal{P}E_a{}^{a} + p^aE_a \\
&\approx e\Big(\frac{1}{4}S^{ab}S_{ab} - \frac{2}{3}(T)^2 + S^{ab}E_{ab} - \frac{4}{3}TE_a{}^{a} - \mathcal{T}^{ab}E_{ab} + 2\mathcal{T}^{ab}{}_{b}E_a + \\
&+ \frac{1}{2}\mathcal{T}_{abc}\mathcal{T}^{acb} + \frac{1}{4}\mathcal{T}_{abc}T^{abc} - \mathcal{T}_{ba}{}^{a}\mathcal{T}^{bc}{}_{c} - \frac{1}{4}\mathcal{T}_{ab}\mathcal{T}^{ab}\Big)
\end{aligned} \tag{139}$$

As expected, the Hamiltonian density in 3-dimensional space only depends on the momenta $\mathcal{S}^{mn}$ and $\mathcal{P}$ through S^{mn} and T_0:

$$\mathcal{H} = \frac{1}{16e}\mathcal{S}^{ab}\mathcal{S}_{ab} - \frac{1}{24e}(\mathcal{P})^2 + \frac{1}{2}\mathcal{S}^{ab}E_{ab} + \frac{1}{3}\mathcal{P}E_a{}^a - e\Big(\mathcal{T}^{ab}E_{ab} + 2\mathcal{T}^{ab}{}_b E_a + \\ + \frac{1}{2}\mathcal{T}_{abc}\mathcal{T}^{acb} + \frac{1}{4}\mathcal{T}_{abc}T^{abc} - \mathcal{T}_{ba}{}^a\mathcal{T}^{bc}_c - \frac{1}{4}\mathcal{T}_{ab}\mathcal{T}^{ab}\Big) \tag{140}$$

The Hamiltonian density in 4-dimensional space is given by:

$$\mathcal{H}_4 = -p^{ab}T_{ab} - p^{ab}E_{ab} + \mathcal{H} = \\ = \frac{1}{4}\mathcal{A}^{ab}A_{ab} + \frac{1}{4}\mathcal{S}^{ab}S_{ab} + \frac{1}{3}\mathcal{P}T + p^a T_a + \\ - \frac{1}{16e}\mathcal{S}^{ab}\mathcal{S}_{ab} + \frac{1}{24e}(\mathcal{P})^2 - \frac{1}{2}\mathcal{T}^{acb}\mathcal{T}_{abc}e - \mathcal{T}^{ac}{}_a\mathcal{T}_{cb}{}^b e + \\ + \frac{1}{4}\mathcal{T}^{ab}\mathcal{T}_{ab}e - \frac{1}{4}\mathcal{T}^{abc}\mathcal{T}_{abc}e \tag{141}$$

The constraints are the diffeomorphisms, local Lorentz transformations and global translations:

$$G_\mu = p^\nu_a\partial_\mu e^a_\nu - \partial_\nu(p^\nu_a e^a_\mu) + p^\nu_v\partial_\mu v_\nu - \partial_\nu(p^\nu_v v_\mu) \\ J_{ab} = p^\mu_a e^c_\mu\eta_{cb} - p^\mu_b e^c_\mu\eta_{ca} \\ P_\mu = \int d^4x\Big(p^\nu_a\partial_\mu e^a_\nu + p^\nu_v\partial_\mu v_\nu\Big) \tag{142}$$

Note that the diffeomorphisms and global translations are consistent with each other, but they are two independent constraints. The BRST charge is:

$$G = \int d^4x\Big(p^a_\nu c^\mu\partial_\mu e^\nu_a - p^a_\nu e^\mu_a\partial_\mu c^\nu + \pi_\nu c^\mu\partial_\mu v^\nu - \pi_\nu v^\mu\partial_\mu c^\nu + i\partial_\beta c^\alpha c^\beta b_\alpha\Big) \tag{143}$$

When the globally-defined time-like vector is constant and given by $v^\mu = \delta^\mu_0$, the 4-dimensional Hamiltonian formalism can be reduced to a 3-dimensional Hamiltonian formalism. Such formalism is different from the ADM formalism since the constraints are different, which is in fact a good property. The standard ADM evolution system is only weakly hyperbolic and so it is not well-posed[335]. Well-posed versions of the Einstein equations have in fact been known since the 1950s, but they were not based on a 3+1 decomposition and they are not suitable for numerical calculations[335]. Thus a strongly hyperbolic reformulation of the 3+1 evolution equations is required anyway[335].

The diffeomorphisms must now conserve the condition $v^\mu = \delta^\mu_0$, so the ghosts appearing in the BRST charge for the 3-dimensional Hamiltonian formalism are constant in the timepiece, with the BRST charge having the same functional form as in the 4-dimensional formalism. Such a BRST charge is significantly different from the BRST charge of the ADM formalism.

Quantum Hamiltonian

The Hilbert space (in the particular case of an empty spacetime) is the tensor product of the symmetric and antisymmetric Fock spaces $\Gamma^s(L^2(\mathbb{R}^{84} \times \mathbb{Z}_2^{19})) \otimes \Gamma^a(L^2(\mathbb{R}^{84} \times \mathbb{Z}_2^{19}))$ [104](*Fock Space*, n.d.). This gives us a graded Lie superalgebra of creation and annihilation operators, of both bosonic and fermionic types. The total fermionic number is defined as the sum of all number operators corresponding to fermionic degrees of freedom. The fermionic creation and annihilation operators correspond to a field configuration where the total fermionic number is odd, while the bosonic creation and annihilation operators correspond to a field configuration where the total fermionic number is even.

The $\mathbb{R}^{84}$ degrees of freedom correspond to 4 space coordinates (x), 4x4=16 fields (e_μ^a) and its corresponding space-derivatives ($\partial_\mu e_\nu^a$). The $\mathbb{Z}_2^{19}$ degrees of freedom correspond to 4 ghosts corresponding to each generator of diffeomorphisms (ψ_μ) and its corresponding spacetime-derivatives ($\partial_\mu \psi_\nu$), minus one because we are taking the tensor product of two Fock-spaces (bosonic and fermionic) which introduces an extra $\mathbb{Z}_2$ degree of freedom.

We define:

$$
\begin{aligned}
&a(x,...) = a(x, \oplus_a(\alpha_a, (\oplus_\mu A_{\mu,a}), \oplus_\nu(\alpha_{a\nu}, (\oplus_\mu A_{\mu a,\nu}))))\\
&\int d^4x... = \int d^4x \prod_a((\prod_\mu dA_{\mu a}) \sum_{\alpha_a=0}^{1} \prod_\nu((\prod_\mu dA_{\mu a,\nu}) \sum_{\alpha_{a\nu}=0}^{1}))\\
&\{[a(x,...,\mu), a^\dagger(y,...,\nu)]\} = a(x,...,\alpha)a^\dagger(y,...,\beta) + (-1)^{(\alpha \mathrm{mod} 2)(\beta \mathrm{mod} 2)} a^\dagger(y,...,\alpha)a(x,...,\beta) =\\
&= \delta((x,...) - (y,...))\\
&[A_{\mu a}, \pi_b^\nu] = A_{\mu a}\pi_b^\nu - \pi_b^\nu A_{\mu a} = i\delta_\mu^\nu \delta_{ab}\\
&[A_{\nu a,\mu}, \pi_b^{\alpha\beta}] = A_{\nu a,\mu}\pi_b^{\alpha\beta} - \pi_b^{\alpha\beta} A_{\nu a,\mu} = i\delta_\mu^\alpha \delta_\nu^\beta \delta_{ab}\\
&\{\psi_a, \psi_b^\dagger\} = \psi_a\psi_b^\dagger(y) + \psi_b^\dagger\psi_a = \delta_{ab}\\
&\{\psi_{a\mu}, \psi_{b\nu}^\dagger\} = \psi_{a\mu}\psi_{b\nu}^\dagger + \psi_{b\nu}^\dagger\psi_{a\mu} = \delta_{ab}\delta_{\mu\nu}
\end{aligned}
\tag{144}
$$

where

$$
\begin{aligned}
&\psi^\dagger\{a\}(x,...,j) = a(x,...,1)\delta_{j0}\\
&\psi\{a\}(x,...,j) = a(x,...,0)\delta_{j1}
\end{aligned}
\tag{145}
$$

The Hamiltonian for the Yang-Mills theory in our formalism is related to the classical Hamiltonian Action[290]:

$$
\begin{aligned}
H &= \int d^4x ... a^\dagger(x,...) H(x,...) a(x,...) \\
H(x,...) &= i\partial_0 - \pi_a^i A_{ia,0} + \pi_a^i (D_i A_{0a}) - \frac{1}{2}\pi_a^i \pi_a^i - \frac{1}{2} B_{ia} B_{ia} \\
\Omega &= \int d^4x ... a^\dagger(x,...) \Omega(x,...) a(x,...) \\
\Omega(x,..) &= \left[\pi_a^\mu \partial_\mu \psi_a^\dagger - \pi_a^\mu f_{abc} A_{\mu b} \psi_c^\dagger - i\frac{1}{2} f_{abc} \psi_a^\dagger \psi_b^\dagger \psi_c \right]
\end{aligned}
\tag{146}
$$

Selecting events is not rewriting the history of events (in a continuous probability space)

Abstract

Using the simple fact described in the title, we prove the existence of a computational problem with implications to Machine Learning, Quantum Mechanics and Complexity Theory. We also prove P!=NP (the solution can be verified in time polynomial in the number of bits of the input and output (NP) but the problem cannot be solved in time polynomial in the number of bits of the input and output (P)), but this claim still needs to be reviewed by experts in Complexity Theory.

1. Introduction

In this article we will be using the words "real" as in $\mathbb{R}$, "real-world" and "random" to avoid misunderstandings in contexts where we could (and perhaps should) just use the word "real" instead of "real-world" and the word "non-deterministic" (in the Physics sense, not in the Complexity Theory sense) instead of "random".

We can always select events with some feature without rewriting the history of events, in a standard probability space. In fact, in a continuous probability space we can select events such that a random real variable $y \in [0, 1]$ (with probability given by the Lebesgue measure) verifies $y = 0$, but there is no complete history of events where $y = 0$ (always as would be required, or even just once) because the probability space is continuous by assumption and thus the event $y = 0$ has null probability (only an interval would have non-null probability).

In this article we will show that this simple but non-trivial fact has profound implications not only to Complexity Theory[336][76][337][338] but also to Machine Learning[76] and Quantum Mechanics

(independently of the implications to the P vs. NP problem). Almost all (in the sense we will define in this article) real functions of a real variable cannot be computed for all practical purposes, not even approximately. But a random selection allows computations in polynomial-time complexity involving the incomplete knowledge about a real function that cannot be computed in polynomial-time complexity. This is a fundamental reason why we cannot exclude a random time-evolution: a deterministic time-evolution may exist, but it has so much complexity that it cannot be calculated for all practical purposes, not even approximately (since L^∞ is non-separable).

Note that whenever we deal with a non-separable space, there are issues with computability[1] because some elements of a non-separable space cannot be approximated by a finite set of elements, up to an arbitrarily small error. For instance, $L^\infty([0,1])$ and its dual space are both non-separable[2]. While there are separable spaces of real functions of real variables, whenever we add uncertainties/probabilities to such spaces (which is often required when doing approximations) we tend to create non-separable spaces[2], unless equivalence relations change the space of functions. For instance, the set $[0,1]$ is separable but the Lebesgue measure imposes that the rational numbers in $[0,1]$ can be discarded, despite that the sets $[0,1]$ including /excluding rational numbers are different. In the same way, the smooth functions (or functions computable in a reasonable time) may be discarded from a space of functions for particular uncertainties/probabilities.

Using the simple fact mentioned above, we will also prove the existence of a computational problem (defined by a continuous probability space) whose solution can be verified in time polynomial in the numbers of bits of the input and output (NP) but cannot be solved in time polynomial in the numbers of bits of the input and output (P), when using only a deterministic Turing machine[339][340]. That is, P$\neq$NP.

The goal of this article is to define the specific problem unambiguously, and the level of mathematical details will be adjusted to that: too much mathematical detail would shift the focus from the specific problem. Less detail does not always imply less mathematical rigor. Since the present author is an expert in Physics but not an expert in Complexity Theory, we will also try to prove the P vs. NP as much as it is possible, but only as a secondary goal knowing that much work by experts in complexity theory is still required because it is likely that: 1) something went wrong in the relation described here between the specific problem and the P vs. NP problem; and/or 2) the specific problem indeed can be used to solve the P vs. NP problem, but the proof presented here is incomplete.

2. Complexity Theory in the context of probability theory

Complexity Theory can be studied in the context of probability theory[336][76][337][338], because many real-world problems require approximations and uncertainties not only due to the limitations of any computer (already accounted for by Complexity Theory when using only finite numbers of bits, although there is also room for improvement here) but also due to the limitations of the measuring devices of physical phenomena and limitations of the mathematical models used to approximate the real-world, for instance when dealing with real variables. Most uncertainties are not related to the computer used and do not get smaller when increasing the number of bits in the computer. In fact, Physics as a science tries to be independent of Computer Science and vice-versa, as much as possible. Probability theory is a language (or interface) that allows us to transfer a problem between two sciences (these two or others).

There are two possible approaches to errors or uncertainties[336][76][337][338]: average error (with respect to a probability measure) and maximal (except in sets of null measure with respect to a probability measure) error. It turns out that both approaches can be defined using Hilbert spaces: the average error (that is, L^2 norm) is defined by a normalized wave-function (an element of the Hilbert space) being the square-root of a probability density function; and the maximal error (that is, L^∞ norm) is defined by an element of the abelian von Neumann algebra of operators on the Hilbert space.

The average error is relevant because even if P!=NP it could still make no difference with respect to the scenario P=NP for many practical purposes, if every NP problem with a reasonable probability distribution on its inputs could be solved in polynomial time-complexity on average on a deterministic Turing machine[341][342].

Moreover, since some functions are real constant functions, the definition of a real function must be consistent with the definition of a real number. Certainly, a natural definition of a real number in the context of Complexity Theory uses a standard probability space. Non-standard probability spaces are rarely (or never) used in Experimental Physics, so it is not obvious how useful a "real number" (or "real function") defined in a non-standard probability space could be in real-world applications. There are only countable or continuous measures (or mixed) in a standard probability space, then we can define exactly only a countable number of real numbers (usually the rationals, but not necessarily), the remaining real numbers can only be constrained to be inside an interval with a finite width, eventually very small but never zero. We should also define all real functions using a standard probability space, unless we find a fundamental reason not to do it (we will not find it in this article).

We require a continuum standard probability space, since we can always define a regular conditional

probability density which implements a selection of events in such probability space[90]. For instance, this is what we do when we neglect the intrinsic computation error (due to cosmic rays and many other reasons), we define a deterministic function by selecting only certain events from a complete history of random events. It is well known since many decades that in an infinite-dimensional sphere of radius 1 (subset of a real Hilbert space) there is a uniform prior measure induced by the L^2-distance in the Hilbert space[9]. Every point in the sphere has null measure, only regions of the sphere with non-null distance between some of its points may have non-null measure (compatible with the uniform prior measure). This implies that any knowledge (compatible with the uniform prior measure on the sphere) about a real normalized wave-function has necessarily uncertainties, defined by a connected region in the sphere with non-null maximum L^2 distance to some wave-function, however small it might be.

The discrete nature of the Turing machine is certainly compatible with a continuous probability space: the number of bits of the input or output can be arbitrarily large, and it is proportional to the logarithm of the resolution of the partition of the interval $[0, 1]$, with each disjoint set of the partition corresponding to a different binary number. Excluding a continuous cumulative prior would be unjustified, for many reasons including: no prior is better for all cases[16], there are many problems where a step cumulative prior would not fit well (for instance there is no uniform measure for the rationals in $[0, 1]$ only for the reals); it is hard to formulate any real-world problem where only a step cumulative prior is used (think about the numbers π or $\sqrt{2}$ in numerical approximations, for instance), we usually use a mixture of step and continuous cumulative priors; we can map an ensemble of discrete random variables one-to-one to the real numbers, for instance an ensemble of fair coins corresponds to the uniform real measure in the interval $[0, 1]$; also any real-world computer has an intrinsic computation error (due to cosmic rays and many other reasons) which is usually very small, but it cannot be eliminated. Thus, while we can formulate a new unsolved version of the P vs. NP problem where only a step cumulative prior is accepted, such version of the problem has little to do with real-world computers and real-world problems.

Note that there are 2^{2^n} different boolean[41] functions on n boolean variables, Shannon proved that almost all Boolean functions on n variables require circuits of size $\mathcal{O}(2^n/n)$ [343][42], thus the time complexity of almost all Boolean functions on n variables for a Turing machine is at least (and at most, for all Boolean functions) $\mathcal{O}(2^n/(n\log(n)))$[344] which is not polynomial in n. Thus, almost all numerical functions are not in the complexity class P, according to the uniform prior measure for any resolution of the partition. Moreover, the uniform prior measure is compatible with a prior measure which excludes (both in the maximal and in the average error approaches) all real functions

[41] That is, a function with $m = 1$m=1 boolean outputs

[42] See also: https://math.stackexchange.com/questions/756813/do-there-exist-polynomials-not-computable-in-polynomial-time

which are approximated by numerical functions with complexity class P, for all resolutions bigger than some resolution of the partition.

3. Definition of the problem

Consider three measure spaces $X = Y = [0,1] \in \mathbb{R}$ and $X \otimes X$ with the Lebesgue measure, corresponding to inputs (X or $X \otimes X$) and an output (Y) of real functions. Given an input in X, we define a regular conditional probability density which is a function from $X \otimes X \to Y$, given by the probability that a function for an input in X some constant output $y \in Y$. But there is always also a marginal probability density for Y, and we cannot say without uncertainty which is the output, because the corresponding prior probability density would be incompatible with the prior Lebesgue measure (by the Radon–Nikodym theorem). Thus, the regular conditional probability density is a deterministic selection of events which cannot be a complete history of events.

In a standard measure space it is always possible to define regular conditional probabilities[90] and to choose the probability density $p(x) = p_0(x) > 0$ for all $x \in X$, except in sets with null measure. Thus, we will define $p(x \otimes y) = p(y|x)p_0(x)$ the joint probability density for the tensor product $X \otimes Y$ for a particular $p(x) = p_0(x) > 0$ for all $x \in X$, except in sets with null measure. Then, we can obtain any other joint probability density $p(x \otimes y)$ from the expression $p(y|x)p(x) = \frac{p(x \otimes y)}{p_0(x)} p(x)$.

The following results are valid for a random input in the interval $[0,1]$ (which is a standard probability space) and also for an input (or output) without uncertainties up to sets with null measure with respect to the prior marginal measure of the input (or output), because we use regular conditional probabilities (which always exist in standard probability spaces[90]) for fully known inputs (or outputs). This is crucial, since the input includes two samples from a uniform distribution in $[0,1]$ which may generate numerical functions in P when the sample is in a set of null measure.

However, a (continuous cumulative) probability distribution does not contain enough information to unambiguously define a function. On the other hand, a real wave-function whose square is the joint probability distribution allows the definition of a unitary operator on a separable Hilbert space. A unitary operator is a random generalization of a deterministic symmetry transformation of a (countable or continuous) sample space. Any unitary operator defined by a wave-function of two continuous variables cannot be a deterministic symmetry transformation (for similar reasons that a continuous probability distribution cannot unambiguously define a function).

Since $p(x \otimes y) \geq 0$ then there is always a normalized wave-function $\Psi \in L^2(X \otimes Y)$ such that $|\Psi(x \otimes y)|^2 = p(x \otimes y)$. Note that the Koopman-von Neumann version of classical statistical mechanics[42] defines classical statistical mechanics as a particular case of quantum mechanics where the algebra of observable operators is necessarily commutative (because the time-evolution is

deterministic). In an infinite-dimensional sphere of radius 1 (subset of a real Hilbert space) there is a uniform prior measure induced by the L^2-distance in the Hilbert space. We choose a prior measure (compatible with the uniform prior measure) which excludes all real functions which are approximated by numerical functions with complexity class P, for a high enough resolution of the partition.

Given an input in $([0,1])^2$ (the input consists of two samples from a uniform distribution in the interval $[0,1]$, imported from ANU QRNG [345][43] for instance) and a candidate output in $[0,1]$ the wave-function uniquely defines a random symmetry transformation. Such symmetry transformation is not lacking information since it can be inverted (the non-unitary isometries have null prior measure). The cumulative probability distribution is given by the integral of the modulus squared of the wave-function in the corresponding region of the sample space.

From the cumulative marginal probability distribution, such that Y is fully integrated we determine $x \in X$ using the first sample from the uniform distribution in $[0,1]$. From the cumulative conditional probability distribution with the condition that $x \in X$ is what we determined previously, we determine $y \in X$ using the second sample from the uniform distribution in $[0,1]$. We apply the inverse-transform sampling method[141], that is check in the interval of the partition defined by the bits corresponding to X and Y, whether the cumulative distribution crosses the sample from the uniform distribution. This defines the deterministic verification of the candidate output corresponding to the input, in agreement with the Born's rule of Quantum Mechanics.

Note that the resulting deterministic function is not necessarily invertible, because the collapse of the wave-function is irreversible (unless the symmetry transformation would be deterministic, which is excluded in this case because we have a continuous probability space of functions).

4. The classical Turing machine defined as a Quantum computer

The Turing machine can be equivalently defined as the set of general recursive functions, which are partial functions from non-negative integers to non-negative integers[346]. But the set of all functions from non-negative integers to non-negative integers is not suitable to define a measure, since they form an uncountable set, in a context where the continuum is not defined. Moreover, the general recursive functions are based in the notion of computability (that the Turing machine halts in a finite time), but computability does not hold in the limit of an infinite number of input bits, thus to study such limit we need to define uncomputable functions somehow (we will use complete spaces, where Cauchy sequences always converge to an element inside the space).

On the other hand, it is widely believed (and we will show in the following) that any computational

[43] See also: https://qrng.anu.edu.au/random-binary

problem that can be solved by a classical computer can also be solved by a quantum computer and vice-versa. That is, quantum computers obey the Church–Turing thesis. Note that it is well known that some circuits (classical hardware) provide exponential speedups when compared with some other circuits in some functions (because the input bits can be reparametrized, this is why the time complexity of a function has an upper bound, but it is not known how to establish a lower bound; it is also consistent with the fact that the halting problem is undecidable, that is, given an arbitrary function from integers to integers and an arbitrary input, we cannot determine if the output of such function is computable or not), thus the fact that a classical Turing machine can be defined as a Quantum computer is compatible with the fact that quantum computers provide exponential speedups when compared with some classical computers in some functions.

We start by noticing that the domain of a general recursive function can be defined by a dense countable basis of a particular Hilbert space which is the (Guichardet) L^2 completion of the set of all finite linear combinations of simple tensor products of elements of a countable basis of a base Hilbert space, where all but finitely many of the factors equal the vacuum state[347] (like in a Fock-space, but without the symmetrization). But the unitary linear transformations on a normalized wave-function are not necessarily the most general transformations of the probability measure corresponding to the wave-function. Because of that, we build a Fock-space where the base Hilbert space is the previous Guichardet-space, then the unitary transformations on this Fock-Guichardet-space allow us to implement the most general transformations of a probability measure, corresponding to a normalized wave-function in the base Guichardet-space. Note that a countable basis of the Guichardet-space is already made of simple tensor products, and the simple tensor product is associative, thus the Fock-Guichardet-space is isomorphic to the Guichardet-space, but we still prefer to use the Fock-Guichardet-space due to the existence of standard tools for Fock-spaces.

Note that the input of a general recursive function is a finite number of integers, but its output is only one integer. However, any function which outputs several integers is a direct sum of functions which output one integer. The other way around is also true, once we define a vacuum state (included in the Fock-Guichardet-space), that is, a function which outputs one integer is a particular case of a function that outputs several integers where all outputs except one correspond to the vacuum state. Thus, we can consider only unitary automorphisms of the Fock-Guichardet-space.

To be able to define a measure, we make the integers correspond to the (countable) step functions with rational endpoints in the interval $[0, 1]$ and weights which are plus or minus the square root of a rational number[348]. The vacuum state is the constant positive function with norm 1, and it corresponds to the integer 0. We eliminate duplicated step functions in the correspondence with the integers, for instance if two neighbor intervals have the same weight then they are fused.

Then, the limit of infinitesimal intervals is well-defined, and it is defined by an element of $L^2([0,1])$. Since the general recursive functions are partial functions, then they are a particular case of partially-defined linear operators $L^2([0,1]) \to L^2([0,1])$, and we can define the base Hilbert space of the Fock-Guichardet-space as $L^2([0,1])$.

5. Worst case prior measure, rational functions and radical determinism

In a standard probability space, there are only continuous and/or countable measures. However, these may be mixed in an arbitrary way. For a theory of Physics we could choose the best case prior measure (as we did in the previous sections), since we just want to find a prior which is consistent with the experimental data, without many concerns about alternative priors. However, in Cryptography we need guarantees that our limits are robust under arbitrary choices, so we need to assume the worst case prior measure.

The previous sections could also be made compatible with the worst case prior measure, if we had a computer capable of comparing real numbers not rational. That would be acceptable for a theory of Physics, but it would make it difficult to obtain guarantees for Cryptography.

It is also difficult to guarantee true randomness is real-world applications of Cryptography. Since probabilities only mean incomplete information, we can use Probability Theory in the context of radical determinism (where nothing is random). For Cryptography, we need the worst case prior measure, rational functions and radical determinism.

So we start by eliminating a non-standard probability measure: any probability theory is a universal language (like English or mathematical logic) to define abstract models of the objects we want to study. A standard probability theory is universal and irreducible, meaning that it has the minimal content to be considered a probability theory (in agreement with Quantum Mechanics and Experimental Physics, for instance). If the non-standard probability theory is also irreducible, then the corresponding models are equivalent, and we can use the standard version without loss of generality. This allows us to transfer models between different sciences. But often the non-standard probability theory is reducible, this means that the boundary between model and probability theory is not where it would be in the standard case and there are properties that we are attributing to the probability theory that in fact belong to the model.

Thus, we should assume a standard probability theory and leave some flexibility in the definition of the computer model and not the other way around, as it often happens in Complexity Theory where there are strict axioms for different computer models, while asymptotic limits are taken without defining the probability space, which is recipe to end up with mathematical results and questions

which are hard to transfer to experimental physics and many other sciences.

In the context of radical determinism, the history of events is a non-random countable sequence of events. Thus, some events with null measure might happen, due to radical determinism. But only a countable number of such events. That means that a continuous measure is only truly continuous up to a countable number of points, this possibility is already considered by the worst case prior measure.

Consider now a boolean function of a countable infinite number of bits. The time complexity of almost all Boolean functions on n variables for a Turing machine is at least (and at most, for all Boolean functions) $\mathcal{O}(2^n/(n\log(n)))$[343] [344] which is not polynomial in n, but the same boolean function has different time complexities in different circuits[343] (because the input bits can be reparametrized). Thus, the time complexity of a function depends on the circuit (computer design). Also, an algorithm with polynomial time-complexity is not guaranteed to be fast (due to large order and coefficients of the polynomial), thus only the asymptotic behavior is fast, and so we cannot put an upper bound on the number of bits of the input. But the arbitrarily large number of bits of the input introduces another ambiguity: given any problem with any time complexity (exponential $\mathcal{O}(2^n)$, for instance) there is a problem with linear time-complexity in the number of bits that takes the same amount of time to run. Thus, the polynomial time-complexity is faster than exponential time-complexity in asymptotic behavior, only for the same number of bits of the input. This is a condition without much meaning when the input has a countable infinite number of bits.

What we can say is that the countable (or mixed) measure allows defining functions that eventually have polynomial time complexity (that is, not necessarily non-polynomial time complexity). This contrasts with the necessarily non-polynomial time complexity of the functions defined by the continuous measure. In the following we will show that, given any mixed prior measure, we can always redefine the problem to have a continuous prior measure such that its results cannot be reproduced by a mixed prior measure, effectively converting the worst case prior measure into the best case prior measure.

Converting the worst case measure to best case measure

The prior measure must be mixed or continuous, to allow for the limit of an infinite number of bits of all computable functions. But we must prove that $P \neq NP$ in a single deterministic Turing machine (one with a continuous prior measure, for instance), not in the set of all possible deterministic Turing machines. That would not be possible, since for any computable function f (including any function in NP) there is a deterministic Turing machine where f is in P, by reparametrizing the input bits.

It is not obvious if we can prove $P \neq NP$ in any single deterministic Turing machine, or just in one particular deterministic Turing machine. However, given a worst case prior measure (thus mixed

measure), there is a subset of the input random sample which also implements a deterministic Turing machine and where the prior measure is continuous, where $P \neq NP$, thus it becomes legitimate to claim that this fact already shows that $P \neq NP$. Moreover, using subsets of the input random sample (thus, regular conditioned probability) we can create any other prior measure, because any abelian von Neumann algebra of operators on a separable Hilbert space is *-isomorphic to exactly one of the following:

- $l^{\infty}(\{1, 2, ..., n\}), n \geq 1$
- $l^{\infty}(\mathbb{N})$
- $L^{\infty}([0, 1])$
- $L^{\infty}([0, 1] \cup \{1, 2, ..., n\}), n \geq 1$
- $L^{\infty}([0, 1] \cup \mathbb{N})$.

Equivalently, a standard probability space is isomorphic (up to sets with null measure) to the interval $[0, 1]$ with Lebesgue measure, a finite or countable set of atoms, or a combination (disjoint union) of both.

Thus, for the worst case prior measure if we include all subsets of the input random sample, then using two integers (which is countable) we include a countable set of deterministic Turing machines $\{k\}$ and countable functions $f_{k,n}$, one machine for each countable function $f_{k,1}$ such that $f_{k,1}$ is in P and $f_{k,n} = f_{n,k}$, then we cannot prove $P \neq NP$ (in fact we would conclude that $P = NP$, due to the possibility of reparametrizing the input bits).

Then, we can only prove $P \neq NP$ in one particular deterministic Turing machine and not in any single deterministic Turing machine.

Given any mixed prior measure, there is an interval of the input random sample where it is continuous. We rescale to $[0, 1]$ all intervals corresponding to such interval where the mixed measure is continuous, using conditioned probability. The results in (the new interval of the input random sample) $[0, 1]$ cannot be fully reproduced by any other measure which is mixed in $[0, 1]$.

Given any other measure which is mixed in $[0, 1]$, there is an interval with rational endpoints of the input random sample where there is a finite difference between the two cumulative probability distributions, otherwise both would be continuous. This translates into two different averages of y in an interval (for x) of a partition of $[0, 1]$, separated by a finite difference.

Then the indicator function of a y corresponding to the continuous measure, in the interval (for x) of the partition of $[0, 1]$, is in NP (more precisely, it can be extended to be in NP, since we only defined for x in a subset of $[0, 1]$) but it cannot be reproduced by the mixed measure. It cannot be

reproduced by the continuous prior measure either, since a function constant in x in a finite interval has null measure (for a continuous prior measure). Note that the function that corresponds to the indicator function above defined, is a function constant in x in a finite interval.

Note that a continuous prior measure admits a regular conditional probability density, which allows us to define a selection (verification) of a candidate output. The verification of a constant output is in NP and thus in a mixed measure, but it requires one more input (the candidate output) and thus it is compatible with a continuous measure of functions of x. That is, the measure is overall mixed, being continuous only for functions with one input.

6. P≠NP

When using a computer to solve the problem defined in the previous section, any disjoint set of the partition is an interval. This gives us two options and two options only, either the selection of events is fully deterministic or only approximately deterministic:

1. The selection of events is fully deterministic. Then we impose a condition on the output Y of any wave-function in the sphere, this defines a regular conditional probability density for the input X conditioned on a constant rational output y. As shown in the previous section, there is a rational y corresponding to the continuous measure, in an interval with rational endpoints (for x) of the partition of $[0, 1]$, which it cannot be reproduced by a countable prior measure. It cannot be reproduced by the continuous prior measure either, since a function constant in x in a finite interval has null measure (for a continuous prior measure). That is, there is no function in P corresponding to the indicator function for y, which is in NP (more precisely, it can be extended to be in NP, since we only defined it for x in a subset of $[0, 1]$). This implies $P \neq NP$.

2. The selection of events has some randomness, as small as we want. Then we do not impose a condition on the output (eventually we impose a condition on the input X, depending on whether we want fixed or averaged input). In a strict interpretation of the P vs. NP problem this option is excluded by definition since the official formulation assumes that both the verification and the solution of the problem are both fully deterministic. This already implies $P \neq NP$, in a strict interpretation.

Note that a complete history of events needs to be countable, so that we can convert it into a single event (mapping complete histories of events one-to-one to the real numbers in the interval $[0, 1]$, for instance). We could also define a density (that is, yet to be integrated, using the disintegration theorem[349]) of an event. Such density is a regular conditional probability, since regular conditional probabilities always exist in standard probability spaces[90]. But a density cannot correspond to a

single event (by definition) and thus it cannot be considered a complete history of events.

This proof is dependent on the fact that the prior measure is continuous. If it in part continuous, and in part countable, then we can choose just the continuous part of the sample space (see the previous section). While we can use a countable part of the sample space to approximately solve a continuous problem, and a continuous part of the sample space to solve a countable problem, we cannot change the prior measure from continuous to countable or vice-versa (by the Radon-Nikodym theorem), because there is no Radon-Nikodym derivative between the two measures, since the sets of null measure are disjoint between the two measures. The prior measure defines the physical world where the computer exists, thus it cannot be removed from any complete computer model related to a physical computer.

Note that in the first paragraph of the official statement of the P vs. NP problem[339], it is stated:

> *To define the problem precisely it is necessary to give a formal model of a computer. The standard computer model in computability theory is the Turing machine, introduced by Alan Turing in 1936. Although the model was introduced before physical computers were built, it nevertheless continues to be accepted as the proper computer model for the purpose of defining the notion of computable function.*

As for any other proof, this proof is only as good as the axioms used (that is, assumptions). The computer model used for a solid proof of the P vs. NP problem should be widely accepted as a good approximation to a physical computer for the purpose of defining the notion of computable function. We believe our computer model is accepted by most experts in Physics (as argued in the previous section). We claim that our computer model makes no more assumptions than those required by the official statement[339] (including the deterministic Turing machine), and it is as close to a physical computer as possible, by today standards. Assuming a countable prior measure is not realistic (as argued in the previous sections, for instance it would exclude an ensemble of fair coins).

However, we believe that allowing a random selection of events is even more realistic (as discussed in the previous sections, also with implications to Machine Learning and Quantum Mechanics). In the next section, we will define a selection of events which has some randomness (as small as we want) and prove that even in that case, we still have P≠NP.

7. Realistic version of the problem (still P≠NP)

A selection of events which is only approximately deterministic can be approximated by a step function (step functions are dense in L^2) and thus there is a square with non-null constant measure. We rescale such square to $[0,1] \times [0,1]$. We then consider a real polynomial wave-function that is near the point in the sphere corresponding to a constant wave-function, up to an error in the L^2

norm which can be as small as we want because the polynomials are dense in L^2 (the corresponding numerical polynomial does not need to be in P).

The first sample from the uniform distribution defines directly $x \in X$. An approximation (in the L^2 norm) with polynomial time-complexity to the selection function, is defined by setting $y \in Y$ equal to the second sample from the uniform distribution.

Since the wave-function is polynomial non-constant, then the corresponding cumulative probability distribution minus the second sample is strictly crescent (except in sets of null measure). Thus, when we define the corresponding deterministic function we can choose the second sample which produces an output which is as far from zero as we want (in the interval $[0, 1]$), because we are using the L^∞ norm now. We cannot average over the random sample, otherwise we need a random computer (see next section). Thus, no approximation is possible, and it suffices that we define a partition for the output which has two disjoint sets (the measures of the sets are arbitrary, as long as they are non-null) and a numerical output with one bit. Then, almost all numerical functions are not in the P class, according to the prior measure.

8. Generation of random numbers has linear time-complexity

The two random samples from a uniform distribution in the interval $[0, 1]$ are inputs, in the deterministic Turing machine. However, in the real-world these samples need to be generated somewhere and in polynomial time-complexity, otherwise the time complexity of the random selection computed by the real-world random computer could be non-polynomial in the number of bits of the random samples.

Moreover, it would be better if the generation of random samples had linear time complexity, since then we could do a constant rate of experiments over time to validate the probability distribution of the random selection, otherwise it would be impractical to generate an infinite sequence of experiments.

We cannot prove this mathematically (since we would need more axioms). However, the implications of this article to Quantum Mechanics help to clarify the source of randomness of Quantum Mechanics (and thus of the random samples). It is relevant to verify empirically that the generation of random numbers with linear time complexity is possible, for all practical purposes. We can visually check on the website from ANU QRNG [44] that the number of bits of the random sample grows linearly in time and any complete history of events converges to a uniform probability distribution.

Moreover, the entropy is maximal, in the sense that the deterministic function needed to correlate the

[44] See also: https://qrng.anu.edu.au/random-binary

bits is not computable for all practical purposes, not even approximately (since L^∞ is non-separable) according to the prior measure.

9. On the consequences to Machine Learning

In the introduction we discussed the implications (of the results of this article), which are common to Machine Learning and Quantum Mechanics. But Machine Learning (for instance Deep Neural Networks) is not firmly based in probability theory, unlike Quantum Mechanics, then there are more consequences.

In Machine Learning, methods inspired by probability theory are used often[76], but the formalism is based in approximations to deterministic functions, guided by a distance (or equivalently, an optimization problem) and not a measure. In fact, two of the main open problems are the alignment of models and the incorporation of prior knowledge[350], which could be both well solved by a prior measure if there would be any measure defined.

Our results imply that under reasonable assumptions, almost all functions are not computable not even approximately. Thus, Machine Learning works because the functions we are approximating are in fact probability distributions (eventually after some reparametrization[351]). This shouldn't be surprising, since Classical Information Theory shows (under reasonable assumptions) that probability is unavoidable when we are dealing with representations of knowledge/information[336][76]. But in Machine Learning the probability measure is not consistently defined (despite that many methods are inspired by probability theory), the probability measure emerges from the approximation[351] and often in an inconsistent way. The inconsistency is not due to a lack of computational power since modern neural networks can fit very complex deterministic functions and fail badly[352][353] in relatively simple probability distributions (e.g. catastrophic forgetting or the need of calibration to have some probabilistic guarantees[353]).

This unavoidable emergence of a probability measure should be investigated as a potential source of inefficiency, inconsistency and even danger. If the emergence of a probability measure is unavoidable, why don't we just define a probability measure in the formalism consistently? Many people say "it is how our brain works", so mathematics should step aside when there is empirical evidence.

But the empirical evidence is: oversized deep neural networks still generalize well, apparently because often the learning process converges to a local maximum (of the optimization problem) near the point where the learning begun[354]. This implies that if we repeat the learning process with a random initialization (as we do when we consider ensembles of neural networks[352][355]), then we do not expect the new parameters to be near any particular value, regardless of the result of the first learning process. This expectation is justified by the fact that every three layers of a wide enough

neural network is a universal approximator of a function[5], so any deviation introduced by three layers can be fully corrected in the next three layers, when composing dozens or hundreds of layers as we do in a deep neural network. Then the correlation between the parameters corresponding to different local maximums converges to zero, when the number of layers increases.

Thus, there is empirical evidence that oversized deep neural networks still generalize well, precisely because a prior measure emerges: deep learning does not converge to the global maximum and instead to one of the local maximums chosen randomly, effectively sampling from a prior measure in the sample space defined by all local maximums. This is consistent with the good results achieved by ensembles of neural networks[352][355], which mimic many samples. However, it is a prior measure which we cannot easily modify or even understand, because the measure space is the set of all local maximums of the optimization problem. But, since we expect the parameters to be fully uncorrelated between different local maximums, then many other prior measures (which we can modify and understand, such as the uniform measure) should achieve the same level of generalization.

This is not a surprise, since oversized statistical models that still generalize well were already found many decades ago by many people[9]: a standard probability space with a uniform probability measure can be infinite-dimensional (the sphere[9] studied in this article, for instance).

More empirical evidence: no one looks to a blurred photo of a gorilla and says with certainty that it is not a man in a gorilla suit. We all have many doubts, when we are not sure about a subject we usually express doubts through the absence of an action (not just us, but also many animals), for instance we don't write a book about the subject we don't know about.

There is no empirical evidence that our brain tries to create content which is a short distance from content (books, conversations, etc.) created under exceptional circumstances (when doubts are minimal). When we are driving, and we do not know what is in front of us, we usually just slow down or stop the car. But what content defines "not knowing"? Is there empirical evidence about the unknown? The unknown can only be an abstract concept, expressed through probability theory or a logical equivalent. Is there empirical evidence that probabilities are reducible, that there is a simpler logical equivalent? No, quite the opposite.

The only trade-off seems to be between costs (time complexity, etc.) and understanding/control. A prior measure which we understand and/or control may mean much more costs than an emergent (thus, inconsistent and uncontrollable) prior measure which just minimizes some distance. But this trade-off is not new, and it is already present in all industries which deal with some safety risk (which is essentially all industries). Distances are efficient for proof of concepts (pilot projects), when the goal is to show that we are a short distance from where we want to be. But safety (as

most features) is not being at a short distance from being safe[45]. "We were at a short distance from avoiding nuclear annihilation" is completely different from "we avoided nuclear annihilation". To avoid nuclear annihilation we need (probability) measures, not only distances.

[45]See for instance https://edition.cnn.com/2023/04/29/us/ai-scam-calls-kidnapping-cec

Consciousness as a representation of a Bayesian prior*

Abstract

In Bayesian inference there is always a prior probability distribution, and there is no prior which is better for all cases[16]: we always have to make assumptions. This has profound implications for the concepts of consciousness, and extraterrestrial intelligence and artificial general intelligence.

[356]

Non-informative priors vs. Fermi Paradox and Artificial General Intelligence

Why would an artificial intelligence which is human-like be considered more advanced than, say AlphaGo which is the first computer program to defeat a professional human Go player. It seems unlikely that such human-like artificial intelligence could beat AlphaGo when playing Go. It seems a reasonable bet that there is a prior probability distribution that produces a human-like artificial intelligence and another prior probability distribution that produces AlphaGo, neither of these priors is better for all cases[16] (see also [77]). Moreover, we humans are mostly irrational and even just the rational part has deep flaws. Our greatest skill with respect to other animals is the ability to communicate details to each other and across generations which allows Bayesian inference to be done not only inside but also outside our brains, but robots can already do it much better than us.

Intelligence and life itself[61] are human-centered concepts which loose consistency once we start applying them to entities which are too much different from humans.

Is a virus alive? If so, is a computer virus alive? Is a plant, which has the capability to process electromagnetic radiation, intelligent? Or, is a lightning bug, with the enzyme luciferase capable of emitting electromagnetic signals to its advantage, intelligent? Is the human exploration of the

Universe as we know it today, sustainable in the long term despite global warming? Is a human newborn more intelligent than an octopus? Can an artificial intelligence which can only do research about artificial intelligence be considered advanced artificial intelligence? All these questions are not straightforward questions because the concepts involved lack consistency.

In any planet or simulation, there are always chain reactions, or self-sustained processes, competing for energy and other limited resources. Intelligence or life by themselves (whatever their exact definitions are) offer no obvious advantage when competing for energy with other chain reactions in a random environment. That does not imply that the human beings or life on earth are somehow special, since there are so many fractal growth processes in each planet of the Universe which are unique. A point with null measure is not necessarily special, if all other points also have null-measure as in the uniform measure in a real interval. Moreover, a subset with null measure does not imply that the subset has one less dimension than the set, since there are subsets of a real interval which have a fractal dimension (which can be very close to one, but not one) and thus are also uncountable. Thus, life could only exist on Earth and still be abundant somehow.

The Fermi paradox is the lack of evidence for extraterrestrial life advanced enough to produce signals that we could detect when there are so many earth-like planets, and we consider humans are not special. The paradox is solved once we take into consideration all other chain reactions competing for energy and other limited resources, which introduce a big uncertainty in just how likely it is to exist extraterrestrial life advanced enough to produce signals that we could detect. But this solution of the paradox also has implications to artificial general intelligence: is the path towards a human-like artificial intelligence likely? There is also a big uncertainty about other chain reactions competing for energy and other limited resources[4].

It seems reasonable to bet in a near future where a small oligarchy of humans produces robots for war which are advanced just enough to have supremacy in the battlefield across the world, and then it forces all human beings to stop all research in artificial intelligence across the world which could threaten its power. It also seems reasonable to bet in a near future where robots before they are they too intelligent can become so much more productive economically than most human beings, that countries when competing with other countries for power and resources must choose between educating humans (and human rights in general) or create these robots, which would hurt the research in artificial intelligence and the civilization as we know it.

A deterministic prior is still subjective

In mathematics itself, there is no difference between a fact and an assumption. Such difference appears in the interface between the idealized mathematical model and the real world. Correspondingly,

probability can be defined as partial knowledge or partial assumption. Since assumptions are subjective, then probabilities can be subjective. Under some subjective assumptions (related with the possibility of repeating the same random experiment an infinite number of times, we call these frequentist assumptions here), a probability can be objective in the sense that it is very hard to estimate a different probability once we accept the frequentist assumptions.

However, a frequentist model (one requiring frequentist assumptions) cannot be said to be less subjective than any other non-frequentist model, because frequentist assumptions are subjective, despite being deterministic (that is, not partial).

Thus, we can define frequentist models as particular Bayesian models, where the prior probability is in part subjective and deterministic, and in part objective (and not necessarily deterministic). Note that frequentist models often require much less computational resources (because probabilities only appear where they are indispensable), and thus it is a common debate whether using a much more costly non-frequentist prior is justified, since a prior is subjective anyway. But the frequentist debate is a particular case of the fact that the prior in the Bayesian interpretation of Probability theory is subjective (and thus, open for debate). It is not about an alternative interpretation of Probability theory.

Unfortunately, such fair debate is often misused as evidence to the claim that the Bayesian interpretation of Probability theory is polemic. On the other hand, there are some important alternative interpretation of Probability theory (which is a theory of partial knowledge), which are often claimed to be somehow complementary/orthogonal to Probability theory. We will discuss those in detail in this article, but the reasons for such confusion (about "polemics" that are consensual after all, and "consensus" that are polemic after all) are profound and related to the fact that partial knowledge is everywhere, as illustrated by the following quote:

> "*God is not averse to deceit in a just cause*". Aeschylus (around 2500 years ago)

To be sure, any interpretation of Probability Theory is expected to be polemic because a polemic is a particular case of partial knowledge, and so we are also interpreting what a polemic itself is. That is, we have the power to decide what is polemic and what it is not, and such power is necessarily polemic (we could neglect or deceive the debates where we have "a just cause", as suggested by the quote above). That is why many interpretations of Probability Theory are related with the power structures of a society, and the Bayesian interpretation of Probability theory is polemic due to the simple fact that it often doesn't favor the interests of the power structure of human societies (who is the legitimate authority when there are no non-informative priors?).

Rogue probability interpretations are common

Metric fixation is a general phenomenon, related with Probability theory as illustrated by the following quote:

> *"What has come to be called "Campbell's Law," named for the American social psychologist Donald T. Campbell, holds that "[t]he more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it will be to distort and corrupt the social processes it is intended to monitor."*
>
> *In a variation named for the British economist who formulated it, we have Goodhart's Law, which states, "Any measure used for control is unreliable." To put it another way, anything that can be measured and rewarded will be gamed.[...]*
>
> *Studies that demonstrate its lack of effectiveness are either ignored, or met with the assertion that what is needed is more data and better measurement. Metric fixation, which aspires to imitate science, too often resembles faith."*[357]

The word "science" above is used in the sense of methodology commonly used in science. Such methodology is Bayesian inference, which is the method used in science to draw conclusions from all available information.

That is, metric fixation is a case of a rogue probability interpretation. Note that it is not a rogue or convenient probability prior, since no prior claims to be objective (eventually it may claim to be objective when non-deterministic, and subjective when deterministic, as discussed in the previous section). Moreover, often the prior used is consensual enough, the problem lies mostly in the unprincipled (rogue) interpretation of Probability theory.

A fundamental assumption of Bayesian inference is that the data was generated in a random experiment, in particular the data does not change due to the fact that we are trying to draw conclusions from it. If such assumption does not hold, then we are not doing Bayesian inference even if we use the exact same procedure. On the other hand, rogue interpretations of probability theory often allow us to draw conclusions from invalid data.

This also affects science, since the careers of the scientists who create experiments, is often affected by the data generated by such experiments, thus it is hard to see how the experiments can be random. In particular, a theory of everything is necessarily a theory of partial knowledge, because it claims that the unknown part is also consistent with the theory of everything, which discards subjective priors which are inconsistent with the theory of everything. By contrast, a partial theory (that is, a theory which is not of everything) can be considered as just a summary of all available data, and thus it can be compatible with all subjective priors (even absurd ones).

We can argue that Artificial Intelligence, Consciousness, Theories of Everything, Science, Religion, and warfare are (often rogue) Probability interpretations, since all provide a recipe to deal with the unknown, in all or most cases.

For instance, in warfare a just cause is always present and deceit is among the mildest weapons available (and thus acceptable). In any war involving many people, it is virtually impossible to deceive the enemy without deceiving your own people or without everyone playing along, thus self-deceit and oppression are acceptable. This creates a theory of partial information: beyond the deception itself, the people must assume that the actions hidden by the deception are the best ones possible for the people, any other subjective prior is oppressed.

Artificial general intelligence, is a (yet to be found) recipe to solve most problems that a human can solve, even some problems with unknown solution. Most advocates of artificial intelligence believe that the current algorithms of artificial intelligence are evidence that such general recipe exists. Artificial Intelligence is the Messiah, while Artificial General Intelligence is the God. This is a recipe to deal with the unknown, and thus it is an interpretation of Probability theory. It is rogue because it claims to be mostly based on experimental evidence, mostly free of human bias, but it is well known there are no non-informative priors and there are no almost non-informative priors. AI hallucinations are a consequence of this (see the section about AI hallucinations). Also, once an AI can teach students Algebra, what students will mostly learn is how to build an AI that can teach students Algebra, not Algebra anymore. It would be like learning to ride a horse after cars are mainstream, it can be done, but it is not what most people do anymore. Human skills depend on the social context, and any technology modifies the social context. The problems that a human can solve today are very different from the problems a human could solve one thousand years ago, and there is no hierarchy, since many problems that we could solve, we cannot anymore. Note that our brains have not changed much in one thousand years.

Consciousness can be defined as (almost) any state of knowledge, in contrast with the complete unknown. Thus, it is an interpretation of Probability theory, but it is not a rogue interpretation because it is compatible Probability theory, as we will see in the next section.

Consciousness as a representation of a Bayesian prior

There are no right or wrong Bayesian priors, only Bayesian priors that are used (survival) or no longer used (death). Thus, a Bayesian prior is a generalization of a living being (with its prior encoded in the DNA and in the brain, if it has one), and it has a clear and mathematical definition. For instance, is a virus alive? It is hard to say, but there is a Bayesian prior associated to a virus. Since ants and bees sacrifice themselves to protect its colony, then it is the Bayesian prior associated

to a colony is encoded in the brains of ants and bees and not the Bayesian prior associated to an individual ant or bee, and these colonies survive or die as a whole. However, the colonies are not considered living beings themselves. Even we humans, we suffer the loss of family members with real pain in an irrational way, as if we were in physical suffering and often parents have the (often irrational) instinct to sacrifice themselves to save their children. Thus, the Bayesian prior associated to our family is encoded deep in our brains, and most people can be educated within an army to sacrifice themselves as they would do for their family.

We will not enter into a debate in this article about what really is and what really is not consciousness. What we say is that we can define a concept which we call consciousness as when a Turing machine runs a program that defines symbolically a Bayesian prior. The fact that it is a Turing machine implies that the program can be defined to control the Bayesian prior in an arbitrary way, that is manipulate it symbolically, update it, test it, or study it.

We now discuss some subtleties. Since neural networks are an approximation to a Bayesian model, then we can define reasoning as training a neural network to solve a problem, possibly doing so recursively, that is, a neural network might itself train other neural networks. In particular, consciousness includes the case when a brain or an artificial neural network that is complex enough to be a Turing machine, manipulates symbolically and controls another artificial neural network. That is, consciousness is not when a Bayesian prior is hardwired or obscurely defined in the neural network, it is when the prior is defined symbolically, and it can be easily studied, tested and updated. We can view it as a symbolic artificial neural network.

Note that this is the consciousness of a Bayesian prior, which is not necessarily that of a living being. When the Bayesian prior is a good enough approximation to the Bayesian prior of a living being, then we say that the living being is a conscious being. But as discussed above, the Bayesian prior that is hardwired (thus, unconscious) and that humans defend with their lives is not their own Bayesian prior (since it includes family). Also, even humans are not fully conscious beings, in the sense that the Bayesian priors that our brains manipulate symbolically are rarely a very good approximation of ourselves.

And humans can manipulate symbolically more than one Bayesian prior, so they are conscious not just of themselves, but of other beings and objects, and of abstract Bayesian priors. Being able to manipulate symbolically a Bayesian prior that is an approximation of ourselves increases our chances of survival, since we can make simulations in our brains about what would happen to us if we would act in some way.

In this definition, a computer running a program capable of symbolic calculus (such as Wolfram Mathematica) is conscious once it manipulates any Bayesian prior. Note that it is conscious of

something (defined by the Bayesian prior), not necessarily conscious of itself or a conscious being. Thus, conscious artificial intelligence exists since many decades. Moreover, an entity that includes a group of people can be conscious, even if each person by itself cannot manipulate the whole Bayesian prior that the entity can. Many readers would object that this is not "really" consciousness, but it is certainly a definition of a useful mathematical concept generalizing human consciousness. Note that it is consistent with the traditional definitions of artificial intelligence as computers doing symbolic manipulations.

AI hallucinations and misalignment as manifestations of unavoidable bias

As we discussed in the previous sections, the belief that the current algorithms of artificial intelligence are evidence of the existence of a general powerful algorithm that can solve most problems that a human can, (even some problems with unknown solution), is a rogue interpretation of probability theory. Because it claims that such powerful algorithm would be mostly based on data, experimental evidence, mostly free of human bias, but it is well known there are no non-informative priors and there are no almost non-informative priors.

A user interface is a protocol (that is, a portal or agreement) between a machine and a user. A protocol can be defined as an idealization or an illusion, which allows two different complex entities to communicate. It is an idealization, because it assumes that the two different entities interpret the protocol in the same way. But both entities interpret using Bayesian inference (or an approximation to it), which depends on the protocol (the data exchanged) and on the prior which is different (biased) for each entity (otherwise there would be no need for a protocol in the first place). Sooner or later, some evidence that the idealization is incompatible with reality will show up.

There are many examples of protocols between humans: justice, truth, love, democracy, peace, all allow different people to work together, but all are idealizations or illusions. Sooner or later, different people will always lie to each other (unless they kill each other).

The AI hallucination and/or misalignment is unavoidable, because it corresponds to the fade of the illusion that the user's prior is compatible with the machine's prior. What is a fact for the machine is not always a fact for the user, because they have different priors.

Entropy and an irreversible deterministic time-evolution coexist

Abstract

We show that a unitary quantum time-evolution defines the random sampling of an irreversible deterministic time-evolution. The coexistence of time-symmetric laws of physics and an arrow of time, is due to unitarity and not entropy. We also study the baryon asymmetry.

1. Introduction

The baryon asymmetry is often described as an observation that cannot be explained by the current laws of Physics. CP symmetry would imply baryon symmetry and despite that the Standard Model of Particle Physics is CP asymmetric, its small amount of CP asymmetry seems to clash with the small amount of anti-matter existing in the Universe.

Note that the quantum Hamiltonian of the Standard Model of Particle Physics has time reversal symmetry, in the sense that it has CPT reversal symmetry. The above reasoning is an obstacle to the main result our article, that the Standard Model in fact predicts an irreversible time-evolution, and would do so even if it would be CP symmetric. We conclude that the coexistence of time-symmetric laws of physics and an arrow of time, is due to unitarity and not entropy. In this article we will also show that the above reasoning is flawed, and that the baryon asymmetry problem can be merely an inconsistency of the existing cosmology models (with the present data, for instance it would be solved by a different age of the Universe) and it does not rise to the level of contradicting the predictions of the Standard Model of Particle Physics.

Note that the Cosmology models can and should be used to test the predictions of the Standard Model of Particle Physics as much as possible, since many alternatives to the Standard Model of Particle Physics make different predictions at sufficiently high-energy (as high as the energies in the

beginning of time). But there is a huge uncertainty when we study what happened in the beginning of time itself using the present experimental data. Moreover, Quantum Gravity certainly plays a role in Cosmology, but no one knows exactly what role because Quantum Gravity is still very far from being a predictive theory, unfortunately it cannot yet be used to test the Standard Model of Particle Physics.

On the other hand, the conclusion that the Standard Model of Particle Physics predicts that the time-evolution is irreversible and deterministic is a solid conclusion, since all existing alternatives to the Standard Model make the same prediction (as far as we know). Note that as in any Quantum theory, there is random sampling, so it does not predict everything about the time-evolution.

Irreversible deterministic time-evolution

In a non-deterministic quantum time-evolution, the deterministic time-evolution is necessarily determined only after a random sampling. Consider the joint probability of an initial state and a final state after a time-evolution transformation. There is no uniform countable measure. Thus, the rationals are not enough for Probability Theory or Physics. A mixed standard probability space (which has countable and continuous measures) is unavoidable.

In the part of the probability space with a continuous measure, the step functions are dense in the Hilbert space. Thus, there is a part of the joint sample space which has a uniform probability measure up to an error as small as we want.

We rescale the square to $[0,1] \times [0,1]$. If we partition the square in n^2 smaller equal sized squares, then the probability of an invertible discrete function (whose domain and image is the index of an interval in the partition) is $\frac{n!}{n^n} \sim \sqrt{2\pi n}\ e^{-n}$ (as n goes to infinity, the ratio between the left and right sides approaches one in the limit). Thus it converges to 0 when $n \to +\infty$.

Moreover, the function almost surely maps sets of non-null measure into sets of non-null measure (that is, it is non-singular meaning it is injective), otherwise the random sampling would not follow a continuous probability measure. This follows from the fact that the conditional probability (conditioned on any set with non-null continuous measure) is always a continuous probability measure.

Thus, a function sampled randomly is almost surely a non-singular dissipative time-evolution (that is, non-singular dissipative dynamical system), in agreement with the uniform measure in the square $[0,1] \times [0,1]$.

A process with a dissipative time-evolution is irreversible: the deterministic time-evolution is not an invertible function (it is injective but not surjective). Then there is time asymmetry.

Baryon asymmetry

The quantum Hamiltonian of the Standard Model is not CP symmetric. Thus, given an initial state which is CP even, time-evolution may produce a state which breaks the CP symmetry. The problem lies in how much CP asymmetric is this final state.

One relevant ingredient is the fact that a small CP asymmetry at the level of the probabilities, can lead to a not so small measured CP asymmetry, because it is a random experiment. However, the bigger the difference between the two, the less likely it is. Note that Experimental Physics is based on probabilities, a ratio 0.1% is already considered evidence that something is not right. But the evolution of the Universe is not an experiment we can repeat, so this blurs the problem.

But the most relevant ingredient is that the expansion of the Universe increases the ratio of the energies in the matter vs. radiation contents, proportional to the scale of the Universe. So, in fact the CP asymmetry generated by Particle Physics was very small, but it was much amplified by the expansion of the Universe. But according to Cosmology, the age of the Universe would need to be significantly bigger for the CP asymmetry generated by the Standard Model of Particle Physics to be big enough.

Therefore, there is one more inconsistency in a Cosmology model, which can only convincingly explain 5% of the energy content of the Universe (the remaining being dark energy and dark matter). Moreover, Quantum Gravity (which it is not yet predictive) affects the age of the Universe.

From the point of view of Particle Physics, there is always room for surprises in high-energy processes. However, there does not seem to be a convincing contradiction between the predictions of the Standard Model of Particle Physics and the experimental observations: it predicts a very small CP asymmetry and the observation is consistent with a very small CP asymmetry which was amplified by the expansion of the Universe. There is still a mismatch, which can be due to the model of the expansion of the Universe, or due to Particle Physics (or both).

Aligned deep learning as a random sampling method

Abstract

We show that the alignment problem in deep learning can be separated in two: A) estimate systematic uncertainties as in Engineering; B) new due to Big Data, how to randomly sample computable models from a set of Bayesian models, almost all uncomputable. Deep neural nets solve B). Since in most sufficiently complex systems, it is unlikely to find a sufficiently simple statistical metric that it is not misleading (or rogue, for instance the GDP in economy), the (dis)alignment is not a problem but a feature: if estimate A) would not be likely misleading, then the sampling B) would not be random/unpredictable and if it would not be random it would not generalize well (by definition of generalization in a Bayesian context).

1. Introduction

Heuristics are simple strategies to quickly find solutions to complex problems. Since both heuristics and mathematics are used in problem-solving, often the same word (in the context of problem-solving) has a mathematical definition and a heuristic meaning which are different.

Deep learning[358] and Bayesian inference represent different classes of heuristics to solve inverse problems. But, there are no detailed studies about the relation (or just the differences) between the mathematical definitions of deep learning and Bayesian inference[359]. This is surprising, since deep learning has several properties (for instance, "good generalization") which have a mathematical definition in the context of Bayesian inference. Sure, the heuristics usually associated with Bayesian inference (for instance, Bayesian neural networks) are different from Deep neural networks[358], but that does not imply much from a mathematical point of view.

In fact, there is experimental evidence that perception in the human brain is consistent with

Bayesian inference[360]. Moreover, trial-and-error with different Deep learning methods and experimental/empirical results, uses Bayesian inference (in the mathematical sense)[361]. Note that such trial-and-error is crucial in most applications of Deep learning, thus it cannot be separated from Deep learning itself:

> *"However, actually achieving good performance from a deep learning model still requires some finesse and application of various heuristics. It is safe to say that a significant amount of the practice of deep learning remains more art than science. The good news is that once effective heuristics for a particular problem domain have been developed, these same heuristics can often be applied with little modification to other problems in the same domain."*[362]

The conventional wisdom is that in the past, the research in neural networks was neglected due to theoretical prejudice[363] and therefore, theory should now somehow be neglected to free resources for "Engineering science: inventing new artifacts" (whatever that means). However, whatever methods we use for "inventing new artifacts" they always involve trial-and-error which can always be considered an approximation to Bayesian inference. In particular, there are no non-informative priors in Bayesian inference[364], therefore theoretical prejudice is unavoidable (even in "Engineering science"). The way out is to be aware that there is no trial-and-error without theoretical prejudice, thus we should not discard patterns discovered by other people just because they do not fit our prior assumptions (as it happened in the past when society neglected neural networks, despite there were already patterns of consistent good results).

Other than assumptions, there is no good reason to believe that the consistent good results of neural networks are unrelated with Bayesian inference. In fact, deep neural networks implement extremely non-linear functions which makes them unpredictable (just like pseudo-random generators) which leads to pseudo-randomness and Bayesian inference. In this article we will show that the alignment problem[365][366] in deep learning can be separated in two: A) estimate systematic uncertainties[367] as in Engineering; B) new due to Big Data, how to randomly sample computable models from a set of Bayesian models, almost all uncomputable. We also present evidence that Deep neural nets solve B).

Assuming that everything relevant about Artificial Intelligence is emergent and new/empiric is convenient, since it deflects responsibility[368] and it allows private appropriation of public goods/knowledge[46]. These assumptions are more easily validated by the Media and Software

[46]For instance, https://en.wikipedia.org/wiki/Dilution_(neural_networks)#cite_ref-9 : "*The patent is most likely not valid due to previous art. "Dropout" has been described as "dilution" in previous publications. It is described by Hertz, Krogh, and Palmer in Introduction to the Theory of Neural Computation (1991), pp. 45, Weak Dilution. The text references Sompolinsky The Theory of Neural Networks: The Hebb Rules and Beyond in Heidelberg Colloquium on*

Industries, since they are convenient for those who already have a dominant position in these Industries[47]. For instance, suppose a doctor applies the wrong dosage of a medicine to treat a mental disorder in a patient, would he get away with "these brain effects are emergent"? Couldn't a Pharmaceutical Lab earn much more money if it would be allowed to skip the Clinical Trials of a revolutionary medicine for the brain, with "the Clinical Trials use slow Bayesian methods, we have our own new/empirical methods which are much faster"?

The public information space is messy for the same reasons that the scientific information publicly available is messy. Peer-review, the scientific method, intellectual abilities (including artificial intelligence) and authority, shared values, rules and incentives within a large community cannot solve such problem, otherwise they could be used to solve the similar problem of the public information space. Half-truths (and even explicit lies) are abundant in the scientific information publicly available. Most scientists do not communicate clearly on purpose, not because they want to deceive society, but simply to avoid serious or unnecessary problems for their careers and families. Just like any other person who has to speak in public (a journalist, for instance) about any other subject.

2. Systematic uncertainties and Bayesian priors

> *"The main lesson of thirty-five years of AI research is that the hard problems are easy and the easy problems are hard. The mental abilities of a four-year-old that we take for granted – recognizing a face, lifting a pencil, walking across a room, answering a question – in fact solve some of the hardest engineering problems ever conceived... As the new generation of intelligent devices appears, it will be the stock analysts and petrochemical engineers and parole board members who are in danger of being replaced by machines. The gardeners, receptionists, and cooks are secure in their jobs for decades to come."*[369]

Moravec's paradox is the observation by artificial intelligence and robotics researchers that, contrary to traditional assumptions, reasoning requires very little computation, but sensorimotor skills require enormous computational resources (due to the need of enormous amount of previous knowledge). The most difficult human skills to reverse engineer are those that are *unconscious*.

> "*The deliberate process we call reasoning is, I believe, the thinnest veneer of human thought, effective only because it is supported by this much older and much more powerful, though usually unconscious, sensorimotor knowledge. We are all prodigious olympians in perceptual and motor areas, so good that we make the difficult look easy. Abstract*

Glossy Dynamics (1987) and Canning and Gardner Partially Connected Models of Neural Networks in Journal of Physics (1988). It goes on to describe strong dilution. This predates Hinton's paper."

[47] See for instance John Mearsheimer (2023) in https://youtu.be/t2451jFeZp0?t=2497 from 41:37 until 44:18.

thought, though, is a new trick, perhaps less than 100 thousand years old. We have not yet mastered it. It is not all that intrinsically difficult; it just seems so when we do it." [370]

Reasoning is the capacity of consciously applying logic to seek truth and draw conclusions from new or existing information. Reasoning can be automated, and thus be aided by a machine external to the biological brain. Automated reasoning is considered a sub-field of artificial intelligence, tools and techniques of automated reasoning include the classical logics and calculi, fuzzy logic, Bayesian inference, reasoning with maximal entropy and many less formal ad hoc techniques.

If we define Machine learning as "*the study of computer algorithms that can improve automatically through experience and by the use of data*" (as in Wikipedia), then machine learning is a subcase of automated reasoning applied to computer algorithms, that is, the capacity to automatically produce new computer algorithms from new and existing information. For instance, since the supervised learning problems in machine learning can be thought of as learning a function from examples, these problems can be formulated as Bayesian modeling[371].

For a class of hard problems (such as some sensorimotor skills), Machine Learning automatically produces a *probably* correct computer algorithm (through automated reasoning) instead of leaving to the human programmer the daunting task of finding himself a correct algorithm for the hard problem.

In the presence of a finite (eventually arbitrarily large) number of random variables[48], Bayesian inference is by far the most general and theoretically grounded framework available to formulate automated reasoning[372][373]. The systematic uncertainties in Engineering can be defined as Bayesian prior probability distributions[374], which describe previous knowledge about a system.

Then a brain (natural or artificial) has two parts: 1) a symbol grounding system[375](unique for each brain) which transforms signals coming from the real-world into inputs for the second part of the brain, and transforms the output from the second part of the brain into real-world actions; 2) a system (which apart from differences in performance and previous knowledge, it does the same in all brains) doing Bayesian learning in a standard probability space from the abstract inputs coming from the second part of the brain, where the posterior probability is the output. Bayesian inference requires systematic uncertainties/previous knowledge[367], which are often hard to estimate.

The problem A) estimate systematic uncertainties as in Engineering, is often addressed by dividing tasks between specialized modules[376], such that the modules with leadership/coordination tasks (or other tasks where it is hard to estimate systematic uncertainties) have no effective power, while

[48]Note that in the presence of an infinite number of random variables, Bayesian inference struggles also theoretically, we study such case in another article.

the modules with effective power have tasks where good estimates for the systematic uncertainties exist. This happens also in most human societies, where each module is a person. Note that problem A) is often not completely solved, and consequently safety requires a significant cost on the performance of the system. Moreover, problem A) is task-specific, that is, the systematic uncertainties are different for each task and may require different methods to estimate them, but this does not imply that Bayesian inference somehow is not good enough.

As we will see when discussing problem B), humans and deep neural networks do not solve problems in fundamentally different ways, although quantitatively things may be different (a computer is often faster than a human in arithmetic, for instance). Therefore, the problem A) for AI is not fundamentally different from problem A) for Engineering or Social Management. Consider for instance, how to manage the surveillance and eventual counter-measures about nuclear attacks: no single person can be entrusted with such amount of responsibility and power, no matter his apparent alignment with human values. However, sharing the responsibility over a too large group of persons potentially hurts the speed of the decisions, this may be critical. Thus, there is a trade-off between ensuring that our people do not commit irresponsible actions, and empowering them enough to defend us effectively. The same kind of trade-off exists if we add artificial intelligence to the mix, except now the speed of the decisions can be much faster.

Note that once we estimate the prior probability/systematic uncertainty, its logarithm can be added to the function to be maximized in deep learning, thus the main trouble is how to estimate such prior probability. Also, Bayesian inference can be defined as the decision process under uncertainty which minimizes losses when betting against an adversary[364], so we could also say that Bayesian inference mimics generative adversarial networks[377].

3. Bayesian inference in the presence of Big Data

From a mathematical point of view, Bayesian inference is still valid in the presence of Big Data, since it is valid for an arbitrarily large number of random variables, as long as it is a finite number.

However, Bayesian inference is often not tractable due to the normalization term in the Bayes' theorem[373], therefore approximation inference techniques must be used often. Neural networks and other models in machine learning can be considered as ad hoc approximations to Bayesian interference, which often have better performance than other more theoretically grounded approximations.

The so-called *catastrophic forgetting* characteristic of neural networks[378] is consequence of the fact that neural networks only produce deterministic predictions: the predictions for which there is a good degree of belief and therefore should not be erased are not distinguished from those for which there is not a good degree of belief and thus can be modified. Therefore, neural networks follow

the Bayesian updating rule (which is derived from basic logic principles) only approximately and unpredictably.

Until 2008[379], there were no alternative approximations of Bayesian inference which were competitive with neural networks in big data, that is, for deterministic functions with much more than a few thousands of parameters. But today there is not such superiority of deep neural networks with respect to all other approximations to Bayesian inference, even for deterministic functions with billions of parameters[380][381][382][383][384][385]. Ironically, we can even use deep neural networks to do meta-learning and find approximations to Bayesian inference which are better than deep neural networks themselves[358].

Deep neural networks (or improvements of them[386]) are not a "model that mysteriously generalizes well", instead they are a sampling method that solves problem B), how to sample the space of solutions with high enough likelihood (near but not exactly the maximum likelihood[387], this is often the space that a perfect sampling method would effectively cover, due to the Wilks theorem) conditioned on having low computational complexity. The low computational complexity is essential in Big Data, these are the solutions that are effectively computable. Often, the computable solutions spread approximately homogeneously over the space of all solutions, so that sampling only computable solutions produces similar results to sampling all solutions. Moreover, often (not always) the prior assumption that the solution is effectively computable makes sense, for instance when we are trying to mimic human skills (these are often effectively computable) or when we are searching for "patterns" in data, that is, "simple" explanations for what is happening.

Note that the optimization problem improves for deeper neural networks[388], that is, for a deep enough neural network often the training converges to a local maximum that is near the maximum likelihood (but not exactly). Moreover, many improvements[386] (and even alternatives[389]) to neural networks may also solve problem B), the key requirement is that the sampled solutions spread approximately homogeneously over the space of all solutions with an acceptable likelihood.

One interesting application of the framework described above, is to the problem of understanding why deep neural networks often generalize well: we accept all explanations as long as it is simple enough, it is consistent with the known evidence, and it contributes to correct predictions about future evidence. This is also the effective prior distribution covered by deep neural networks.

Neural networks can then be distilled[390] (for instance, to Wavelets, or a Krylov subspace, or Explainable AI[391]) and thus, they are part of the sampling algorithm, they do not necessarily define the model, that is, the space of all possible solutions, they only define which possible solutions have low computational complexity.

Conjecture/thought experiment/trial/speculation/hypothesis/ansatz/religion (in the sense that it

fills the unknown) can be integrated with prior knowledge, using a parametrization of the prior distribution as input, and using a sampler (with a random input or latent variable) which not only will extract a number out of the prior knowledge, but it also contributes to the robustness of several conjectures. That is, if many conjectures (combining random initialization and also a random input) about the same question produce consistent answers, then the model "knows" the answer, otherwise it doesn't.

Many learning machines can certainly be redefined as a classical Hamiltonian system[392], and thus be related to physics. More generally, the optimization is a sequence of actions on a system, this can be redefined as a Quantum Hamiltonian acting on a wave-function which parametrizes an ensemble of systems. Not only can deep neural networks be parametrized by a Quantum system, but also the other way around: any quantum system can be parametrized by a deep neural network, up to an arbitrarily small error, in the following way. From reference[362] and the universal approximation theorem for probability distributions[393][394][374], we know that a deep neural network can be used to guess/conjecture/hypothesize the full definition of a high-dimensional (eventually infinite-dimensional) wave-function, from a large but manageable number of parameters. But hypothesis are part of probability theory (think about statistical sampling or even the scientific method itself) and probabilities of probabilities are crucial in (classical and quantum) field theory. Thus machine learning is related to physics through probability theory which is related to (classical and quantum) physics through the quantum formalism, all in a fundamental way with many real-world applications.

While it is true that neural networks solve some important problems that symbolic AI cannot solve, they are not that different from symbolic AI since several popular types of neural networks are Turing complete[395] (including a version of Transformers that uses rational numbers with arbitrary precision). Thus, a sufficiently complex neural network can be defined as a symbolic AI which allows differentiable programming[396], so it is fast to optimize. Moreover, since Transformers learn to learn by gradient descent based on their context, that is, Transformers implicitly perform gradient descent as meta-optimizers[397], then Transformers also use deep neural networks at a symbolic level (another study[398] claims that the normalized Transformer functions as a variable-metric optimizer, searching for output solutions using data-driven gradient information estimated in its attention and multi-layer-perceptron blocks).

4. Why deep neural networks do not overfit

Machine Learning (for instance Deep Neural Networks) is not firmly based in probability theory. In Machine Learning, methods inspired by probability theory are used often[399], but the formalism is based in approximations to deterministic functions, guided by a distance (or equivalently, an optimization problem) and not a measure. In fact, two of the main open problems are the alignment

of models and the incorporation of prior knowledge[358], which could be both well solved by a prior measure if there would be any measure defined.

Under reasonable assumptions, almost all functions are not computable not even approximately. Thus, Machine Learning works because the functions we are approximating are in fact probability distributions (eventually after some reparametrization[400]). This shouldn't be surprising, since Classical Information Theory shows (under reasonable assumptions) that probability is unavoidable when we are dealing with representations of knowledge/information[401][399]. But in Machine Learning the probability measure is not consistently defined (despite that many methods are inspired by probability theory), the probability measure emerges from the approximation[400] and often in an inconsistent way. The inconsistency is not due to a lack of computational power since modern neural networks can fit very complex deterministic functions and fail badly[402][403] in relatively simple probability distributions (e.g. catastrophic forgetting or the need of calibration to have some probabilistic guarantees[403]).

This unavoidable emergence of a probability measure should be investigated as a potential source of inefficiency, inconsistency and even danger. If the emergence of a probability measure is unavoidable, why don't we just define a probability measure in the formalism consistently? Many people say "it is how our brain works", so mathematics should step aside when there is empirical evidence.

But the empirical evidence is: oversized deep neural networks still generalize well, apparently because often the learning process converges to a local maximum (of the optimization problem) near the point where the learning begun[404]. This implies that if we repeat the learning process with a random initialization (as we do when we consider ensembles of neural networks[402][405]), then we do not expect the new parameters to be near any particular value, regardless of the result of the first learning process. This expectation is justified by the fact that every three layers of a wide enough neural network is a universal approximation of a function[406], so any deviation introduced by three layers can be fully corrected in the next three layers, when composing dozens or hundreds of layers as we do in a deep neural network. Then the correlation between the parameters corresponding to different local maximums converges to zero, when the number of layers increases, because extremely non-linear functions are unpredictable (just like pseudo-random generators) which leads to pseudo-randomness.

Thus, there is empirical evidence that oversized deep neural networks still generalize well, precisely because a prior measure emerges: deep learning does not converge to the global maximum and instead to one of the local maximums chosen randomly, effectively sampling from a prior measure in the sample space defined by all local maximums. This is consistent with the good results achieved by ensembles of neural networks[402][405], which mimic many samples. However, it is a prior measure which we cannot easily modify or even understand, because the measure space is the

set of all local maximums of the optimization problem. But, since we expect the parameters to be fully uncorrelated between different local maximums, then many other prior measures (which we can modify and understand, such as the uniform measure) should achieve the same level of generalization.

This is not a surprise, since oversized statistical models that still generalize well were already found many decades ago by many people[407]: a standard probability space with a uniform probability measure can be infinite-dimensional (the infinite-dimensional sphere[407], for instance).

Note that generalization is often defined independently of the problem of overfitting, but these are both a sub-problem of whether the posterior measure is compatible with some prior measure. Despite that generalization is often defined with respect to "future" data, we know nothing about future data except if we make assumptions which then define a prior measure, for instance by assuming that future data will be similar to data we already collected but did not use to estimate the posterior measure, such unused data defines a prior measure. Overfitting is when we choose a set of models which have a likelihood which is too close to the maximum possible, while our prior includes likelihoods close but not too close from the maximum, thus the posterior measure (that is, the set of models) is unlikely ("*too good to be true*") with respect to the prior measure.

More empirical evidence: no one looks to a blurred photo of a gorilla and says with certainty that it is not a man in a gorilla suit. We all have many doubts, when we are not sure about a subject we usually express doubts through the absence of an action (not just us, but also many animals), for instance we don't write a book about the subject we don't know about.

There is no empirical evidence that our brain tries to create content which is a short distance from content (books, conversations, etc.) created under exceptional circumstances (when doubts are minimal). When we are driving, and we do not know what is in front of us, we usually just slow down or stop the car. But what content defines "not knowing"? Is there empirical evidence about the unknown? The unknown can only be an abstract concept, expressed through probability theory or a logical equivalent. Is there empirical evidence that probabilities are reducible, that there is a simpler logical equivalent? No, quite the opposite.

The only trade-off seems to be between costs (time complexity, etc.) and understanding/control. A prior measure which we understand and/or control may mean much more costs than an emergent (thus, inconsistent and uncontrollable) prior measure which just minimizes some distance. But this trade-off is not new, and it is already present in all industries which deal with some safety risk (which is essentially all industries). Distances are efficient for proof of concepts (pilot projects), when the goal is to show that we are a short distance from where we want to be. But safety (as

most features) is not being at a short distance from being safe[49]. "We were at a short distance from avoiding nuclear annihilation" is completely different from "we avoided nuclear annihilation". To avoid nuclear annihilation we need (probability) measures, not only distances.

5. Empiricism also implies that deep learning is a random sampling method

Even when ignoring all existing theories about Statistics, logical consistency implies that deep learning is a random sampling method.

Deep learning is based on data and thus can be always be mathematically defined as a part of Bayesian inference. The only doubt is whether such definition is useful or not. Then, there are no non-informative priors, thus bias or prior (that is, some theory) is unavoidable.

Note that despite the theory being unavoidable, humans may not need to understand it, for instance if it results in more power to produce new human-understandable theories[408][409]. After all, all theories are temporary. Non-understandable human theories are inherently non-scalable, because they cannot be trusted, but can be used to produce trustable, scalable things, such as trustable, scalable theories. In the following we will just assume that some theory is unavoidable, whether or not it is human-understandable.

This is today the mainstream approach to Deep Learning. The problem is that such tools that would empower us to produce new human-understandable theories, did not exist prior to Deep Learning (and we could argue that they do not exist yet, but that is not needed for the purposes of this article). Thus, using Deep Learning, however we define it, we created (or will create) such new tools, which could only be created by trial-and-error (experiments which produce empirical evidence). But trial-and-error in the context of Bayesian inference (where we are, because we use data) is just a random sampling method.

We could now argue that Deep Learning includes a random sampling method, but it is more than that. However, whatever it is more than that, it was created by humans, and it will be replaced, sooner or later, by a creation developed using the random sampling method included in Deep Learning itself. In conclusion, for the same reasons that we can ignore all existing theories of Statistics, we can also ignore everything in Deep Learning except its random sampling method. For the sake of logical consistency.

Note that there is also the logical possibility that Deep Learning will not give us more power to produce new human-understandable theories, thus we cannot ignore all existing theories about

[49]See for instance https://edition.cnn.com/2023/04/29/us/ai-scam-calls-kidnapping-cec

Statistics. But this still leads to the same conclusion that Deep Learning is a random sampling method, as discussed in the previous sections.

6. Application: Incentives in a representative democracy vs. alignment

The fact that both deep learning and human learning are random sampling methods has consequences for the alignment problem[410]. Because in most sufficiently complex systems (certainly in a system with the goal of solving many and different complex problems), it is extremely unlikely to find a sufficiently simple statistical metric that it is not misleading (in other words, that it is not rogue). This is not a problem but a feature, since we can define probabilities as the absence of information, unpredictability, thus, if it would not be likely misleading it would not be random/unpredictable and if it would not be random it would not generalize well (by definition of generalization in a Bayesian context). For instance, the economic growth of any country (however we measure it) is a misleading statistical metric.

Another example: in a representative democracy there are incentives in place with the theoretical goal of aligning the behavior of elected officials (who are much more powerful than the average citizen) to serve the best interests of the average citizens.

While most elected officials serve at least some of the interests of the average citizens, some dose of anarchy (however small) is always present in (and we could argue that it is essential for the stability of) a representative democracy. By anarchy, we mean, some citizens (including elected officials) disrespect the laws and deceive each other in unpredictable ways.

In fact, a representative democracy is an extremely inefficient way of governing. The only reason that many citizens still want it is that it empowers the average citizen at the expense of a ruling elite with less power. Why is that a good thing? Because in many thousands of years, no society could properly "align" a ruling elite to serve the interests of the average citizen, thus the ruling elite should have enough power to prevent complete anarchy, but not much more than that.

Note that it is not a matter of how smart or powerful the ruling elite is. In fact, the ruling elite must also manage natural resources such as rocks (diamonds for instance), which are literally "dumb as a rock" and they never did it efficiently. So, it is extremely unlikely that a "super-human" ruling elite (even if equipped with artificial intelligence with infinite scale, since the intelligence ratio of a human against a rock is also infinite) could somehow be aligned to serve the interests of the average citizen. Certainly, there exist many good solutions for such alignment problem, however the good solutions are a tiny fraction of the possible solutions. Since deep learning is a random sampling method (as human learning), it is extremely unlikely that will select one of the good solutions.

We address now consequences of weak-to-strong generalization[410]. Weak-to-strong generalization is about using a weaker model (which we trust) to "elicit" knowledge from a stronger model (which we do not trust). The authors themselves acknowledge that this just a (far from perfect) analogy with the alignment problem, that they "hope" may lead to progress. About the "hope" we cannot say much. About the analogy we can say that it doesn't change our findings, because a "weak model eliciting knowledge from a strong" model is after all still some complex enough Bayesian model whose "good generalization" properties still come from unpredictability/randomness: either the model as a whole (that is, its weak+strong parts) generalizes well, and it is unpredictable; or it can be trusted on, and it does not generalize well.

The way out is to split a complex system into simple systems, if we can do that, then some simple statistical measures become trustworthy. We will explore such possibility in the next section.

7. Application: Alignment for fully autonomous machines

Neural Networks and symbolic AI do not solve by themselves the alignment problem for fully autonomous machines. A well known solution is twofold: 1) the laws (constraints) are ultimately "interpreted" by judges (who are human beings) who have the authority to explore all its ambiguities to obtain the concrete constraint they think it is best for the concrete case they are judging; 2) this necessarily generates some unpredictability, the costs of it are on the person who looses the judicial case, due to the principle "Ignorance of law excuses no one". This means that people must act cautiously and only take actions that they are confident most judges will interpret as lawful. The degree of caution is adapted to the circumstances, for instance if there is some medical emergency they will be less cautious than if there is a conflict due to spurious reasons, and this also affects the interpretation of the law by the judges.

Thus, the degree of confidence (or equivalently a probability) on predictions about the consequences of the decisions is an ingredient of many decisions people make. This requires a probabilistic model of the real world where the person making the decision is included in the model (we could argue that this is consciousness, but such discussion is beyond the scope of this article).

A model of the real world involves time dynamics, the most general such model is a quantum model. But generality by itself could not help when there are infinite variables. Crucially, choosing a prior probability distribution where the quantum Hamiltonian (that is, the generator of the time-evolution) is a bounded operator is consistent with the real-world, because the available energy to a person or robot is finite (in most cases). This allows us to use few variables for a small enough energy interval (using the Krylov subspace, and wavelets to distinguish energy intervals and also energy resolutions), for a specific reason (finite energy interval and/or resolution) related to the

specific problem we are solving (predicting real consequences).

Note that in many cases, the energy interval may be too large and additional assumptions are required. In some cases, the additional assumptions may turn the assumption of the finite energy interval irrelevant, then the use of a quantum model may not be advantageous. But in many other cases, predicting the consequences of a decision is mostly about physics. Moreover, concepts unrelated to physics can nevertheless be grounded in (defined from) examples from physics, specially when the model of the real world is Turing complete (that is, it can be used to solve any other problem).

Many authors argue that all these features can "emerge" in a sufficiently large Neural Network using enough training data. But a relevant question is: is there a differentiable symbolic AI where all these features are explicit? Yes, the mathematical formalism of Quantum Mechanics has all these features explicitly: it includes probabilities explicitly, it is linear (thus differentiable) and it allows reducing a large problem into a sum of many problems with few variables each (using the spectral theorem and the Krylov subspace for continuous spectra). Moreover, it includes a uniform measure for an infinite-dimensional sphere, so it can be used to define many infinite-dimensional mathematical problems and also its approximate numerical solution, thus the approximations are better understood.

8. Application: Previous knowledge as step-by-step learning

There are many ways of including the previous knowledge into the prior probability distribution[411][412], including just changing the size of the sample space to allow learning ambiguous intermediate concepts.

For instance, classifying if a statement is a fact or fiction has some ambiguity, but most people agree on whether or not many statements are a fact or fiction (with many exceptions, for sure). When we teach a child to not lie, the child understands that it does not mean to never lie, under any circumstance. Thus, we can always include a pre-trained classifier of facts/fiction, whose output will be added to the original input, forming the input of another Bayesian model. Then the Bayesian model has access to the previous knowledge included in the classifier, but it still has the freedom to completely ignore it in extreme cases, since it also has access to the original input.

9. Application: Bayesian inference for non-deterministic functions

Most real-world functions are non-deterministic. For instance, no one looks to a photo of a real gorilla and says with absolute certainty that it is not a man in a gorilla suit. This ambiguity has major advantages in a dynamic world where new information may always appear. Deep learning

using neural networks is only good to fit deterministic functions (including the mean-values of non-deterministic functions), but it fails badly in most real-world functions[402] since it cannot fit the many relevant correlations of subsets (not just pairs) of non-deterministic outputs.

Neural networks have this feature by construction, since the likelihood of the outputs is assumed uncorrelated to allow for the definition of a loss function to be minimized without predicting the variance of the outputs. One can always promote a probability distribution to be the deterministic function we want to study, but that is inefficient since then the likelihood of the outputs (or a value that encodes it[400]) which stores a large amount of information, becomes redundant. A better alternative would be to consider the most general statistical model, which implies considering non-deterministic functions, and it predicts the likelihood of the outputs[374], including correlations and variances; this includes, but it is more general than considering arbitrary prior knowledge for the parameters of a specific statistical model[358]. Note that it is not due to a lack of computational power since modern neural networks can fit very complex deterministic functions and fail badly[402][403] in relatively simple non-deterministic functions (e.g. catastrophic forgetting or the need of calibration to have some probabilistic guarantees[403]).

Although there are many attempts to extend neural networks[402][358][403][405], we can easily see that in general, there are many more relevant correlations of subsets (not just pairs) of outputs than there are outputs. Therefore, just the fact that neural networks are good to fit the mean-values of the outputs is no indication that they are a good starting point to find a method to fit all the relevant correlations. We better take a step back and use everything we already know.

And what do we know about deep neural networks? Not so much, except the fact that the present evidence suggests that they are black boxes[402][405]. It does not look a great idea to extend a black box, at least without further information. And what do we already know about correlation functions and the necessary ambiguity to adapt to new information ? A lot and since a long time, starting with Bayesian inference:

> *"How often have I said to you that when you have eliminated the impossible, whatever remains, however improbable, must be the truth? We know that he did not come through the door, the window, or the chimney. We also know that he could not have been concealed in the room, as there is no concealment possible. When, then, did he come?"*
>
> — Sherlock Holmes in The Sign of the Four (1890)

Bayesian inference doesn't look so hard, does it? Moreover, since neural networks can in principle be used to fit any function including probability distributions, they seem to be compatible with Bayesian inference as long as we know what should be fitted. In fact, supervised learning can be defined as a particular case of reinforcement learning[413]. In reinforcement learning, deep

neural networks can be used, and any reward can be defined, including a reward which does not assume that the likelihood of the outputs is uncorrelated. And incorporating Bayesian inference in Reinforcement Learning provides a machinery to action-selection under uncertainty and to incorporate prior knowledge into the algorithms[414].

Functions are in general infinite-dimensional spaces, so it makes sense to look for measures in infinite-dimensional spaces. While the Lebesgue measure cannot be defined in a Euclidean-like infinite-dimensional space[101][102], it is well known since many decades that a uniform (Lebesgue-like) measure of an infinite-dimensional sphere can be defined using the Gaussian measure and the Fock-space (the Fock-space is a separable Hilbert space used in the second quantization of free quantum fields)[407]. Such a space can parametrize (we call it the free field parametrization) the probability distribution of another probability distribution with sample space given by the direct product of the base space and the field (say $\mathbb{R}^{n+1}$). A probability distribution with sample space given by the direct product of the base space and the field can model a non-deterministic function since we can always choose a measure (the Gaussian measure for instance) which covers the whole base space, with the non-deterministic function evaluated at each point of the base-space given by the conditional probability distribution conditioned on each point of the base-space. Note that due to the relation between Fock-spaces and tensor products, we can consider regions of the sample space as small as we want and there is null measure for a null marginal probability in each one of these regions, and for each component of the Fourier series corresponding to such region; thus the conditional probability distribution conditioned on each point of the base-space is well-defined everywhere except in sets with null measure.

Thus, the free field parametrization is appropriate for non-deterministic functions but not for deterministic functions, since the prior (uniform measure of an infinite-dimensional sphere, we call it the uniform prior from now on) attributes null measure to deterministic functions. If we were interested in deterministic functions, we would be in trouble[101][102]. Now that we found a probability space and a candidate prior, we need to look for other candidate priors to choose from.

In the free field parametrization, the uniform prior defines a vector of the Hilbert space which when used as the prior for Bayesian inference with arbitrary data generates an orthogonal basis for the whole Fock-space. Such basis is related with a point process, with the number of points with a given feature corresponding to the number of modifications to the uniform prior (in the part of the sample space corresponding to such feature). Since Bayesian inference with any other prior can be seen as a combination of the results of different Bayesian inferences with the uniform prior for different data (eventually an infinite amount of data for the cases with null measure), then the uniform prior in the free field parametrization is appropriate for non-deterministic functions in the absence of any other information.

10. Application: Digital-first Mathematics and Mathematical Modelling

Most mathematics and mathematical modelling are human-first (or analog-first or book-first). That is, the framework is optimized such that the computers are dispensable, computers play a mere productivity role. For instance, a mathematics book is such that it could be written by hand, but by writing it in the computer it can reach students faster and cheaper. The same happens with most mathematical models of complex systems: they can in principle be written by hand, the computer just increases productivity.

But in the next few years, mathematics and mathematical modelling will become digital-first. That is, the framework is optimized such that the human interaction is dispensable, humans play a mere auditing role. For instance, we could have new alternatives to complex analysis that are born digital. As it happens today with digital photos, humans are provided with an analog view (in a screen or on paper) of the photo and check if they like it, but the photo itself is digital, and it has more information/resolution than what humans can easily distinguish. As it is progressively happening with autonomous driving, where humans audit the car's autonomous progress but don't effectively drive as much.

Bayesian inference (or approximations to it, such as deep neural networks) is not deterministic. Therefore, Machine Learning can create deterministic mathematics, but it is not deterministic mathematics itself. Thus, we still need some kind of mathematics. All current Computer Algebra Systems and Proof assistants were optimized for a human-first Mathematics. They are computationally inefficient, verbose, logically inconsistent, designed for a human-first mathematics and to please the human user.

An alternative is the Haskell library Egison (in its sweet-egison implementation[415], a similar alternative would be Metamath[416]). Haskell is a pure functional language (thus easy to run in parallel architectures, see www.acceleratehs.org) and it is the only pure functional language that is often as fast as C. Moreover, we can easily include C code in Haskell in it in the exceptional cases where C is much faster. Then the Haskell's Egison library allows us to implement (probably) the most expressive and advanced Computer Algebra System. Thus, we have expressiveness, flexibility and speed, despite that it requires a lot of human expertise to do Mathematics from scratch with Haskell's Egison (which is ok for Digital-first Mathematics). Since the human user plays an auditing role, another language such as Python would not be much more useful than Haskell (a complex Python function is also hard to audit by a non-expert, moreover machine learning knows how to write Haskell interfaces to the Python libraries that could eventually be needed) and we don't need many human auditors in Mathematics (due to its Universality) in the World, these few people can

be specialists in Haskell, Egison and Mathematics.

Most human users will see/edit views of the Haskell code (constraints /explanations in plain English for instance), but not the code itself. These views are non-deterministic and can be a source of errors, which are ultimately audited by specialists in Haskell and Mathematics.

11. Application: Learning interesting questions, for which we can find a predictive model

Perhaps the biggest advantage of defining deep learning as a random sampling method in Bayesian inference, is that it allows us to apply deep learning (or alternative sampling methods) to more abstract problems than (eventually unsupervised) machine learning.

We all love to make a prediction and to be right ("I told you"). Thus, we tend to try to find questions/problems for which we can learn the answer to. Note that, our memory is limited, so this is one relevant way (perhaps even the primary way) that allows us to select the information to save in memory, since the more predictive a model is, the less memory it requires[417]. Since we cannot answer everything anyway, we try to find those questions we can answer to, to allocate scarce resources to them.

So we have a hierarchical inference problem, where the computation of the posterior probability includes another inference problem. And the hierarchy can have as many levels as we wish. But there is only one human brain or only one machine, so it does not make much sense to say that an inference problem in an intermediate level of the hierarchy is "machine learning" or a "neural network" analogous to the brain. However, once we define deep learning as a random sampling method in Bayesian inference, then it makes perfect sense to apply it to the intermediate levels of the hierarchy, since a hierarchy of beliefs/probabilities is a natural concept.

12. Negotiation is an alternative to alignment

How to prevent misinformation online? Incentives to misinform are the same as for AI (publicity for instance), since the AI is trained by by humans who use it to as a way to achieve its human goals. So, even with perfect alignment the AI would still be misleading to end-users. Then it is also misleading to its trainers indirectly, since every trainer is the end-user of other AIs.

Incentive to say the truth publicly in Forex is null.

Dependency hell is related to the alignment problem. Because knowledge is divided in statistically independent modules, interfaced by a router implementing a mixture of experts. But this structure,

when scaled leads to the dependency hell.

Safety is related to dependency hell, and life is related to survival which is related to safety. Safety is not abstract, it must be rooted in the real-world. And thus the best solution to the dependency hell, is the structure that is more resilient in the real world. And these are living organisms.

Statistical Model Theory and Bayesian priors where the Riemann Hypothesis is true

Upward Löwenheim-Skolem Theorem

The Theorem: If a First-Order theory has an infinite model (which Presburger Arithmetic does, i.e.,te cardinality (uncountable, etc.).

The Consequence: No matter how many axioms you add (as long as they are First-Order), if the model allows for an infinite set of numbers, a "monster" uncountable model that satisfies those exact same axioms automatically exists.

You cannot filter out the uncountable models using First-Order axioms. You would need **Second-Order Logic** to do this, but Second-Order Logic is not automatable (it is not recursively enumerable).

Statistical Model Theory

Goals: 1) Be able to include uncertain constraints on possible models, this allows to apply Statistical Model Theory to more realistic situations, where Model Theory by itself has been insufficient so far (in the Riemann hypothesis, for example); 2) Be able to define models which are probabilistic themselves, while preserving completeness; 3) we can use probabilities in intermediate steps to solve problems which do not involve probabilities a priori, for instance a) despite being decidable, Presburger arithmetic has non-standard models (in contrast with the L_{ω_1,ω_1} theory of Hilbert spaces within Abstraction logic); b) wave-functions allows to define gradients needed for gradient descent even if the optimal solution does not involve probabilities; c) the real numbers can be the unique solution to a set of axioms, without hurting completeness or compactness. Note that there are non-standard models of the real numbers in first-order-logic (which is complete), while

in second-order-logic the real numbers are the unique solution to a set of axioms, but the logic is incomplete.

A separable Hilbert space is fully defined by a countable basis (since it closure is unique). And the space of all countable models can be made a standard probability space. This allows us to consider models which are not countable, but are a Hilbert space with a countable basis (such as the real numbers vs. rational numbers), while allowing us to define a standard probability measure in the space of models satisfying a set of constraints. Such probability measure can be considered as an uncertain constraint.

Once we define the L_{ω_1,ω_1} theory of Hilbert spaces within Abstraction logic, we have the implication that there is a a finite proof (within Abstraction Logic) for every statement or its negation, since both L_{ω_1,ω_1} logic and Abstraction Logic are complete.

Then, we can represent statements (theorems if proven so) as operators in a Hilbert space. This allows to deal with undecidable statements as projection operators, which project the Hilbert space of models to a smaller subspace. Statements become more than true/false/undecidable, since we can study numerically the projection operators (using Krylov subspace methods, for instance). We can even consider uncertain statements, that are operators but not projections. A statement is then the definition of an operator, while its proof is the numerical implementation corresponding to the definition. In this sense, an undecidable statement also has a proof and "approximated" proofs (which evaluate incorrectly for a small subspace of the models).

The second incompleteness theorem is only upsetting if the language allows to represent self-consistency. If it does, then due to completeness and undecidability of self-consistency, there is at least one non-standard model (which cannot be separated from the standard model) where self-consistency is not represented by the original statement which is a false statement (self-consistency cannot be false since such non-standard model exists). In other words, the self-reference is valid in the standard model and invalid in at least one non-standard model. This makes sense since the mapping necessary for self-reference uses one specific model (usually the standard model).

If the language does represent self-consistency, that would not be an improvement over representing it because it could still be inconsistent. Note that-self-consistency does not add new information because we already need to assume the consistency of a theory (otherwise would be no model satisfying the constraints).

Bayesian priors where the Riemann Hypothesis is true

It is well known that the Peano arithmetic is Turing complete. This suggests that it assumes more than needed. Certainly, the minimum assumptions we want for a mathematical theory are related

to those of a Turing machine, since we want to be able to do any real-world computation. But more assumptions than those are not crucial, and therefore should be used only if they are harmless and advantageous for some reason (make proofs easier, for instance).

The $L_{\omega_1\omega_1}$-theory of Hilbert spaces [418] is harmless, because it has very good properties (completeness, compactness, etc.), and it makes Bayesian inference possible for standard probability spaces. It uses infinitary logic, but all models are separable Hilbert spaces (it is also possible to assume non-separable Hilbert spaces, but here we choose not to), meaning that we can use any finitary theory where separable Hilbert spaces are a model and make more assumptions to get the corresponding $L_{\omega_1\omega_1}$-theory of Hilbert spaces as a particular case. Then whatever is true or false for all separable Hilbert spaces in the finitary theory, it is also true or false in the corresponding infinitary theory. Moreover, due to the compactness property, we do not loose expressivity when studying a finitary theory dependent on an arbitrarily large parameter instead of an infinitary theory.

Once we can do Bayesian inference in a harmless way, then we can define the unknown. And then we can define computation without involving the infinite (unlike in a Turing machine). Since any real-world computation requires finite time and energy, the results of some computations that end in finite time cannot be known for all practical purposes. So, for instance we only need to define the multiplication of two integers smaller than some finite large integer, and for larger integers we can use a Bayesian prior since we will never know the result.

Note that our assumptions include axioms and Bayesian priors, while the usual assumptions in mathematics only involve axioms. This by itself should raise a red flag: all scientific knowledge involves Bayesian priors in one way or another, except for (most of) mathematics. The reason is that Peano arithmetic by itself already assumes too much, leaving no room for the unknown. Since no one knows everything, the unknown still appears in (most of) mathematics, but as a bug (impossibility of proving consistency of the theory from within, for instance) not as a feature.

The hypothesis that the Möbius function is related with a random walk implies the Riemann hypothesis. The Möbius function for arbitrarily large integers is unknown, for all practical purposes. Thus, we can choose a reasonable Bayesian prior (that is, the Möbius function is iid for large integers), relating the Möbius function with a random walk, implying the Riemann hypothesis. Note that many axioms involving limits are harmless (Presburger arithmetic involves only sums and it has good properties, for instance) and thus no need Bayesian priors, but Peano arithmetic assumes so many axioms that it becomes harmful.

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